

The Portsmouth Papers



A Hands-On Tour of PostgreSQL Beyond the Relational Model

*"PostgreSQL is not a database with extensions.
It is an extensible data platform that happens to speak SQL."*

About This Book

Most PostgreSQL tutorials end where the interesting work begins.

Once you know how to `SELECT`, `JOIN`, and `GROUP BY`, you have mastered perhaps twenty percent of what PostgreSQL can do. The remaining eighty percent — geospatial queries, semantic search, real-time notifications, fuzzy matching, vector embeddings, distributed coordination, and more — lives in a ecosystem of extensions, index types, and language features that most practitioners never discover.

This book is a guided tour of that other eighty percent.

Each chapter is built around a concrete engineering problem faced by the fictional city of **Portsmith** and its data platform team. We store business directories as semi-structured JSON documents. We route emergency services using geospatial proximity queries. We build a job queue with no message broker. We search municipal records with fuzzy matching that survives typos and OCR errors. We expose the entire platform as a REST API with zero application code.

Every chapter follows the same structure: synthetic data is generated first, giving you a realistic dataset to work against, and then a series of exercises walks you from first principles to production-ready technique. The exercises are written to be done — not just read.

By the end, you will see PostgreSQL not as a place to store rows, but as a programmable data infrastructure layer capable of doing work that most teams reach for separate specialized systems to handle.

What you will need:

- A working PostgreSQL 16 installation, plus two disposable Docker containers (PostgreSQL 19 and 18) for the final chapters — setup for all three covered in Appendix A
- Python 3.12+
- A Debian-based Linux environment
- Familiarity with basic SQL (`SELECT`, `INSERT`, `JOIN`, `GROUP BY`)
- Curiosity about what else is in there

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Chapter 1 — JSONB: Semi-Structured Data Without a Schema Tax

“A schema is a prediction about the future. JSONB lets you hedge.”

Background

Relational databases enforce a contract: every row in a table has exactly the same columns. That contract is usually a feature — it catches mistakes and makes queries predictable. But sometimes you genuinely don't know what columns you'll need until the data arrives. A business directory is a good example. A restaurant has hours, cuisine, and a reservations policy. A hardware shop has trade accounts and parking. A hotel has star ratings and room types. Forcing them all into the same set of columns means drowning in NULLs or building a rats' nest of one-to-many extension tables.

PostgreSQL's `JSONB` type lets you store arbitrary JSON documents in a column while keeping the rest of your row fully relational. It is not a NoSQL escape hatch — it is an indexed, queryable, patchable document field that sits inside a SQL table. You can `WHERE` on it, join against it, aggregate over it, and update individual keys without rewriting the whole document.

This chapter works through all the operators and patterns you'll reach for most often. The exercises are short. Run every one, then try a variation of your own.

The Scenario

Portsmouth's city council maintains a public business directory. Every business gets a short relational record — an ID, a name, an address, and a neighbourhood — but the supplementary detail varies so widely by business type that a single fixed schema would be unworkable.

The `businesses` table therefore stores all of that variable metadata in a `details` JSONB column. Each business category contributes its own keys:

Category	JSONB keys present
restaurant	cuisine, price_range, rating, hours, tags, accepts_reservations, outdoor_seating
retail	subcategory, rating, hours, tags, payment_methods, has_parking

Category	JSONB keys present
service	subcategory, rating, hours, tags, appointment_required, specialties
accommodation	subcategory, star_rating, rating, hours, amenities, room_types, price_per_night
entertainment	subcategory, rating, hours, tags, live_music, age_restriction

The `hours` value is itself a nested object keyed by day of week (`mon–sun`). Each day is either `null` (closed) or `{"open": "HH:MM", "close": "HH:MM"}`. Several businesses also carry optional keys that appear only for their category, like `social` (social media handles), `happy_hour`, or `approved_brands`.

This heterogeneity is intentional — it is what makes the data a good vehicle for learning JSONB operators.

Exercise Goals

By the end of this chapter you will be able to:

- Extract values from a JSONB document using `->`, `->>`, and `#>>`, and explain why the operator choice changes the result type.
 - Filter rows using the containment operator `@>` and the key-existence operators `?`, `?|`, and `?&`.
 - Create a GIN index on a JSONB column and confirm that PostgreSQL uses it.
 - Update a document in place with `jsonb_set` and the `||` merge operator without rewriting the whole column.
 - Expand a JSONB array into a set of rows with `jsonb_array_elements` and aggregate the results.
 - Write `jsonb_path_query` expressions to filter on nested values.
-

Installation

1 — PostgreSQL

If PostgreSQL 16 is not already installed:

```
sudo apt update
sudo apt install -y postgresql-16 postgresql-client-16
```

Confirm the server is running:

```
pg_lsclusters
```

You should see a cluster at version 16 in the `online` state.

2 — Python 3.12 and psycpg

```
sudo apt install -y python3.12 python3.12-venv
```

Create a virtual environment in the book's working directory and install the PostgreSQL driver:

```
python3.12 -m venv .venv
source .venv/bin/activate
pip install "psycpg[binary]"
```

Note: psycpg (version 3) is the modern Python driver for PostgreSQL. It is not available as an apt package for Python 3.12, so we install it via pip into the virtual environment. The [binary] extra bundles a pre-compiled C extension so you don't need libpq-dev.

Loading the Data

Create the database

```
# Create a role matching your OS user (skip if it already exists)
sudo -u postgres createuser --createdb "$(whoami)"

# Create the portsmith database
createdb portsmith
```

Run the seed script

From the book/ directory, with the virtual environment active:

```
python data/ch01_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Creating schema ...
Inserting 48 businesses ...
Done - 48 rows in businesses.
```

Verify the load

Open a psql session:

```
psql portsmith
```

Run these checks. If all three pass, the data is correct.

Check 1 — Row count and column types:

```
\d businesses
```

Table "public.businesses"				
Column	Type	Collation	Nullable	Default
id	integer		not null	nextval('businesses_id_seq'::regclass)
name	text		not null	
address	text		not null	
neighbourhood	text		not null	
details	jsonb		not null	

Check 2 — Counts by neighbourhood:

```
SELECT neighbourhood, COUNT(*) AS businesses
FROM   businesses
GROUP  BY neighbourhood
ORDER  BY neighbourhood;
```

neighbourhood	businesses
Harbour District	9
Industrial Port	7
Northgate	9
Old Town	9
Riverside	9
University Quarter	5

(6 rows)

Check 3 — Counts by category:

```
SELECT details->>'category' AS category, COUNT(*) AS businesses
FROM   businesses
GROUP  BY category
ORDER  BY category;
```

category	businesses
accommodation	5
entertainment	6
restaurant	15
retail	12
service	10

(5 rows)

If all three match, proceed to the exercises.

Exercises

Exercise 1 — The Three Extraction Operators

JSONB has two families of accessor. The arrow operators (`->` and `->>`) navigate one key at a time. The path operator (`#>>`) jumps straight to a deeply nested value. The difference between them is the **type they return**.

1.1 — Arrow operators

Run these two queries and compare the result types:

```
-- -> returns JSONB
SELECT name,
       details -> 'rating' AS rating_jsonb
FROM   businesses
LIMIT 5;

-- ->> returns TEXT
SELECT name,
       details ->> 'rating' AS rating_text
FROM   businesses
LIMIT 5;
```

Look carefully at the column headings in `psql` — one will show `jsonb`, the other `text`. This distinction matters the moment you try to do arithmetic:

```
-- This works: cast text to numeric
SELECT name,
       (details ->> 'rating')::numeric AS rating
FROM   businesses
ORDER BY rating DESC
LIMIT 5;
```

name	rating
Lighthouse Bookshop	4.9
Finch & Sons Barbers	4.9
Portsmouth Vet. Clinic	4.8
River Bend Bakery	4.8
Quarter Note Jazz Club	4.8

(5 rows)

Key point: Use `->` when you want to keep working with JSONB (e.g., to navigate deeper). Use `->>` when you need the final value as text for comparisons, casting, or display.

1.2 — The path operator `#>>`

Navigating nested keys with chained `->` quickly becomes unreadable. The path operator takes an array of keys and returns the leaf as text in one step.

```

-- Chained arrows (verbose)
SELECT name,
       details -> 'hours' -> 'fri' ->> 'close' AS friday_close
FROM   businesses
WHERE  details ->> 'category' = 'restaurant'
LIMIT 8;

-- Path operator (equivalent, cleaner)
SELECT name,
       details #>> '{hours,fri,close}' AS friday_close
FROM   businesses
WHERE  details ->> 'category' = 'restaurant'
LIMIT 8;

```

Both produce the same result. Notice that some rows return `NULL` — those restaurants are closed on Fridays (the `fri` key is JSON `null`).

name	friday_close
The Gilded Clam	22:30
Anchor & Oar Tavern	01:00
Bella Napoli	23:00
Le Petit Bistro	14:30
Dragon Palace	23:00
Spice Garden	23:00
Sol y Mar	23:00
Mango Bay Caribbean	22:30

(8 rows)

Exercise 2 — Filtering with Containment and Key Existence

2.1 — Containment: @>

The containment operator checks whether the left JSONB document contains all the key-value pairs in the right document. It is the most common way to filter on JSONB values.

Find all waterfront businesses:

```

SELECT name, neighbourhood
FROM   businesses
WHERE  details @> '{"tags": ["waterfront"]}'
ORDER BY neighbourhood, name;

```

name	neighbourhood
Anchor & Oar Tavern	Harbour District
Harbour Inn	Harbour District
Harbour View Theater	Harbour District
Mariners Rest B&B	Harbour District
Portsmouth Fish Market	Harbour District

Saltbox Gallery	Harbour District
The Gilded Clam	Harbour District
Tidal Wave Surf Shop	Harbour District
Old Brewery Tap	Industrial Port
The Rusty Anchor	Industrial Port

(10 rows)

Why this works: The @> operator checks whether the tags *array* on the left contains the tags array ["waterfront"] on the right — array containment is supported natively.

2.2 — Key existence: ?

The ? operator tests whether a key exists in a JSONB object (or a value exists in a JSONB array). Unlike @>, it does not check the value — only presence.

Find all businesses that publish social media handles:

```
SELECT name,
       details -> 'social' AS social_links
FROM   businesses
WHERE  details ? 'social'
ORDER BY name;
```

name	social_links
Bella Napoli	{"instagram": "@bellanapoli_portsmith"}
Le Petit Bistro	{"instagram": "@lepetitbistro"}
Quarter Note Jazz Club	{"instagram": "@quarternote_portsmith"}
Spice Garden	{"instagram": "@spicegarden_portsmith"}
The Gilded Clam	{"facebook": "thegildedclam", "instagram": "@gildedclam"}
The Riverside Vegan	{"instagram": "@riverside_vegan"}

(6 rows)

2.3 — Any-key and all-key existence: ?| and ?&

?| returns true if *at least one* of the listed keys exists. ?& returns true only if *all* listed keys exist.

```
-- Businesses that offer either delivery OR takeaway
SELECT name
FROM   businesses
WHERE  details ?| ARRAY['delivery', 'takeaway']
ORDER BY name;

-- Businesses that offer BOTH delivery AND takeaway
SELECT name
FROM   businesses
WHERE  details ?& ARRAY['delivery', 'takeaway']
ORDER BY name;
```

Run both and note the difference in result count. The ?& set is a strict subset of the ?| set.

Exercise 3 — Missing Keys vs. NULL Values

JSONB has two distinct ways to represent “no value”: a JSON `null` and a **missing key**. They behave differently in queries.

In our data, a business closed on Monday can be represented two ways:

- `"mon": null` — the key exists, the value is JSON `null`
- the `mon` key is absent entirely

The seed data uses `"mon": null` for closures, so we can observe this.

3.1 — The difference matters for ?

```
-- ? checks key EXISTENCE, not value
SELECT name
FROM   businesses
WHERE  (details -> 'hours') ? 'sun'    -- key exists (even if null)
ORDER BY name;
```

Run it. You should get **43 rows** — every business that uses a day-keyed hours object. The five accommodation businesses are excluded because their hours look like `{"reception": "07:00-23:00"}` and have no `sun` key.

Now try what seems like a natural follow-up:

```
-- △ This does NOT do what you might expect
SELECT name
FROM   businesses
WHERE  (details -> 'hours' -> 'sun') IS NOT NULL
ORDER BY name;
```

Run it. You still get **43 rows** — identical to the first query.

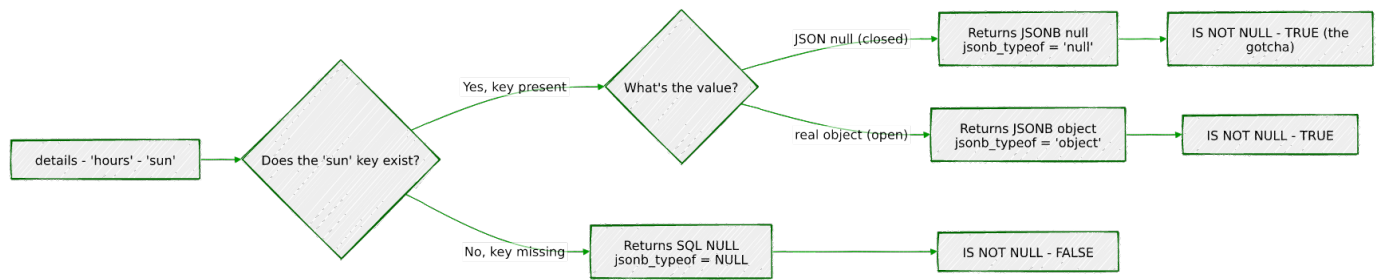
Why? This is one of JSONB’s most important gotchas:

JSONB `null` is not SQL `NULL`.

When the seed script stores `"sun": None` in Python, `json.dumps` produces `"sun": null` — a JSON *null value* under an existing key. When PostgreSQL evaluates `details -> 'hours' -> 'sun'` on such a row, it returns the JSONB value `null`. That is a real value — not the absence of a value — so `IS NOT NULL` evaluates to `TRUE`.

SQL `NULL` only appears when the key itself is **missing** from the object. That is what happens for the five accommodation businesses: `details -> 'hours'` returns `{"reception": "..."}` , and then `-> 'sun'` on that object finds no such key and returns SQL `NULL`, so `IS NOT NULL` is `FALSE`.

The same logic, traced as a flowchart instead of read as prose:



To correctly distinguish the three states, use `jsonb_typeof()`:

State	details -> 'hours' -> 'sun'	jsonb_typeof(...)	IS NOT NULL
Key missing (accommodation)	SQL NULL	SQL NULL	FALSE
Key present, closed (null)	JSONB null	'null'	TRUE ← gotcha
Key present, open (object)	JSONB object	'object'	TRUE

The correct query for “businesses actually open on Sunday”:

```
SELECT name
FROM businesses
WHERE jsonb_typeof(details -> 'hours' -> 'sun') = 'object'
ORDER BY name;
```

This returns **25 rows** — only the businesses with a real hours object for Sunday. `jsonb_typeof` returns 'object' only for JSON objects, 'null' for JSON null, and SQL NULL for a missing key (which then fails the = test).

Rule of thumb: Never use `IS NOT NULL` to test whether a JSONB navigation result is a “real” value. Use `jsonb_typeof(...)` to check the actual JSON type, or navigate one level deeper (e.g., check `details #>> '{hours,sun,open}'`) — the `#>>` operator converts JSON null to SQL NULL, so a deeper path naturally filters it out.

3.2 — Handling accommodation differently

Hotels and hostels store hours as a single "reception": "HH:MM-HH:MM" string rather than a day-keyed object. This means the ? 'sun' test returns false for them even though they may be open 24/7. Write a query that finds all businesses that are either:

- Open on Sunday (day-keyed hours with a non-null sun), **or**
- An accommodation with 24/7 reception

```

SELECT name, neighbourhood,
CASE
    WHEN jsonb_typeof(details -> 'hours' -> 'sun') = 'object'
        THEN (details #>> '{hours,sun,open}') || '-' ||
            (details #>> '{hours,sun,close}')
    WHEN details #>> '{hours,reception}' = '24/7'
        THEN '24/7'
    ELSE details #>> '{hours,reception}'
END AS sunday_availability
FROM businesses
WHERE jsonb_typeof(details -> 'hours' -> 'sun') = 'object'
    OR (details -> 'hours') ? 'reception'
ORDER BY neighbourhood, name;

```

Exercise 4 — GIN Indexes

Without an index, every JSONB query is a sequential scan — PostgreSQL reads every row and evaluates the expression. For a 48-row toy dataset that is unnoticeable, but with millions of rows it becomes a bottleneck.

A **GIN** (Generalised Inverted Index) index over a JSONB column indexes every key and value inside every document. It accelerates `@>`, `?`, `?|`, and `?&` queries.

4.1 — Observe the plan before indexing

```

EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM businesses
WHERE details @> '{"tags": ["waterfront"]}';

```

With 48 rows you will see `seq scan` (sequential scan, i.e. reading every row) — that is expected. On a larger table this would say `seq scan` with a high cost estimate.

4.2 — Create the GIN index

```

CREATE INDEX idx_businesses_details_gin
ON businesses
USING GIN (details);

```

4.3 — Observe the plan after indexing

```

EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM businesses
WHERE details @> '{"tags": ["waterfront"]}';

```

On a small table PostgreSQL may still choose a sequential scan (the planner knows the table fits in memory). Force it to use the index to see the mechanism:

```

SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM businesses
WHERE details @> '{"tags": ["waterfront"]}';

SET enable_seqscan = on;  -- always restore this

```

You should now see Bitmap Index Scan on `idx_businesses_details_gin` in the plan. The GIN index will pay for itself in production as the table grows.

What GIN indexes: The default `jsonb_ops` operator class indexes every key and value in the document, supporting `@>`, `?`, `?|`, `?&`. A `jsonb_path_ops` class indexes only values (not keys), producing a smaller index that is faster for `@>` but cannot accelerate `?` queries. Use `jsonb_path_ops` when you only ever query by value containment and index size matters.

```

-- Alternative: value-only index (smaller, faster @>, no ? support)
CREATE INDEX idx_businesses_details_pathops
ON businesses
USING GIN (details jsonb_path_ops);

```

Exercise 5 — Updating Documents in Place

JSONB columns are immutable at the document level — you cannot change a single key without rewriting the whole value. PostgreSQL gives you two tools to do that rewrite concisely.

5.1 — `jsonb_set`: replace or add one value

The Lighthouse Bookshop just received a flood of five-star reviews. Update its rating to 5.0:

```

UPDATE businesses
SET details = jsonb_set(details, '{rating}', '5.0')
WHERE name = 'Lighthouse Bookshop';

```

Verify:

```

SELECT name, details ->> 'rating' AS rating
FROM businesses
WHERE name = 'Lighthouse Bookshop';

```

`jsonb_set(target, path, new_value)` — the path is a text array of keys. If the key already exists it is replaced; if it does not exist it is added (by default — there is a fourth parameter to suppress creation).

5.2 — `jsonb_set` on a nested key

The Gilded Clam is now closing an hour later on Sundays:

```
UPDATE businesses
SET     details = jsonb_set(
            details,
            '{hours,sun,close}',
            '"21:00"'      -- note: a JSON string must be double-quoted
        )
WHERE   name = 'The Gilded Clam';

SELECT name,
       details #>> '{hours,sun,close}' AS new_sunday_close
FROM   businesses
WHERE  name = 'The Gilded Clam';
```

5.3 — || merge operator: add or overwrite multiple keys at once

The || operator merges two JSONB objects, with the right side winning on conflicts. Use it to add a verified flag and update review_count in a single statement:

```
UPDATE businesses
SET     details = details || '{"verified": true, "review_count": 999}'
WHERE   name = 'Anchor & Oar Tavern';

SELECT name,
       details ->> 'verified'      AS verified,
       details ->> 'review_count' AS reviews
FROM   businesses
WHERE  name = 'Anchor & Oar Tavern';
```

Caution: || does a *shallow* merge. If details contains a nested object like hours and you merge another object that also has hours, the entire hours value is replaced — not deep-merged. Use jsonb_set for targeted nested updates.

Exercise 6 — Expanding Arrays with jsonb_array_elements

The tags key in most business documents is a JSON array. To query across tags — for example, to find the most common ones across the whole directory — you need to expand the array into individual rows.

6.1 — Expand one business's tags

```
SELECT name,
       jsonb_array_elements_text(details -> 'tags') AS tag
FROM   businesses
WHERE  name = 'The Gilded Clam';
```

name	tag
-----+-----	

```
The Gilded Clam | waterfront
The Gilded Clam | seafood
The Gilded Clam | romantic
The Gilded Clam | outdoor_seating
(4 rows)
```

`jsonb_array_elements_text` is a set-returning function: each element of the array becomes its own row. The result has more rows than the input.

6.2 — Count tags across all businesses

```
SELECT tag, COUNT(*) AS occurrences
FROM   businesses,
       jsonb_array_elements_text(details -> 'tags') AS tag
GROUP BY tag
ORDER BY occurrences DESC
LIMIT 15;
```

tag	occurrences
waterfront	10
takeaway	5
family_friendly	4
family_owned	4
organic	3
...	

(15 rows)

Your exact numbers will differ — this shows the query pattern. The comma between `businesses` and the `jsonb_array_elements_text(...)` call is a **lateral join** (PostgreSQL expands the function for each input row automatically when used in the `FROM` clause this way).

6.3 — Only businesses with multiple specific tags

Find restaurants that are both `vegan_options` and `takeaway`:

```
SELECT name, details ->> 'cuisine' AS cuisine
FROM   businesses
WHERE  details @> '{"tags": ["vegan_options"]}'
      AND details @> '{"tags": ["takeaway"]}';
```

Because each `@>` call is independent, both conditions must hold. This is more efficient than expanding the array when the GIN index is in place.

Exercise 7 — Path Queries with `jsonb_path_query`

The `@?` operator and `jsonb_path_query()` function implement **SQL/JSON path language** — a mini-query language for navigating and filtering within a JSONB document. It is more expressive than chained arrow operators for conditional navigation.

7.1 — Simple path existence

Find businesses where the social links include Instagram:

```
SELECT name
FROM   businesses
WHERE  details @? '$.social.instagram';
```

`$.social.instagram` means: start at the root (`$`), descend to `social`, then to `instagram`. `@?` returns true if the path resolves to at least one value.

7.2 — Filter on a nested value

Find restaurants with a rating above 4.5:

```
SELECT name,
       details ->> 'rating' AS rating
FROM   businesses
WHERE  details @? '$ ? (@.category == "restaurant" && @.rating > 4.5)';
```

name	rating
Bella Napoli	4.6
Le Petit Bistro	4.7
River Bend Bakery	4.8
Spice Garden	4.6
The Gilded Clam	4.5

(5 rows)

The `?` (filter) syntax inside a path expression is equivalent to a `WHERE` clause inside the document.

7.3 — Find businesses open late on Friday

Businesses that close at or after 21:00 on Friday (using string comparison, which works correctly for 24-hour `HH:MM` strings):

```
SELECT name,
       details #>> '{hours,fri,close}' AS friday_close
FROM   businesses
WHERE  details @? '$.hours.fri.close ? (@ >= "21:00")'
ORDER BY friday_close DESC, name;
```

name	friday_close
Bella Napoli	23:00
Dragon Palace	23:00
Harbour View Theater	23:00
The Hungry Scholar	23:00
Spice Garden	23:00
Sol y Mar	23:00
Mango Bay Caribbean	22:30
The Gilded Clam	22:30
Thai Orchid	22:00
The Riverside Vegan	22:00
Northgate Grocers	21:00
Lotus Spa & Wellness	20:00

(12 rows)

Limitation to notice: Businesses that close *after midnight* — Anchor & Oar, The Clocktower Pub, The Rusty Anchor, and others — show a `close` of "01:00" or "02:00". Lexicographically, "02:00" < "21:00", so these businesses are *excluded* from the above results even though they are open very late. This is a design trade-off of storing times as plain strings. One solution is to store closing times past midnight as "25:00", "26:00", etc. — an unusual but practical convention for 24-hour time arithmetic.

7.4 — Extract values with `jsonb_path_query`

`@?` is a boolean test. `jsonb_path_query()` returns the actual matched values:

```
SELECT name,
       jsonb_path_query(details, '$.hours.fri.close') AS friday_close_json
FROM   businesses
WHERE  details @? '$.hours.fri'
ORDER BY name
LIMIT 8;
```

The returned values are JSONB (note the quotes around the time strings). Use `jsonb_path_query_first(...) #>> '{}'` to extract a single scalar as text without the surrounding quotes.

Summary — What You Should Now Know

You have worked through the core JSONB toolkit. Here is what each operator and function you used actually does:

Tool

`-> 'key'`

What it does

Navigate one level; returns **JSONB**

Tool	What it does
<code>->> 'key'</code>	Navigate one level; returns text
<code>#>> '{a,b,c}'</code>	Navigate a path; returns text ; JSON null becomes SQL NULL
<code>@> '{"k":"v"}'</code>	Containment test; accelerated by GIN
<code>? 'key'</code>	Key exists test (regardless of value); accelerated by GIN
<code>? / ?&</code>	Any-key / all-keys exist; accelerated by GIN
<code>jsonb_typeof(val)</code>	Returns 'object', 'array', 'string', 'number', 'boolean', or 'null'; SQL NULL for a missing key
<code>jsonb_set(col, path, val)</code>	Replace or insert a nested value
<code>col \ '{"k":"v"}'</code>	Shallow-merge a document
<code>jsonb_array_elements_text(col)</code>	Expand a JSON array into rows
<code>@? 'path'</code>	SQL/JSON path existence test
<code>jsonb_path_query(col, 'path')</code>	SQL/JSON path — return matched values

Remember: JSONB `null` $\neq$ SQL NULL. Use `jsonb_typeof()` — not `IS NOT NULL` — when you need to distinguish a missing key from a key that exists but holds a JSON null value.

The key design insight from this chapter is that JSONB lets you keep a relational spine — the columns your queries always need (`id`, `name`, `neighbourhood`) — while storing everything else in a document whose shape can vary row by row. The GIN index means you do not pay a query performance penalty for that flexibility.

In the next chapter you will add point geometry to the `businesses` table and use PostGIS to answer spatial questions: which businesses are within 500 metres of the harbour, and which neighbourhood does each one belong to.

Going further: PostgreSQL 14+ supports the `jsonpath` type natively. The `jsonb_path_query_array` and `jsonb_path_query_first` variants are useful for pagination and single-value extraction. For write-heavy workloads, profile whether JSONB or a computed stored column (Chapter 16) gives better INSERT/UPDATE throughput on your hardware.

Chapter 2 — PostGIS: Geospatial Queries on Real Geometry

“A city is not a list of rows. It is a shape on the ground.”

Background

Every interesting question about a city is ultimately a spatial question. Which businesses are near the harbour? Which neighbourhood is this address in? How large is the industrial waterfront? Relational databases answer these badly when location is stored as a text field or a pair of float columns — there is no native concept of “within”, “contains”, or “distance”.

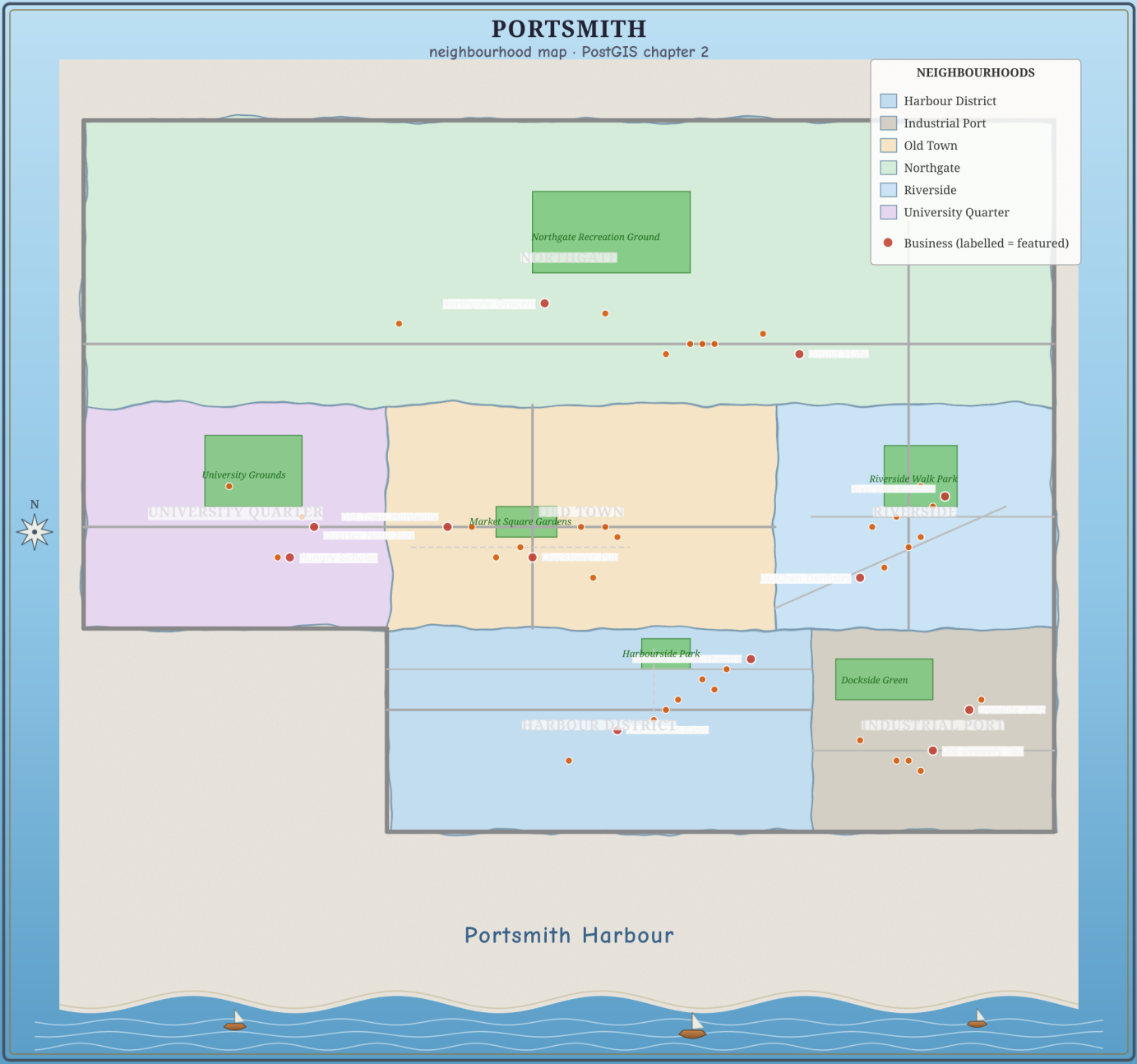
PostGIS is a PostgreSQL extension that adds first-class geometry and geography types, plus several hundred spatial functions and operators. It turns PostgreSQL into a full spatial database: you can store points, lines, and polygons; index them with GIST; and answer proximity, containment, and area queries in SQL without an external GIS system.

This matters in practice far beyond mapping applications. Address geocoding, logistics routing, fraud detection (is this login coming from the expected region?), real estate valuation, and urban planning all reach for spatial queries. PostGIS is the standard tool for all of them in the PostgreSQL ecosystem.

The Scenario

The Portsmouth business directory from Chapter 1 stores each business’s neighbourhood as a plain text column. That works for simple filtering, but it cannot answer *where* questions: it cannot find businesses near a given coordinate, cannot verify that a business address actually falls inside its declared neighbourhood, and cannot measure distances.

This chapter adds a point geometry to every business record, then introduces three new spatial tables:



Portsmouth’s six neighbourhoods, parks, and featured businesses — rendered directly from this chapter’s own neighborhoods, parks, and businesses tables via `utils/render_map.py`. Every shape and dot on this map is a row you’re about to query.

Table	Geometry type	What it holds
neighborhoods	POLYGON	Boundary polygons for Portsmouth’s six neighbourhoods
parks	POLYGON	Six public parks and green spaces
city_infrastructure	LINESTRING	Twelve named road segments

All coordinates are in WGS-84 (SRID 4326), the same coordinate system used by GPS and most web mapping APIs.

Exercise Goals

By the end of this chapter you will be able to:

- Store and inspect `POLYGON` geometry using WKT (Well-Known Text) and `ST_GeomFromText`.
 - Run proximity searches with `ST_DWithin`, and understand why the `::geography` cast matters for distance accuracy.
 - Perform spatial joins using `ST_Within` and `ST_Contains`, and explain the difference between them.
 - Compute polygon areas in square kilometres using `ST_Area` with the `geography` type.
 - Find the nearest feature for every row using `ST_Distance` and a `CROSS JOIN LATERAL`.
 - Create a GIST spatial index and confirm that PostgreSQL uses it.
-

Installation

1 — PostGIS server package

PostGIS is a separate package from PostgreSQL. On Debian/Ubuntu:

```
sudo apt install -y postgresql-16-postgis-3
```

2 — Enable PostGIS in the database

Connect to the `portsmith` database and enable the extension:

```
psql portsmith  
  
CREATE EXTENSION IF NOT EXISTS postgis;
```

Verify it loaded:

```
SELECT postgis_full_version();
```

You should see a long string beginning with `POSTGIS="3.x.x"`. If you see an error about the extension not existing, the server package is not installed.

Loading the Data

Prerequisites

Chapter 1's seed script must have been run first — the `businesses` table must exist. If it does not:

```
python data/ch01_seed.py
```

Run the Chapter 2 seed

```
python data/ch02_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Enabling PostGIS extension ...
Applying DDL ...
Inserting 6 neighbourhoods ...
Inserting 6 parks ...
Inserting 12 road segments ...
Updating 48 business locations ...
```

Done:

```
businesses with geometry : 48
neighbourhoods           : 6
parks                    : 6
road segments            : 12
```

Verify the load

Open `psql portsmith` and run these four checks.

Check 1 — businesses now has a geometry column:

```
\d businesses
```

You should see a new `geom` column of type `geometry(Point,4326)`.

Check 2 — neighbourhood table structure and row count:

```
SELECT name, population,
       ST_AsText(geom) AS wkt_preview
FROM   neighborhoods
ORDER BY name;
```

name	population	wkt_preview
Harbour District	4200	POLYGON((-1.805 50.69,...))

Industrial Port		2100		POLYGON((-1.77 50.69,...))
Northgate		18500		POLYGON((-1.83 50.732,...))
Old Town		6800		POLYGON((-1.805 50.71,...))
Riverside		9300		POLYGON((-1.773 50.71,...))
University Quarter		11200		POLYGON((-1.83 50.71,...))

(6 rows)

Check 3 — parks and roads:

```
SELECT COUNT(*) FROM parks;
SELECT COUNT(*) FROM city_infrastructure;
```

Both should return 6 and 12 respectively.

Check 4 — SRID on all tables:

```
SELECT f_table_name, f_geometry_column, srid, type
FROM geometry_columns
WHERE f_table_schema = 'public'
ORDER BY f_table_name;
```

f_table_name		f_geometry_column		srid		type
businesses		geom		4326		POINT
city_infrastructure		geom		4326		LINESTRING
neighborhoods		geom		4326		POLYGON
parks		geom		4326		POLYGON

(4 rows)

If all four pass, proceed to the exercises.

Exercises

Exercise 1 — Neighbourhood Polygons and WKT

Geometry in PostGIS is often loaded from **Well-Known Text** (WKT), a human-readable representation of shapes. Understanding WKT lets you read geometry from SQL output, write it in queries, and reason about what you stored.

1.1 — Read a polygon in WKT

```
SELECT name, ST_AsText(geom) AS wkt
FROM neighborhoods
WHERE name = 'Harbour District';
```

name		wkt
Harbour District		POLYGON((-1.805 50.69,-1.77 50.69,-1.77 50.71,...))

WKT for a polygon is `POLYGON((x1 y1, x2 y2, ...))`. In geographic coordinates the convention is *longitude latitude* (x then y), matching the X/Y convention in maths. The ring must **close** — the last coordinate must equal the first.

1.2 — Inspect the bounding box

`ST_Envelope` returns the minimum bounding rectangle for any geometry:

```
SELECT name,
       ST_XMin(ST_Envelope(geom)) AS west,
       ST_XMax(ST_Envelope(geom)) AS east,
       ST_YMin(ST_Envelope(geom)) AS south,
       ST_YMax(ST_Envelope(geom)) AS north
FROM   neighborhoods
ORDER BY name;
```

Use this output to confirm each polygon sits in the right part of the coordinate space (longitudes around -1.75 to -1.83 , latitudes around 50.69 to 50.76).

1.3 — Count vertices

```
SELECT name,
       ST_NPoints(geom) AS vertex_count
FROM   neighborhoods
ORDER BY name;
```

Each neighbourhood polygon has five vertices (four corners plus the closing duplicate). A real city boundary imported from an OS/census shapefile might have thousands.

1.4 — Understand the SRID

`ST_SRID` returns the spatial reference identifier stored with the geometry:

```
SELECT name, ST_SRID(geom) AS srid
FROM   neighborhoods
LIMIT 3;
```

SRID 4326 is the WGS-84 system used by GPS. Every geometry in these tables carries the same SRID, which means they can be compared and joined directly. If SRIDs differ, PostGIS will return an error — a deliberate safety check.

The geometry/geography distinction: In PostGIS, `geometry` stores coordinates in whatever units the SRS defines. For SRID 4326 that means degrees. When you ask for a distance between two `geometry` points in 4326, you get a value in *degrees* — nearly useless for human-scale distances. The `geography` type, by contrast, always works on a

spheroid and returns distances in *metres*. You can cast any SRID-4326 geometry to geography with `::geography` to get metre-based calculations. The exercises use this cast throughout.

Exercise 2 — Proximity Search with `ST_DWithin`

`ST_DWithin(a, b, distance)` returns true when the distance between `a` and `b` is at most `distance`. With geography inputs the distance is in metres.

2.1 — Find businesses within 500 m of Portsmouth Pier

Portsmouth's main pier entrance sits at approximately `(-1.785, 50.700)`. Find every business within 500 metres of it:

```
SELECT b.name,
       b.neighbourhood,
       ROUND(
         ST_Distance(b.geom::geography,
                     ST_Point(-1.785, 50.700)::geography)::numeric
       ) AS distance_m
FROM   businesses b
WHERE  ST_DWithin(b.geom::geography,
                  ST_Point(-1.785, 50.700)::geography,
                  500)
ORDER BY distance_m;
```

name	neighbourhood	distance_m
The Gilded Clam	Harbour District	70
Anchor & Oar Tavern	Harbour District	179
Tidal Wave Surf Shop	Harbour District	307
Harbour Inn	Harbour District	437
Portsmouth Fish Market	Harbour District	484

(5 rows)

All five results are in the Harbour District — exactly what you would expect for a search centred on the pier.

2.2 — Why `::geography` matters

Try the same query without the cast. Note that `ST_Point(...)` without an explicit SRID gets SRID 0, which PostGIS refuses to compare against SRID-4326 geometry — so you must use `ST_SetSRID`:

```
SELECT b.name,
       ROUND(ST_Distance(b.geom,
                         ST_SetSRID(ST_Point(-1.785, 50.700), 4326))::numeric,
       6) AS distance_degrees
FROM   businesses b
```

```

WHERE ST_DWithin(b.geom,
                ST_SetSRID(ST_Point(-1.785, 50.700), 4326),
                500)
ORDER BY distance_degrees;

```

The `WHERE` clause now has a threshold of 500 — but in *degrees*. One degree of latitude is about 111 km, so this matches businesses within roughly 55,000 km: all 48 of them, from the whole city. The `distance_degrees` values in the output are tiny fractions like 0.001000, not useful numbers.

rows returned: 48 (the entire city, not 5)

Rule: For any distance query on SRID-4326 data, always cast to `::geography` in `ST_DWithin` and `ST_Distance`. The cast has negligible performance cost at the scales used in city-level data.

2.3 — Try different radii

How many businesses fall within 1 km of the pier? 2 km?

```

SELECT radius_m,
       COUNT(*) AS business_count
FROM   (VALUES (500), (1000), (2000)) AS radii(radius_m)
CROSS JOIN LATERAL (
  SELECT 1
  FROM    businesses b
  WHERE   ST_DWithin(b.geom::geography,
                    ST_Point(-1.785, 50.700)::geography,
                    radius_m)
) AS matches
GROUP BY radius_m
ORDER BY radius_m;

```

radius_m	business_count
500	5
1000	8
2000	17

(3 rows)

The pier sits in the southern Harbour District. Doubling the radius to 2 km pulls in more of the central neighbourhoods, but the northern Northgate businesses are nearly 5 km away — a reminder that Portsmouth stretches a significant distance north from the waterfront.

Exercise 3 — Spatial Joins with `ST_Within` and `ST_Contains`

A **spatial join** links rows from two tables based on a geometric relationship rather than a key match. The most common are containment tests: does point A fall inside polygon B?

3.1 — Which neighbourhood is each business in?

```
SELECT b.name,  
       b.neighbourhood      AS declared_neighbourhood,  
       n.name               AS postgis_neighbourhood  
FROM   businesses b  
JOIN   neighborhoods n ON ST_Within(b.geom, n.geom)  
ORDER BY n.name, b.name;
```

ST_Within(a, b) returns true when geometry a lies completely inside geometry b. The query should return all 48 businesses, each matched to its neighbourhood.

Confirm the declared neighbourhood column always matches postgis_neighbourhood:

```
SELECT COUNT(*) AS mismatches  
FROM   businesses b  
JOIN   neighborhoods n ON ST_Within(b.geom, n.geom)  
WHERE  b.neighbourhood <> n.name;
```

```
mismatches  
-----  
          0  
(1 row)
```

Zero mismatches: the text column from Chapter 1 and the geometry are consistent.

3.2 — ST_Contains is the inverse

ST_Contains(a, b) returns true when polygon a contains geometry b — it is the mirror of ST_Within. These two queries produce identical results:

```
-- Point inside polygon  
SELECT b.name FROM businesses b  
JOIN   neighborhoods n ON ST_Within(b.geom, n.geom)  
WHERE  n.name = 'Old Town'  
ORDER BY b.name;  
  
-- Polygon contains point  
SELECT b.name FROM businesses b  
JOIN   neighborhoods n ON ST_Contains(n.geom, b.geom)  
WHERE  n.name = 'Old Town'  
ORDER BY b.name;
```

```
      name  
-----  
Bella Napoli  
Finch & Sons Barbers  
Le Petit Bistro  
Old Town Hardware  
Portsmouth Accountancy Ltd.  
Portsmouth Arms Hotel  
Portsmouth Legal Group
```

Portsmith Tailors
The Clocktower Pub
(9 rows)

Both return the same 9 Old Town businesses.

The subtle difference: `ST_Contains(A, B)` requires that no point of B lies on the boundary of A. A point sitting exactly *on* a polygon edge would fail `ST_Contains` but pass `ST_Covers`. In practice, coordinates rarely land precisely on a boundary, so the two functions behave identically for most point-in-polygon work. Reach for `ST_Covers` if you need to include boundary-touching cases explicitly.

3.3 — Check for gaps: businesses in no neighbourhood

This query finds any businesses whose geometry falls outside every neighbourhood polygon — useful for catching data quality problems:

```
SELECT b.name, b.neighbourhood
FROM   businesses b
WHERE  NOT EXISTS (
    SELECT 1
    FROM   neighborhoods n
    WHERE  ST_Within(b.geom, n.geom)
);
```

```
name | neighbourhood
-----+-----
(0 rows)
```

All 48 businesses are inside a neighbourhood polygon.

Exercise 4 — Computing Area in Square Kilometres

`ST_Area` returns the area of a polygon. Called on `geometry` (degrees), it returns a value in square degrees — meaningless to most people. Called on `geography`, it returns square metres.

4.1 — Incorrect: area in square degrees

```
SELECT name,
    ROUND(ST_Area(geom)::numeric, 6) AS area_sq_degrees
FROM   neighborhoods
ORDER BY area_sq_degrees DESC;
```

The numbers look tiny (around 0.0007). They are geometrically correct but impossible to interpret as real-world area because degrees are not uniform units.

4.2 — Correct: area in square metres, then kilometres

```

SELECT name,
       ROUND((ST_Area(geom::geography) / 1e6)::numeric, 2) AS area_km2
FROM   neighborhoods
ORDER BY area_km2 DESC;

```

name	area_km2
Northgate	17.59
Old Town	5.53
Harbour District	5.50
University Quarter	4.32
Riverside	3.98
Industrial Port	3.14

(6 rows)

Northgate is by far the largest neighbourhood — it spans the entire northern width of the city. Industrial Port is the smallest, a tight strip of dockside land. Old Town and Harbour District are almost identical in area despite having very different characters.

Why the numbers differ slightly from manual estimates: ST_Area on geography uses the WGS-84 spheroid, which accounts for the fact that the Earth is not a perfect sphere. At latitude 50.7° the correction is small (under 0.3%) but present.

4.3 — Population density

Combine the computed area with the population column:

```

SELECT name,
       population,
       ROUND((ST_Area(geom::geography) / 1e6)::numeric, 2) AS area_km2,
       ROUND(
         (population / (ST_Area(geom::geography) / 1e6))::numeric
       ) AS pop_per_km2
FROM   neighborhoods
ORDER BY pop_per_km2 DESC;

```

name	population	area_km2	pop_per_km2
University Quarter	11200	4.32	2592
Riverside	9300	3.98	2340
Old Town	6800	5.53	1230
Northgate	18500	17.59	1052
Harbour District	4200	5.50	763
Industrial Port	2100	3.14	668

(6 rows)

The University Quarter and Riverside are the most densely populated neighbourhoods — student housing and riverside apartments pack a lot of people into compact areas. The Industrial Port is the least dense despite its small size; much of it is warehouse and dock rather than residential.

Exercise 5 — Nearest Park Using `ST_Distance` and a Lateral Join

Finding the nearest feature from another table requires a **lateral join** — a subquery that can reference columns from the outer query row by row.

5.1 — Distance from one business to one park

`ST_Distance` between two geography values returns metres:

```
SELECT b.name           AS business,
       p.name           AS park,
       ROUND(ST_Distance(b.geom::geography,
                          p.geom::geography)::numeric) AS distance_m
FROM   businesses b,
       parks p
WHERE  b.name = 'The Gilded Clam'
ORDER BY distance_m;
```

business	park	distance_m
The Gilded Clam	Harbourside Park	682
The Gilded Clam	Dockside Green	1315
The Gilded Clam	Market Square Gardens	2143
The Gilded Clam	Riverside Walk Park	2899
The Gilded Clam	University Grounds	3060
The Gilded Clam	Northgate Recreation Ground	5006

(6 rows)

The nearest park to The Gilded Clam is Harbourside Park, 682 m away. The `ST_Distance` to a polygon returns the distance from the point to the nearest point on the polygon’s boundary — zero when the point is inside the polygon.

5.2 — Nearest park for a single business (lateral pattern)

The canonical pattern for “nearest one thing” is `ORDER BY ... LIMIT 1` inside a lateral subquery:

```
SELECT b.name,
       nearest.park_name,
       nearest.distance_m
FROM   businesses b
CROSS JOIN LATERAL (
    SELECT p.name           AS park_name,
           ROUND(ST_Distance(b.geom::geography,
                              p.geom::geography)::numeric) AS distance_m
    FROM   parks p
    ORDER BY b.geom::geography <-> p.geom::geography
    LIMIT 1
) AS nearest
WHERE  b.name = 'Quarter Note Jazz Club';
```

name	park_name	distance_m
Quarter Note Jazz Club	University Grounds	233

(1 row)

The <-> operator is the KNN (k-nearest-neighbour) distance operator. When used in an ORDER BY clause inside a lateral join, PostGIS can accelerate it with the GIST index (which you will create in Exercise 6). Using <-> in ORDER BY is preferred over ORDER BY ST_Distance(...) for this reason.

5.3 — Nearest park for every business

Remove the WHERE filter to run across all 48 businesses:

```
SELECT b.name AS business,
       b.neighbourhood,
       nearest.park_name,
       nearest.distance_m
FROM   businesses b
CROSS JOIN LATERAL (
    SELECT p.name AS park_name,
           ROUND(ST_Distance(b.geom::geography,
                             p.geom::geography)::numeric) AS distance_m
    FROM   parks p
    ORDER BY b.geom::geography <-> p.geom::geography
    LIMIT 1
) AS nearest
ORDER BY b.neighbourhood, b.name;
```

Scan the results. You should see a clean pattern: each neighbourhood’s businesses point to the park in that same neighbourhood. Notice that three Riverside businesses (Portsmouth Pharmacy, Portsmouth Veterinary Clinic, Riverside Cinema) and University Bookshop all show distance_m = 0 — their coordinates fall *inside* the park polygon. ST_Distance to a polygon returns zero when the point is contained within it.

5.4 — Aggregate: average walking distance to a park, by neighbourhood

```
SELECT b.neighbourhood,
       ROUND(AVG(nearest.distance_m)::numeric) AS avg_distance_m
FROM   businesses b
CROSS JOIN LATERAL (
    SELECT ROUND(ST_Distance(b.geom::geography,
                             p.geom::geography)::numeric) AS distance_m
    FROM   parks p
    ORDER BY b.geom::geography <-> p.geom::geography
    LIMIT 1
) AS nearest
GROUP BY b.neighbourhood
ORDER BY avg_distance_m;
```

This gives a rough “walkability to green space” metric per neighbourhood. Old Town ranks first — Market Square Gardens sits centrally within the neighbourhood. Northgate comes last despite having the largest park, because the park is in the far north of the district while businesses cluster near the southern edge.

Exercise 6 — GIST Indexes and Spatial Query Plans

Without an index, every spatial query is a sequential scan: PostgreSQL reads every row, applies the geometry function, and discards non-matching rows. For a 48-row dataset that is instant, but at 48 million rows it is not.

PostGIS spatial queries are accelerated by **GIST** (Generalised Search Tree) indexes. A GIST index on a geometry column builds a tree of bounding boxes, allowing the planner to quickly eliminate large portions of the table.

6.1 — Observe the plan before indexing

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM   businesses
WHERE  ST_DWithin(geom::geography,
                ST_Point(-1.785, 50.700)::geography,
                500);
```

On the 48-row businesses table you will see something like:

```
Seq Scan on businesses  (cost=0.00..4.60 rows=1 width=...) ...
  Filter: (st_dwithin(...))
  Rows Removed by Filter: 43
```

A sequential scan is expected on a tiny table — the planner knows the overhead of an index lookup exceeds the cost of reading all 48 rows.

6.2 — Create GIST indexes on the geometry columns

```
CREATE INDEX idx_businesses_geom
  ON businesses USING GIST (geom);

CREATE INDEX idx_neighborhoods_geom
  ON neighborhoods USING GIST (geom);

CREATE INDEX idx_parks_geom
  ON parks USING GIST (geom);

CREATE INDEX idx_city_infrastructure_geom
  ON city_infrastructure USING GIST (geom);
```

6.3 — The geometry/geography index split

With `enable_seqscan` off, verify which plan the proximity query uses:

```
SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM   businesses
WHERE  ST_DWithin(geom::geography,
                  ST_Point(-1.785, 50.700)::geography,
                  500);

SET enable_seqscan = on;
```

You will see — perhaps surprisingly — that the planner *still* uses a sequential scan, even though `idx_businesses_geom` exists:

```
Seq Scan on businesses  (cost=10000000000.00...) ...
  Filter: st_dwithin((geom)::geography, ...)
```

Why? The GIST index was built on `geom` (type `geometry`). The query filters on `geom::geography` (type `geography`). These are different types — the index is not usable for geography operations.

To accelerate geography-based distance queries you need a **functional index** on the cast:

```
CREATE INDEX idx_businesses_geom_geography
ON   businesses USING GIST (CAST(geom AS geography));
```

Now with `enable_seqscan` off:

```
SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM   businesses
WHERE  ST_DWithin(geom::geography,
                  ST_Point(-1.785, 50.700)::geography,
                  500);

SET enable_seqscan = on;
```

```
Index Scan using idx_businesses_geom_geography on businesses
  Index Cond: ((geom)::geography &&
               _st_expand('...', '500'))
  Filter: st_dwithin((geom)::geography, ...)
  Rows Removed by Filter: 2
```

The index is now used. The index condition uses `&&` (bounding-box overlap on the spheroid) to quickly discard most rows, then `st_dwithin` rechecks the exact distance for the survivors.

6.4 — Verify the spatial join plan

The containment join uses geometry-to-geometry operations, so `idx_businesses_geom` (the plain geometry index) *is* used there:

```
SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT b.name, n.name AS neighbourhood
FROM   businesses b
JOIN   neighborhoods n ON ST_Within(b.geom, n.geom);

SET enable_seqscan = on;
```

Nested Loop ...

```
-> Seq Scan on neighborhoods n  (6 rows - tiny table)
-> Index Scan using idx_businesses_geom on businesses b
    Index Cond: (geom @ n.geom)
    Filter: st_within(geom, n.geom)
```

For each neighbourhood polygon, `idx_businesses_geom` is used to find businesses whose bounding box falls inside the neighbourhood's bounding box (`geom @ n.geom`), then `st_within` rechecks the exact containment.

The index rule: A GIST index on `geom` geometry accelerates geometry-to-geometry operations (`ST_Within`, `ST_Contains`, `ST_Intersects`, &&). A GIST index on `CAST(geom AS geography)` accelerates geography operations (`ST_DWithin(...::geography, ...), <-> ON geography`). If you query both ways, create both indexes — they coexist happily on the same table.

Summary — What You Should Now Know

You have worked through the core PostGIS toolkit for points, polygons, and linear features. Here is a reference for everything used:

Function / operator

`ST_GeomFromText(wkt, srid)`

`ST_AsText(geom)`

`ST_SetSRID(ST_MakePoint(lon, lat), srid)`

`ST_Point(lon, lat)::geography`

What it does

Parse WKT into a geometry with the given SRID

Format geometry as WKT for display

Construct a point geometry

Construct a point and cast to geography

Function / operator

`geom::geography`

`ST_SRID(geom)`

`ST_NPoints(geom)`

`ST_Envelope(geom)`

`ST_XMin/XMax/YMin/YMax(geom)`

`ST_DWithin(a, b, d)`

`ST_Distance(a, b)`

`a::geography <-> b::geography`

`ST_Within(a, b)`

`ST_Contains(a, b)`

`ST_Area(geom::geography)`

`CROSS JOIN LATERAL (... LIMIT 1)`

`USING GIST (geom)`

What it does

Cast a 4326 geometry to geography (distances now in metres)

Return the SRID stored with a geometry

Count vertices in a geometry

Return the bounding-box rectangle

Extract bounding-box extents

True when the distance between a and b is $\leq d$

Distance between two geometries (metres for geography)

KNN distance operator; use in ORDER BY for index acceleration

True when a lies completely inside b

True when a contains b (inverse of ST_Within)

Area in square metres on the spheroid

Nearest-neighbour join pattern

Create a GIST spatial index

Geometry vs geography in one sentence: Use `geometry` for storage and when working with projected coordinate systems where units are already metres or feet. Cast to `geography` any time you need distance, area, or proximity results in real-world units from SRID-4326 data.

The coordinates added in this chapter will carry forward. Chapter 12 extends the `city_infrastructure` road network into a graph and uses a recursive CTE to find the shortest path between two intersections.

Going further: PostGIS supports many more geometry types — `MULTIPOLYGON` for areas with holes, `GEOMETRYCOLLECTION` for mixed types, and 3D geometries with a Z coordinate for elevation data. For routing specifically, `pgRouting` builds on PostGIS to provide Dijkstra, A, and turn-restriction-aware shortest-path algorithms over road network graphs. For importing real boundary data, `shp2pgsql` converts ESRI Shapefiles directly into PostGIS-compatible `INSERT` statements, and `ogr2ogr` handles GeoJSON, KML, GeoPackage, and dozens of other formats.*

Chapter 3 — Job Queues: FOR UPDATE SKIP LOCKED

“A queue is just a table that everyone is racing to read.”

Background

Sooner or later almost every application needs a queue: a list of work items that a pool of workers processes one at a time, safely, without two workers ever grabbing the same item. The reflexive answer is to reach for a message broker — Redis, RabbitMQ, SQS, Kafka. Those tools earn their keep at serious scale. But if your data already lives in PostgreSQL, running a second system just to hand out rows to workers is often unnecessary complexity: another service to deploy, monitor, and keep consistent with the database.

PostgreSQL can do this job itself. The `FOR UPDATE SKIP LOCKED` row-locking clause, combined with an ordinary table, gives you an atomic “claim the next item and don’t let anyone else touch it” primitive — the same guarantee a dedicated queue product sells you, built out of two SQL keywords. This chapter builds a job queue from scratch: the schema, the atomic claim query, concurrent worker behaviour, stalled-job recovery, and a dead-letter path for jobs that keep failing.

This is not a toy exercise. This exact pattern — a status column, a claim query, a heartbeat, a dead-letter table — is what libraries like [river](#), [oban](#) (Elixir), and countless in-house job runners implement on top of PostgreSQL in production.

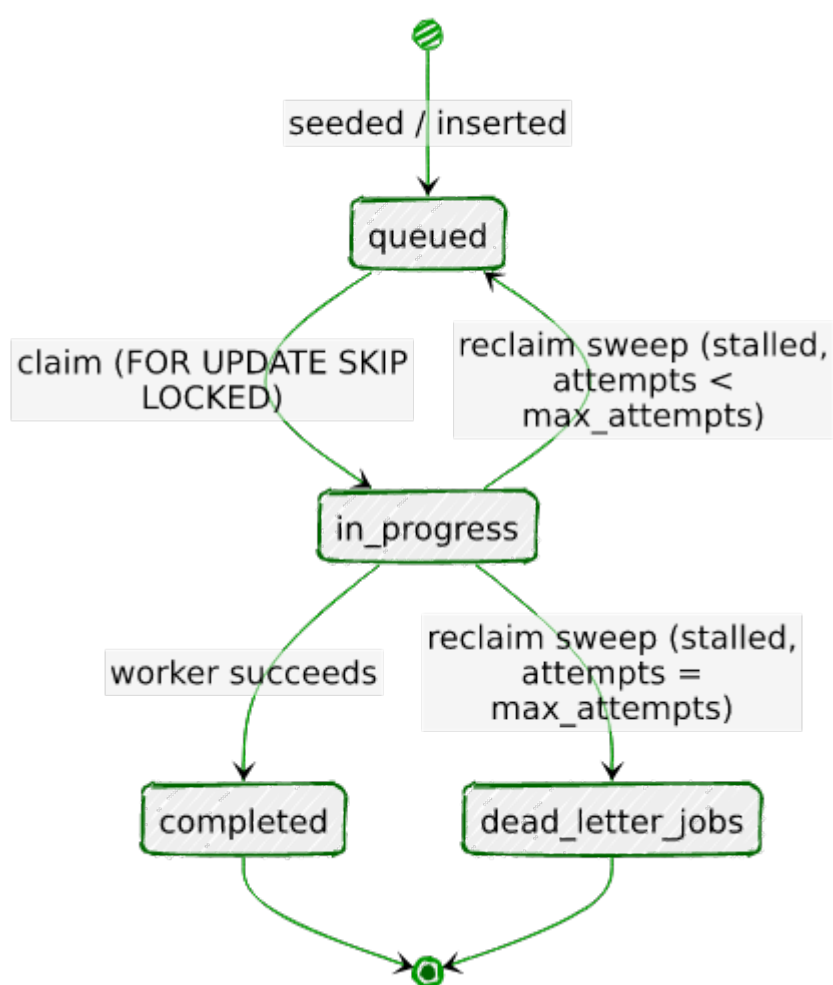
The Scenario

Portsmouth’s permitting office processes a steady stream of permit applications: building work, business licenses, public events, signage, and demolitions. Each application needs to move through a review pipeline, and the office wants that processing to happen asynchronously and reliably — work should never be lost, never double-processed, and a crashed reviewer process shouldn’t leave an application stuck in limbo forever.

The `jobs` table models this as a queue. Every row is one permit application awaiting review. A `status` column tracks its life cycle, a `priority` column lets safety-critical work (demolitions) jump ahead of routine work (sign permits), and a retry counter with a companion `dead_letter_jobs` table handles applications whose processing keeps failing.

Column	Purpose
status	queued → in_progress → completed (or back to queued, or dead-lettered)
priority	1 (most urgent — demolitions) to 5 (least urgent — sign permits)
payload	JSONB — the permit application details
attempts / max_attempts	Retry bookkeeping
claimed_by / claimed_at	Which worker has the job, and since when
heartbeat_at	Updated periodically by the worker while it holds the job

The status column's life cycle, drawn out — this is what Exercises 2 through 5 each implement one piece of:



Every arrow here is a specific query you'll write by hand later in this chapter — there's no hidden state machine enforcing this, just the `status` column, the claim query, and the reclaim sweep agreeing on what each value means.

No extensions are required for this chapter — everything here is built on core PostgreSQL locking semantics.

Exercise Goals

By the end of this chapter you will be able to:

- Design a queue table schema, including a partial index tuned for the claim query's access pattern.
 - Write the atomic `UPDATE ... FROM (SELECT ... FOR UPDATE SKIP LOCKED)` claim query, and explain why naive two-step approaches race.
 - Observe, with two concurrent `psql` sessions, that `SKIP LOCKED` lets workers proceed past rows another worker is holding — and contrast that with the blocking behaviour of plain `FOR UPDATE`.
 - Implement a heartbeat/timeout mechanism that reclaims jobs abandoned by a crashed worker.
 - Route jobs that exhaust their retries into a dead-letter table.
 - Benchmark claim throughput at different concurrency levels with `pgbench`.
-

Installation

This chapter needs nothing beyond what Chapter 1 already set up: PostgreSQL 16 and a Python 3.12 virtual environment with `psycopg`. If you skipped Chapter 1, see its Installation section. You will also use `pgbench` for Exercise 6, which ships with the standard PostgreSQL client tools (`postgresql-client-16` on Debian/Ubuntu).

Loading the Data

Run the seed script

From the `book/` directory, with the virtual environment active:

```
python data/ch03_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Creating schema ...
Inserting 45 jobs ...
Done - 45 rows in jobs, all queued.
```

The seed script is self-contained — it does not depend on Chapter 1 or 2's data.

Verify the load

Open psql portsmith and run these checks.

Check 1 — table structure:

```
\d jobs

Table "public.jobs"
  Column      |      Type      | Collation | Nullable |      Default
-----+-----+-----+-----+-----
id            | bigint         |           | not null |
nextval('jobs_id_seq'::regclass)
job_type     | text           |           | not null |
payload      | jsonb          |           | not null |
status       | text           |           | not null | 'queued'::text
priority     | smallint       |           | not null | 5
attempts     | integer        |           | not null | 0
max_attempts | integer        |           | not null | 3
created_at   | timestamp with time zone |           | not null | clock_timestamp()
claimed_at   | timestamp with time zone |           |           |
claimed_by   | text           |           |           |
heartbeat_at | timestamp with time zone |           |           |
completed_at | timestamp with time zone |           |           |
last_error   | text           |           |           |

Indexes:
  "jobs_pkey" PRIMARY KEY, btree (id)
  "idx_jobs_claim_order" btree (priority, created_at, id) WHERE status = 'queued'::text
  "idx_jobs_status" btree (status)

Check constraints:
  "jobs_status_check" CHECK (status = ANY (ARRAY['queued'::text, 'in_progress'::text, 'completed'::text, 'failed'::text]))
```

Check 2 — job counts by type and priority:

```
SELECT job_type, priority, COUNT(*) AS jobs
FROM   jobs
GROUP BY job_type, priority
ORDER BY priority;

 job_type      | priority | jobs
-----+-----+-----
demolition_permit |         1 |    4
business_license |         2 |   12
building_permit  |         3 |   15
event_permit     |         4 |    8
sign_permit      |         5 |    6
(5 rows)
```

Check 3 — everything starts queued:

```
SELECT status, COUNT(*) FROM jobs GROUP BY status;
```

```
status | count  
-----+-----  
queued |      45  
(1 row)
```

If all three match, proceed to the exercises.

Note: If you re-run `ch03_seed.py` at any point during the exercises to reset to a clean state, it drops and recreates both `jobs` and `dead_letter_jobs`.

Exercises

Exercise 1 — Designing the Queue Schema

1.1 — Why a partial index

The claim query (which you'll write in Exercise 2) only ever looks at rows where `status = 'queued'`, ordered by `priority` then `created_at`. As the queue runs, the vast majority of rows will end up `completed` — a normal B-tree index on `(priority, created_at, id)` would faithfully index every one of those settled rows even though the claim query never looks at them.

`idx_jobs_claim_order` is a **partial index** — it only indexes rows matching `WHERE status = 'queued'`:

```
CREATE INDEX idx_jobs_claim_order  
  ON jobs (priority, created_at, id)  
  WHERE status = 'queued';
```

This keeps the index small regardless of how many historical jobs pile up in `completed` or `failed` state, because settled rows are never in it.

1.2 — Why `id` is part of the sort key, not just `created_at`

You might expect `ORDER BY priority, created_at` to be enough — oldest job in the highest-priority bucket goes first. But timestamps are not always unique. If two jobs are inserted in the same transaction, both can get an identical `created_at` (more on this below), and `ORDER BY` over tied values has no defined order. Appending the primary key, `id`, as a final tiebreaker guarantees a deterministic order even when timestamps collide:

```
ORDER BY priority ASC, created_at ASC, id ASC
```

A gotcha worth knowing: `now()` returns the **transaction's** start time, not the current wall-clock time — every call to `now()` inside the same transaction returns the same

value. The seed script originally used `created_at TIMESTAMPTZ DEFAULT now()`, and because all 45 rows were inserted in one transaction, every single row ended up with an *identical* `created_at`. The fix is `clock_timestamp()`, which returns the actual current time at the moment it's evaluated, differing row to row even within one transaction. `jobs.created_at` uses `clock_timestamp()` for exactly this reason. This is the same family of surprise as the JSONB `null` vs. SQL `NULL` gotcha from Chapter 1: a function name that looks interchangeable with another is not.

1.3 — Confirm the index is used

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT id
FROM   jobs
WHERE  status = 'queued'
ORDER BY priority ASC, created_at ASC, id ASC
FOR UPDATE SKIP LOCKED
LIMIT 1;
```

```
Limit  (cost=0.14..6.17 rows=1 width=24) (actual time=0.022..0.022 rows=1 loops=1)
  Buffers: shared hit=3
  -> LockRows  (cost=0.14..12.20 rows=2 width=24) (actual time=0.021..0.021 rows=1 loops=1)
        Buffers: shared hit=3
        -> Index Scan using idx_jobs_claim_order on jobs  (cost=0.14..12.18 rows=2 width=24)
            (actual time=0.010..0.010 rows=1 loops=1)
                Filter: (status = 'queued'::text)
                Buffers: shared hit=2
```

Now drop the index and run the identical query again inside a transaction you roll back (so the drop doesn't stick):

```
BEGIN;
DROP INDEX idx_jobs_claim_order;
```

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT id
FROM   jobs
WHERE  status = 'queued'
ORDER BY priority ASC, created_at ASC, id ASC
FOR UPDATE SKIP LOCKED
LIMIT 1;
```

```
ROLLBACK;
```

```
Limit  (cost=9.51..9.53 rows=1 width=24) (actual time=0.071..0.072 rows=1 loops=1)
  -> LockRows  (cost=9.51..9.54 rows=2 width=24) (actual time=0.070..0.071 rows=1 loops=1)
        -> Sort  (cost=9.51..9.52 rows=2 width=24) (actual time=0.067..0.068 rows=1 loops=1)
              Sort Key: priority, created_at, id
              Sort Method: quicksort  Memory: 27kB
              -> Bitmap Heap Scan on jobs  (cost=4.16..9.50 rows=2 width=24) (actual
time=0.018..0.027 rows=45 loops=1)
                    Recheck Cond: (status = 'queued'::text)
```

```
-> Bitmap Index Scan on idx_jobs_status (cost=0.00..4.16 rows=2 width=0)
(actual time=0.013..0.013 rows=45 loops=1)
      Index Cond: (status = 'queued'::text)
```

Without `idx_jobs_claim_order`, PostgreSQL still finds the queued rows (via `idx_jobs_status`), but it must then **sort all of them** to find the one with the lowest (`priority`, `created_at`, `id`) — an extra sort node. With the tailored partial index, the rows are already stored in claim order, so PostgreSQL walks the index and stops at the first match. On a 45-row table the difference is invisible; on a busy production queue with a deep backlog, eliminating the sort on every single claim matters a great deal.

Exercise 2 — The Atomic Claim Query

2.1 — Why a naive two-step claim races

The tempting first approach is to `SELECT` a candidate row, then `UPDATE` it in a second statement:

```
-- Δ Do not do this - it has a race condition
SELECT id FROM jobs WHERE status = 'queued' ORDER BY priority, created_at, id LIMIT 1;
-- ... application reads id = 7 ...
UPDATE jobs SET status = 'in_progress' WHERE id = 7;
```

Between the `SELECT` and the `UPDATE`, nothing stops a second worker from running the exact same `SELECT`, reading the same `id = 7`, and also issuing the `UPDATE`. Both workers now believe they own job 7. This is a classic **check-then-act race condition** — the gap between reading and acting is exactly where two workers can interleave.

2.2 — `FOR UPDATE` closes the gap, but blocks

Locking the row as part of the `SELECT` closes the race:

```
SELECT id FROM jobs WHERE status = 'queued' ORDER BY priority, created_at, id
FOR UPDATE LIMIT 1;
```

Now a second worker running the same query, in a separate transaction, **blocks** until the first worker's transaction commits or rolls back — correct, but it means every worker but one sits idle waiting for a lock instead of moving on to a different job. Section 3.1 demonstrates this.

2.3 — `SKIP LOCKED` lets workers move past each other

Adding `SKIP LOCKED` tells PostgreSQL: if the next candidate row is already locked by another transaction, don't wait for it — skip it and consider the row after it.

```
SELECT id FROM jobs WHERE status = 'queued' ORDER BY priority, created_at, id
FOR UPDATE SKIP LOCKED LIMIT 1;
```


This is the piece that makes PostgreSQL usable as a concurrent queue: N workers can run this query at the same instant and each will walk away with a *different* row, with no blocking and no double-claims.

2.4 — The full atomic claim: lock, update, and return in one statement

Locking the row is only half the job — you still need to mark it `in_progress` before releasing the lock, and you want the whole thing to happen as a single round trip. Combine the `SELECT ... FOR UPDATE SKIP LOCKED` with the `UPDATE` using a CTE:

```
WITH next_job AS (
  SELECT id
  FROM   jobs
  WHERE  status = 'queued'
  ORDER BY priority ASC, created_at ASC, id ASC
  FOR UPDATE SKIP LOCKED
  LIMIT 1
)
UPDATE jobs
SET    status      = 'in_progress',
       claimed_at   = now(),
       claimed_by   = 'demo-worker',
       heartbeat_at = now(),
       attempts     = attempts + 1
FROM   next_job
WHERE  jobs.id = next_job.id
RETURNING jobs.id, jobs.job_type, jobs.payload ->> 'application_id' AS application_id,
          jobs.priority, jobs.attempts, jobs.status;
```

id	job_type	application_id	priority	attempts	status
1	demolition_permit	DP-2024-0001	1	1	in_progress

(1 row)

This single statement is the entire claim operation: find the best candidate, skip anything locked, mark it claimed, and hand back its data — atomically, with no window for a race. This is exactly the query `data/ch03_worker.py` runs (see `CLAIM_SQL`). Run it again and the demolition permit at `id = 2` comes back next — `id = 1` is now `in_progress`, so it's no longer a candidate.

Roll this back if you ran it directly in `psql` and want to keep the queue clean for later exercises: wrap it in `BEGIN; ... ROLLBACK;`.

Exercise 3 — Simulating Concurrent Workers

3.1 — `SKIP LOCKED` vs. plain `FOR UPDATE`, in two `psql` sessions

Open two terminals with `psql portsmith` in each. In **Session A**, start a transaction, claim 5 rows, and hold the transaction open (don't commit yet):

```
-- Session A
BEGIN;
SELECT id, job_type FROM jobs WHERE status = 'queued'
ORDER BY priority, created_at, id
FOR UPDATE SKIP LOCKED LIMIT 5;
```

id	job_type
1	demolition_permit
2	demolition_permit
3	demolition_permit
4	demolition_permit
5	business_license

(5 rows)

Leave that transaction open. In **Session B**, run the identical query:

```
-- Session B (Session A is still open, holding locks on ids 1-5)
BEGIN;
SELECT id, job_type FROM jobs WHERE status = 'queued'
ORDER BY priority, created_at, id
FOR UPDATE SKIP LOCKED LIMIT 5;
```

id	job_type
6	business_license
7	business_license
8	business_license
9	business_license
10	business_license

(5 rows)

Session B returns immediately with a **different** set of rows — it silently skipped 1–5 because they were locked, and moved on to the next five unlocked candidates. Commit or roll back both sessions to release the locks:

```
COMMIT; -- run in both sessions
```

3.2 — Now try it with plain `FOR UPDATE` (no `SKIP LOCKED`)

Repeat the same two-session experiment, but drop `SKIP LOCKED`:

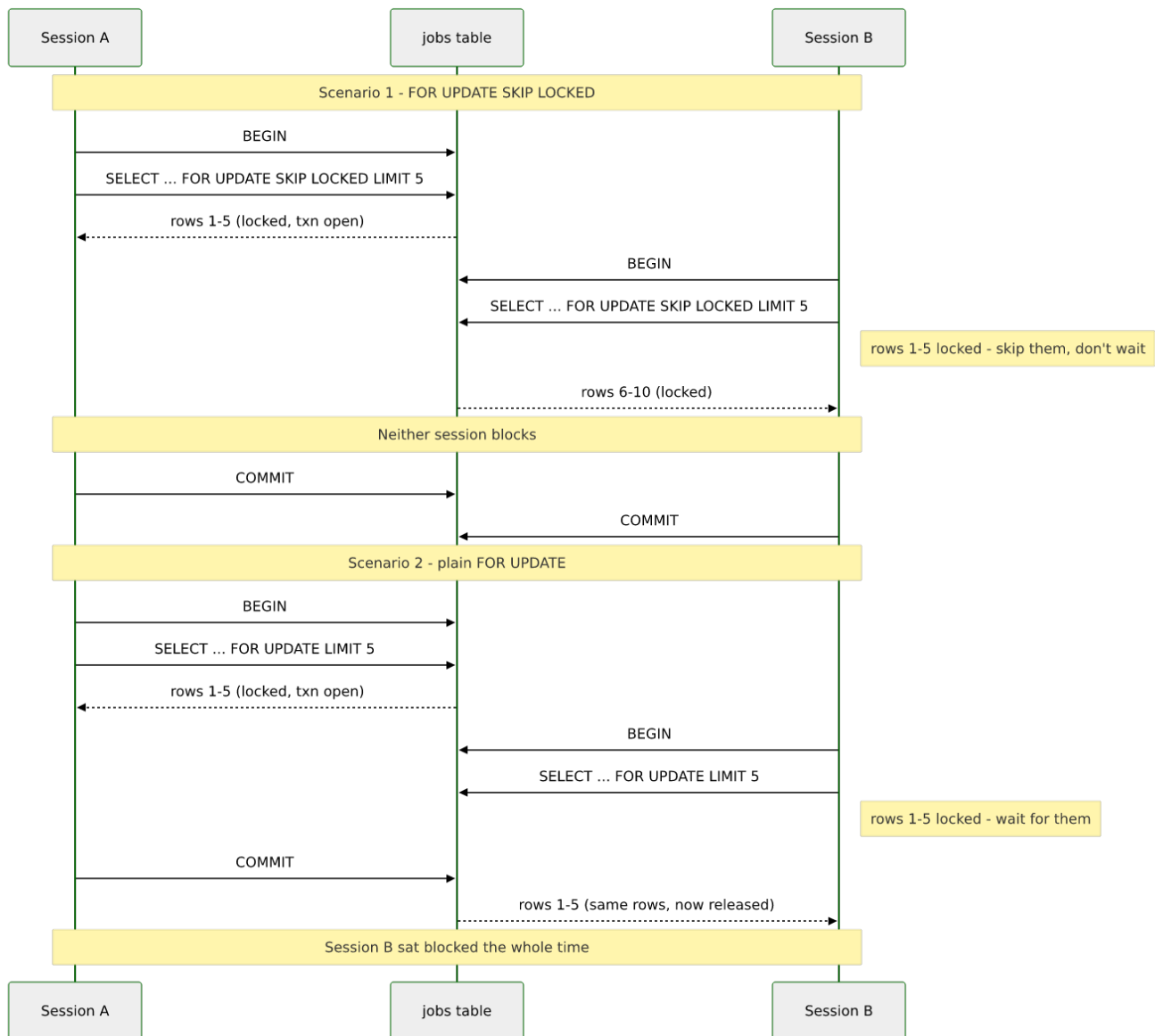
```
-- Session A
BEGIN;
SELECT id, job_type FROM jobs WHERE status = 'queued'
ORDER BY priority, created_at, id
FOR UPDATE LIMIT 5;
-- (leave this transaction open)
```

```
-- Session B - this will hang
BEGIN;
SELECT id, job_type FROM jobs WHERE status = 'queued'
ORDER BY priority, created_at, id
FOR UPDATE LIMIT 5;
```

Session B does not return. It is blocked, waiting for Session A's row locks to be released. Only once Session A runs COMMIT (or ROLLBACK) does Session B's query complete — and when it does, it returns the *same* five rows Session A had, now that they're unlocked again:

```
-- (Session B, after Session A commits)
id |      job_type
----+-----
 1 | demolition_permit
 2 | demolition_permit
 3 | demolition_permit
 4 | demolition_permit
 5 | business_license
(5 rows)
```

This is the difference in one sentence: **plain FOR UPDATE serializes workers through the lock; SKIP LOCKED lets them fan out across the table.** For a job queue, you always want the latter — a blocked worker is a wasted worker. Both runs side by side, as a timeline:



3.3 — Real concurrent workers with `ch03_worker.py`

Reset the data (`python data/ch03_seed.py`) if you ran the manual claim in Exercise 2.4 without rolling it back. Then launch two workers in the background, each capped to 10 jobs so they finish quickly:

```
python data/ch03_worker.py --worker-id w1 --max-jobs 10 --fail-rate 0 &
python data/ch03_worker.py --worker-id w2 --max-jobs 10 --fail-rate 0 &
wait
```

Each prints a running log as it claims and completes jobs, e.g.:

```
[w1] claimed job 2 (demolition_permit, attempt 1/3): DP-2024-0002 - processing for 0.09s
[w1] job 2 completed
[w2] claimed job 1 (demolition_permit, attempt 1/3): DP-2024-0001 - processing for 0.12s
[w2] job 1 completed
```

Once both finish, confirm the split was clean — 20 jobs completed total, no job claimed by both workers:

```
SELECT claimed_by, COUNT(*) FROM jobs WHERE status = 'completed'
GROUP BY claimed_by ORDER BY claimed_by;
```

claimed_by	count
w1	10
w2	10

(2 rows)

```
SELECT status, COUNT(*) FROM jobs GROUP BY status;
```

status	count
completed	20
queued	25

(2 rows)

10 + 10 = 20, matching exactly the `--max-jobs 10` cap on each worker — no overlaps, no double-processing, no lost jobs.

Exercise 4 — Heartbeat and Stalled-Job Recovery

Claiming a job is only half the reliability story. What happens if the worker that claimed a job crashes, gets OOM-killed, or loses its network connection halfway through? The row is stuck at `status = 'in_progress'` forever unless something notices and puts it back.

4.1 — The heartbeat column

`heartbeat_at` exists for exactly this. A well-behaved worker updates it periodically while it holds a job (see the `while` loop in `process_job()` in `ch03_worker.py`, which sends a heartbeat roughly every 2 seconds during simulated processing). If a job has been `in_progress` for a long time *and* its heartbeat has gone stale, that's a strong signal the worker holding it is dead — a live worker would have updated it.

4.2 — Simulate a crashed worker

Pick any queued job and manually walk it through what a real claim would do, then simulate a crash by never sending a heartbeat again:

```
UPDATE jobs
SET     status = 'in_progress', claimed_at = now(),
        claimed_by = 'worker-crashed', heartbeat_at = now(),
        attempts = attempts + 1
WHERE  id = 21;
```

```
-- Simulate 5 minutes of silence from the "crashed" worker
```

```
UPDATE jobs SET heartbeat_at = now() - interval '5 minutes' WHERE id = 21;
```

4.3 — Run the reclaim sweep

```
python data/ch03_reclaim.py --timeout 30
```

Connecting to: dbname=portsmith

job 21 (building_permit) stalled – requeued (attempt 1)

Done: 1 requeued, 0 dead-lettered.

The sweep looks for `in_progress` rows whose `heartbeat_at` is older than `--timeout` seconds (`FIND_STALLED_SQL` in `ch03_reclaim.py`), and since this job’s attempts (1) is still below `max_attempts` (3), it goes back to `queued`:

```
SELECT id, status, attempts, claimed_by, heartbeat_at, last_error
FROM   jobs WHERE id = 21;
```

id	status	attempts	claimed_by	heartbeat_at	last_error
21	queued	1			stalled: no heartbeat since ... (last claimed by worker-crashed)

Notice `attempts` stayed at 1 — the reclaim did not reset it. The attempt the crashed worker used is still counted; the job doesn’t get a free retry just because the worker that lost it never got to report failure honestly.

4.4 — Why the timeout, not an outright deadline

A fixed timeout on the *last heartbeat* (rather than on total processing time) lets jobs run arbitrarily long as long as they keep proving they’re alive. A 10-minute report-generation job with a 30-second heartbeat interval will never be mistaken for stalled, while a worker that dies mid-task is detected within one missed heartbeat window. Run `ch03_reclaim.py` again immediately — it correctly finds nothing to do, since the reclaimed job now has a fresh `queued` state and no stale `in_progress` row exists:

No jobs stalled beyond 30s – nothing to do.

Exercise 5 — Dead-Lettering Exhausted Jobs

Some jobs will never succeed no matter how many times you retry them — a malformed application, a permanently invalid address, a bug that only triggers on one particular payload. Retrying forever wastes worker time and can mask a real problem. Once a job has used its last attempt, it belongs in `dead_letter_jobs`: out of the active queue, but preserved for a human to inspect.

5.1 — The `dead_letter_jobs` schema

```
\d dead_letter_jobs
```

Table "public.dead_letter_jobs"				
Column	Type	Collation	Nullable	Default
id	bigint		not null	
job_type	text		not null	
payload	jsonb		not null	
priority	smallint		not null	
attempts	integer		not null	
max_attempts	integer		not null	
created_at	timestamp with time zone		not null	
last_error	text			
failed_at	timestamp with time zone		not null	now()

It mirrors `jobs` but adds `failed_at` and drops the in-flight columns (`status`, `claimed_by`, `heartbeat_at`) that no longer mean anything once a job is out of the active queue.

5.2 — Push a job past its retry limit

Continuing with job 21 from Exercise 4 (now back to `queued` with `attempts = 1`), fast-forward it to its last attempt and let it stall again:

```
UPDATE jobs
SET     status = 'in_progress', claimed_at = now(),
        claimed_by = 'worker-crashed-2', heartbeat_at = now() - interval '5 minutes',
        attempts = max_attempts
WHERE  id = 21;
```

```
python data/ch03_reclaim.py --timeout 30
```

```
Connecting to: dbname=portsmith
```

```
job 21 (building_permit) exhausted retries - dead-lettered
```

```
Done: 0 requeued, 1 dead-lettered.
```

This time `attempts` (3) is no longer less than `max_attempts` (3), so the sweep's `DEAD_LETTER_SQL` runs instead: it `DELETES` the row from `jobs` and `INSERTS` it into `dead_letter_jobs` in one CTE, so the row is never visible in neither-place or both-places, even under concurrent access.

```
SELECT id FROM jobs WHERE id = 21;
```

```
id
----
(0 rows)
```

```
SELECT id, job_type, payload ->> 'application_id' AS application_id,
       attempts, max_attempts, last_error, failed_at
FROM   dead_letter_jobs WHERE id = 21;
```

```
id | job_type | application_id | attempts | max_attempts |
last_error | failed_at
```

```

-----+-----+-----+-----+-----
+-----+-----+-----+-----+-----
21 | building_permit | BP-2024-0005 | 3 | 3 | stalled: no heartbeat
since ... (last claimed ...) | 2026-07-12 22:15:45.613583-04
(1 row)

```

The exact same dead-lettering logic runs inline inside `ch03_worker.py` when a job fails organically (not via a stall) on its last attempt — see `DEAD_LETTER_SQL` in that script. Whether a job dies from an explicit failure or from going silent, it ends up in the same place, with a record of why.

Exercise 6 — Benchmarking Claim Throughput with `pgbench`

6.1 — A benchmark script that doesn’t drain the queue

`pgbench` repeatedly runs a SQL script against the database for a fixed duration, from any number of concurrent client connections — exactly the tool for measuring how the claim query holds up under contention. The catch: if the script just claims a job and leaves it claimed, 45 rows disappear from the pool in well under a second and every worker after that finds nothing to do. `data/ch03_claim_bench.sql` works around this by claiming a job and then **immediately releasing it back to queued** in the same transaction:

```

BEGIN;

WITH next_job AS (
    SELECT id
    FROM   jobs
    WHERE  status = 'queued'
    ORDER BY priority ASC, created_at ASC, id ASC
    FOR UPDATE SKIP LOCKED
    LIMIT 1
)
UPDATE jobs
SET    status      = 'in_progress',
       claimed_at   = now(),
       claimed_by   = 'bench-' || :client_id,
       heartbeat_at = now()
FROM   next_job
WHERE  jobs.id = next_job.id
RETURNING jobs.id AS claimed_id \gset

UPDATE jobs
SET    status = 'queued', claimed_at = NULL, claimed_by = NULL, heartbeat_at = NULL
WHERE  id = :claimed_id;

COMMIT;

```


`:client_id` is a `pgbench` built-in variable — the number of the simulated client running this script. `\gset` captures the claimed row's `id` into a `pgbench` variable (`:claimed_id`) so the release step can target it. This keeps the pool at a constant 45 claimable rows for the whole benchmark run, so throughput reflects lock contention rather than the queue running dry.

6.2 — Run at increasing concurrency

```
pgbench -n -c 1 -j 1 -T 5 -f data/ch03_claim_bench.sql portsmith
pgbench -n -c 4 -j 4 -T 5 -f data/ch03_claim_bench.sql portsmith
pgbench -n -c 16 -j 16 -T 5 -f data/ch03_claim_bench.sql portsmith
```

`-c` is the number of simulated concurrent workers (clients), `-j` the number of `pgbench` threads driving them, and `-T 5` runs each test for 5 seconds. Results on the machine used to write this chapter:

Clients (-c)	tps	Avg. latency
1	799	1.25 ms
4	1,202	3.33 ms
16	2,570	6.23 ms

Your numbers will differ with your hardware — the shape of the result is what matters. Throughput climbs with concurrency because `SKIP LOCKED` lets additional workers keep finding unlocked rows to claim instead of queueing up behind each other; latency per transaction also climbs because more workers are contending for the same 45-row table and for CPU. Try dropping `idx_jobs_claim_order` (inside a `BEGIN; ... ROLLBACK;` block, as in Exercise 1.3) and rerunning the `-c 16` case — you should see tps fall, since every claim now pays for a sort over all queued rows instead of an index walk.

6.3 — Where this stops being enough

`pgbench` here is exercising claim contention on 45 rows — a stand-in for “the working set the claim query actually scans.” A real permitting-office queue might hold a very different shape of data (thousands of queued rows, heavy skew toward one `priority` bucket), and the honest way to benchmark your own workload is to seed a table that resembles it, not a 45-row toy. Chapter 20 (`pg_stat_statements` and Query Performance) picks this thread back up: rather than a synthetic benchmark, it measures and fixes real slow queries observed in production traffic.

Summary — What You Should Now Know

You built a working, concurrency-safe job queue entirely out of core PostgreSQL features. Here is a reference for the pieces:

Tool	What it does
<code>FOR UPDATE</code>	Locks selected rows; other transactions selecting the same rows block until released
<code>FOR UPDATE SKIP LOCKED</code>	Locks selected rows; other transactions skip already-locked rows instead of blocking
<code>WITH next_job AS (... FOR UPDATE SKIP LOCKED LIMIT 1) UPDATE ... FROM next_job</code>	The atomic claim pattern: find, lock, and mark a row in one round trip
<code>Partial index WHERE status = 'queued'</code>	Keeps the claim-path index small regardless of how many settled rows accumulate
<code>clock_timestamp()</code> VS. <code>now()</code>	<code>now()</code> freezes at transaction start; <code>clock_timestamp()</code> reflects real wall-clock time on every call
<code>heartbeat_at + timeout</code> sweep	Detects and recovers jobs abandoned by a crashed worker
<code>dead_letter_jobs</code>	Removes permanently-failing jobs from the active queue while preserving them for inspection
<code>pgbench -f script.sql</code>	Benchmarks a custom SQL workload at a chosen concurrency level

The key design insight from this chapter is that a reliable queue is not one clever query — it’s the combination of an atomic claim, a way to detect workers that go silent, and a place for work that genuinely cannot succeed. PostgreSQL’s row-locking primitives give you the first two almost for free; the dead-letter table is just a normal table.

The `jobs` table you built here is reused directly in later chapters: Chapter 13 adds a trigger that fires `NOTIFY` on status changes so listeners can react to queue activity in real time, Chapter 14 uses advisory locks alongside it for leader-election patterns, and Chapter 19 schedules the reclaim sweep you wrote by hand in Exercise 4 to run automatically with `pg_cron`.

Going further: the pattern in this chapter — status column, atomic claim, heartbeat, dead-letter — is exactly what PostgreSQL-backed job-queue libraries like `river` (Go) and `oban` (Elixir) implement, with more polish around scheduling, uniqueness constraints, and observability. If you outgrow a single-database queue — cross-database work distribution, guaranteed ordering at very high throughput, or consumer groups — that’s the point at which a dedicated broker starts to earn its operational cost. For most applications below that scale, the table you just built is enough.

Chapter 4 — Full-Text Search: `tsvector`, Stopwords, and Ranking

“Grep finds characters. Full-text search finds meaning — or at least gets a lot closer.”

Background

LIKE `'%harbour%'` finds the literal substring “harbour”. It will not find “Harbour”, “harbours”, or a document about “the harbor” (different spelling) unless you handle every variation yourself, and it cannot tell you whether a match in the title is more relevant than a match buried in paragraph six. As soon as an application needs to search prose — meeting minutes, support tickets, product descriptions, articles — substring matching stops being enough. The usual reflex is to reach for Elasticsearch or a similar dedicated search engine.

PostgreSQL has had a full-text search engine built in since version 8.3. It tokenizes text into words, reduces those words to normalized root forms (*stemming*), discards low-information words like “the” and “of” (*stopwords*), and stores the result in a specialized `tsvector` type that a GIN index can search in milliseconds. A companion `tsquery` type lets you express boolean and phrase searches, and a family of ranking functions score how well each match fits the query. None of this requires an extension — it is core PostgreSQL, the same engine `websearch_to_tsquery`-powered search boxes on production sites are built on.

This chapter builds a search feature over a small document archive: how text becomes a `tsvector`, what stopwords removal actually throws away, how to keep the vector in sync as rows change, how query syntax differs between a developer-facing boolean query language and a plain-text search box, and how to rank and highlight results. The final exercise customizes the stopwords list itself — because in a *city* government’s document archive, the words “city” and “council” appear so often they stop being useful signals, exactly the kind of domain-specific tuning full-text search is designed to support.

The Scenario

Portsmouth’s city clerk publishes an archive of public records: council meeting minutes, zoning ordinances, and public notices. Residents need to search this archive by keyword — “what happened with the Riverside dog park?”, “which ordinances mention Canal Road?” — and get

back the most relevant documents first, with a highlighted snippet showing why each one matched.

The `city_documents` table holds all three document types in one place, with a `body` column of plain prose text and a few relational columns for filtering:

Column	Purpose
<code>doc_type</code>	<code>council_minutes</code> , <code>zoning_ordinance</code> , Or <code>public_notice</code>
<code>department</code>	Which city department published it (City Council, Public Works, ...)
<code>title</code>	Short document title
<code>body</code>	Full plain-text document body — the field full-text search runs over
<code>published_date</code>	When the document was published

The documents share recurring topics on purpose — the Riverside dog park, the Harbour District waterfront rezoning, the Canal Road bike lane — so that searches in the exercises return more than one hit, and ranking has something real to sort between. No extensions are required for this chapter: `tsvector`, `tsquery`, and every function used below are core PostgreSQL.

Exercise Goals

By the end of this chapter you will be able to:

- Convert text to a `tsvector` and read the lexeme:position notation it produces.
 - Explain exactly what stopwords removal and stemming do to a document, using `ts_debug` to see PostgreSQL's tokenizer classify each word.
 - Maintain a `tsvector` column automatically with a trigger, and index it with GIN for sub-millisecond searches.
 - Write boolean queries with `to_tsquery` and understand why `plainto_tsquery` is the safer choice for a plain search box.
 - Rank results with `ts_rank` and `ts_rank_cd`, and generate highlighted snippets with `ts_headline`.
 - Build a custom text search configuration that treats domain-specific words as stopwords, and see it change real search results.
-

Installation

This chapter needs nothing beyond what Chapter 1 already set up: PostgreSQL 16 and a Python 3.12 virtual environment with `psycopg`. If you skipped Chapter 1, see its Installation section. Full-text search is core PostgreSQL — there is no extension to enable.

Loading the Data

Run the seed script

From the `book/` directory, with the virtual environment active:

```
python data/ch04_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Creating schema ...
Inserting 30 documents ...
Done - 30 rows in city_documents.
```

The seed script is self-contained — it does not depend on any earlier chapter’s data. Note that it deliberately does **not** create a `tsvector` column or index; you build both by hand in Exercise 3, which is the point of the chapter.

Verify the load

Open `psql portsmith` and run these checks.

Check 1 — table structure:

```
\d city_documents

              Table "public.city_documents"
  Column      | Type      | Collation | Nullable | Default
-----+-----+-----+-----+-----
 id           | integer   |           | not null | nextval('city_documents_id_seq'::regclass)
 doc_type     | text      |           | not null |
 department   | text      |           | not null |
 title        | text      |           | not null |
 body         | text      |           | not null |
 published_date | date      |           | not null |

Indexes:
    "city_documents_pkey" PRIMARY KEY, btree (id)
    "idx_city_documents_doc_type" btree (doc_type)
```

```
"idx_city_documents_published_date" btree (published_date)
```

Check constraints:

```
"city_documents_doc_type_check" CHECK (doc_type = ANY (ARRAY['council_minutes'::text,  
'zoning_ordinance'::text, 'public_notice'::text]))
```

Check 2 — counts by document type:

```
SELECT doc_type, COUNT(*) AS documents  
FROM    city_documents  
GROUP BY doc_type  
ORDER BY doc_type;
```

doc_type	documents
council_minutes	10
public_notice	10
zoning_ordinance	10

(3 rows)

Check 3 — counts by department:

```
SELECT department, COUNT(*) AS documents  
FROM    city_documents  
GROUP BY department  
ORDER BY department;
```

department	documents
City Council	9
Finance	1
Parks & Recreation	3
Planning & Zoning	12
Public Works	5

(5 rows)

If all three match, proceed to the exercises.

Exercises

Exercise 1 — Converting Text to `tsvector`

1.1 — A basic conversion

`to_tsvector` is the function that turns plain text into PostgreSQL's searchable representation. Try it on one document's body:

```
SELECT title FROM city_documents WHERE id = 1;
```

title

Council Minutes – Harbour District Waterfront Renovation Budget

```
SELECT to_tsvector('english', body) FROM city_documents WHERE id = 1;
```

```
'along':38 'approv':67 'begin':79 'boardwalk':34 'bond':53 'budget':10 'citi':3  
'construct':76 'conven':5 'council':4,41,61 'cover':28 'debat':56 'director':18  
'discuss':43 'district':14 'expect':77 'fall':82 'first':69 'fund':44,74  
'grant':50 'harbour':13,40,72 'includ':46 'infrastructur':49 'light':36  
'measur':54 'member':42 'new':32 'one':65 'pedestrian':33 'phase':26,70  
'plan':27 'portsmith':2 'present':22 'propos':9 'public':20 'renov':16,73  
'repair':30 'review':7 'seawal':29 'six':63 'sourc':45 'state':48  
'three':25 'three-phas':24 'timelin':59 'upgrad':37 'vote':62  
'waterfront':15 'work':21
```

1.2 — Reading the format

Each entry is 'lexeme':position,position, A **lexeme** is a normalized word form, not the literal text — notice 'vote':62 even though the source text says “voted”, 'approv':67 for “approve”, and 'renov':16,73 for both “renovation” (position 16) and “renovation” again later (position 73, from “harbour **renovation** funding”). PostgreSQL stems words to their root so that a search for “vote” also matches documents containing “voted”, “votes”, or “voting” — you don’t have to enumerate every inflection yourself.

The **positions** are the word’s ordinal location in the original text (1st word, 2nd word, ...). They exist for two reasons: `ts_rank` uses them to compute how spread out or clustered a document’s matches are, and phrase search (`<->`, used in Exercise 4) uses them to require words to appear adjacent to each other, not just anywhere in the document.

1.3 — Case and punctuation are already handled

Note that “Council” (capitalized, position 4) and “council” (lowercase, positions 41 and 61) both collapsed into the single lexeme 'council' with three positions — `to_tsvector` lowercases everything and strips punctuation as part of tokenization, before stemming ever runs.

Exercise 2 — What Stopword Removal Actually Throws Away

2.1 — Compare `simple` and `english` configurations

PostgreSQL ships several **text search configurations** — named bundles of tokenizer + dictionary rules. `simple` lowercases and tokenizes but does *no* stemming and drops *no* stopwords; `english` does both. Run the same body through each and count the resulting lexemes:

```
SELECT array_length(regexp_split_to_array(body, '\s+'), 1) AS raw_words  
FROM city_documents WHERE id = 1;
```

```

raw_words
-----
      80

SELECT array_length(tsvector_to_array(to_tsvector('simple', body)), 1) AS lexemes_simple,
       array_length(tsvector_to_array(to_tsvector('english', body)), 1) AS lexemes_english
FROM   city_documents WHERE id = 1;

lexemes_simple | lexemes_english
-----+-----
              59 |                  49

```

80 raw whitespace-separated words collapse to 59 distinct lexemes under `simple` (repeated words like “the” and “council” count once each, but nothing is removed or stemmed) and to 49 under `english` — ten fewer, because `english` additionally discards stopwords like “the”, “a”, “of”, “to”, “in”, and “and” entirely. They carry no search-relevant meaning on their own, and indexing them would only bloat the index with entries that match nearly every document.

2.2 — Watch the tokenizer classify each word with `ts_debug`

`ts_debug` is the diagnostic function for seeing exactly what a configuration does to a piece of text, word by word — useful whenever a search isn’t matching what you expect and you need to know why. Run it on a short sentence:

```

SELECT alias, token, dictionaries, lexemes
FROM   ts_debug('english', 'The council voted to approve the budget for the harbour
renovation.')
WHERE  alias <> 'blank';

```

alias	token	dictionaries	lexemes
asciiword	The	{english_stem}	{}
asciiword	council	{english_stem}	{council}
asciiword	voted	{english_stem}	{vote}
asciiword	to	{english_stem}	{}
asciiword	approve	{english_stem}	{approv}
asciiword	the	{english_stem}	{}
asciiword	budget	{english_stem}	{budget}
asciiword	for	{english_stem}	{}
asciiword	the	{english_stem}	{}
asciiword	harbour	{english_stem}	{harbour}
asciiword	renovation	{english_stem}	{renov}

Every token gets classified (`asciiword` here — a plain word) and routed to the `english_stem` dictionary. Content words come back with a stemmed lexeme; stopwords (“The”, “to”, “the”, “for”, “the”) come back with an **empty** `lexemes` array — recognized, looked up, and explicitly discarded. That empty array is the entire mechanism: a word is a stopwords precisely when its dictionary entry maps it to nothing.

2.3 — When you’d reach for `simple` instead

`english` is the right default for prose. `simple` is useful for columns where stemming would be actively wrong — product SKUs, tag lists, or anything where “waterfront” and “waterfronts” should *not* be treated as the same token. `city_documents.body` is prose, so the rest of this chapter uses `english`.

Exercise 3 — A Maintained Column and a GIN Index

Computing `to_tsvector(body)` at query time works, but it means recomputing the same tokenization on every single search, over every row, every time. The standard pattern is to store the vector in its own column, keep it current as rows change, and index it.

3.1 — Add the column and backfill it

```
ALTER TABLE city_documents ADD COLUMN search_vector tsvector;

UPDATE city_documents
SET    search_vector = to_tsvector('english', title || ' ' || body);
```

Indexing `title` alongside `body` means a search for a word that only appears in the title (not repeated in the body text) still matches.

3.2 — Keep it current with a trigger

A column populated once goes stale the moment anyone edits `title` or `body`. A `BEFORE` trigger recomputes it on every write:

```
CREATE FUNCTION city_documents_search_vector_update() RETURNS trigger AS $$
BEGIN
    NEW.search_vector := to_tsvector('english', NEW.title || ' ' || NEW.body);
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_city_documents_search_vector
    BEFORE INSERT OR UPDATE OF title, body ON city_documents
    FOR EACH ROW
    EXECUTE FUNCTION city_documents_search_vector_update();
```

`OF title, body` scopes the trigger so it only fires when one of the two source columns actually changes — an `UPDATE` that only touches `published_date` doesn’t pay for a needless re-tokenization.

Note — the manual way vs. the modern way: This trigger pattern is how every PostgreSQL version has supported derived `tsvector` columns, and it’s still exactly right when the derived value depends on more than one column, as it does here (`title` **and**

body). PostgreSQL 12 added **generated columns** (`GENERATED ALWAYS AS (...) STORED`), which handle the common case — a `tsvector` derived from a single column — with less boilerplate and no trigger function to maintain by hand. Chapter 16 revisits this exact table and replaces this trigger with a generated column once `body` alone is the source. Both approaches produce an identical `tsvector`; the trigger is simply the general-purpose tool that works whenever the derivation is more than one column deep.

3.3 — Index it with GIN

```
CREATE INDEX idx_city_documents_search_vector
ON city_documents USING GIN (search_vector);

\d city_documents
```

Table "public.city_documents"					
Column	Type	Collation	Nullable	Default	
id	integer		not null	nextval('city_documents_id_seq'::regclass)	
doc_type	text		not null		
department	text		not null		
title	text		not null		
body	text		not null		
published_date	date		not null		
search_vector	tsvector				

```
Indexes:
    "city_documents_pkey" PRIMARY KEY, btree (id)
    "idx_city_documents_doc_type" btree (doc_type)
    "idx_city_documents_published_date" btree (published_date)
    "idx_city_documents_search_vector" gin (search_vector)
```

```
Check constraints:
    "city_documents_doc_type_check" CHECK (doc_type = ANY (ARRAY['council_minutes'::text,
'zoning_ordinance'::text, 'public_notice'::text]))
```

```
Triggers:
    trg_city_documents_search_vector BEFORE INSERT OR UPDATE OF title, body ON city_documents
FOR EACH ROW EXECUTE FUNCTION city_documents_search_vector_update()
```

3.4 — Confirm the index is used

The `@@` operator tests whether a `tsvector` matches a `tsquery`:

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT id, title
FROM city_documents
WHERE search_vector @@ to_tsquery('english', 'harbour & waterfront');
```

QUERY PLAN	
Seq Scan on city_documents (cost=0.00..8.38 rows=1 width=36) (actual time=0.025..0.047 rows=5 loops=1)	
Filter: (search_vector @@ ''harbour'' & ''waterfront''::tsquery)	

```
Rows Removed by Filter: 25
Buffers: shared hit=8
```

On 30 rows the planner reasonably decides a sequential scan is cheaper than an index lookup — same behaviour you saw with the JSONB GIN index in Chapter 1. Force the index to confirm it works, exactly as in that chapter:

```
SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT id, title
FROM   city_documents
WHERE  search_vector @@ to_tsquery('english', 'harbour & waterfront');

SET enable_seqscan = on;  -- always restore this
```

Query Plan

```
-----
Bitmap Heap Scan on city_documents  (cost=12.98..16.99 rows=1 width=36) (actual
time=0.024..0.033 rows=5 loops=1)
  Recheck Cond: (search_vector @@ '''harbour'' & '''waterfront''':tsquery)
  Heap Blocks: exact=5
  Buffers: shared hit=10
  -> Bitmap Index Scan on idx_city_documents_search_vector  (cost=0.00..12.98 rows=1 width=0)
(actual time=0.014..0.014 rows=5 loops=1)
    Index Cond: (search_vector @@ '''harbour'' & '''waterfront''':tsquery)
    Buffers: shared hit=5
```

Bitmap Index Scan on idx_city_documents_search_vector confirms the GIN index is doing the work. This is what makes full-text search viable at scale: instead of tokenizing every row's text on every query, PostgreSQL looks up each query lexeme directly in the index.

Exercise 4 — to_tsquery VS. plainto_tsquery

4.1 — to_tsquery: a boolean query language

to_tsquery expects an expression using explicit operators: & (AND), | (OR), ! (NOT), and <-> (phrase — “immediately followed by”):

```
SELECT id, title
FROM   city_documents
WHERE  search_vector @@ to_tsquery('english', 'harbour & waterfront')
ORDER BY id;
```

id	title
1	Council Minutes – Harbour District Waterfront Renovation Budget
8	Council Minutes – Harbour District Food Truck Permits
12	Zoning Ordinance – Harbour District Waterfront Height Variance

22 | Public Notice – Public Hearing on Harbour District Waterfront Rezoning
30 | Public Notice – Annual Fireworks Display and Street Closures
(5 rows)

! excludes a term. This finds documents about flooding that are *not* about the flood-mitigation infrastructure program specifically:

```
SELECT id, title
FROM   city_documents
WHERE  search_vector @@ to_tsquery('english', 'flood & !mitigation')
ORDER BY id;
```

id	title
-----+	
5	Council Minutes – Riverside Road Resurfacing Program

(1 row)

That single hit mentions “flooding” (which stems to `flood`) while discussing drainage damage, but never says “mitigation” — exactly what the query asked for.

4.2 — `to_tsquery` has no forgiveness for plain text

Feed it a raw phrase with no operator between the words, and it errors instead of guessing what you meant:

```
SELECT id, title
FROM   city_documents
WHERE  search_vector @@ to_tsquery('english', 'flood mitigation');
```

ERROR: syntax error in tsquery: "flood mitigation"

This is `to_tsquery`’s defining trade-off: it is a precise query language for code that constructs queries deliberately, and it is the wrong function to hand raw user input from a search box — a user typing “flood mitigation” with no operators will get an error page instead of results.

4.3 — `plainto_tsquery`: safe for a search box

`plainto_tsquery` takes plain text, tokenizes and stems it exactly like `to_tsvector` does, drops stopwords, and **ANDs** everything that survives. Feed it the same two words that made `to_tsquery` error out in 4.2, but typed the way an actual user would type them — with a stray “the” in front:

```
SELECT plainto_tsquery('english', 'the flood mitigation');
```

plainto_tsquery

'flood' & 'mitig'

“the” never makes it into the query — `plainto_tsquery` drops it as a stopword, exactly the way `to_tsvector` would drop it from a document, and **ANDs** together whatever content words are left. Run it against the table:

```
SELECT id, title
FROM city_documents
WHERE search_vector @@ plainto_tsquery('english', 'the flood mitigation')
ORDER BY id;
```

id	title
10	Council Minutes – Special Session on Flood Mitigation Infrastructure
25	Public Notice – City Council Special Session on Flood Mitigation

(2 rows)

No error this time, and no operators to get wrong — `plainto_tsquery` took a sentence fragment with a stopword in it and produced exactly the query 4.2 had to write by hand. The trade-off runs the other way from `to_tsquery`: you get safety and stopword handling for free, but you lose the ability to express OR, NOT, or phrase search — `plainto_tsquery` always ANDs every surviving word together.

4.4 — | for OR searches

```
SELECT to_tsquery('english', 'dog | bike');
```

to_tsquery
'dog' 'bike'

```
SELECT id, title
FROM city_documents
WHERE search_vector @@ to_tsquery('english', 'dog | bike')
ORDER BY id;
```

id	title
5	Council Minutes – Riverside Road Resurfacing Program
6	Council Minutes – Riverside Dog Park Funding
9	Council Minutes – Canal Road Bike Lane Expansion
15	Zoning Ordinance – Reduced Parking Minimums Near Transit Corridors
24	Public Notice – Riverside Dog Park Ribbon-Cutting Event
28	Public Notice – Canal Road Bike Lane Construction Schedule
30	Public Notice – Annual Fireworks Display and Street Closures

(7 rows)

Document 5 (road resurfacing) and 15 (parking minimums) show up because each mentions “bike” in passing — the resurfacing minutes note the work being coordinated with “the planned Canal Road bike lane construction,” and the parking ordinance references “the city’s ongoing bike lane expansion.” OR genuinely means *either*, which is exactly why it’s useful and why it returns more, looser matches than AND.

Exercise 5 — Ranking with `ts_rank` and `ts_rank_cd`, and `ts_headline` Snippets

`@@` tells you whether a document matches — it says nothing about *how well*. A user searching a 30-document archive can eyeball every hit, but a user searching 30,000 documents needs the best matches first.

5.1 — `ts_rank`: weight by term frequency

```
SELECT id, title, round(ts_rank(search_vector, query)::numeric, 4) AS rank
FROM   city_documents, to_tsquery('english', 'harbour & waterfront') AS query
WHERE  search_vector @@ query
ORDER BY rank DESC;
```

id	title	rank
12	Zoning Ordinance – Harbour District Waterfront Height Variance	0.1986
1	Council Minutes – Harbour District Waterfront Renovation Budget	0.1959
22	Public Notice – Public Hearing on Harbour District Waterfront Rezoning	0.1895
8	Council Minutes – Harbour District Food Truck Permits	0.0999
30	Public Notice – Annual Fireworks Display and Street Closures	0.0992

(5 rows)

`ts_rank` scores primarily on how often the query terms appear relative to the document’s overall length — it does not care where in the document they appear or how close together they are. Documents 12, 1, and 22 rank highest because “harbour” and “waterfront” are central, repeated themes in short documents; 8 and 30 rank lower because each mentions the harbour only in passing within a longer document about something else.

5.2 — `ts_rank_cd`: weight by proximity too (“cover density”)

```
SELECT id, title,
       round(ts_rank(search_vector, query)::numeric, 4) AS rank,
       round(ts_rank_cd(search_vector, query)::numeric, 4) AS rank_cd
FROM   city_documents, to_tsquery('english', 'harbour & waterfront') AS query
WHERE  search_vector @@ query
ORDER BY rank_cd DESC;
```

id	title	rank	rank_cd
1	Council Minutes – Harbour District Waterfront Renovation Budget	0.1959	0.1107
22	Public Notice – Public Hearing on Harbour District Waterfront Rezoning	0.1895	0.1053
12	Zoning Ordinance – Harbour District Waterfront Height Variance	0.1986	0.0700
30	Public Notice – Annual Fireworks Display and Street Closures	0.0992	0.0545
8	Council Minutes – Harbour District Food Truck Permits	0.0999	0.0542

(5 rows)

The order changes: document 12 had the *highest* `ts_rank` but drops to third under `ts_rank_cd`. `ts_rank_cd` additionally rewards matched terms that cluster close together in the text (“cover

density”) — in documents 1 and 22, “harbour” and “waterfront” appear right next to each other (“Harbour District **Waterfront** Renovation...”); in document 12 they occur further apart. Neither function is universally “more correct” — `ts_rank` is the standard choice; `ts_rank_cd` is worth trying when word proximity itself signals relevance, as it often does for short phrase-like queries.

5.3 — Highlighted snippets with `ts_headline`

A ranked list of titles is useful; showing *why* a document matched, with the query terms highlighted in context, is what users actually expect from a search results page:

```
SELECT ts_headline('english', body, to_tsquery('english', 'harbour & waterfront'),
                'StartSel=**, StopSel=**, MaxWords=25, MinWords=10')
FROM   city_documents
WHERE  id = 1;
```

```
**Harbour** District **waterfront** renovation. The Director of Public Works presented
```

`ts_headline` re-scans the original text (not the `tsvector`) looking for the query terms, extracts a window around the best-matching fragment bounded by `MinWords/MaxWords`, and wraps each match in `StartSel/StopSel` markers — `**...**` here, but in a web application you’d use `<mark>...</mark>` or similar. This is genuinely expensive compared to a `tsvector` lookup (it re-tokenizes the source text on every call), so use it only on the page of results you’re actually displaying, never inside a `WHERE` clause.

Exercise 6 — A Custom Text Search Configuration for Domain Stopwords

6.1 — The problem: some “content” words carry no information here

This is a city document archive. The words “Portsmouth”, “city”, and “council” appear constantly — they are structurally present in almost every record, the way “the” is present in almost every English sentence. Standard `english` stopwords removal has no way to know that, because it is tuned for general English, not this specific corpus:

```
SELECT count(*) FROM city_documents
WHERE  search_vector @@ to_tsquery('english', 'city & council');
```

```
count
-----
      8
```

Eight of thirty documents — over a quarter of the entire archive — “match” a query on words that describe almost nothing about what makes any one of them relevant. Left alone, these words dilute ranking scores and inflate plain-text queries with noise terms the searcher didn’t mean to require.

6.2 — Build a stopwords file that extends the standard list

A text search configuration's stopwords list is a plain text file, one word per line, that PostgreSQL reads from its `tsearch_data` directory. Start from the existing `english.stop` file so you keep every standard English stopwords, then append the domain-specific ones:

```
PG_SHAREDIR=$(pg_config --sharedir)
sudo bash -c "cat '$PG_SHAREDIR/tsearch_data/english.stop' > '$PG_SHAREDIR/tsearch_data/
portsmith_english.stop'"
sudo bash -c "printf 'portsmith\ncity\ncouncil\n' >> '$PG_SHAREDIR/tsearch_data/
portsmith_english.stop'"
```

Placing files here requires root, since `tsearch_data` lives under PostgreSQL's shared install directory rather than anything database-owned. If your platform's PostgreSQL package stores it elsewhere, `pg_config --sharedir` will always point at the right location.

6.3 — Wire the file into a dictionary and a configuration

A stopwords file alone does nothing — it has to be attached to a **text search dictionary**, and that dictionary mapped into a **text search configuration** that queries can actually reference:

```
CREATE TEXT SEARCH DICTIONARY portsmith_stem (
    TEMPLATE = snowball,
    LANGUAGE = english,
    STOPWORDS = portsmith_english
);

CREATE TEXT SEARCH CONFIGURATION public.portsmith_english (COPY = pg_catalog.english);

ALTER TEXT SEARCH CONFIGURATION portsmith_english
    ALTER MAPPING FOR asciiword, asciihword, hword_asciipart, word, hword, hword_part
    WITH portsmith_stem;
```

`TEMPLATE = snowball` reuses PostgreSQL's standard English stemming algorithm — only the stopwords list changes, not the stemming rules. `COPY = pg_catalog.english` starts the new configuration as an exact clone of `english` (same tokenizer, same handling of numbers, emails, URLs, ...), and the `ALTER MAPPING` line is the one substitution: word tokens now route through `portsmith_stem` instead of the stock `english_stem` dictionary.

6.4 — Confirm the noise words are gone

```
SELECT to_tsvector('portsmith_english', body) FROM city_documents WHERE id = 25;

'affect':37 'area':27 'attend':44 'basin':59 'comment':47 'conven':11 'develop':61
'discuss':16 'drain':56 'encourag':42 'engin':53 'flood':17,40,68 'given':4
'herebi':3 'infrastructur':19 'last':65 'mitig':18 'northgat':24 'notic':1
'open':31 'present':50 'propos':60 'provid':46 'public':34 'resid':36
```



```
'respons':63 'retain':25 'retent':58 'riversid':21 'session':14,29 'special':13
'spring':39 'staff':48 'storm':55 'wall':26 'year':66
```

Compare to `to_tsvector('english', body)` on the same row, which still carries `'citi':8,52`, `'council':9`, and `'portsmith':7` — three entries present under `english` and absent under `portsmith_english`. Everything else is untouched, because only the stopwords list changed.

6.5 — Watch a noise-only query collapse to nothing

```
SELECT to_tsquery('portsmith_english', 'city & council');
```

NOTICE: text-search query contains only stop words or doesn't contain lexemes, ignored
to_tsquery

(1 row)

Under `portsmith_english`, “city” and “council” are *both* stopwords, so a query built from nothing else has nothing left to search for — PostgreSQL says so explicitly instead of silently matching everything (or nothing) for an unclear reason.

6.6 — See it fix a real plain-text search

This is the exercise’s payoff: a resident searching for “Portsmouth dog park council” — a completely natural way to phrase it — under `english`:

```
SELECT plainto_tsquery('english', 'Portsmouth dog park council');
```

plainto_tsquery

'portsmith' & 'dog' & 'park' & 'council'

```
SELECT id, title FROM city_documents
WHERE search_vector @@ plainto_tsquery('english', 'Portsmouth dog park council')
ORDER BY id;
```

id	title
24	Public Notice – Riverside Dog Park Ribbon-Cutting Event

(1 row)

Only one hit. The council minutes that actually approved funding for the same dog park (document 6) never happens to use the literal word “Portsmouth,” so requiring it as an AND term silently excludes the single most relevant record. Under `portsmith_english`:

```
SELECT plainto_tsquery('portsmith_english', 'Portsmouth dog park council');
```

plainto_tsquery

'dog' & 'park'

```
SELECT id, title FROM city_documents
WHERE search_vector @@ plainto_tsquery('portsmith_english', 'Portsmith dog park council')
ORDER BY id;
```

```
id | title
---+-----
 6 | Council Minutes – Riverside Dog Park Funding
24 | Public Notice – Riverside Dog Park Ribbon-Cutting Event
(2 rows)
```

Stripped down to the two words that actually distinguish this search — `dog` and `park` — both relevant documents come back. This is the concrete argument for a custom configuration: it is not a cosmetic tweak, it changes which documents a real user actually finds.

Note: the `search_vector` column built in Exercise 3 was populated with the stock `english` configuration, so it still contains `'citi'`, `'council'`, and `'portsmith'` lexemes. The comparisons above work because `@@` only needs the *query's* surviving lexemes to be a subset of the document's — extra lexemes in the document that the query no longer asks for are simply irrelevant. To fully adopt `portsmith_english` as the table's search configuration going forward, you would update the trigger from Exercise 3 to call `to_tsvector('portsmith_english', ...)` and rebuild `search_vector` for existing rows.

Summary — What You Should Now Know

Tool	What it does
<code>to_tsvector(config, text)</code>	Tokenizes, lowercases, stems, and removes stopwords, producing a searchable <code>lexeme:position</code> vector
<code>to_tsquery(config, expr)</code>	Parses a boolean query language (<code>&</code> , <code>\ </code> , <code>!</code> , <code><-></code>); errors on plain text with no operators
<code>plainto_tsquery(config, text)</code>	Tokenizes plain text the same way as <code>to_tsvector</code> , then ANDs every surviving lexeme — safe for a search box
<code>@@</code>	Tests whether a <code>tsvector</code> matches a <code>tsquery</code>
<code>ts_debug(config, text)</code>	Shows exactly how each token was classified and which lexeme (if any) it produced — the tool for “why didn’t this match?”
GIN index on a <code>tsvector</code> column	Turns <code>@@</code> from a sequential scan into an index lookup
<code>ts_rank / ts_rank_cd</code>	Score match quality by term frequency, or by term frequency and proximity (“cover density”)
<code>ts_headline(config, text, query, options)</code>	Re-scans the source text and returns a highlighted snippet — expensive; use only on displayed results

Tool

What it does

CREATE TEXT SEARCH

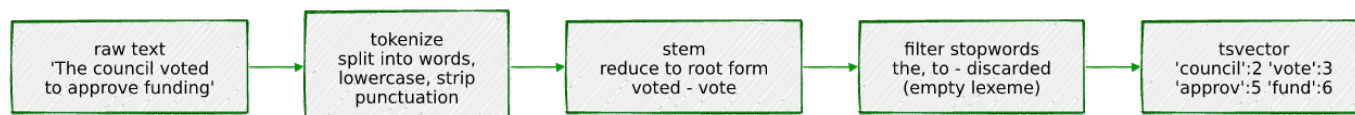
DICTIONARY ... (STOPWORDS

= ...) + CREATE TEXT SEARCH

CONFIGURATION

Build a custom configuration with domain-specific stopwords

The key design insight from this chapter is that full-text search is not one function call — it's a pipeline:



You can inspect every stage of that pipeline with `ts_debug`, store its output with a maintained column and a GIN index, query against it with two very different trade-offs (`to_tsquery` for precision, `plainto_tsquery` for safety), and tune for your specific corpus by swapping the stopwords list — exactly the kind of tuning a generic search engine bolted on top of your database can't do without shipping your schema knowledge to it.

The `city_documents` table you built here is reused directly in later chapters: Chapter 6 (`pgvector`) adds embeddings to the same table for semantic search and builds a hybrid search combining `ts_rank` with cosine distance, and Chapter 16 (Generated Columns) replaces the hand-written trigger from Exercise 3 with a `GENERATED ALWAYS AS (...) STORED` column.

Going further: `websearch_to_tsquery` (PostgreSQL 11+) is worth knowing about even though this chapter didn't use it — it accepts the same quoted-phrase and `-exclude` syntax as a typical web search box (e.g. "flood mitigation" -infrastructure) while remaining as forgiving as `plainto_tsquery`, making it the best default for a real user-facing search field. For very large document sets, look into PostgreSQL's built-in support for weighted vectors (`setweight()`, ranking title matches above body matches) and consider whether a dedicated search engine becomes worthwhile once ranking quality and query latency requirements outgrow what a GIN index over a single table can deliver — the honest answer, for most applications, is later than you'd expect.

Chapter 5 — Fuzzy Matching: `pg_trgm`

“Every registry that outlives a year of real data entry ends up with the same person in it twice, spelled two different ways.”

Background

Full-text search, from the last chapter, is built for a specific kind of imprecision: the same *word*, inflected differently (“vote”, “voted”, “voting”). It does nothing for a different, equally common kind of imprecision — the same word, **spelled** differently. “Portsmith” typed as “Portsmith” versus “Portsmyth”. “McAllister” versus “MacAllister”. A resident registry filled in by hand, twice, by two different clerks, on two different days. Stemming cannot fix a typo, because a typo isn’t a grammatical variant of the correct word — it’s a different string that a human reader recognizes as “close enough” and a computer, by default, does not.

`pg_trgm` is PostgreSQL’s answer: it breaks strings into overlapping three-character sequences (**trigrams**) and measures how many trigrams two strings share. Two spellings of the same name share most of their trigrams even when several letters differ; two unrelated strings usually share almost none. That single idea — measured, thresholded, and indexed — is enough to power “did you mean?” search boxes, deduplicate messy records, and accelerate substring `LIKE` queries that would otherwise force a sequential scan.

This chapter builds a small deduplication and search tool around that idea: what a trigram actually is, how `similarity()` scores a pair of strings, how to turn that into an index instead of an $O(n^2)$ scan, and — the harder, more honest part — where trigram matching’s judgment calls are, because no similarity threshold gets every case right.

The Scenario

Portsmith’s resident registry has been filled in over years by different staff at different counters, and it shows: the same person appears twice under two spellings of their name often enough that nobody trusts a raw `COUNT(*)` on it anymore. Separately, the city’s 311 line fields phone searches for local businesses, and callers rarely spell a business name correctly on the first try.

Two tables model this:

Table	Purpose
<code>residents</code>	Synthetic residents, including intentional near-duplicate entries — the same person, typed twice, with a typo, transposition, or variant spelling the second time
<code>business_names</code>	A flat (<code>business_id</code> , <code>name</code>) lookup extending the <code>businesses</code> table from Chapter 1, used for “did you mean?” search

`residents` includes a `true_duplicate_of` column that a real registry would **not** have — you would not know in advance which rows are duplicates; that is the entire problem fuzzy matching exists to solve. It’s here purely so the exercises can check whether your queries found the right answers. The dataset also includes two pairs that are deliberately *not* marked as duplicates, on purpose — more on those in Exercise 2.

Exercise Goals

By the end of this chapter you will be able to:

- Explain what a trigram is and compute `similarity()` between two strings by hand for a short example.
 - Use the `%` operator to find near-matches at a configurable similarity threshold, and explain why no single threshold is perfectly correct.
 - Use `word_similarity()` to find a short query as a strong partial match inside a longer string, and explain how it differs from `similarity()`.
 - Create a GIN trigram index and confirm it accelerates both `%` and `LIKE '%term%'` queries that would otherwise force a sequential scan.
 - Build a “did you mean?” query using a GiST trigram index and `ORDER BY ... <->`, and explain why GiST — not GIN — is the right index for that access pattern.
 - Compare trigram matching against full-text search on the same kind of short, keyword-style query, and know which one to reach for.
-

Installation

`pg_trgm` ships as part of the standard `postgresql-16` package on Debian/Ubuntu — unlike PostGIS in Chapter 2, there is no separate package to install. Enable it in the database:

```
CREATE EXTENSION IF NOT EXISTS pg_trgm;
```

Loading the Data

Prerequisites

Chapter 1’s seed script must have been run first — `business_names` is populated directly from the `businesses` table:

```
python data/ch01_seed.py
```

Run the Chapter 5 seed

```
python data/ch05_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Creating schema ...
Inserting 58 residents ...
Populating business_names from businesses ...
Done - 58 rows in residents, 48 rows in business_names.
```

Verify the load

Open `psql portsmith` and run these checks.

Check 1 — table structure:

```
\d residents

Table "public.residents"
  Column      | Type   | Collation | Nullable | Default
-----+-----+-----+-----+-----
 id           | integer |           | not null | nextval('residents_id_seq'::regclass)
 full_name    | text    |           | not null |
 neighbourhood | text    |           | not null |
 true_duplicate_of | integer |           |          |
Indexes:
    "residents_pkey" PRIMARY KEY, btree (id)
Foreign-key constraints:
    "residents_true_duplicate_of_fkey" FOREIGN KEY (true_duplicate_of) REFERENCES residents(id)
Referenced by:
    TABLE "residents" CONSTRAINT "residents_true_duplicate_of_fkey" FOREIGN KEY (true_duplicate_of) REFERENCES residents(id)
```

Check 2 — every marked duplicate points at a real canonical row:

```
SELECT COUNT(*) AS duplicate_pairs
FROM residents
WHERE true_duplicate_of IS NOT NULL;
```

```
duplicate_pairs
-----
12
```

Check 3 — business_names mirrors businesses 1:1:

```
SELECT COUNT(*) FROM business_names;
```

```
count
-----
48
```

If all three match, proceed to the exercises.

Exercises

Exercise 1 — What a Trigram Is, and `similarity()`

1.1 — Break a string into trigrams

`show_trgm()` returns the actual set of trigrams PostgreSQL generates for a string:

```
SELECT show_trgm('Eleanor');
```

```
show_trgm
-----
{"  e", " e1", ano, ean, ele, lea, nor, "or "}
```

Before splitting, PostgreSQL pads the string with two leading spaces and one trailing space — " eleanor " — then takes every overlapping run of three characters: " e", " e1", "ele", "lea", "ean", "ano", "nor", "or ". The padding matters: it means the first and last letters of a word each get their own distinguishing trigram (" e" marks “starts with e”; "or " marks “ends with or”), so two strings that only differ at the very start or end still register as different rather than accidentally looking identical in the middle.

padded string: two leading spaces, one trailing space

.	.	e	l	e	a	n	o	r	.
---	---	---	---	---	---	---	---	---	---

. . e l e a n o r . " e" window 1
. . e l e a n o r . " el" window 2
. . e l e a n o r . ele window 3
. . e l e a n o r . lea window 4
. . e l e a n o r . ean window 5
. . e l e a n o r . ano window 6
. . e l e a n o r . nor window 7
. . e l e a n o r . "or " window 8

10 characters, 8 overlapping windows of 3 ($10 - 3 + 1$)

1.2 — similarity(): how much overlap, as a fraction

```
SELECT similarity('Eleanor Whitmore', 'Elenor Whitmore');
```

```
similarity
-----
0.7368421
```

similarity() compares the trigram sets of both strings and returns roughly the fraction that overlap — mostly shared trigrams (both names are 16-17 characters, differing by one dropped letter) means a high score close to 1. Two unrelated strings — “Eleanor Whitmore” and, say, “Ironside Auto” — share almost no trigrams and score close to 0. The scale is intuitive by construction: 1.0 is identical, 0.0 is nothing alike.

1.3 — All twelve duplicate pairs, scored

```
SELECT a.full_name AS canonical, b.full_name AS duplicate_entry,
       round(similarity(a.full_name, b.full_name)::numeric, 3) AS sim
FROM   residents a
JOIN   residents b ON b.true_duplicate_of = a.id
ORDER BY a.id;
```

canonical	duplicate_entry	sim
Eleanor Whitmore	Elenor Whitmore	0.737

Jonathan Castellano		Jonathon Castellano		0.739
Priyanka Deshmukh		Priyanka Deshmuk		0.842
Bartholomew Okonkwo		Bartholemew Okonkwo		0.739
Marguerite Delacroix		Marguerite Delacroix		0.792
Siobhan McAllister		Siobhan MacAllister		0.773
Theodore Vance		Theodor Vance		0.813
Anastasia Volkov		Anastassia Volkov		0.842
Desmond Okafor		Desmund Okafor		0.667
Genevieve Laurent		Genevieve Lorent		0.667
Mikhail Petrenko		Mikail Petrenko		0.737
Fitzgerald Osei		Fitzgerld Osei		0.722

Every genuine duplicate in this dataset scores well above 0.6, regardless of whether the typo was a dropped letter, a swapped letter, or a spelling variant. That range is the working intuition Exercise 2 turns into an actual threshold.

Exercise 2 — The % Operator, and Why One Threshold Isn’t Enough

2.1 — %: “similar enough,” using a session-wide threshold

Computing `similarity()` for every pair by hand doesn’t scale. The `%` operator wraps it into a boolean test against a configurable cutoff:

```
SHOW pg_trgm.similarity_threshold;

pg_trgm.similarity_threshold
-----
0.3
```

0.3 is the default. Self-join `residents` against itself to find every pair of *different* rows that clears it — this is the actual “find likely duplicates” query:

```
SELECT a.id, a.full_name, b.id, b.full_name,
       round(similarity(a.full_name, b.full_name)::numeric, 3) AS sim
FROM   residents a
JOIN   residents b ON a.id < b.id
WHERE  a.full_name % b.full_name
ORDER BY sim DESC, a.id;
```

id	full_name	id	full_name	sim
35	Priyanka Deshmukh	36	Priyanka Deshmuk	0.842
45	Anastasia Volkov	46	Anastassia Volkov	0.842
43	Theodore Vance	44	Theodor Vance	0.813
39	Marguerite Delacroix	40	Marguerite Delacroix	0.792
41	Siobhan McAllister	42	Siobhan MacAllister	0.773
57	Nadia Kowalski	58	Nadia Kowalska	0.765
33	Jonathan Castellano	34	Jonathon Castellano	0.739
37	Bartholomew Okonkwo	38	Bartholemew Okonkwo	0.739

31 Eleanor Whitmore	32 Elenor Whitmore	0.737
51 Mikhail Petrenko	52 Mikail Petrenko	0.737
53 Fitzgerald Osei	54 Fitzgerld Osei	0.722
47 Desmond Okafor	48 Desmund Okafor	0.667
49 Genevieve Laurent	50 Genevieve Lorent	0.667
55 Robert Ashworth	56 Bobby Ashworth	0.409

(14 rows)

`a.id < b.id` keeps each pair once instead of twice (A-vs-B and B-vs-A are the same comparison). Notice: **fourteen** rows came back, not twelve. Check `true_duplicate_of` on every row involved and you'll find twelve of these pairs marked as genuine duplicates and two that are not — "Nadia Kowalski" / "Nadia Kowalska" and "Robert Ashworth" / "Bobby Ashworth". They're in this dataset on purpose, and — this is the point of the exercise — look at *where* they landed in the ranking.

2.2 — A false positive sitting in the middle of the true positives

"Nadia Kowalski" and "Nadia Kowalska" are two unrelated Riverside residents who happen to share a first name and a Polish surname that differs only in its masculine/feminine ending. Their similarity, 0.765, isn't an edge case at the bottom of the list — it sits **between** "Siobhan McAllister" / "Siobhan MacAllister" (0.773) and "Bartholomew Okonkwo" / "Bartholemew Okonkwo" (0.739), two genuine duplicates. `similarity()` is doing exactly what it's designed to do: these two strings really are that close. Whether that means "probably the same person" is a judgment call the function cannot make on its own — the central limitation of fuzzy matching. It finds **candidates**, not verified duplicates. A production dedup pipeline treats a % match as "flag for review" or "compare a second field too" (a shared phone number or address), never as "merge automatically."

2.3 — Prove no threshold fixes it

Try to set a threshold that excludes the Kowalski pair:

```
SET pg_trgm.similarity_threshold = 0.77;
```

Re-run the query from 2.1. "Nadia Kowalski" / "Nadia Kowalska" (0.765) is gone — but so are seven of the twelve genuine duplicates, everything scoring below 0.77. There is no number you can put in that `SET` statement that keeps all twelve real duplicates and excludes the Kowalski pair, because a real duplicate ("Bartholomew Okonkwo" / "Bartholemew Okonkwo", 0.739) scores *lower* than the false positive you're trying to exclude. Put the threshold back before continuing:

```
SET pg_trgm.similarity_threshold = 0.3;
```

2.4 — A different failure: the false negative you'd never even see

"Robert Ashworth" and "Bobby Ashworth" are, in this dataset's backstory, the same person — entered once with a nickname. Notice they only made the results list at all (0.409) because of the shared surname "Ashworth"; as given names alone, "Robert" and "Bobby" share nothing:

```
SELECT similarity('Robert', 'Bobby');
```

```
similarity
-----
0
```

A resident with a *less* distinctive shared surname and a nicknamed given name would score near zero overall and never appear in a % results list at any threshold — not a borderline case to review, just silently absent. `similarity()` measures string closeness, not identity; it has no way to know “Bobby” is short for “Robert” unless something else tells it, such as a synonym table consulted alongside it.

2.5 — The honest takeaway

Between the Kowalski pair and the Ashworth pair, this dataset has one false positive that no threshold can cleanly exclude without also losing real duplicates, and one true duplicate that scores so low it would never surface as a candidate in the first place. Tune `pg_trgm.similarity_threshold` to trade recall against precision for your own data — but go in expecting a trade-off, not a number that gets both sides to zero.

Exercise 3 — `word_similarity()` for Partial Matches

`similarity()` compares two whole strings — it penalizes a short query against a long target just for being different lengths, which makes it a poor fit for “does this short input appear as a strong match somewhere inside this longer string?”

3.1 — Compare the two functions on the same query

```
SELECT name, round(similarity('bake', name)::numeric, 3) AS full_sim
FROM   business_names
ORDER BY full_sim DESC, name
LIMIT 3;
```

name	full_sim
River Bend Bakery	0.222
Campus Bike & Sports	0.091
Finch & Sons Barbers	0.091

```
SELECT name, round(word_similarity('bake', name)::numeric, 3) AS word_sim
FROM   business_names
ORDER BY word_sim DESC, name
LIMIT 3;
```

name	word_sim
River Bend Bakery	0.800

```
Bay Street Electronics | 0.400
Finch & Sons Barbers   | 0.400
```

Same query, same top result, very different score: 0.222 versus 0.800. `word_similarity()` finds the best-matching *substring extent* inside the target — effectively “if I could crop this longer string down to the part that best matches my query, how similar would that piece be?” — rather than scoring the query against the target’s full length. “bake” against “River Bend **Bakery**” scores high because “Bake” is a near-perfect fragment match, even though “bake” is a small fraction of the full business name.

3.2 — When to reach for which

Use `similarity()` when you’re comparing two values that *should* represent the same whole thing — two spellings of one name, as in Exercises 1 and 2. Use `word_similarity()` when a short query is expected to be a fragment of a longer field — autocomplete-style search-as-you-type, or matching a partial business name a caller remembers.

Exercise 4 — GIN Trigram Indexes

4.1 — Without an index

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT name FROM business_names WHERE name LIKE '%Bakery%';
```

QUERY PLAN

```
-----
Seq Scan on business_names (cost=0.00..25.88 rows=1 width=32) (actual time=0.008..0.009
rows=1 loops=1)
  Filter: (name ~~ '%Bakery% '::text)
  Rows Removed by Filter: 47
  Buffers: shared hit=1
```

A plain B-tree index cannot help here — `%Bakery%` has no fixed prefix, so there’s nothing for a B-tree to seek to. This is exactly the case `pg_trgm` was built for: it indexes every trigram in every row, so a `LIKE` pattern (itself broken into trigrams) can be looked up directly.

4.2 — Create the index

```
CREATE INDEX idx_business_names_trgm
ON business_names
USING GIN (name gin_trgm_ops);
```

4.3 — Confirm it’s used

As in earlier chapters, force the index on a table this small to see the mechanism:

```

SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT name FROM business_names WHERE name LIKE '%Bakery%';

SET enable_seqscan = on;  -- always restore this

```

QUERY PLAN

```

-----
Bitmap Heap Scan on business_names  (cost=21.82..25.83 rows=1 width=32) (actual
time=0.024..0.025 rows=1 loops=1)
  Recheck Cond: (name ~~ '%Bakery% '::text)
  Heap Blocks: exact=1
  Buffers: shared hit=10
    -> Bitmap Index Scan on idx_business_names_trgm  (cost=0.00..21.82 rows=1 width=0) (actual
time=0.016..0.017 rows=1 loops=1)
      Index Cond: (name ~~ '%Bakery% '::text)
      Buffers: shared hit=9

```

Bitmap Index Scan on idx_business_names_trgm — the same GIN index also accelerates %, word_similarity() %>, and both LIKE and ILIKE with leading wildcards, all from one index.

Exercise 5 — A “Did You Mean?” Query with GiST

5.1 — The pattern: order by trigram distance, take the top few

pg_trgm defines a distance operator, <-> — the complement of similarity (smaller means more alike) — which makes “closest matches first” a plain ORDER BY ... LIMIT:

```

SELECT name, round(similarity(name, 'Ironsyde Auto')::numeric, 3) AS sim
FROM   business_names
ORDER BY name <-> 'Ironsyde Auto', name
LIMIT 5;

```

name	sim
Ironside Auto	0.647
AutoFix Portsmouth	0.143
Harbour Inn	0.040
The Art Depot	0.037
Riverside Cinema	0.033

A caller who typed “Ironsyde Auto” (transposed letters) gets “Ironside Auto” back as the clear top match, well clear of the noise below it — this is the entire “did you mean?” feature.

5.2 — Why the GIN index from Exercise 4 doesn’t help here

```

EXPLAIN (ANALYZE)
SELECT name FROM business_names

```

```
ORDER BY name <-> 'Ironsyde Auto'
LIMIT 5;
```

QUERY PLAN

```
Limit (cost=2.40..2.41 rows=5 width=36) (actual time=0.182..0.183 rows=5 loops=1)
-> Sort (cost=2.40..2.52 rows=48 width=36) (actual time=0.181..0.182 rows=5 loops=1)
    Sort Key: ((name <-> 'Ironsyde Auto')::text)
    Sort Method: top-N heapsort  Memory: 25kB
    -> Seq Scan on business_names (cost=0.00..1.60 rows=48 width=36) (actual
time=0.040..0.157 rows=48 loops=1)
```

Even with `idx_business_names_trgm` in place, this plan is a full scan plus a sort — GIN accelerates *lookups* (“which rows contain trigrams matching this pattern”) but has no notion of a distance ordering. Nearest-neighbor queries need an index that natively understands “closest,” which is what **GiST** provides:

```
CREATE INDEX idx_business_names_trgm_gist
ON business_names
USING GIST (name gist_trgm_ops);
```

```
EXPLAIN (ANALYZE)
SELECT name FROM business_names
ORDER BY name <-> 'Ironsyde Auto'
LIMIT 5;
```

QUERY PLAN

```
Limit (cost=0.14..1.07 rows=5 width=36) (actual time=0.130..0.143 rows=5 loops=1)
-> Index Scan using idx_business_names_trgm_gist on business_names (cost=0.14..9.09
rows=48 width=36) (actual time=0.129..0.141 rows=5 loops=1)
    Order By: (name <-> 'Ironsyde Auto')::text)
```

Order By: `(name <-> 'Ironsyde Auto')::text` inside an Index Scan — the GiST index walks straight to the nearest rows instead of scoring and sorting every row in the table. **Rule of thumb: GIN for %/LIKE membership lookups, GiST for ORDER BY ... <-> ... LIMIT n nearest-match queries.** It’s common to keep both, as this table now does, if an application needs both access patterns.

5.3 — One more example

```
SELECT name, round(similarity(name, 'Portsmouth Veterinary Clinic')::numeric, 3) AS sim
FROM business_names
ORDER BY name <-> 'Portsmouth Veterinary Clinic', name
LIMIT 5;
```

name	sim
Portsmouth Veterinary Clinic	0.833
AutoFix Portsmouth	0.286
Portsmouth Pharmacy	0.286

Exercise 6 — Trigram Matching vs. Full-Text Search, Head to Head

Chapter 4 built full-text search over `city_documents`; this chapter built trigram matching over `business_names`. Both can answer “find rows matching this word” — they are good at it for different, almost opposite, reasons.

6.1 — A spelling variant: trigram wins

Portsmouth’s own documents consistently use the British spelling “harbour.” A caller who types the American spelling “harbor” gets nothing from full-text search — stemming normalizes *inflection* (“harbors” → “harbor”), not *spelling*:

```
SELECT to_tsvector('english', 'Harbour View Theater') @@ to_tsquery('english', 'harbor');

?column?
-----
f
```

Trigram similarity doesn’t care that they’re “different words” — it only sees how many three-letter fragments overlap, and “harbor”/“harbour” share almost all of theirs:

```
SELECT similarity('harbor', 'harbour');

similarity
-----
0.5
```

6.2 — A short, correctly-spelled keyword: full-text search wins

Run a short, exact keyword against `business_names` both ways:

```
SELECT name, round(similarity(name, 'bay')::numeric, 3) AS sim
FROM business_names
ORDER BY sim DESC, name
LIMIT 6;
```

name	sim
Mango Bay Caribbean	0.200
Bay Street Electronics	0.174
River Bend Bakery	0.105
Finch & Sons Barbers	0.095
Bella Napoli	0.063
Le Petit Bistro	0.053

A three-letter query like “bay” barely produces any trigrams of its own, so the scores are all low and close together — “River Bend Bakery” and “Finch & Sons Barbers” show up with real, if small, similarity despite having nothing to do with “bay.” There’s no clean gap between signal and noise. Full-text search, which matches on whole tokens rather than character fragments, has no such problem:

```
SELECT name FROM business_names
WHERE to_tsvector('english', name) @@ plainto_tsquery('english', 'bay')
ORDER BY name;
```

```
name
-----
Bay Street Electronics
Mango Bay Caribbean
(2 rows)
```

Exactly the two relevant matches, no noise, no threshold to tune.

6.3 — The rule of thumb

Short queries made of correctly-spelled whole words belong to full-text search — it was built to match tokens precisely and cheaply via a GIN index over lexemes. Queries that might be misspelled, transposed, or OCR-damaged belong to trigram matching — it was built to tolerate exactly that kind of noise, at the cost of weaker signal on very short inputs. Many real search boxes run both: try full-text search first, and fall back to a trigram “did you mean?” only when it returns nothing.

Summary — What You Should Now Know

Tool	What it does
<code>show_trgm(text)</code>	Shows the padded, overlapping 3-character sequences a string breaks into
<code>similarity(a, b)</code>	Fraction of shared trigrams between two whole strings — best for “are these two values the same thing, spelled differently?”
<code>word_similarity(a, b)</code>	Best-matching substring extent of a inside b — best for “is a a fragment somewhere inside b?”
<code>a % b</code>	Boolean test: does <code>similarity(a, b)</code> clear <code>pg_trgm.similarity_threshold</code> (default 0.3)?
<code>a <-> b</code>	Trigram distance (<code>1 - similarity</code>) — sort by this for nearest-match-first results
<code>GIN (col gin_trgm_ops)</code>	Accelerates <code>%</code> and <code>LIKE/ILIKE</code> membership-style lookups

Tool	What it does
<code>GIST (col gist_trgm_ops)</code>	Accelerates <code>ORDER BY col <-> query LIMIT n</code> nearest-neighbor lookups
<code>pg_trgm.similarity_threshold</code>	Session-level cutoff for %; tune it, but expect trade-offs, not a perfect split

The key design insight from this chapter is that fuzzy matching produces *candidates*, not verdicts. The % operator surfaced all twelve real duplicate pairs in the `residents` table — and two pairs of genuinely different, or differently-named, people right alongside them, at every threshold tried. That isn’t a shortcoming to engineer away; it’s the nature of measuring string similarity instead of identity. Build fuzzy matching into a review queue or a secondary confirmation step, not an automatic merge.

The `business_names` table you built here is reused in Chapter 10 (PostgREST), which exposes the “did you mean?” query from Exercise 5 as a public RPC endpoint.

Going further: `pg_trgm` also provides `%>` and `<->>` variants tuned for `word_similarity()` rather than `similarity()`, useful for indexing substring-style “did you mean?” search the same way Exercise 5 indexed whole-string search. For very large tables, GiST trigram indexes are typically smaller and faster to build than GIN but slower for pure membership lookups — benchmark both on your actual data distribution rather than assuming one is strictly better. And if deduplication is a recurring, business-critical process rather than an occasional query, look at dedicated record-linkage tools (e.g. the `dedupe` Python library), which combine trigram-style string similarity with other fields — address, phone number, date of birth — and a trained classifier, instead of a single threshold on a single column.

Chapter 6 — Vector Search: `pgvector` for Embeddings

“Full-text search finds documents that use your words. Vector search finds documents that share your meaning, whether or not they share a single word with you.”

Background

Chapter 4 matched documents by shared vocabulary. Chapter 5 matched strings by shared spelling. Neither technique can find a document about “the city’s dog park” when a resident searches “somewhere my pet can run off-leash” — there isn’t a misspelling to correct or a stemmed word in common; the two phrases are simply *about the same thing*, expressed in unrelated words. No amount of stemming, stopword tuning, or trigram threshold-adjusting closes that gap, because all three earlier techniques operate on the text itself. Closing it requires a representation of what the text *means*.

That representation is an **embedding**: a machine learning model reads a piece of text and outputs a vector — literally an array of a few hundred floating-point numbers — positioned in a high-dimensional space such that texts with similar meaning end up as nearby points and unrelated texts end up far apart. “Somewhere my pet can run off-leash” and “the city’s dog park” land close together in that space even though they share zero words. This chapter never trains such a model — it uses a small, well-established, freely available one (`all-MiniLM-L6-v2`) to turn text into 384-number vectors, and PostgreSQL’s `pgvector` extension to store, index, and search them.

The two hard problems `pgvector` solves are storage (a native `vector` type, instead of smuggling floats through a JSONB array or a comma-separated text column) and **search at scale**. Finding the closest vector to a query by brute force means comparing it against every single row — fine for the thirty documents in Chapter 4’s table, hopeless for a million-row photo library. This chapter builds up to two different **approximate nearest neighbor (ANN)** index types, `IVFFlat` and `HNSW`, that trade a small amount of accuracy for a large amount of speed, and it’s honest about exactly how much accuracy each one actually costs you at its defaults, because the answer is more surprising than most introductions let on.

The Scenario

Two tables carry this chapter, and they get their vectors two very different ways, on purpose.

`city_documents` — the council minutes, zoning ordinances, and public notices from Chapter 4 — gets a real `embedding` column computed by actually running `all-MiniLM-L6-v2` over each document’s title and body. These are genuine embeddings: semantic search over them in Exercise 5 finds real conceptual matches, not a scripted demo.

`city_photos` is different, and says so in its own name: it’s a synthetic stand-in for a photo library’s image embeddings (the kind a real vision model like CLIP would produce for actual photographs), used purely to give Exercises 3 and 4 a table large enough — 5,000 rows across ten categories — for approximate indexing to be worth doing at all. Thirty rows, like `city_documents` has, will never be large enough to make an ANN index pay for itself; a sequential scan over thirty rows is already about as fast as computation gets. There is no real photo behind any row in `city_photos` — each one is a random point clustered around one of ten category “anchors” in 384-dimensional space, which reproduces the *shape* of a real embedding space (similar things cluster; different things don’t) without needing a single actual image.

Table	Vectors	Used for
<code>city_documents</code>	Real — <code>all-MiniLM-L6-v2</code> embeddings of each document’s title + body	Semantic search, hybrid search (Exercises 5-6)
<code>city_photos</code>	Synthetic — 5,000 rows, 10 clustered categories, no real images	ANN indexing at scale (Exercises 3-4)

Exercise Goals

By the end of this chapter you will be able to:

- Store embeddings in a native `vector` column and understand what the numbers in it actually represent.
- Compute exact nearest neighbors with `<->` (L2), `<#>` (negative inner product), and `<=>` (cosine distance), and know when the choice between them actually changes your results.
- Build an IVFFlat index, and correctly interpret its `lists` and `probes` parameters instead of trusting the defaults.
- Build an HNSW index, and make an informed build-time-vs-recall trade-off between it and IVFFlat instead of guessing.
- Run real semantic search: find documents by meaning even when they share no vocabulary with the query.

- Build hybrid search that blends a keyword score (`ts_rank`, from Chapter 4) with a semantic score into one ranking.
-

Installation

This chapter has two dependencies, and they're independent of each other — you need both, but you're installing two unrelated things for two unrelated reasons.

1 — pgvector

```
sudo apt install -y postgresql-16-pgvector
```

Enable the extension. **Unlike `pg_trgm` in Chapter 5, `pgvector 0.6.x` is not a “trusted” extension** — its control file doesn't set `trusted = true`, so a regular database-owning role cannot enable it with a plain `CREATE EXTENSION`. It has to be done once, by a superuser:

```
sudo -u postgres psql portsmith -c "CREATE EXTENSION vector;"
```

Confirm it's there:

```
SELECT extversion FROM pg_extension WHERE extname = 'vector';
```

```
extversion
-----
0.6.0
```

2 — sentence-transformers, CPU-only

The embedding model runs locally, in Python, via the `sentence-transformers` library — no API key, no network calls at query time. It depends on PyTorch, and installing PyTorch's default package pulls in a full NVIDIA CUDA toolkit (several gigabytes) even on a machine with no GPU, which is pure waste for embedding a few dozen short documents on CPU. Install the CPU-only build explicitly, then `sentence-transformers` on top of it:

```
python3.12 -m venv .venv # if you haven't already, from earlier chapters
source .venv/bin/activate
pip install torch --index-url https://download.pytorch.org/whl/cpu
pip install sentence-transformers pgvector
```

Note: this installs roughly 3GB into your virtual environment (PyTorch and its numerical dependencies are not small, even CPU-only) and the first time any script loads `all-MiniLM-L6-v2`, it downloads about 90MB of model weights from Hugging Face and caches them in `~/.cache/huggingface`. Every run after that is fully offline. If

you see a Warning: You are sending unauthenticated requests to the HF Hub line the first time, that's expected and harmless — it's just Hugging Face reminding you that an account would raise your download rate limit.

Loading the Data

Prerequisites

Chapter 4's seed script must have been run first:

```
python data/ch04_seed.py
```

ch04_seed.py alone is **not** enough, though — this chapter's hybrid search (Exercise 6) needs `city_documents.search_vector`, and that column only gets created by hand-running Chapter 4's **Exercise 3** SQL; it is not part of `ch04_seed.py`'s schema. (If you re-run `ch04_seed.py` after already having done Exercise 3 — say, to reset the document data — it drops `city_documents` and rebuilds it *without* `search_vector`, silently undoing that exercise.) `ch06_seed.py` (below) checks for the column and will stop with the exact fix if it's missing; if you'd rather do it now, here it is verbatim from Chapter 4:

```
ALTER TABLE city_documents ADD COLUMN search_vector tsvector;
```

```
UPDATE city_documents
SET    search_vector = to_tsvector('english', title || ' ' || body);
```

```
CREATE OR REPLACE FUNCTION city_documents_search_vector_update() RETURNS trigger AS $$
BEGIN
    NEW.search_vector := to_tsvector('english', NEW.title || ' ' || NEW.body);
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;
```

```
CREATE TRIGGER trg_city_documents_search_vector
    BEFORE INSERT OR UPDATE OF title, body ON city_documents
    FOR EACH ROW
    EXECUTE FUNCTION city_documents_search_vector_update();
```

```
CREATE INDEX idx_city_documents_search_vector
    ON city_documents USING GIN (search_vector);
```

Run the Chapter 6 seed

This adds the (initially empty) embedding column to `city_documents` and creates and populates `city_photos`. It needs only `psycpg`, `numpy`, and `pgvector` — not `sentence-transformers` — since `city_photos`'s vectors are synthetic:

```
python data/ch06_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Applying DDL ...
Generating 5000 synthetic photo embeddings ...
Done - 5000 rows in city_photos across 10 categories.
city_documents.embedding added but left NULL - run `python data/ch06_embed_documents.py` next.
```

Compute real embeddings for `city_documents`

This is the step that actually loads and runs the model:

```
python data/ch06_embed_documents.py
```

Expected output:

```
Connecting to: dbname=portsmith
Loading all-MiniLM-L6-v2 (first run downloads ~90MB of model weights) ...
Embedding 30 documents ...
Done - 30 rows in city_documents now have an embedding.
```

Verify the load

Open `psql portsmith` and run these checks.

Check 1 — `city_photos` structure:

```
\d city_photos
```

Table "public.city_photos"					
Column	Type	Collation	Nullable	Default	
id	integer		not null	nextval('city_photos_id_seq'::regclass)	
category	text		not null		
neighbourhood	text		not null		
caption	text		not null		
embedding	vector(384)		not null		

Indexes:

```
"city_photos_pkey" PRIMARY KEY, btree (id)
```

Check 2 — 500 photos per category:

```
SELECT category, COUNT(*) FROM city_photos GROUP BY category ORDER BY category;
```

category	count
community_event	500
harbour_waterfront	500
historic_architecture	500
industrial_dock	500
infrastructure_construction	500
municipal_building	500
public_park	500
residential_street	500
street_market	500
wildlife_nature	500

(10 rows)

Check 3 — every city_documents row has an embedding:

```
SELECT COUNT(*) AS total, COUNT(embedding) AS with_embedding FROM city_documents;
```

total	with_embedding
30	30

If all three match, proceed to the exercises.

Exercises

Exercise 1 — The `vector` Type

1.1 — Literals and dimensions

A `vector` literal is a bracketed list of numbers:

```
SELECT '[1,2,3]':vector;
```

vector
[1,2,3]

A `vector(384)` column, like `city_documents.embedding`, enforces its dimension count at write time — every row's vector must have exactly 384 numbers. There is no meaningful way to compare a 384-dimensional embedding to a 3-dimensional one, so PostgreSQL simply won't let a mismatched value in:

```
SELECT '[1,2,3]':vector(384);
```

ERROR: expected 384 dimensions, not 3

1.2 — What’s actually in the column

`city_documents.embedding` isn’t hand-written — it came from `ch06_embed_documents.py` calling `a11-MiniLM-L6-v2` on each document’s title and body. Look at a real one:

```
SELECT id, title, embedding FROM city_documents WHERE id = 1;
```

The embedding value prints as 384 comma-separated floating-point numbers between roughly -1 and 1 — not something a human reads directly, but exactly what every operator and index in this chapter operates on. What those particular 384 numbers *mean* isn’t individually interpretable (no single dimension corresponds to a concept like “is about harbours”); what matters is only their *position relative to other documents’ vectors*, which is what the distance operators in Exercise 2 measure.

Exercise 2 — Exact Nearest Neighbors: <->, <#>, <=>

`pgvector` gives you three distance operators. They agree more often than you’d expect, and the case where they disagree is worth understanding before it surprises you in production.

2.1 — Three tiny vectors, three operators

```
SELECT '[1,0,0]':vector <-> '[0,1,0]':vector AS l2_distance;
```

```
l2_distance
-----
1.4142135623730951
```

<-> is **Euclidean (L2) distance** — straight-line distance between the two points, exactly like distance in ordinary geometry. Two orthogonal unit vectors are $\sqrt{2}$ apart.

```
SELECT '[1,0,0]':vector <#> '[0,1,0]':vector AS neg_inner_product;
```

```
neg_inner_product
-----
-0
```

<#> is the **negative inner product** — the dot product of the two vectors, negated (`pgvector` negates it so that, consistent with the other two operators, *smaller means closer*). Orthogonal vectors have a dot product of zero.

```
SELECT '[1,0,0]':vector <=> '[0,1,0]':vector AS cosine_distance;
```

```
cosine_distance
-----
1
```


`<=>` is **cosine distance** ($1 - \text{cosine similarity}$) — the angle between the two vectors, completely ignoring their length. Orthogonal vectors have maximum cosine distance, 1.

2.2 — Where they disagree: magnitude

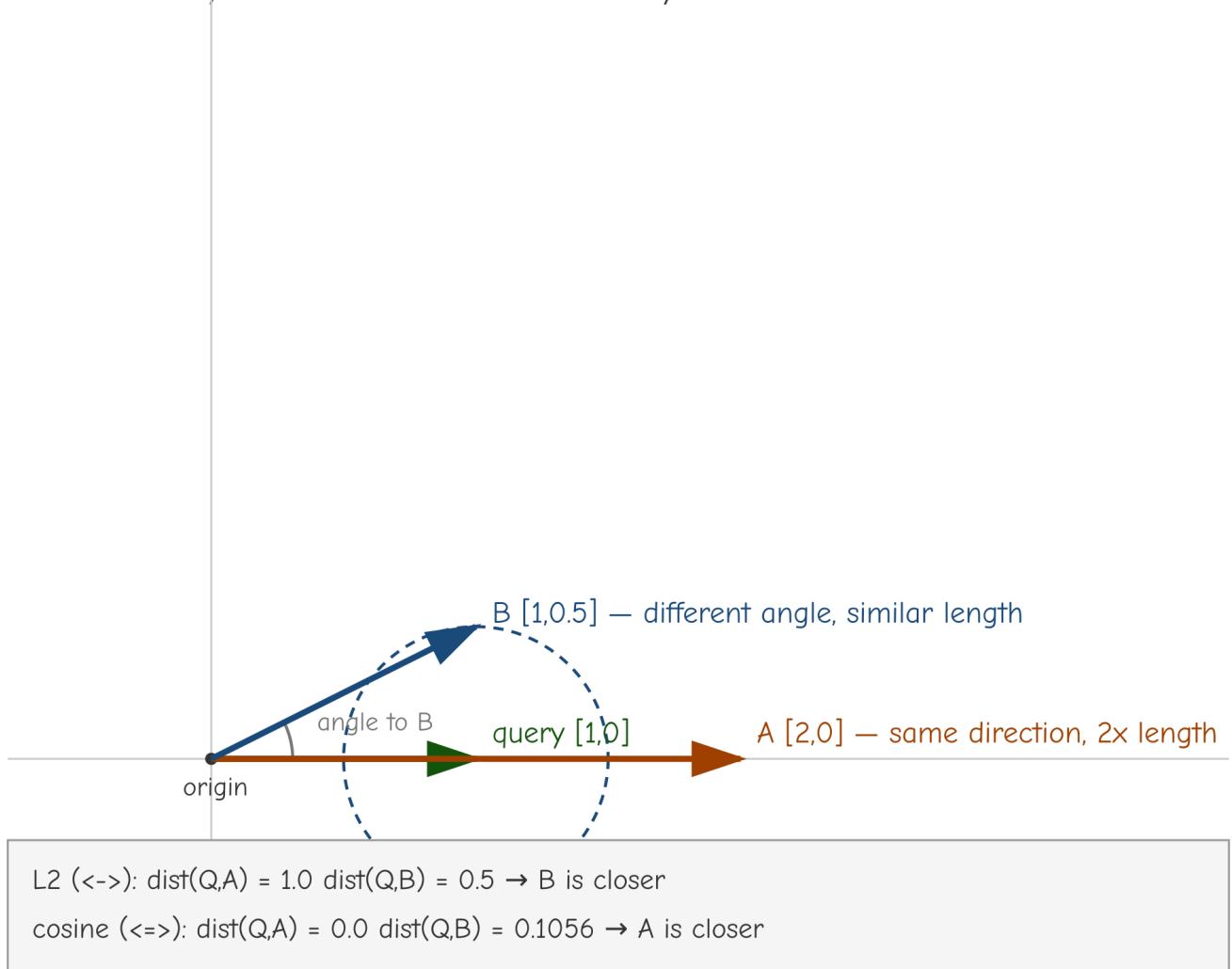
The three operators can rank the *same* pair of candidates in a *different* order the moment the vectors involved aren't the same length. Compare a query vector against two candidates — one that points in exactly the same direction but is twice as long, and one that points in a slightly different direction but is almost the same length:

```
SELECT
  '[1,0]':vector <-> '[2,0]':vector AS l2_to_a,
  '[1,0]':vector <-> '[1,0.5]':vector AS l2_to_b,
  round(('[1,0]':vector <=> '[2,0]':vector)::numeric, 4) AS cos_to_a,
  round(('[1,0]':vector <=> '[1,0.5]':vector)::numeric, 4) AS cos_to_b;

l2_to_a | l2_to_b | cos_to_a | cos_to_b
-----+-----+-----+-----
      1 |      0.5 |    0.0000 |    0.1056
```

The ranking flips. By `<->` (L2), candidate B ([1, 0.5], distance 0.5) is closer than candidate A ([2, 0], distance 1). By `<=>` (cosine), it's the reverse — A is distance 0 (identical direction) while B is 0.1056 away. Neither operator is “wrong”; they’re answering different questions. L2 asks “how far apart are these points.” Cosine asks “how similarly are these vectors oriented, regardless of scale.”

L2 distance says B is closer. Cosine distance says A is closer.



A sits exactly on the query's own line (the overlap in the diagram above is "same direction, twice the length"), which is why its cosine distance is zero. The dashed circle is centered on the query with radius equal to B's L2 distance — B sits right on that boundary, while A sits well outside it, which is the straight-line sense in which B is "closer."

2.3 — Why this chapter normalizes, and what that buys you

`ch06_embed_documents.py` calls `model.encode(texts, normalize_embeddings=True)` — every stored embedding is scaled to unit length before it's written. Once every vector in a comparison has the same length, the magnitude difference that caused Exercise 2.2's disagreement simply can't occur, and all three operators produce the *same ranking* (just on different numeric scales). Embedding the query "waterfront redevelopment funding" (Exercise 5 shows how) and checking the top 3 `city_documents` rows by each operator confirms it:

$\leftrightarrow$ (L2)	$\langle\# \rangle$ (neg. inner product)	$\Leftrightarrow$ (cosine)
Rank 1 doc 1 — 0.8605	doc 1 — -0.6298	doc 1 — 0.3702
Rank 2 doc 22 — 1.0546	doc 22 — -0.4439	doc 22 — 0.5561

`<-> (L2)` `<#> (neg. inner product)` `<=> (cosine)`

Rank 3 doc 5 — 1.0646 doc 5 — -0.4333 doc 5 — 0.5667

Same three documents, same order, every time — only the numbers' scale differs between columns. The practical rule: **normalize your embeddings, and pick whichever operator has an index you can build** — for normalized vectors it stops being a search-quality decision and becomes a performance one, covered next.

Exercise 3 — IVFFlat: Approximate Search with `lists` and `probes`

3.1 — Exact search doesn't scale

`city_photos` has 5,000 rows — small by real standards, but already enough to feel the cost of brute-force search. Find the 10 nearest neighbors of photo `id = 1` by comparing it against every other row:

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT id, category, embedding <-> (SELECT embedding FROM city_photos WHERE id = 1) AS dist
FROM   city_photos
WHERE  id != 1
ORDER BY embedding <-> (SELECT embedding FROM city_photos WHERE id = 1)
LIMIT 10;
```

QUERY PLAN

```
-----
Limit  (cost=1439.32..1439.35 rows=10 width=30) (actual time=4.164..4.167 rows=10 loops=1)
-> Sort  (cost=1431.02..1443.52 rows=4999 width=30) (actual time=4.163..4.164 rows=10
loops=1)
    Sort Key: ((city_photos.embedding <-> $0))
    Sort Method: top-N heapsort  Memory: 26kB
    -> Seq Scan on city_photos  (cost=0.00..1323.00 rows=4999 width=30) (actual
time=0.024..3.381 rows=4999 loops=1)
```

Every one of the 4,999 other rows gets its distance computed and sorted — about 4.6ms here. That cost grows linearly with table size; at real photo library scale (millions of rows) it stops being viable.

3.2 — Build the index

```
CREATE INDEX idx_city_photos_ivfflat
ON city_photos
USING ivfflat (embedding vector_l2_ops)
WITH (lists = 50);
```

```
CREATE INDEX
Time: 152.129 ms
```

lists = 50 partitions the 5,000 vectors into 50 clusters (via k-means at index-build time) and stores each vector under its nearest cluster centroid. A rule of thumb from the pgvector docs: $lists \approx rows / 1000$ for up to roughly a million rows.

3.3 — The planner picks it up on its own

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT id, category, embedding <-> (SELECT embedding FROM city_photos WHERE id = 1) AS dist
FROM   city_photos
WHERE  id != 1
ORDER BY embedding <-> (SELECT embedding FROM city_photos WHERE id = 1)
LIMIT 10;
```

QUERY PLAN

```
-----
Limit  (cost=61.55..62.35 rows=10 width=30) (actual time=0.145..0.168 rows=10 loops=1)
  -> Index Scan using idx_city_photos_ivfflat on city_photos  (cost=53.25..455.00 rows=4999
width=30) (actual time=0.145..0.166 rows=10 loops=1)
    Order By: (embedding <-> $0)
```

No `enable_seqscan = off` trick needed this time, unlike the small tables in earlier chapters — at 5,000 rows the planner's own cost estimate already favors the index: **4.6ms down to about 0.2ms.**

3.4 — The catch: default probes gives up a lot of accuracy

An IVFFlat search only examines the clusters nearest the query vector, not all 50 — controlled by `ivfflat.probes`, which **defaults to 1**. Compare the approximate top-10 against the true exact top-10:

```
-- Exact (force a sequential scan to get true ground truth):
SET enable_indexscan = off;
SELECT array_agg(id) FROM (
  SELECT id FROM city_photos WHERE id != 1
  ORDER BY embedding <-> (SELECT embedding FROM city_photos WHERE id = 1) LIMIT 10
) t;
```

array_agg

```
-----
{146,357,485,468,500,261,384,397,185,226}
```

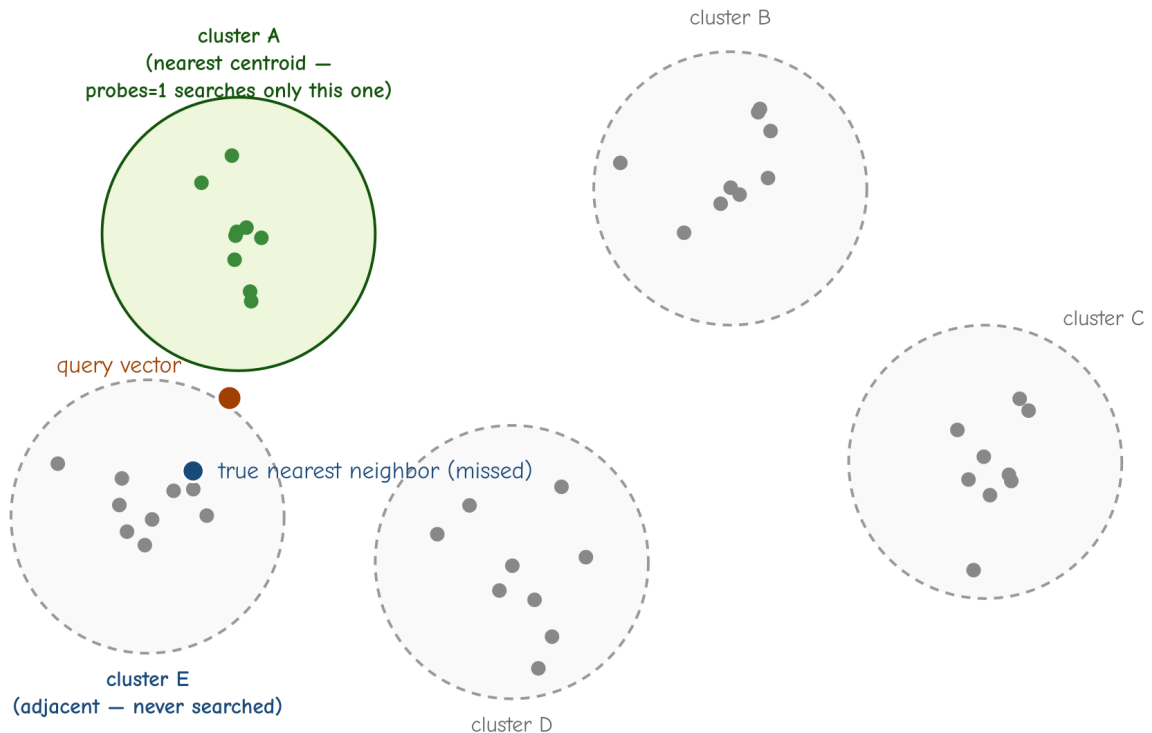
```
-- Approximate, default probes = 1:
RESET enable_indexscan;
SELECT array_agg(id) FROM (
  SELECT id FROM city_photos WHERE id != 1
  ORDER BY embedding <-> (SELECT embedding FROM city_photos WHERE id = 1) LIMIT 10
) t;
```

array_agg

```
-----
{485,261,185,226,385,22,274,433,56,327}
```

Only **4 of the 10** results match (485, 261, 185, 226) — **40% recall**, at the default setting, on a real query. This is not a contrived worst case; it's what `ivfflat.probes = 1` actually does: it searches only the single cluster closest to the query and simply never looks at the other 49, even though some of the true nearest neighbors live in a different cluster.

`probes=1` only searches the nearest cluster, even when a true neighbor sits just across the boundary



IVFFlat clusters vectors at index-build time. `probes=1` checks only the nearest centroid (A) — a real neighbor one cluster over is invisible to the search.

A note on reproducing these exact numbers: IVFFlat's clustering step uses k-means with a random initialization at *index build* time — even against `city_photos`'s fixed, seeded synthetic data, rebuilding the same `CREATE INDEX ... USING ivfflat` statement can shift which photos land in which of the 50 clusters, and therefore which specific IDs and exact recall percentage `probes = 1` produces. Rerunning this exercise on your own machine will very likely show different IDs and a different percentage than the ones printed here. The pattern that reproduces reliably is the *shape*: `probes = 1` is measurably, often substantially, incomplete.

3.5 — Raise `probes`, watch recall recover

```
SET ivfflat.probes = 2;
```

```
array_agg
```

```
{146, 357, 485, 468, 500, 261, 384, 397, 185, 226}
```

At `probes = 2` the result already matches the exact set exactly — **100% recall** on this query. Timing at a slightly more conservative `probes = 5`, and at `probes = 50` (every list — the most exhaustive an IVFFlat search can be):

<code>probes</code>	Recall (this query)	Query time
1 (default)	40%	~0.4ms
5	100%	~0.6ms
50 (all lists)	100%	~0.6ms

Timings vary run to run more than the recall figures do (expect noise in the tenths of a millisecond on a table this size) — the shape that matters is that `probes = 1` is consistently fastest *and* the least reliable, while anything from `probes = 5` up delivers full recall on this query without a further, proportional cost from probing still more lists; the IVFFlat index structure keeps even an “examine every list” search meaningfully faster than the 4.6ms plain sequential scan from Exercise 3.1. Reset before continuing:

```
SET ivfflat.probes = 1;
```

Two probes happened to be enough for this particular query; it isn’t a guarantee for every query, which is why a small safety margin (a starting point of `probes ≈ √lists`, so `√50 ≈ 7`) is the honest recommendation rather than the bare minimum that worked once. The overall lesson: **IVFFlat’s out-of-the-box defaults are fast and unreliable at the same time** — `lists` and `probes` are not optional tuning, they’re the whole feature.

Exercise 4 — HNSW: A Different Trade-off

4.1 — Build it, and time the build

```
CREATE INDEX idx_city_photos_hnsw
ON city_photos
USING hnsw (embedding vector_l2_ops);
```

```
CREATE INDEX
Time: 513.963 ms
```

Compare against IVFFlat’s build time from Exercise 3.2: **152ms**. HNSW (Hierarchical Navigable Small World) builds a multi-layer graph structure connecting each vector to its approximate neighbors, which costs roughly **3x longer to build** than IVFFlat’s k-means clustering, on this table.

4.2 — Recall at HNSW’s default settings

Drop (or temporarily disable) the IVFFlat index so the planner has no choice but to use HNSW, then repeat the exact same top-10 query from Exercise 3.4:

```
BEGIN;
DROP INDEX idx_city_photos_ivfflat;

SELECT array_agg(id) FROM (
  SELECT id FROM city_photos WHERE id != 1
  ORDER BY embedding <-> (SELECT embedding FROM city_photos WHERE id = 1) LIMIT 10
) t;

ROLLBACK; -- put idx_city_photos_ivfflat back
```

array_agg

{146,357,485,468,500,261,384,397,185,226}

Exact match with the true top-10 — 100% recall, at HNSW’s default settings, no tuning required. Query time for this was about 0.63ms. Repeating the comparison against a second, independent query point (`id = 1001`, a `public_park` photo) confirms it isn’t a fluke: HNSW again returned the exact top-10, while IVFFlat at default `probes = 1` again recovered only 3 of 10 (30% recall — as Exercise 3.4 noted, the exact percentage shifts between index rebuilds; what doesn’t shift is that it’s consistently well short of complete). HNSW’s default, across every test in this chapter, has not missed a single true neighbor.

4.3 — The trade-off, stated plainly

	IVFFlat	HNSW
Build time (5,000 rows)	152ms	514ms (~3x)
Recall at defaults	30-40% in this chapter’s tests, varies by build (needs <code>probes</code> tuned up)	~100% (no tuning needed)
Query time (well-tuned)	Comparable	Comparable
Tuning burden	On you, every session (<code>probes</code> is a session-level SET)	Minimal

This matches the general guidance in the pgvector documentation: **HNSW is the better default choice for most workloads** — better recall out of the box, no per-session parameter to remember to set. IVFFlat’s advantages are real but narrower: faster to build (relevant if you rebuild the index often, e.g. after large bulk loads) and somewhat lower memory overhead at very large scale. Choose IVFFlat deliberately for those reasons, not by default.

Exercise 5 — Semantic Search: Finding Meaning, Not Words

5.1 — A query with no vocabulary overlap

`data/ch06_semantic_search.py` embeds a query string with the same model used to build the column, and searches `city_documents` by cosine similarity:

```
python data/ch06_semantic_search.py "stray dogs and pet-friendly spaces"

id      sim  title
6    0.4447 Council Minutes - Riverside Dog Park Funding
24   0.3927 Public Notice - Riverside Dog Park Ribbon-Cutting Event
16   0.2683 Zoning Ordinance - Riverside Short-Term Rental Restrictions
...
```

Not one word of the query — “stray,” “dogs,” “pet-friendly,” “spaces” — appears in document 6’s title, and full-text search (Chapter 4) would return zero rows for it; there’s no shared lexeme to match on, misspelled or otherwise, so Chapter 5’s trigram matching wouldn’t help either. The embedding model has no notion of exact words at all — it read “the city’s dog park” and “somewhere my pet can run off-leash” as *about the same thing*, which is precisely the gap described in this chapter’s Background.

5.2 — A stronger example

```
python data/ch06_semantic_search.py "waterfront redevelopment funding"

id      sim  title
1    0.6298 Council Minutes - Harbour District Waterfront Renovation Budget
22   0.4439 Public Notice - Public Hearing on Harbour District Waterfront Rezoning
5    0.4333 Council Minutes - Riverside Road Resurfacing Program
12   0.4207 Zoning Ordinance - Harbour District Waterfront Height Variance
9    0.4101 Council Minutes - Canal Road Bike Lane Expansion
...
```

5.3 — Reading the gap, not just the ranking

Look at the jump between rank 1 (0.6298) and rank 2 (0.4439) — a much bigger drop than between ranks 2 through 5. That gap is meaningful: it says document 1 isn’t merely the *best available* match, it’s a *genuinely strong* one, while the rest are progressively looser associations. Semantic search has a property full-text search doesn’t: **it never returns zero results**. Every document has *some* cosine similarity to any query — the model will confidently rank all thirty documents even for a nonsense query, it just won’t find any of them close. A real application needs a similarity-score cutoff (or a “confidence” gap check like this one) to distinguish “found it” from “here’s our least-bad guess,” a problem keyword search’s honest zero-rows sidesteps entirely.

Exercise 6 — Hybrid Search: Blending Keyword and Semantic Signals

Semantic search and keyword search fail differently. Semantic search misses exact terms that matter — a specific ordinance number, a street name — treating them as just more words to average into a general impression of meaning. Keyword search misses paraphrases entirely, as Exercise 5 just showed. Hybrid search runs both and blends the scores.

6.1 — Keyword-only ranking

```
SELECT id, title, round(ts_rank(search_vector, plainto_tsquery('english', 'bike
lane'))::numeric,4) AS kw_rank
FROM   city_documents
WHERE  search_vector @@ plainto_tsquery('english', 'bike lane')
ORDER BY kw_rank DESC;
```

id	title	kw_rank
-----+-----		-----+-----
9	Council Minutes – Canal Road Bike Lane Expansion	0.2928
28	Public Notice – Canal Road Bike Lane Construction Schedule	0.1960
30	Public Notice – Annual Fireworks Display and Street Closures	0.1759
5	Council Minutes – Riverside Road Resurfacing Program	0.0992
15	Zoning Ordinance – Reduced Parking Minimums Near Transit Corridors	0.0991

6.2 — Semantic-only ranking, same query

```
python data/ch06_semantic_search.py "bike lane" --top 8
```

id	sim	title
9	0.5513	Council Minutes – Canal Road Bike Lane Expansion
28	0.4433	Public Notice – Canal Road Bike Lane Construction Schedule
15	0.4117	Zoning Ordinance – Reduced Parking Minimums Near Transit Corridors
5	0.3965	Council Minutes – Riverside Road Resurfacing Program
21	0.2754	Public Notice – Bay Street Water Main Repair Road Closure
6	0.2688	Council Minutes – Riverside Dog Park Funding
30	0.2653	Public Notice – Annual Fireworks Display and Street Closures
10	0.2359	Council Minutes – Special Session on Flood Mitigation Infrastructure

Both agree on the top 2 (documents 9 and 28 — a real, strong signal either way finds them). They disagree further down: document 30 ranks **3rd** by keyword (it happens to use the phrase “biking along the Canal Road bike lane” once, in passing) but only **7th** by semantic similarity, while document 15 ranks **5th** by keyword but **3rd** semantically (it’s substantively about bike-lane-adjacent parking policy, using different phrasing throughout).

6.3 — Blend both into one score

```
python data/ch06_semantic_search.py "bike lane" --hybrid --top 8
```

id	kw	sem	hybrid	title
9	0.2928	0.5513	0.4479	Council Minutes – Canal Road Bike Lane Expansion
28	0.1960	0.4433	0.3444	Public Notice – Canal Road Bike Lane Construction Schedule

15	0.0991	0.4117	0.2867	Zoning Ordinance – Reduced Parking Minimums Near Transit Corridors
5	0.0992	0.3965	0.2776	Council Minutes – Riverside Road Resurfacing Program
30	0.1759	0.2653	0.2295	Public Notice – Annual Fireworks Display and Street Closures
21	0.0000	0.2754	0.1652	Public Notice – Bay Street Water Main Repair Road Closure
6	0.0000	0.2688	0.1613	Council Minutes – Riverside Dog Park Funding
10	0.0000	0.2359	0.1415	Council Minutes – Special Session on Flood Mitigation Infrastructure

The underlying SQL (HYBRID_SQL in ch06_semantic_search.py) is a LEFT JOIN between a keyword-matched candidate set and a semantic top-10 candidate set, so documents 21, 6, and 10 — never matched by the keyword query at all (kw = 0.0000) — still appear, carried in purely by semantic similarity:

```
hybrid_score = 0.4 * COALESCE(kw_score, 0) + 0.6 * COALESCE(sem_score, 0)
```

Document 15 moves from 5th (keyword) up to 3rd (hybrid); document 30 moves from 3rd (keyword) down to 5th (hybrid) — the blend genuinely changes the ranking, not just the score.

6.4 — The honest caveat about the weights

0.4 and 0.6 here are fixed constants chosen because ts_rank and cosine similarity happen to land in roughly comparable numeric ranges on *this* dataset — they are not principled, and they are not portable to a different corpus or a different embedding model without re-checking that assumption. A production hybrid-search system typically normalizes each signal (min-max or z-score, over the actual candidate set returned) before blending, rather than trusting that two arbitrary scoring functions happen to land in the same numeric neighborhood. Treat the weights here as a teaching simplification, not a formula to copy into production untested.

Summary — What You Should Now Know

Tool	What it does
vector(N)	A fixed-dimension floating-point array column type; enforces N at write time
<->	L2 (Euclidean) distance — straight-line distance between points
<#>	Negative inner product — dot product, negated so smaller means closer
<=>	Cosine distance — angle between vectors, ignoring magnitude
normalize_embeddings=True	Makes all three operators agree on ranking; without it, they can disagree

Tool	What it does
<code>USING ivfflat (col vector_l2_ops) WITH (lists = N)</code>	Clusters vectors into <code>N</code> buckets; fast to build, needs probes tuned up from its default to get usable recall
<code>SET ivfflat.probes = N</code>	How many clusters an IVFFlat search actually examines — defaults to 1, often too low
<code>USING hnsw (col vector_l2_ops)</code>	Graph-based ANN index; ~3x slower to build than IVFFlat here, but near-exact recall by default
<code>ts_rank(...) + (1 - cosine_distance) weighted blend</code>	Hybrid search — keyword precision plus semantic recall in one ranking

The key design insight from this chapter is that “approximate” in “approximate nearest neighbor” is not a rounding error — at IVFFlat’s default settings, this chapter repeatedly measured 30-40% recall on real queries, which means 6-7 of the top 10 *true* nearest neighbors were silently missing from the results, with no error, no warning, nothing to indicate anything was wrong. Every ANN index trades recall for speed on a dial you control (`probes`, `ef_search`); the failure mode to avoid is not knowing where that dial is set.

The `city_documents` table’s embedding column is reused in Chapter 10 (PostgREST), which exposes the semantic search query from Exercise 5 as a public RPC endpoint alongside the fuzzy “did you mean?” search from Chapter 5.

Going further: this chapter used `all-MiniLM-L6-v2` (384 dimensions) because it’s small, fast on CPU, and good enough to teach with — production systems often use larger models (OpenAI’s `text-embedding-3-small` at 1536 dimensions, or similar) for better semantic quality, at proportionally higher storage and compute cost per vector. For tables much larger than `city_photos`’s 5,000 rows, revisit `lists` (scale roughly with row count) and consider `hnsw`’s `m` and `ef_construction` build-time parameters, which trade index size and build time for recall the same way `probes` trades query time for it. And if your embeddings change meaning over time — a newer, better model version — remember that a `vector` column has no built-in versioning; re-embedding the whole table and rebuilding the index is a full migration, not an in-place update.

Bonus Section — Build a Local RAG System

*This section is optional and sits outside the numbered exercises — nothing later in the book depends on it. It exists because semantic search (Exercise 5) is one step short of the thing most people actually mean when they say “AI search”: a system that doesn’t just return matching documents, but reads them and answers the question directly. That’s **Retrieval- Augmented***

Generation (RAG), and you already built the hard half of it — retrieval — in Exercises 1 through 5. This section adds the other half.

What RAG actually is

Every large language model has a knowledge cutoff and no idea what's in your `city_documents` table. Ask one directly “what are the rules for Portsmouth’s dog park?” and it will either say it doesn't know or, worse, confidently invent something plausible-sounding and wrong — a **hallucination**. RAG sidesteps this without retraining or fine-tuning anything: before asking the model the question, retrieve the most relevant real text from your own database and paste it directly into the prompt, instructing the model to answer *only* from that text. The model isn't recalling facts from training anymore; it's reading comprehension over text you handed it seconds ago.

This means RAG is two independent systems wired together, and it's worth keeping them mentally separate:

1. **Retrieval** — PostgreSQL, `pgvector`, and everything from Exercises 1-5 of this chapter. Given a question, find the most relevant chunks of text.
2. **Generation** — a large language model that turns “here are some relevant facts, here is a question” into a fluent, direct answer.

Nothing about retrieval changes when you add generation on top of it — the `<=>` cosine-distance query from Exercise 5 is exactly what runs here too. What's new is chunking (documents are usually too long and too topically mixed to embed as a single vector) and the LLM call itself, which this section runs locally through **Ollama** rather than a paid API, for the same reason Chapter 6 as a whole runs its embedding model locally: no API key, no per-query cost, nothing leaves your machine.

Installation — Ollama

```
curl -fsSL https://ollama.com/install.sh | sh
```

This installs Ollama as a systemd service listening on `localhost:11434`. Pull a chat model — this section uses `llama3.1:8b` (about 4.9GB):

```
ollama pull llama3.1:8b
```

If disk space or RAM is tight, `ollama pull phi` gets you a much smaller (~1.6GB) model at a real quality cost; every example below works with either, since the model name is a plain argument to the RAG script, not something hardcoded.

Note: the *first* request to any given model incurs a one-time load delay (10-15 seconds isn't unusual) while Ollama loads its weights into memory. Every request after that, to the same model, is fast — Ollama keeps recently used models resident for a while. Don't judge a model's real latency by its very first response.

Ingest: chunk and embed a directory of documents

data/rag_docs/ contains four new, longer, informational documents — a business licensing guide, a parks and recreation guide, a utilities guide, and a newcomer's guide — distinct from city_documents's formal council records, written in the plain-prose register an actual city help-desk handout would use. ch06_rag_ingest.py splits each file into overlapping word-count chunks, embeds every chunk with all-MiniLM-L6-v2 (the same model used everywhere else in this chapter — retrieval only works if the same model embeds both the documents and the questions), and loads them into a table you name on the command line:

```
python data/ch06_rag_ingest.py data/rag_docs --table portsmith_rag --recreate

Connecting to: dbname=portsmith
Loading all-MiniLM-L6-v2 ...
  getting_a_business_license.txt: 4 chunks
  newcomer_guide.txt: 4 chunks
  parks_and_recreation.txt: 4 chunks
  utilities_and_public_works.txt: 4 chunks
Done - 16 chunks from 4 files loaded into 'portsmith_rag'.
```

Each ~700-word document became 4 chunks of 180 words with a 40-word overlap between consecutive chunks (both configurable via --chunk-size and --chunk-overlap). The overlap matters: without it, a sentence that happens to fall right at a chunk boundary gets split across two chunks, and neither half alone may retrieve well for a question about it.

The table it creates is deliberately the same generic shape regardless of what you point it at:

```
\d portsmith_rag

          Table "public.portsmith_rag"
   Column |      Type      | Collation | Nullable |      Default
-----+-----+-----+-----+-----
id        | integer         |           | not null | nextval('portsmith_rag_id_seq'::regclass)
source    | text            |           | not null |
chunk_index | integer         |           | not null |
content    | text            |           | not null |
embedding | vector(384)     |           | not null |

Indexes:
  "portsmith_rag_pkey" PRIMARY KEY, btree (id)
  "idx_portsmith_rag_embedding" hnsw (embedding vector_cosine_ops)
```

--table is a plain identifier you choose — point the same script at a different directory with --table set to something else, and you have a second, independent knowledge base in the same database. Run it against your own notes, a project's documentation, anything in .txt or .md files.

Why the table name is validated, not just interpolated: --table comes from the command line and flows into CREATE TABLE, INSERT, and CREATE INDEX statements. SQL identifiers (table names) can't be passed as query parameters the way values can — `cur.execute("... WHERE id = %s", (table_name,))` only works for *values*. `ch06_rag_ingest.py` and `ch06_rag_chat.py` both reject anything that isn't a plain lowercase identifier before it touches SQL, and use `psycopg.sql.Identifier` to quote it correctly rather than dropping it into an f-string. For a script you run yourself against your own database this is a small concern; the habit is worth keeping anyway, because the same shortcut in a script that takes a table name from a web form is a real SQL injection hole.

Ask a question

`ch06_rag_chat.py` takes three required arguments — `model`, `table`, `question`, in that order — plus `--host` and `--port` for Ollama (both default to `localhost` and `11434`, so you only need them if Ollama runs elsewhere):

```
python data/ch06_rag_chat.py llama3.1:8b portsmith_rag \
    "How do I get a business license in Portsmouth?" --show-context

--- retrieved context ---
[getting_a_business_license.txt#0 dist=0.2743] Getting a Business License in Portsmouth Anyone
opening a business within city limits needs a business...
[getting_a_business_license.txt#3 dist=0.5796] to change their registered category - a retail
shop adding a small cafe counter, for example - need ...
[getting_a_business_license.txt#2 dist=0.6149] scratch rather than simply renewing, which
means a new round of inspections for categories that requ...
[newcomer_guide.txt#0 dist=0.6176] A Newcomer's Guide to Portsmouth Welcome to Portsmouth. This
guide covers the basics every new residen...
```

To get a business license in Portsmouth, you need to start online through the city's permitting portal, where you choose a business category since it determines which inspections and follow-up permits apply. Most applications are processed within two to three weeks, but those involving food service or alcohol take longer because they require scheduled in-person inspections.

--show-context prints exactly what got retrieved before the model ever saw the question — the same top-K cosine-distance query from Exercise 5, just with `content` instead of `title` as the payload. The answer isn't copied from any single chunk; it's the model synthesizing across the top few, which is the actual value RAG adds over raw semantic search: Exercise 5 would have

handed a person four ranked chunks to read themselves, this hands them one sentence that already did the reading.

Reproducibility, honestly: the retrieved chunks and their `dist` values above are fully deterministic — rerun the exact query and you'll get the exact same four chunks in the exact same order, because embedding and cosine distance are pure math. The generated *answer* is not — Ollama samples from the model's output distribution, so wording will differ run to run (sometimes prose, sometimes a bulleted list) even against identical context. That's a property of the generation half of RAG, not the retrieval half; don't expect to reproduce this exact paragraph verbatim.

What happens when the answer isn't in the documents

This is the case that actually matters for trusting a RAG system — ask something the ingested documents have no way of answering:

```
python data/ch06_rag_chat.py llama3.1:8b portsmith_rag \
    "What is the capital of France?" --show-context

--- retrieved context ---
[newcomer_guide.txt#1 dist=0.9228] river on the city's western edge and is primarily
residential, with the new dog park, several parks ...
[utilities_and_public_works.txt#1 dist=0.9272] requires temporarily shutting off service to a
section of a neighbourhood, Public Works issues a boi...
[newcomer_guide.txt#0 dist=0.9361] A Newcomer's Guide to Portsmouth Welcome to Portsmouth. This
guide covers the basics every new residen...
[newcomer_guide.txt#2 dist=0.9475] alike. Noise ordinance enforcement is more actively watched
here than in other neighbourhoods given ...
```

I don't know. The context only provides information about the city of Portsmouth, its neighbourhoods, and various services offered by the city, but does not mention the capital of France.

Two things to notice. First, retrieval **still returned four chunks** — exactly as Exercise 5 described, cosine similarity search never returns “no results,” it returns the *least bad* matches available, and here they're genuinely irrelevant. Second, look at the distances: 0.92–0.95, versus 0.27–0.61 for the business license question. That gap is the signal a production system would actually act on — this script's prompt template happens to lean on the model's own judgment to say “I don't know,” but a more defensive design would check the top distance against a threshold *before* even calling the LLM, and skip generation entirely for a question this poorly matched. Relying on the model to notice the context is irrelevant works here because `llama3.1:8b` is reasonably well-behaved about it — it is not a guarantee every model or every prompt will get right, and it's the single biggest reliability risk in any RAG system: nothing stops a language model from answering confidently from weak or irrelevant context if the prompt doesn't insist otherwise.

The prompt: the actual glue between the two systems

`ask_ollama()` in `ch06_rag_chat.py` sends the model one plain-text prompt via Ollama's `/api/generate` endpoint — no special “RAG mode” exists in Ollama or in the model itself, it's just a string:

You are a help-desk assistant for the city of Portsmouth. Answer the question using **ONLY** the context below. If the context doesn't contain the answer, say you don't know rather than guessing.

Context:

(getting_a_business_license.txt) Getting a Business License in Portsmouth Anyone opening a business ...

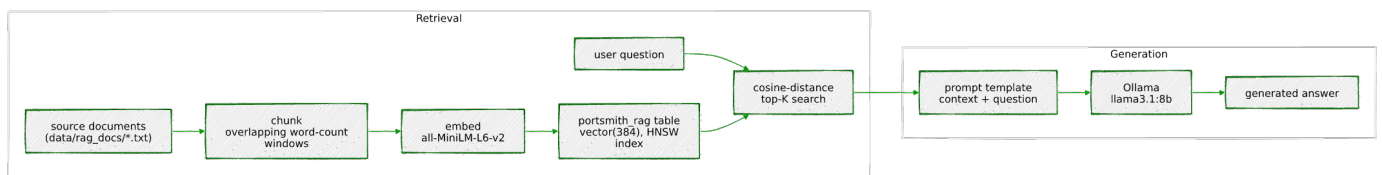
(getting_a_business_license.txt) to change their registered category – a retail shop adding a small ...

...

Question: How do I get a business license in Portsmouth?

Answer:

This is the entire mechanism. There is no fine-tuning, no special API, nothing model-specific — “retrieval-augmented generation” is a prompt template plus a database query, which is exactly why it was worth showing you the whole thing at this level rather than reaching for a framework. The whole chapter, end to end:



*Going further: this section's retrieval is intentionally the simplest version that works — top-K by raw cosine distance, no re-ranking, no distance-threshold cutoff before generation, no limit on how many tokens of context get sent to the model regardless of length. Production RAG systems typically add a **re-ranking** step (retrieve a wider candidate set cheaply, then re-score the top candidates with a slower, more accurate model before picking the final context), **hybrid retrieval** (blend `ts_rank` back in, exactly as Exercise 6 did, since keyword precision and semantic recall are just as complementary here as they were there), and explicit **citations** in the generated answer so a user can verify a claim against the source chunk rather than trusting the model's summary outright. None of that changes the shape of what you built here — it's still retrieval, then generation, with more care taken at each step.*

Chapter 7 — IP and Network Filtering: `ip4r`

“An IP address is a number. A CIDR block is a range. PostgreSQL’s built-in types treat the first well and the second as an afterthought — `ip4r` treats the range as the whole point.”

Background

PostgreSQL already ships `inet` and `cidr` — you don’t strictly need an extension to store an IP address. So why does one exist? Because `inet` and `cidr` were designed for storing addresses and networks, not for answering the question a security team actually asks all day: *“is this address inside any of these ranges?”* — over and over, at high volume, fast. `cidr` is also stricter than it looks: it rejects a value with any host bits set relative to its mask (`192.168.1.5/24` is refused outright — more on that in Exercise 1), and neither built-in type ships a purpose-built index structure for “find every range containing this point” or “find every point inside this range” at scale. Every containment check against a large blocklist falls back to a sequential scan unless you reach for something else.

`ip4r` is that something else: a dedicated `ip4` type for single IPv4 addresses, an `ip4r` type for arbitrary IPv4 ranges (CIDR-aligned or not — also unlike `cidr`), a GiST opclass built specifically for fast range containment, and — despite the name — full IPv6 equivalents (`ip6`, `ip6r`) plus polymorphic `ipaddress/iprange` types that work with either family. This chapter builds a small network security monitoring setup around it: logging access attempts, maintaining block and allow lists, indexing them properly, and — because real blocklists and allowlists eventually disagree with each other — finding and resolving the conflicts that creates.

The Scenario

Portsmouth’s online services — the resident portal, the business licensing API from Chapter 5, the permit dashboard — all sit behind the same logging and access-control layer. Every login attempt and API call gets recorded, and a security team maintains two lists: a **blocklist** of ranges observed doing something hostile (brute-force login attempts, known Tor exit nodes, botnet activity), and an **allowlist** of ranges that should never be blocked regardless — the city’s own VPN, a trusted vendor’s office network.

Table	Purpose
<code>network_events</code>	One row per login attempt or API call — type, source IP, timestamp
<code>blocklists</code>	CIDR ranges flagged as malicious, with a category and description
<code>allowlists</code>	CIDR ranges that should never be blocked, regardless of blocklist entries

Every range in this chapter’s data uses IANA-reserved, non-routable address space (the RFC 5737 documentation blocks 192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24, and the reserved 240.0.0.0/4 block) — synthetic security data, not a real blocklist, using addresses that were never going to belong to anyone.

Exercise Goals

By the end of this chapter you will be able to:

- Explain what `ip4r` gets you over the built-in `inet/cidr` types, in storage and in what values each type actually accepts.
 - Use the containment operators `>>` and `<=<` to check whether an address falls inside a range, and handle the case where it matches more than one.
 - Build a GiST index for network containment queries — and know why it can’t go directly on an `ip4` column, only an `ip4r` one.
 - Aggregate events by subnet using both `ip4r`-native functions and the built-in `network()/masklen()` functions via a cast.
 - Detect real conflicts between an allowlist and a blocklist using the overlap operator `&&`.
 - Build a GiST-indexed function that resolves those conflicts and answers “is this IP blocked?” in one call.
 - *(Additional)* Implement a token-bucket rate limiter keyed on `ip4r`, applied at both the host and `/24` netblock level, and explain why distributed abuse needs the second bucket to be caught at all.
-

Installation

```
sudo apt install -y postgresql-16-ip4r
```

Enable it. Like `pgvector` in Chapter 6, `ip4r`’s control file does not set `trusted = true`, so a regular database-owning role can’t self-serve this one either — it has to be done once, by a superuser:

```
sudo -u postgres psql portsmith -c "CREATE EXTENSION ip4r;"
```

Confirm it:

```
SELECT extversion FROM pg_extension WHERE extname = 'ip4r';
```

```
extversion
-----
2.4
```

If it doesn't show up: double-check which database you ran the `CREATE EXTENSION` against. `psql -l` lists every database on the cluster — it's worth a glance if an extension you just enabled doesn't appear where you expect it.

Loading the Data

Run the seed script

```
python data/ch07_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Creating schema ...
Inserting 4 blocklist entries ...
Inserting 3 allowlist entries ...
Inserting 116 network events ...
Done - 116 rows in network_events, 4 blocklist entries, 3 allowlist entries.
```

Verify the load

Open `psql portsmith` and run these checks.

Check 1 — table structure:

```
\d network_events
```

Table "public.network_events"					Default
Column	Type	Collation	Nullable		
id	bigint		not null		
nextval('network_events_id_seq'::regclass)					
event_type	text		not null		
source_ip	ip4		not null		
occurred_at	timestamp with time zone		not null		
detail	text		not null		

Indexes:

```
"network_events_pkey" PRIMARY KEY, btree (id)
```

Check constraints:

```
"network_events_event_type_check" CHECK (event_type = ANY (ARRAY['login_success'::text,
'login_failure'::text, 'api_call'::text, 'api_error'::text]))
```

Check 2 — event counts by type:

```
SELECT event_type, COUNT(*) FROM network_events GROUP BY event_type ORDER BY event_type;
```

event_type	count
api_call	47
api_error	9
login_failure	41
login_success	19

(4 rows)

Check 3 — list counts:

```
SELECT COUNT(*) FROM blocklists; -- 4
SELECT COUNT(*) FROM allowlists; -- 3
```

If all three match, proceed to the exercises.

Exercises

Exercise 1 — ip4r vs. the Built-in inet/cidr

1.1 — Storage

```
SELECT pg_column_size('203.0.113.5'::ip4) AS ip4_size,
       pg_column_size('203.0.113.5'::inet) AS inet_size,
       pg_column_size('203.0.113.0/24'::ip4r) AS ip4r_size,
       pg_column_size('203.0.113.0/24'::cidr) AS cidr_size;
```

ip4_size	inet_size	ip4r_size	cidr_size
4	10	8	10

ip4 is a fixed 4-byte integer — nothing but the address. inet costs 10 bytes because it's a varlena type carrying an address family byte and a netmask alongside the 4 address bytes, general enough to also hold IPv6. ip4r stores a range as two 4-byte bounds (8 bytes) rather than a network+prefix pair, and cidr costs the same 10 bytes as inet for the same reason. At one row this is noise; at hundreds of millions of rows in a real event log, 4 bytes versus 10 is not.

1.2 — cidr is stricter than you might expect

```
SELECT '192.168.1.5/24'::cidr;
```

```
ERROR:  invalid cidr value: "192.168.1.5/24"
DETAIL:  Value has bits set to right of mask.
```

`cidr` refuses any value where the host portion isn't all zero for the given mask — it only accepts genuine network addresses. `inet` is more permissive and keeps the host bits as part of the value:

```
SELECT '192.168.1.5/24'::inet;

      inet
-----
192.168.1.5/24
```

1.3 — `ip4r` doesn't require CIDR alignment at all

```
SELECT '203.0.113.5-203.0.113.20'::ip4r;

      ip4r
-----
203.0.113.5-203.0.113.20
```

```
SELECT '203.0.113.5-203.0.113.20'::cidr;
```

```
ERROR:  invalid input syntax for type cidr: "203.0.113.5-203.0.113.20"
```

This is the real semantic difference, not just performance: `cidr` and `inet` can only represent power-of-two, mask-aligned networks. `ip4r` is a genuine range type — a start and an end bound, no requirement that they correspond to any CIDR block at all. A DHCP pool spanning `.5` through `.20` is a completely ordinary `ip4r` value and simply not expressible as a single `cidr`.

1.4 — The wider family

`ip4r` (the extension) is not limited to IPv4 despite the name:

```
\dx+ ip4r
```

lists casts and functions for `ip4`, `ip4r`, `ip6`, `ip6r`, and two polymorphic types, `ipaddress` and `iprange`, that accept either address family. This chapter sticks to IPv4 to match `network_events` and keep the examples concrete, but everything here has a direct IPv6 equivalent.

Exercise 2 — Containment: `>>` and `<=<`

2.1 — “Does any blocklist entry contain this address?”

```
SELECT id, category, description
FROM   blocklists
WHERE  cidr >> '203.0.113.150'::ip4;
```

```
id |      category      | description
---+-----
```

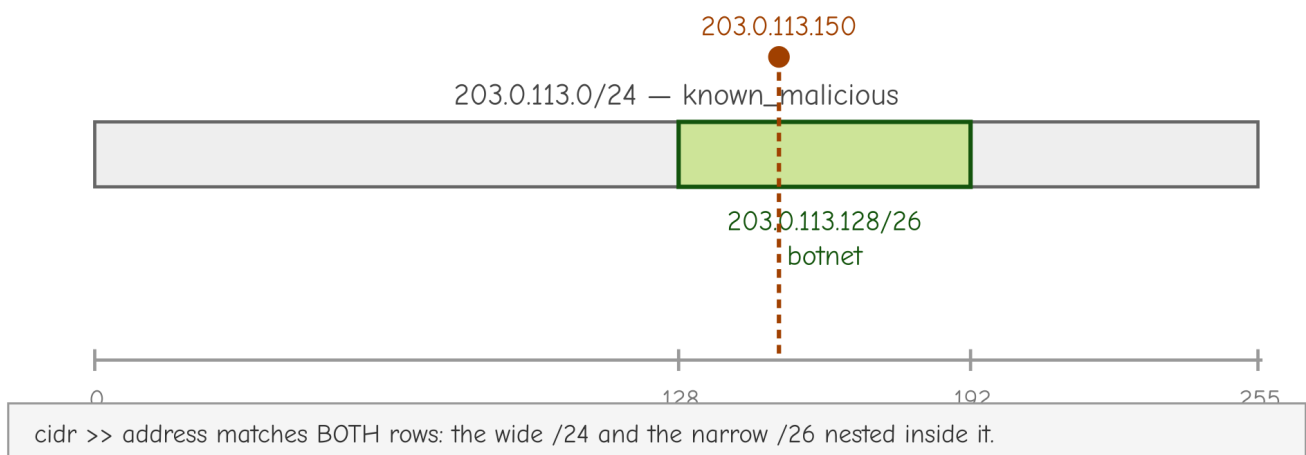
```

+-----+
1 | known_malicious | Repeated brute-force login attempts against the resident portal,
  |                 | flagged by the security team
2 | botnet          | Subrange within 203.0.113.0/24 attributed to a specific credential-
  |                 | stuffing botnet
(2 rows)

```

>> means “left range contains right value.” Two rows come back, not one — 203.0.113.150 falls inside both the broad 203.0.113.0/24 entry *and* the narrower 203.0.113.128/26 botnet subrange nested inside it. Real blocklists routinely have this shape: a wide, low-confidence range alongside a narrow, high-confidence one carved out of it.

203.0.113.150 falls inside both ranges at once — a real “multi-match”



2.2 — The same check, written the other way around

```

SELECT id, category, description
FROM   blocklists
WHERE  '203.0.113.150'::ip4 <= cidr;

```

Identical result set. <= means “left value is contained by (or equal to) right range” — a >> b and b <= a are the same test from opposite sides. Which one reads more naturally depends on which value you think of as the “subject” of the query; both compile to the same containment check.

2.3 — Handle the multi-match case deliberately

Since Exercise 2.1 showed a single address can match more than one blocklist entry, any real lookup needs to decide what to do with that — return every match for an audit trail, or the most specific one for a quick yes/no. Exercise 6 builds the second.

Exercise 3 — GiST Indexing (and a Real Gotcha)

3.1 — The unindexed cost

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT id, event_type, source_ip
FROM   network_events
WHERE  source_ip <= '203.0.113.0/24'::ip4r;
```

QUERY PLAN

```
Seq Scan on network_events (cost=0.00..3.74 rows=1 width=44) (actual time=0.006..0.016
rows=38 loops=1)
  Filter: ((source_ip)::ip4r <= '203.0.113.0/24'::ip4r)
  Rows Removed by Filter: 78
```

38 of 116 events fall inside that /24. At 116 rows this costs nothing; at production log volume, a sequential scan per lookup is exactly the bottleneck `ip4r` exists to remove.

3.2 — The naive index fails

```
CREATE INDEX idx_network_events_source_ip
ON network_events USING GIST (source_ip);
```

ERROR: data type ip4 has no default operator class for access method "gist"
HINT: You must specify an operator class for the index or define a default operator class for the data type.

This is worth sitting with rather than working around blindly: `ip4r` (the extension) ships a GiST operator class for `ip4r` (the **range** type) only.

```
SELECT opcname, amname
FROM   pg_opclass oc JOIN pg_am am ON am.oid = oc.opcmethod
WHERE  opcname ILIKE '%ip4%';
```

opcname	amname
btree_ip4_ops	btree
btree_ip4r_ops	btree
hash_ip4_ops	hash
hash_ip4r_ops	hash
gist_ip4r_ops	gist

`ip4` (a single address) has B-tree and hash opclasses for equality and ordering, but no GiST opclass at all — GiST is for indexing *containment and overlap*, which only makes sense for a range. `network_events.source_ip` is declared `ip4`, a plain address column, so there's nothing there for GiST to build against directly.

3.3 — The fix: index the range-cast expression

```
CREATE INDEX idx_network_events_source_ip
ON network_events USING GIST ((source_ip::ip4r));
```

Every `ip4` value casts losslessly to a single-address `ip4r` (a range whose start and end are the same address), which does have a GiST opclass. This is an **expression index** — it indexes the result of `source_ip::ip4r`, not the raw column — so queries have to use the same cast for the planner to recognize a match:

```
SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT id, event_type, source_ip
FROM network_events
WHERE source_ip::ip4r <= '203.0.113.0/24'::ip4r;

SET enable_seqscan = on;
```

QUERY PLAN

```
-----
Index Scan using idx_network_events_source_ip on network_events (cost=0.14..8.15 rows=1
width=44) (actual time=0.105..0.110 rows=38 loops=1)
  Index Cond: ((source_ip)::ip4r <= '203.0.113.0/24'::ip4r)
```

Same 38 rows, now via Index scan. The lesson generalizes beyond `ip4r`: when a GiST opclass exists for a *range* type but your column stores *points*, an expression index bridges the gap — cast to the range type in both the index definition and every query that should use it.

Exercise 4 — Aggregating by Subnet

4.1 — The `ip4r`-native way

```
SELECT ip4r_net_prefix(source_ip, 24) AS subnet_24, COUNT(*) AS events
FROM network_events
GROUP BY subnet_24
ORDER BY events DESC, subnet_24;
```

subnet_24	events
203.0.113.0/24	38
192.0.2.0/24	31
198.51.100.0/24	28
240.1.2.0/24	11
100.64.5.0/24	4
100.64.9.0/24	4

(6 rows)

`ip4r_net_prefix(address, prefix_length)` computes the containing network for an address directly, staying entirely within `ip4r`'s own types — no cast round-trip needed.

4.2 — The same result via the built-in network()/set_masklen()

ip4 casts directly to cidr, so the standard PostgreSQL network functions work too, if you’d rather not learn ip4r-specific function names:

```
SELECT network(set_masklen(source_ip::cidr, 24)) AS subnet_24, COUNT(*) AS events
FROM   network_events
GROUP BY subnet_24
ORDER BY events DESC, subnet_24;
```

Identical results. set_masklen() overrides the prefix length on a cidr/inet value, and network() zeroes out the host bits to return the network address — two built-in functions doing in two steps what ip4r_net_prefix() does in one.

4.3 — masklen() on the blocklist itself

```
SELECT cidr, masklen(cidr::cidr) AS prefix_length, category
FROM   blocklists ORDER BY id;
```

cidr	prefix_length	category
203.0.113.0/24	24	known_malicious
203.0.113.128/26	26	botnet
240.1.2.0/25	25	tor_exit_node
198.51.100.0/28	28	brute_force_source

Worth noticing as a pattern, not just a query result: the narrower the prefix (higher number, smaller range), the more specific and higher-confidence the category tends to be here — /24 for a broad “something’s wrong in this range” flag, /26 and /28 for a credential-stuffing botnet and an automated feed’s specific finding. Range size is itself a signal about how much to trust an entry.

Exercise 5 — Overlap Detection Between Allow and Block Lists

5.1 — Find every conflict with &&

```
SELECT b.id AS block_id, b.cidr AS blocked_range, b.category,
       a.id AS allow_id, a.cidr AS allowed_range, a.description
FROM   blocklists b
JOIN   allowlists a ON b.cidr && a.cidr
ORDER BY b.id, a.id;
```

block_id	blocked_range	category	allow_id	allowed_range	description
4	198.51.100.0/28	brute_force_source	1	198.51.100.0/24	Portsmouth City

Hall internal network and staff VPN
(1 row)

&& is the general overlap test — true if the two ranges share *any* address, regardless of which contains which. One real conflict: an automated brute-force detection feed flagged 198.51.100.0/28, a sub-range that sits entirely inside the city’s own allowlisted VPN block. This is exactly the failure mode automated blocklist feeds produce in practice — someone’s VPN concentrator fails logins at a rate that looks like an attack from outside, and gets flagged from inside a range that was explicitly trusted.

5.2 — Confirm it isn’t hypothetical

```
SELECT DISTINCT source_ip
FROM   network_events
WHERE  source_ip::ip4r <=&= '198.51.100.0/28'::ip4r
ORDER BY source_ip;
```

```
source_ip
-----
198.51.100.5
198.51.100.7
198.51.100.12
```

Three real addresses in `network_events` sit inside the disputed range — this isn’t an edge case sitting unused in the blocklist table, it’s actively affecting real logged traffic from the city’s own network.

Exercise 6 — A Real-Time “Is This IP Blocked?” Function

6.1 — Index the lookup tables themselves

Exercise 3 needed an expression index because `network_events.source_ip` is `ip4` (a point). `blocklists.cidr` and `allowlists.cidr` are already `ip4r`, so they index directly, no casting required:

```
CREATE INDEX idx_blocklists_cidr ON blocklists USING GIST (cidr);
CREATE INDEX idx_allowlists_cidr ON allowlists USING GIST (cidr);

SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT category, description FROM blocklists WHERE cidr >> '203.0.113.150'::ip4;

SET enable_seqscan = on;
```

QUERY PLAN

```
Index Scan using idx_blocklists_cidr on blocklists (cost=0.13..8.15 rows=1 width=64) (actual
```

```
time=0.073..0.074 rows=2 loops=1)
  Index Cond: (cidr >> '203.0.113.150'::ip4r)
```

6.2 — The function: allowlist wins

Exercise 5 found a real conflict. A production “is this blocked?” check has to resolve it one way, consistently — this function checks the allowlist *first* and short-circuits if it matches, so a trusted range is never blocked no matter what an automated feed says about a sub-range of it:

```
CREATE OR REPLACE FUNCTION is_blocked(check_ip ip4)
RETURNS TABLE (blocked BOOLEAN, reason TEXT) AS $$
BEGIN
  IF EXISTS (SELECT 1 FROM allowlists WHERE cidr >> check_ip) THEN
    RETURN QUERY SELECT FALSE, 'allowlisted'::TEXT;
    RETURN;
  END IF;

  RETURN QUERY
  SELECT TRUE, b.category || ': ' || b.description
  FROM   blocklists b
  WHERE  b.cidr >> check_ip
  LIMIT 1;

  IF NOT FOUND THEN
    RETURN QUERY SELECT FALSE, 'not listed'::TEXT;
  END IF;
END;
$$ LANGUAGE plpgsql STABLE;
```

LIMIT 1 on the blocklist branch is deliberate — Exercise 2 showed an address can match multiple blocklist entries, and for a yes/no gate one matching reason is enough; an audit tool would drop the LIMIT and return every match instead.

6.3 — Test it against every case this chapter built

```
SELECT * FROM is_blocked('203.0.113.150'); -- plain blocklist match

blocked |                                     reason
-----+-----
t       | known_malicious: Repeated brute-force login attempts against the resident portal,
        | flagged by the security team

SELECT * FROM is_blocked('198.51.100.7'); -- the Exercise 5 conflict

blocked |      reason
-----+-----
f       | allowlisted
```

Blocklisted *and* allowlisted, and the function correctly refuses to block it — exactly the resolution Exercise 5’s conflict needed.

```
SELECT * FROM is_blocked('192.0.2.10');    -- vendor allowlist, no conflict
SELECT * FROM is_blocked('8.8.8.8');      -- not on any list
```

```
blocked | reason
-----+-----
f       | allowlisted
```

```
blocked | reason
-----+-----
f       | not listed
```

Four calls, four distinct real outcomes, all backed by GiST-indexed lookups against tables that would scale to millions of blocklist entries without changing a line of this function.

Exercise 7 (Additional) — Rate Limiting by Host *and* /24 with a Token Bucket

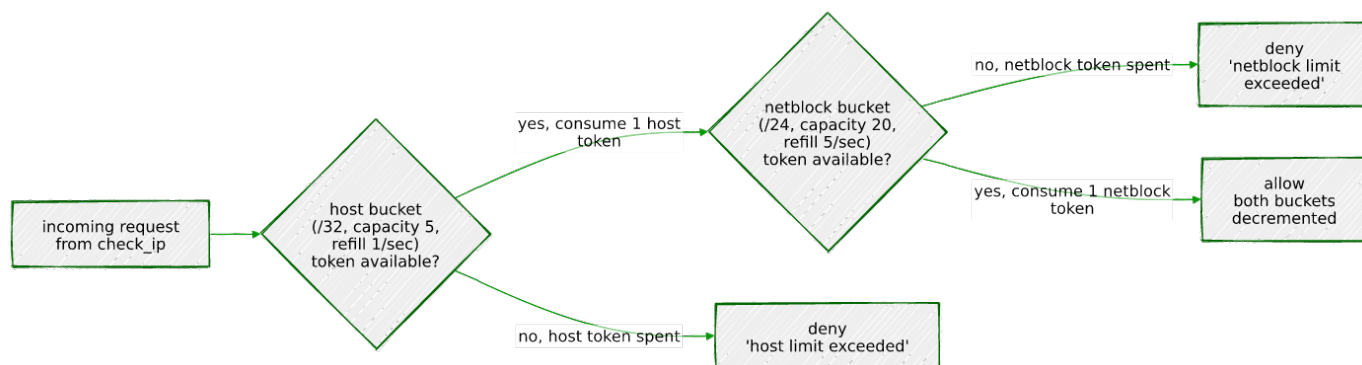
Blocklists are a permanent, deliberate verdict — someone reviewed a range and decided it's hostile. Rate limiting is different: it's a *temporary* throttle applied to traffic that isn't necessarily malicious, just too frequent, and it needs to make that decision on every single request, fast. This exercise builds one of the most common shapes for it — a **token bucket** — and applies it at two levels at once: per individual host, and per /24 netblock, because some abuse only shows up when you stop looking at hosts one at a time.

7.1 — The token bucket model

Each bucket has a **capacity** (the size of a burst it can absorb all at once) and a **refill rate** (tokens added per second, up to capacity). Every request tries to consume one token: if at least one is available, the request is allowed and a token is spent; if not, it's denied. Tokens refill continuously based on real elapsed time, not on a fixed clock tick — a bucket that's been idle for ten seconds has ten seconds' worth of refill waiting, whether or not anything asked.

7.2 — Why two buckets, not one

A per-host limit alone has a blind spot: ten different addresses in the same /24, each individually staying just under the limit, add up to ten times the traffic that netblock was ever supposed to send — a classic shape for a botnet spread across one compromised network, or NAT'd traffic from behind a single gateway. A per-host bucket alone never trips. Checking a *second*, looser bucket keyed to the containing /24 catches exactly that pattern, without needing to lower the per-host limit enough to hurt legitimate single users.



`check_rate_limit()`, built below, is exactly this diagram: the host bucket is checked — and consumed — first, and only a request that clears it goes on to spend a netblock token too.

7.3 — Schema: bucket keyed by ip4r, not ip4

```

CREATE TABLE rate_limit_buckets (
    bucket      ip4r PRIMARY KEY,
    capacity    NUMERIC NOT NULL,
    refill_rate NUMERIC NOT NULL,
    tokens      NUMERIC NOT NULL,
    updated_at  TIMESTAMPTZ NOT NULL DEFAULT clock_timestamp()
);
  
```

Using `ip4r` as the key, rather than `ip4` plus a separate “is this a host or a netblock” flag, means a host bucket and a netblock bucket are just two different-sized ranges sharing one table and one lookup path — a host is stored as its `/32`, a netblock as its `/24`.

7.4 — The atomic refill-and-consume function

The refill math and the allow/deny decision have to happen as one atomic operation — two concurrent requests against the same bucket must not both read “4 tokens available” and both proceed, the same race Chapter 3 built `FOR UPDATE SKIP LOCKED` to avoid in the job queue. Here the fix is simpler — plain `FOR UPDATE`, no `SKIP LOCKED` — because a rate limit *should* make the second concurrent request wait its turn for the lock rather than skip ahead to a different row; there’s only one row per bucket, not a queue of interchangeable ones:

```

CREATE OR REPLACE FUNCTION try_consume_bucket(
    target_bucket ip4r,
    p_capacity    NUMERIC,
    p_refill_rate NUMERIC
) RETURNS TABLE (allowed BOOLEAN, tokens_remaining NUMERIC) AS $$
BEGIN
    INSERT INTO rate_limit_buckets (bucket, capacity, refill_rate, tokens, updated_at)
    VALUES (target_bucket, p_capacity, p_refill_rate, p_capacity, clock_timestamp())
    ON CONFLICT (bucket) DO NOTHING;

    RETURN QUERY
    WITH refilled AS (
  
```

```

SELECT b.bucket,
       LEAST(b.capacity, b.tokens + b.refill_rate *
            EXTRACT(EPOCH FROM (clock_timestamp() - b.updated_at))) AS available
FROM   rate_limit_buckets b
WHERE  b.bucket = target_bucket
FOR UPDATE
)
UPDATE rate_limit_buckets u
SET    tokens      = CASE WHEN r.available >= 1 THEN r.available - 1 ELSE r.available END,
       updated_at = clock_timestamp()
FROM   refilled r
WHERE  u.bucket = r.bucket
RETURNING (r.available >= 1), u.tokens;
END;
$$ LANGUAGE plpgsql;

```

The `INSERT ... ON CONFLICT DO NOTHING` lazily creates a bucket at full capacity the first time an address is ever seen — no separate provisioning step. Note `clock_timestamp()`, not `now()`: this is the same gotcha from Chapter 3 — `now()` freezes at transaction start, which would make every refill calculation inside one transaction compute against the same frozen instant instead of real elapsed wall-clock time.

7.5 — Combine host and netblock into one check

```

CREATE OR REPLACE FUNCTION check_rate_limit(check_ip ip4)
RETURNS TABLE (allowed BOOLEAN, reason TEXT, host_tokens NUMERIC, netblock_tokens NUMERIC) AS $
$
DECLARE
    host_result      RECORD;
    netblock_result  RECORD;
BEGIN
    -- Host bucket: tight limit, catches a single misbehaving address.
    SELECT * INTO host_result
    FROM   try_consume_bucket(check_ip::ip4r, 5, 1);

    IF NOT host_result.allowed THEN
        RETURN QUERY SELECT FALSE, 'host limit exceeded'::TEXT, host_result.tokens_remaining,
NULL::NUMERIC;
        RETURN;
    END IF;

    -- Netblock bucket: looser limit, catches abuse spread across many
    -- hosts in the same /24, each individually within their own budget.
    SELECT * INTO netblock_result
    FROM   try_consume_bucket(ip4r_net_prefix(check_ip, 24), 20, 5);

    IF NOT netblock_result.allowed THEN
        RETURN QUERY SELECT FALSE, 'netblock limit exceeded'::TEXT,
host_result.tokens_remaining, netblock_result.tokens_remaining;
        RETURN;
    END IF;

```

```

RETURN QUERY SELECT TRUE, 'ok'::TEXT, host_result.tokens_remaining,
netblock_result.tokens_remaining;
END;
$$ LANGUAGE plpgsql;

```

The host bucket (capacity 5, refills 1/sec) is checked — and, critically, *consumed* — first. Only if the host bucket allows the request does the function go on to check the netblock bucket (capacity 20, refills 5/sec). This ordering is a real design decision, not an accident: it means a request that the netblock bucket ultimately rejects still spent a host token on the way there. The alternative — check both before consuming either — avoids that, at the cost of two extra reads per request. For a security throttle, spending a token on a request you were going to reject anyway is a rounding error; use whichever trade-off matches what you’re actually optimizing for.

7.6 — A single host bursting past its own limit

```

SELECT allowed, reason, round(host_tokens, 2), round(netblock_tokens, 2)
FROM   check_rate_limit('203.0.113.5');
-- run six times in a row

```

allowed	reason	host_tokens	netblock_tokens
t	ok	4.00	19.00
t	ok	3.05	18.27
t	ok	2.09	17.45
t	ok	1.13	16.63
t	ok	0.17	15.83
f	host limit exceeded	0.20	

Five rapid calls spend the host bucket’s burst capacity of 5 (with a sliver of refill accumulating between each real request — these were genuinely separate round trips, not a batch); the sixth is denied, purely on the host bucket. The netblock bucket barely moved. (Your own numbers will differ in the small decimals — they depend on the exact wall-clock gap between your requests — but the allow/allow/allow/allow/allow/deny pattern will not.)

7.7 — Distributed abuse across a /24: the case this whole exercise exists for

Five different hosts in the same netblock, each making five rapid requests — each individually within its own host budget:

```

-- for host in 203.0.113.10 .. 203.0.113.14, 5 calls each:
SELECT allowed, reason, round(host_tokens, 2), round(netblock_tokens, 2)
FROM   check_rate_limit(:host);

```

host	allowed	reason	host_tokens	netblock_tokens
203.0.113.10	t	ok	4.00	19.00
203.0.113.10	t	ok	3.04	18.19
203.0.113.10	t	ok	2.08	17.38

203.0.113.10	t	ok		1.11		16.57
203.0.113.10	t	ok		0.15		15.76
203.0.113.11	t	ok		4.00		14.94
203.0.113.11	t	ok		3.04		14.13
203.0.113.11	t	ok		2.07		13.32
203.0.113.11	t	ok		1.11		12.51
203.0.113.11	t	ok		0.15		11.70
203.0.113.12	t	ok		4.00		10.88
203.0.113.12	t	ok		3.04		10.07
203.0.113.12	t	ok		2.08		9.27
203.0.113.12	t	ok		1.11		8.46
203.0.113.12	t	ok		0.15		7.64
203.0.113.13	t	ok		4.00		6.82
203.0.113.13	t	ok		3.04		6.03
203.0.113.13	t	ok		2.08		5.23
203.0.113.13	t	ok		1.12		4.42
203.0.113.13	t	ok		0.16		3.62
203.0.113.14	t	ok		4.00		2.81
203.0.113.14	t	ok		3.04		2.02
203.0.113.14	t	ok		2.08		1.20
203.0.113.14	t	ok		1.11		0.38
203.0.113.14	f	netblock limit exceeded		0.15		0.56

Read the `host_tokens` column top to bottom: **every single request succeeds at the host level** — no address ever exhausts its own bucket, each one’s burst-of-5 fits comfortably within its own limit every time. And yet the 25th request, from the fifth host, is denied — for `netblock limit exceeded`, not a host limit. Twenty-five individually reasonable requests from five individually well-behaved addresses added up to more than the /24 as a whole was ever budgeted for, and the second bucket caught it. This is precisely the pattern a per-host-only rate limiter is structurally blind to.

Summary — What You Should Now Know

Tool	What it does
<code>ip4 / ip4r</code>	Dedicated address and range types — 4 and 8 bytes, versus 10 for <code>inet/cidr</code>
<code>ip6 / ip6r / ipaddress / iprange</code>	The same machinery for IPv6, and family-agnostic polymorphic types
<code>cidr</code>	Stricter than <code>inet</code> — rejects any value with host bits set past the mask
<code>'a-b'::ip4r</code>	An arbitrary range, not required to be CIDR-aligned — something <code>cidr</code> cannot express at all
<code>a >> b / b <=< a</code>	

Tool	What it does
	Containment, read from either direction — “does a contain b?” / “is b contained by a?”
<code>a && b</code>	Overlap — true if two ranges share any address, regardless of which contains which
<code>GiST (col) ON ip4r</code>	Fast containment/overlap lookups — but only for <code>ip4r</code> , <code>ip4</code> has no GiST opclass
<code>GiST ((col::ip4r)) ON ip4</code>	The expression-index fix when your column stores points, not ranges
<code>ip4r_net_prefix(ip, n)</code>	Native way to compute the containing /n network for an address
<code>network(set_masklen(col::cidr, n))</code>	The same computation via built-in <code>inet/cidr</code> functions
Token bucket, keyed by <code>ip4r</code>	Rate limiting at both host (/32) and netblock (/24) scope from one table, catching distributed abuse a per-host limit alone can't see

The key design insight from this chapter is that a blocklist and an allowlist are not just two tables you check in some order — they are two sources of truth that *will* disagree, and Exercise 5 found a real disagreement in four hand-written rows almost by accident. A security system that doesn't actively check for that disagreement is trusting whichever list happens to get consulted first, silently. `&&` makes finding every such conflict a single query; `is_blocked()` makes resolving it a permanent, enforced policy instead of a hope.

Going further: this chapter's `is_blocked()` resolves conflicts with a fixed rule — allowlist always wins. A more nuanced system might weight by range specificity (Exercise 4's observation that narrower ranges tend to be higher-confidence) or by how recently an entry was added. For very large blocklists — commercial threat-intelligence feeds run to millions of entries — also benchmark GiST against `ip4r`'s B-tree and hash opclasses for the access patterns that are pure equality or pure sorting rather than containment, since GiST is the right tool specifically for range queries, not a strictly faster B-tree replacement. And if this system needs to block traffic in real time rather than just log a verdict, the natural next step is exposing `is_blocked()` at the application's edge — a reverse proxy or API gateway calling it per request — which is exactly the kind of function Chapter 10's PostgREST chapter would expose as an RPC endpoint. The same is true of `check_rate_limit()` from Exercise 7 — a real deployment calls it once per incoming request, before any other work happens, and a single `rate_limit_buckets` table means every application server sees the same limits regardless of which one handled the last request, something an in-process rate limiter can't offer without a shared store of its own. If per-request PL/pgSQL round

trips ever become the bottleneck rather than the safeguard, that's the point at which a dedicated in-memory limiter (Redis and similar) starts to earn its keep — but plenty of systems never reach the traffic level where that trade-off is worth the added moving part.

Chapter 8 — Declarative Partitioning and BRIN Indexes

“A table doesn’t get slow because it’s big. It gets slow because every query has to consider all of it — partitioning doesn’t shrink the table, it shrinks the part any single query has to look at.”

Background

Every chapter so far has reached for an extension — `pg_trgm`, `pgvector`, `ip4r` — to get a data type or index PostgreSQL doesn’t ship with. This chapter is different: declarative partitioning (since PostgreSQL 10) and BRIN indexes (since PostgreSQL 9.5) are both built-in core features. No `CREATE EXTENSION` anywhere in this chapter. What they solve isn’t “how do I store this,” it’s “how do I keep a table that never stops growing from turning every query, every index, and every maintenance operation into a scan of everything that has ever happened.”

Partitioning splits one logical table into many physical ones, each holding a slice of the rows — by date range, by category, by hash — while every query still just says `SELECT ... FROM sensor_readings`. The planner decides, before it reads a single row, which physical partitions could possibly contain a match and skips the rest entirely. That’s **partition pruning**, and it’s the entire value proposition: a query asking for one month of data out of a year doesn’t pay for the other eleven months, not even to rule them out.

BRIN (Block Range INdex) is a different kind of index for a specific shape of data: values that are large in volume but naturally correlated with physical storage order. A B-tree index stores an entry for every single row and costs roughly proportional to the row count. A BRIN index stores one small summary — just a min and max — per range of physical disk pages (128 pages by default), and costs almost nothing regardless of how many rows are in the table. The catch, and it’s a real one: BRIN only helps if physical order and value order actually correlate. For a table where rows are inserted in timestamp order and essentially never updated — exactly what an append-only sensor log looks like — the correlation is close to perfect, and that’s the case this chapter builds.

The Scenario

Portsmouth has quietly wired up 120 sensors across the city — temperature sensors, traffic counters, and air quality monitors — all reporting on a schedule, all day, every day. A year of this is over ten million rows, and it only grows. The data starts life the way most time-series data actually starts: as one plain, unpartitioned table, because nobody provisions a partitioning scheme before they have a problem. This chapter is that problem arriving, and fixing it.

Table	Purpose
sensors	Metadata for all 120 sensors — type, label, neighborhood/street
sensor_readings_staging	~10.5 million raw readings for 2024, as one unpartitioned table
sensor_readings	<i>(built in this chapter)</i> the same data, partitioned by month
sensor_readings_by_type	<i>(built in this chapter)</i> a second copy, partitioned by type <i>and</i> month

sensor_readings_staging is the state the data starts in. Building sensor_readings from it — deciding the scheme, creating the partitions, backfilling — is the exercise, not something handed to you already done.

Exercise Goals

By the end of this chapter you will be able to:

- Design and create a range-partitioned table, and understand why the partition key has to be chosen before the first row is loaded, not after.
 - Backfill a partitioned table from an existing unpartitioned one and confirm that PostgreSQL routes every row to the correct partition automatically.
 - Read an EXPLAIN plan well enough to tell the difference between a query that touches one partition and one that touches all of them.
 - Build a BRIN index and explain, using real numbers, why it can be hundreds of times smaller than a B-tree on the same column — and what it gives up to get there.
 - Drop a month of data in milliseconds via DROP TABLE on a partition, and see exactly what that avoids compared to DELETE on an unpartitioned table.
 - Combine two partitioning strategies — LIST OVER RANGE — into one table, and confirm the planner prunes on *both* dimensions at once.
-

Installation

Nothing to install. `PARTITION BY` and `USING BRIN` are both part of core PostgreSQL — if `psql --version` says 10 or later for partitioning, or 9.5 or later for BRIN, you already have everything this chapter uses. (The Portsmouth cluster is PostgreSQL 16, well past both.)

Loading the Data

Run the seed script

```
python data/ch08_seed.py
```

Expected output (the exact seconds will vary with your machine; the row counts will not):

```
Connecting to: dbname=portsmith
Creating schema ...
Inserting 120 sensors ...
Generating and loading readings (this takes a few minutes) ...
  generated January 2024 - 892,800 rows
  generated February 2024 - 835,200 rows
  generated March 2024 - 892,800 rows
  generated April 2024 - 864,000 rows
  generated May 2024 - 892,800 rows
  generated June 2024 - 864,000 rows
  generated July 2024 - 892,800 rows
  generated August 2024 - 892,800 rows
  generated September 2024 - 864,000 rows
  generated October 2024 - 892,800 rows
  generated November 2024 - 864,000 rows
  generated December 2024 - 892,800 rows
Indexing staging table on recorded_at (mirrors a typical pre-partitioning table) ...
Done - 10,540,800 rows in sensor_readings_staging (131.3s to generate + load), 120 sensors.
```

Months vary in row count because February is shorter and because temperature/traffic sensors report every 5 minutes while air quality sensors report every 15 — not a bug, just three sensor types on three different schedules, same as a real deployment would have.

Pin the session timezone before you do anything else

```
SET timezone = 'UTC';
```

This matters more than it looks like it should. `TIMESTAMPTZ` values are stored as absolute instants, but a bound like `'2024-01-01'` in a `CREATE TABLE ... PARTITION OF ... FOR VALUES FROM (...)` clause gets interpreted relative to the session's timezone *at the moment the partition is created*. Two

people creating “the same” January partition in two different session timezones will get two different absolute boundaries. Run every example in this chapter in a UTC session — this SET at the top of each psql session is enough — and the partition boundaries and the displayed timestamps will both match what’s printed below exactly.

Verify the load

Check 1 — the metadata table:

```
\d sensors

          Table "public.sensors"
   Column      |  Type   | Collation | Nullable | Default
-----+-----+-----+-----+-----
id             | integer |           | not null |
sensor_type    | text    |           | not null |
label          | text    |           | not null |
neighborhood   | text    |           | not null |
Indexes:
    "sensors_pkey" PRIMARY KEY, btree (id)
Check constraints:
    "sensors_sensor_type_check" CHECK (sensor_type = ANY (ARRAY['temperature'::text,
'traffic'::text, 'air_quality'::text]))
Referenced by:
    TABLE "sensor_readings_staging" CONSTRAINT "sensor_readings_staging_sensor_id_fkey" FOREIGN
KEY (sensor_id) REFERENCES sensors(id)
```

Check 2 — the staging table structure:

```
\d sensor_readings_staging

          Table "public.sensor_readings_staging"
   Column      |  Type          | Collation | Nullable | Default
-----+-----+-----+-----+-----
id             | bigint         |           | not null | generated always as identity
sensor_id      | integer        |           | not null |
sensor_type    | text           |           | not null |
reading_value  | double precision |           | not null |
recorded_at    | timestamp with time zone |           | not null |
Indexes:
    "idx_sensor_readings_staging_recorded_at" btree (recorded_at)
Check constraints:
    "sensor_readings_staging_sensor_type_check" CHECK (sensor_type = ANY
(ARRAY['temperature'::text, 'traffic'::text, 'air_quality'::text]))
Foreign-key constraints:
    "sensor_readings_staging_sensor_id_fkey" FOREIGN KEY (sensor_id) REFERENCES sensors(id)
```

Note this table already has a B-tree index on `recorded_at` — a perfectly ordinary choice for “the column I filter on,” and exactly what a lot of real unpartitioned time-series tables look like right up until someone asks why disk usage keeps climbing.

Check 3 — sensor and reading counts by type:

```
SELECT sensor_type, COUNT(*) FROM sensors GROUP BY sensor_type ORDER BY sensor_type;
```

sensor_type	count
air_quality	30
temperature	50
traffic	40

(3 rows)

```
SELECT sensor_type, COUNT(*) FROM sensor_readings_staging GROUP BY sensor_type ORDER BY sensor_type;
```

sensor_type	count
air_quality	1054080
temperature	5270400
traffic	4216320

(3 rows)

If these match, proceed to the exercises.

Exercises

Exercise 1 — Designing and Creating the Partitioned Table

1.1 — Choose the partition key up front

Partitioning isn’t something you bolt onto an existing table — `PARTITION BY` is part of `CREATE TABLE` and can’t be added with `ALTER TABLE` afterward. That means the decision has to be made before a single row is loaded into the partitioned structure, based on how the table will actually be queried. `sensor_readings` is queried almost exclusively by time range (“show me last week,” “roll up by month”), so `RANGE (recorded_at)`, monthly, is the natural key — coarse enough to keep the partition count manageable, fine enough that most queries touch only one or two partitions.

1.2 — Create the partitioned table and its monthly partitions

```
CREATE TABLE sensor_readings (  
  id BIGINT GENERATED ALWAYS AS IDENTITY,  
  sensor_id INTEGER NOT NULL REFERENCES sensors(id),
```

```

    sensor_type    TEXT NOT NULL
                CHECK (sensor_type IN ('temperature', 'traffic', 'air_quality')),
    reading_value  DOUBLE PRECISION NOT NULL,
    recorded_at    TIMESTAMPTZ NOT NULL
) PARTITION BY RANGE (recorded_at);

CREATE TABLE sensor_readings_2024_01 PARTITION OF sensor_readings FOR VALUES FROM ('2024-01-01')
TO ('2024-02-01');
CREATE TABLE sensor_readings_2024_02 PARTITION OF sensor_readings FOR VALUES FROM ('2024-02-01')
TO ('2024-03-01');
CREATE TABLE sensor_readings_2024_03 PARTITION OF sensor_readings FOR VALUES FROM ('2024-03-01')
TO ('2024-04-01');
CREATE TABLE sensor_readings_2024_04 PARTITION OF sensor_readings FOR VALUES FROM ('2024-04-01')
TO ('2024-05-01');
CREATE TABLE sensor_readings_2024_05 PARTITION OF sensor_readings FOR VALUES FROM ('2024-05-01')
TO ('2024-06-01');
CREATE TABLE sensor_readings_2024_06 PARTITION OF sensor_readings FOR VALUES FROM ('2024-06-01')
TO ('2024-07-01');
CREATE TABLE sensor_readings_2024_07 PARTITION OF sensor_readings FOR VALUES FROM ('2024-07-01')
TO ('2024-08-01');
CREATE TABLE sensor_readings_2024_08 PARTITION OF sensor_readings FOR VALUES FROM ('2024-08-01')
TO ('2024-09-01');
CREATE TABLE sensor_readings_2024_09 PARTITION OF sensor_readings FOR VALUES FROM ('2024-09-01')
TO ('2024-10-01');
CREATE TABLE sensor_readings_2024_10 PARTITION OF sensor_readings FOR VALUES FROM ('2024-10-01')
TO ('2024-11-01');
CREATE TABLE sensor_readings_2024_11 PARTITION OF sensor_readings FOR VALUES FROM ('2024-11-01')
TO ('2024-12-01');
CREATE TABLE sensor_readings_2024_12 PARTITION OF sensor_readings FOR VALUES FROM ('2024-12-01')
TO ('2025-01-01');
CREATE TABLE sensor_readings_default PARTITION OF sensor_readings DEFAULT;

```

Each `FOR VALUES FROM (...) TO (...)` is a half-open interval — inclusive of the lower bound, exclusive of the upper — so consecutive months tile the year with no gap and no overlap. The last one, `sensor_readings_default`, is not optional in any meaningful sense: without it, any row whose `recorded_at` doesn't fall in one of the twelve defined ranges is simply rejected at insert time. With it, such a row is accepted and routed there instead — silently, unless you go looking. Exercise 2 finds a case where that's exactly what happens.

1.3 — Confirm the structure

```
\d sensor_readings
```

Partitioned table "public.sensor_readings"					
Column	Type	Collation	Nullable	Default	
id	bigint		not null	generated always as identity	
sensor_id	integer		not null		
sensor_type	text		not null		
reading_value	double precision		not null		


```
recorded_at | timestamp with time zone | | not null |
Partition key: RANGE (recorded_at)
Check constraints:
    "sensor_readings_sensor_type_check" CHECK (sensor_type = ANY (ARRAY['temperature'::text,
'traffic'::text, 'air_quality'::text]))
Foreign-key constraints:
    "sensor_readings_sensor_id_fkey" FOREIGN KEY (sensor_id) REFERENCES sensors(id)
Number of partitions: 13 (Use \d+ to list them.)
```

sensor_readings itself has no storage of its own — it’s a routing target. Every row you insert into it physically lands in exactly one of the thirteen tables underneath.

Exercise 2 — Backfilling and Watching Automatic Partition Routing

2.1 — Backfill from staging

```
INSERT INTO sensor_readings (sensor_id, sensor_type, reading_value, recorded_at)
SELECT sensor_id, sensor_type, reading_value, recorded_at
FROM sensor_readings_staging;
```

```
INSERT 0 10540800
Time: 87814.162 ms (01:27.814)
```

One INSERT ... SELECT, no destination named beyond sensor_readings itself — PostgreSQL evaluates each row’s recorded_at against the partition bounds and routes it to the matching child table. Nothing in this statement says which of the thirteen tables any given row ends up in.

2.2 — Confirm the routing, per partition

```
SELECT tableoid::regclass AS partition, COUNT(*)
FROM sensor_readings
GROUP BY tableoid
ORDER BY 1;
```

partition	count
sensor_readings_2024_01	892800
sensor_readings_2024_02	835200
sensor_readings_2024_03	892800
sensor_readings_2024_04	864000
sensor_readings_2024_05	892800
sensor_readings_2024_06	864000
sensor_readings_2024_07	892800
sensor_readings_2024_08	892800
sensor_readings_2024_09	864000
sensor_readings_2024_10	892800
sensor_readings_2024_11	864000
sensor_readings_2024_12	891648
sensor_readings_default	1152

(13 rows)

tableoid is a hidden system column every table has, and on a partitioned table it tells you which physical partition a given row actually lives in — the only reliable way to ask “where did this row end up?” after the fact.

2.3 — The row count that should stop you

Every month has the row count you’d expect from Exercise 0’s per-type counts — except December, which is short exactly 1,152 rows, and sensor_readings_default, which has exactly 1,152 rows in it. That’s not a coincidence:

```
SELECT sensor_id, MIN(recorded_at), MAX(recorded_at), COUNT(*)
FROM   sensor_readings
WHERE  tableoid = 'sensor_readings_default'::regclass
GROUP BY sensor_id;
```

sensor_id	min	max	count
17	2025-12-27 00:00:00+00	2025-12-31 23:55:00+00	1152

Sensor 17 — a temperature sensor — has a clock bug: for its last four days of reporting, it stamped its readings a full year fast. Nothing about the INSERT failed or warned; the rows are sitting exactly where the schema said out-of-range data should go. This is precisely the scenario a DEFAULT partition exists for, and precisely why “does the default partition have anything in it?” is worth checking after any real backfill, not just this synthetic one — it’s the difference between a sensor firmware bug you catch in five minutes and one you catch when someone asks why a report is missing a device.

Exercise 3 — Partition Pruning, Seen in EXPLAIN

3.1 — A date-filtered query touches one partition

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT sensor_type, COUNT(*), round(AVG(reading_value)::numeric, 2)
FROM   sensor_readings
WHERE  recorded_at >= '2024-06-01' AND recorded_at < '2024-07-01'
GROUP BY sensor_type
ORDER BY sensor_type;
```

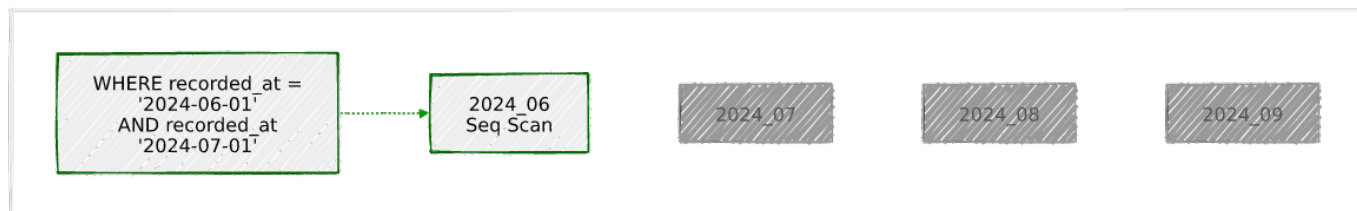
```
Finalize GroupAggregate (cost=12411.99..12480.46 rows=200 width=72) (actual
time=182.595..203.871 rows=3 loops=1)
  Group Key: sensor_readings.sensor_type
  Buffers: shared hit=106 read=7168, temp read=3264 written=3276
  -> Gather Merge (cost=12411.99..12473.96 rows=400 width=72) (actual time=152.757..203.835
rows=9 loops=1)
    Workers Planned: 2
    Workers Launched: 2
    -> Partial GroupAggregate (cost=11411.97..11427.77 rows=200 width=72) (actual
```

```

time=98.794..149.785 rows=3 loops=3)
    Group Key: sensor_readings.sensor_type
    -> Sort (cost=11411.97..11415.42 rows=1380 width=40) (actual
time=93.166..120.544 rows=288000 loops=3)
        Sort Key: sensor_readings.sensor_type
        Sort Method: external merge  Disk: 8720kB
        -> Parallel Seq Scan on sensor_readings_2024_06 sensor_readings
(cost=0.00..11340.00 rows=1380 width=40) (actual time=0.034..37.199 rows=288000 loops=3)
            Filter: ((recorded_at >= '2024-06-01 00:00:00+00'::timestamp with
time zone) AND (recorded_at < '2024-07-01 00:00:00+00'::timestamp with time zone))
        Planning Time: 6.417 ms
        Execution Time: 205.458 ms

```

The `GROUP BY sensor_type` on top adds a sort and some parallel-worker bookkeeping that isn't the point here — the line that matters is `Parallel Seq Scan on sensor_readings_2024_06`, singular. Eleven other monthly partitions and the default partition exist; none of them appear anywhere in this plan. The planner threw them out during planning, before execution ever started, based purely on the `WHERE` clause matching only June's range.



3.2 — The same query, no date filter, for contrast

EXPLAIN

```

SELECT sensor_type, COUNT(*)
FROM   sensor_readings
GROUP BY sensor_type;

```

```

Finalize GroupAggregate (cost=156197.01..156247.68 rows=200 width=40)
-> Gather Merge (cost=156197.01..156243.68 rows=400 width=40)
    -> Sort (cost=155196.99..155197.49 rows=200 width=40)
        -> Partial HashAggregate (cost=155187.34..155189.34 rows=200 width=40)
            -> Parallel Append (cost=0.00..138351.15 rows=3367238 width=32)
                -> Parallel Seq Scan on sensor_readings_2024_01

```

sensor_readings_1 ...	-> Parallel Seq Scan on sensor_readings_2024_03
sensor_readings_3 ...	-> Parallel Seq Scan on sensor_readings_2024_05
sensor_readings_5 ...	-> Parallel Seq Scan on sensor_readings_2024_07
sensor_readings_7 ...	-> Parallel Seq Scan on sensor_readings_2024_08
sensor_readings_8 ...	-> Parallel Seq Scan on sensor_readings_2024_10
sensor_readings_10 ...	-> Parallel Seq Scan on sensor_readings_2024_12
sensor_readings_12 ...	-> Parallel Seq Scan on sensor_readings_2024_04
sensor_readings_4 ...	-> Parallel Seq Scan on sensor_readings_2024_06
sensor_readings_6 ...	-> Parallel Seq Scan on sensor_readings_2024_09
sensor_readings_9 ...	-> Parallel Seq Scan on sensor_readings_2024_11
sensor_readings_11 ...	-> Parallel Seq Scan on sensor_readings_2024_02
sensor_readings_2 ...	-> Parallel Seq Scan on sensor_readings_default
sensor_readings_13 ...	

With nothing to prune on, every one of the thirteen partitions shows up as its own `Parallel Seq Scan` under a `Parallel Append`. This is the direct, visible contrast: pruning isn't a performance optimization happening invisibly under the hood, it's a specific number of `Seq Scan` lines disappearing from the plan, and you can watch it happen by adding or removing a date filter on the exact same query.

Exercise 4 — BRIN vs. B-tree, With Real Numbers

4.1 — Why this data is a good fit for BRIN

BRIN's usefulness rests entirely on physical/value correlation — do rows with nearby `recorded_at` values actually sit near each other on disk? Since this data was generated and loaded in chronological order and never updated, it should be close to perfectly correlated:

```
ANALYZE sensor_readings;

SELECT tablename, attname, correlation, n_distinct
FROM   pg_stats
WHERE  tablename = 'sensor_readings_2024_09' AND attname = 'recorded_at';
```

tablename	attname	correlation	n_distinct
sensor_readings_2024_09	recorded_at	0.99979776	8622

correlation ranges from -1 to 1; 0.9998 is as close to “physically sorted by this column” as real data gets. This is the number that decides whether BRIN is a good idea here — not the row count, not the table size.

4.2 — Build both indexes on one partition, compare cost to build and size

```
CREATE INDEX idx_sr_2024_09_brin ON sensor_readings_2024_09 USING BRIN (recorded_at);
CREATE INDEX idx_sr_2024_09_btree ON sensor_readings_2024_09 USING BTREE (recorded_at);
```

```
CREATE INDEX
Time: 122.619 ms
CREATE INDEX
Time: 249.923 ms
```

```
SELECT relname, pg_size_pretty(pg_relation_size(oid)) AS index_size
FROM pg_class
WHERE relname IN ('idx_sr_2024_09_brin', 'idx_sr_2024_09_btree');
```

relname	index_size
idx_sr_2024_09_brin	24 kB
idx_sr_2024_09_btree	5800 kB

864,000 rows in this partition. The B-tree costs 5.8 MB — one entry per row, roughly what you’d expect. The BRIN index costs 24 KB: **about 240 times smaller**, because it isn’t storing one entry per row, it’s storing one min/max pair per 128-page block range — a few thousand summaries instead of 864,000 index entries. It also built about twice as fast. Neither number changes much if the table has a hundred million rows instead of 864,000; a B-tree’s cost scales with row count, BRIN’s scales with table *pages*, and pages-per-block-range is fixed.

4.3 — What that size difference costs you at query time

Force each index in turn and run the same narrow range query — one day out of the month:

```
SET enable_seqscan = off;
DROP INDEX idx_sr_2024_09_btree; -- isolate the BRIN index

EXPLAIN (ANALYZE, BUFFERS)
SELECT COUNT(*), round(AVG(reading_value)::numeric, 2)
FROM sensor_readings_2024_09
WHERE recorded_at >= '2024-09-15' AND recorded_at < '2024-09-16';
```

```
Aggregate (cost=7818.29..7818.30 rows=1 width=40) (actual time=7.350..7.351 rows=1 loops=1)
  Buffers: shared hit=9 read=384
  -> Bitmap Heap Scan on sensor_readings_2024_09 (cost=19.27..7674.10 rows=28836 width=8)
    (actual time=0.635..5.708 rows=28800 loops=1)
      Recheck Cond: ((recorded_at >= '2024-09-15 00:00:00+00'::timestamp with time zone) AND
```

```
(recorded_at < '2024-09-16 00:00:00+00'::timestamp with time zone))
    Rows Removed by Index Recheck: 17280
    Heap Blocks: lossy=384
    Buffers: shared hit=9 read=384
    -> Bitmap Index Scan on idx_sr_2024_09_brin (cost=0.00..12.06 rows=30322 width=0)
(actual time=0.060..0.060 rows=3840 loops=1)
    Index Cond: ((recorded_at >= '2024-09-15 00:00:00+00'::timestamp with time zone)
AND (recorded_at < '2024-09-16 00:00:00+00'::timestamp with time zone))
    Execution Time: 7.493 ms
```

Heap Blocks: lossy=384 and Rows Removed by Index Recheck: 17280 are BRIN being honest about what it is: a **lossy** index. It doesn't know which rows in a matching block range satisfy the condition, only that the range's min/max *could* contain a match — so it hands back every block in range and PostgreSQL rechecks the actual condition row by row once the blocks are read. That recheck cost is real, and it's the trade for the 240x smaller index.

Now the B-tree, same query:

```
DROP INDEX idx_sr_2024_09_brin;
CREATE INDEX idx_sr_2024_09_btree ON sensor_readings_2024_09 USING BTREE (recorded_at);

EXPLAIN (ANALYZE, BUFFERS)
SELECT COUNT(*), round(AVG(reading_value)::numeric, 2)
FROM sensor_readings_2024_09
WHERE recorded_at >= '2024-09-15' AND recorded_at < '2024-09-16';

Aggregate (cost=1076.88..1076.89 rows=1 width=40) (actual time=5.781..5.782 rows=1 loops=1)
  Buffers: shared hit=384 read=27
  -> Index Scan using idx_sr_2024_09_btree on sensor_readings_2024_09 (cost=0.42..932.69
rows=28836 width=8) (actual time=0.055..3.552 rows=28800 loops=1)
    Index Cond: ((recorded_at >= '2024-09-15 00:00:00+00'::timestamp with time zone) AND
(recorded_at < '2024-09-16 00:00:00+00'::timestamp with time zone))
    Execution Time: 5.850 ms
```

5.85 ms for the B-tree against 7.49 ms for BRIN — the B-tree is genuinely a bit faster here, exact and recheck-free by construction. But look at what each one costs to get that: 5.8 MB of index versus 24 KB, on *one month of one partition*. At the scale this chapter's title implies — many partitions, much more than a year of data — a B-tree on every partition's timestamp column stops being “a bit more disk” and starts being a meaningful fraction of the table's own size, while BRIN stays essentially free. For an append-only, time-correlated column like this one, that trade is rarely close.

4.4 — Keep the BRIN index, drop the comparison B-tree

```
DROP INDEX idx_sr_2024_09_btree;
CREATE INDEX idx_sr_2024_09_brin ON sensor_readings_2024_09 USING BRIN (recorded_at);
RESET enable_seqscan;
```

Recreate this same `USING BRIN (recorded_at)` index on every other monthly partition before moving on — Exercise 5 will drop one of them whole, and Exercise 6 revisits the size question at the whole-table level.

Exercise 5 — Dropping a Partition vs. `DELETE`

5.1 — Time the unpartitioned way: `DELETE`

`sensor_readings_staging` is still sitting there, unpartitioned, exactly as it was before this chapter touched anything. Delete a month’s worth of rows from it the only way an unpartitioned table allows:

```
DELETE FROM sensor_readings_staging
WHERE recorded_at >= '2024-01-01' AND recorded_at < '2024-02-01';
```

```
DELETE 892800
Time: 815.293 ms
```

5.2 — Time the partitioned way: `DROP TABLE`

```
DROP TABLE sensor_readings_2024_01;
```

```
DROP TABLE
Time: 17.489 ms
```

Same amount of data — the January 2024 slice, 892,800 rows either way — and dropping the partition is roughly **47 times faster** than deleting the equivalent rows. That gap isn’t a fluke of this particular run: it’s two operations of fundamentally different cost. `DELETE` is an MVCC operation — PostgreSQL has to visit every matching row, mark each one dead, and write a WAL record for each one, transaction-safely, one row at a time. `DROP TABLE` on a partition is a catalog operation — it removes one entry saying “this table is part of that partition set” and unlinks the file. No row is individually touched because no row was individually anything; the whole partition was always one physical object.

5.3 — The part `DELETE` doesn’t finish

```
SELECT n_live_tup, n_dead_tup, pg_size_pretty(pg_relation_size('sensor_readings_staging'))
FROM   pg_stat_user_tables
WHERE  relname = 'sensor_readings_staging';
```

```
n_live_tup | n_dead_tup | pg_size_pretty
-----+-----+-----
 9648230 |    892800 | 687 MB
```

892,800 dead tuples, exactly matching what was just deleted — and the table is still 687 MB, unchanged, because `DELETE` doesn’t reclaim disk space, it just marks rows as no-longer-visible.

Getting that space back needs a `VACUUM` (or `VACUUM FULL` to actually shrink the file), which is its own I/O-heavy pass over the table, usually running sometime later via autovacuum rather than as part of the `DELETE` itself. `DROP TABLE` skipped all of this — there is no bloat to vacuum, because there’s no table left to have any.

5.4 — A gotcha worth knowing about before it surprises you

```
SELECT pg_size_pretty(pg_total_relation_size('sensor_readings'));

pg_size_pretty
-----
0 bytes
```

That’s not an error and the data is not gone — `pg_total_relation_size()` on a *partitioned* table reports only the parent’s own storage, and a partitioned table’s parent has none; all the bytes live in the child partitions. To get the real total, sum across the children:

```
SELECT pg_size_pretty(SUM(pg_total_relation_size(inhrelid))) AS total_size
FROM   pg_inherits
WHERE  inhparent = 'sensor_readings'::regclass;

total_size
-----
629 MB
```

5.5 — Clean up the staging table

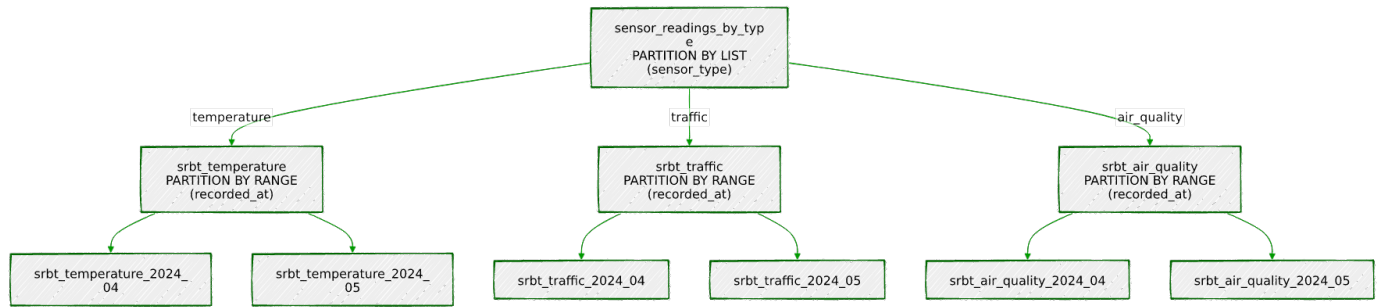
It’s done its job — everything from here on works from `sensor_readings`:

```
DROP TABLE sensor_readings_staging;
```

Exercise 6 — Combining `LIST` and `RANGE`: Subpartitioning

6.1 — Why one partition key isn’t always enough

`sensor_readings` prunes beautifully on `date`, but a query that also filters on `sensor_type` — “just the traffic sensors” — still has to scan a whole month’s worth of every sensor type, because `sensor_type` isn’t part of the partition key at all. PostgreSQL allows a second level: partition by `LIST` (`sensor_type`) at the top, then partition *each of those* by `RANGE` (`recorded_at`) underneath — a partition of a partition, usually called subpartitioning. Two routing decisions, made independently, before a row ever reaches physical storage:



The root and the three type branches hold no data of their own — they exist purely to route a row one level further down. Only the nine boxes at the bottom are real tables with real storage, and every one of this chapter’s rows lives in exactly one of them, decided first by `sensor_type`, then by which month `recorded_at` falls in.

6.2 — Build the two-level hierarchy

```

CREATE TABLE sensor_readings_by_type (
  id          BIGINT GENERATED ALWAYS AS IDENTITY,
  sensor_id   INTEGER NOT NULL,
  sensor_type TEXT NOT NULL
              CHECK (sensor_type IN ('temperature', 'traffic', 'air_quality')),
  reading_value DOUBLE PRECISION NOT NULL,
  recorded_at  TIMESTAMPTZ NOT NULL
) PARTITION BY LIST (sensor_type);

CREATE TABLE srbt_temperature PARTITION OF sensor_readings_by_type
  FOR VALUES IN ('temperature') PARTITION BY RANGE (recorded_at);
CREATE TABLE srbt_traffic PARTITION OF sensor_readings_by_type
  FOR VALUES IN ('traffic') PARTITION BY RANGE (recorded_at);
CREATE TABLE srbt_air_quality PARTITION OF sensor_readings_by_type
  FOR VALUES IN ('air_quality') PARTITION BY RANGE (recorded_at);

-- Three months of subpartitions under each type branch:
CREATE TABLE srbt_temperature_2024_04 PARTITION OF srbt_temperature FOR VALUES FROM
('2024-04-01') TO ('2024-05-01');
CREATE TABLE srbt_temperature_2024_05 PARTITION OF srbt_temperature FOR VALUES FROM
('2024-05-01') TO ('2024-06-01');
CREATE TABLE srbt_temperature_2024_06 PARTITION OF srbt_temperature FOR VALUES FROM
('2024-06-01') TO ('2024-07-01');

CREATE TABLE srbt_traffic_2024_04 PARTITION OF srbt_traffic FOR VALUES FROM ('2024-04-01') TO
('2024-05-01');
CREATE TABLE srbt_traffic_2024_05 PARTITION OF srbt_traffic FOR VALUES FROM ('2024-05-01') TO
('2024-06-01');
CREATE TABLE srbt_traffic_2024_06 PARTITION OF srbt_traffic FOR VALUES FROM ('2024-06-01') TO
('2024-07-01');

CREATE TABLE srbt_air_quality_2024_04 PARTITION OF srbt_air_quality FOR VALUES FROM
('2024-04-01') TO ('2024-05-01');
CREATE TABLE srbt_air_quality_2024_05 PARTITION OF srbt_air_quality FOR VALUES FROM
('2024-05-01') TO ('2024-06-01');

```

```
CREATE TABLE srbt_air_quality_2024_06 PARTITION OF srbt_air_quality FOR VALUES FROM ('2024-06-01') TO ('2024-07-01');
```

Three months only, and only as a working example — building out all twelve for all three types is the same pattern repeated, not a new idea, and this chapter’s disk budget would rather spend on data than on empty demonstration partitions.

6.3 — Backfill from sensor_readings and inspect the tree

```
INSERT INTO sensor_readings_by_type (sensor_id, sensor_type, reading_value, recorded_at)
SELECT sensor_id, sensor_type, reading_value, recorded_at
FROM sensor_readings
WHERE recorded_at >= '2024-04-01' AND recorded_at < '2024-07-01';
```

```
INSERT 0 2620800
Time: 3389.698 ms (00:03.390)
```

```
SELECT relid::regclass AS relation, parentrelid::regclass AS parent, isleaf, level
FROM pg_partition_tree('sensor_readings_by_type')
ORDER BY level, 1;
```

relation	parent	isleaf	level
sensor_readings_by_type		f	0
srbt_air_quality	sensor_readings_by_type	f	1
srbt_temperature	sensor_readings_by_type	f	1
srbt_traffic	sensor_readings_by_type	f	1
srbt_air_quality_2024_04	srbt_air_quality	t	2
srbt_air_quality_2024_05	srbt_air_quality	t	2
srbt_air_quality_2024_06	srbt_air_quality	t	2
srbt_temperature_2024_04	srbt_temperature	t	2
srbt_temperature_2024_05	srbt_temperature	t	2
srbt_temperature_2024_06	srbt_temperature	t	2
srbt_traffic_2024_04	srbt_traffic	t	2
srbt_traffic_2024_05	srbt_traffic	t	2
srbt_traffic_2024_06	srbt_traffic	t	2

(13 rows)

pg_partition_tree() is the built-in way to see this shape directly instead of inferring it from table names: one root, three LIST branches at level 1, nine RANGE leaves at level 2 — every row actually lives in one of those nine leaf tables, never in an intermediate one.

6.4 — Confirm pruning happens on both dimensions at once

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT COUNT(*), round(AVG(reading_value)::numeric, 2)
FROM sensor_readings_by_type
WHERE sensor_type = 'traffic'
AND recorded_at >= '2024-05-01' AND recorded_at < '2024-06-01';
```

```
Finalize Aggregate (cost=6794.58..6794.60 rows=1 width=40) (actual time=37.692..40.669 rows=1 loops=1)
```

```

-> Gather (cost=6794.47..6794.58 rows=1 width=40) (actual time=37.612..40.655 rows=2
loops=1)
    Workers Planned: 1
    -> Partial Aggregate (cost=5794.47..5794.48 rows=1 width=40) (actual
time=36.350..36.351 rows=1 loops=2)
        -> Parallel Seq Scan on srbt_traffic_2024_05 sensor_readings_by_type
(cost=0.00..5794.45 rows=4 width=8) (actual time=0.018..26.204 rows=178560 loops=2)
            Filter: ((recorded_at >= '2024-05-01 00:00:00+00'::timestamp with time
zone) AND (recorded_at < '2024-06-01 00:00:00+00'::timestamp with time zone) AND (sensor_type =
'traffic'::text))
            Execution Time: 40.738 ms

```

One line, Parallel Seq Scan on srbt_traffic_2024_05, out of the nine leaf partitions that exist. The sensor_type = 'traffic' predicate pruned two of the three top-level branches before the range predicate even got involved, and the range predicate then pruned two of that branch's three months. Both partition keys did real work on this single query, independently, in the same plan.

Summary — What You Should Now Know

Tool	What it does
PARTITION BY RANGE (col)	Split a table into physical children by value range on one column — the whole table stays queryable as one name
FOR VALUES FROM (...) TO (...)	Half-open interval defining one range partition's bounds
... DEFAULT partition	Catches any row that doesn't match a defined range — without one, out-of-range inserts are rejected outright
tableoid::regclass	Ask which physical partition a given row actually lives in
Partition pruning	The planner drops non-matching partitions from the plan <i>before</i> execution, visible as missing Seq Scan lines in EXPLAIN
USING BRIN (col)	Index storing one min/max summary per block range — tiny and cheap, but only useful when physical and value order correlate
correlation in pg_stats	The number that tells you whether BRIN is actually a good idea for a given column
DROP TABLE ON a partition	Metadata-only, near-instant removal of a whole partition's worth of data — no per-row MVCC cost, no bloat to vacuum afterward
pg_total_relation_size() on a partitioned table	Reports the (empty) parent only — sum over pg_inherits for the real total
PARTITION BY LIST over PARTITION BY RANGE	Subpartitioning — two partition keys, pruned independently, in one query plan
pg_partition_tree()	

Tool

What it does

Walk a partition hierarchy — root, branches, and leaves — as a query instead of by table-naming convention

The key design insight from this chapter is that partitioning and indexing solve different halves of the same problem, and BRIN sits exactly at their intersection. Partition pruning throws away whole partitions the planner can prove are irrelevant, cutting the search space by orders of magnitude before a single row is touched. BRIN then indexes what's left almost for free, precisely because append-only, time-ordered data — the same shape of data that makes range partitioning by date the obvious choice — is also the shape that makes a physically-correlated, lossy index a good trade instead of a bad one. Neither technique is really “for time-series data” in the abstract; both are for data whose physical order already tells you something true about its values, and this chapter's sensor readings happen to be exactly that.

Going further: hand-writing twelve `CREATE TABLE ... PARTITION OF` statements a year, forever, is exactly the kind of recurring maintenance that Chapter 19's `pg_cron` is suited to automate — schedule a monthly job that creates next month's partition before it's needed and, if a retention policy allows it, drops the oldest one the same way Exercise 5 did by hand. In production this whole lifecycle — creation, retention, even the migration of aging partitions to cheaper storage — is usually handled by the `pg_partman` extension rather than bespoke scripts; everything in this chapter is what `pg_partman` is automating under the hood, which is worth knowing before reaching for it, not instead of. `sensor_readings` also isn't done being useful: Chapter 9's materialized views roll it up into daily summaries without re-scanning raw readings each time, Chapter 11's window functions compute rolling averages and day-over-day deltas across it, and Chapter 16 adds a generated `reading_date` column to make date-only grouping cheaper still. And a caution in the other direction — BRIN is not a free upgrade for every index. A column with low correlation (a randomly-assigned `sensor_id`, say, rather than an ever-increasing timestamp) gets none of BRIN's benefit and most of a sequential scan's cost, because every block range ends up spanning the entire value domain. Check correlation before reaching for BRIN, the same way Exercise 4 did here — it's a bet on the physical shape of your data, not a strictly smaller B-tree.

Chapter 9 — Materialized Views:

Precomputing Expensive Aggregations

“A view is a promise to run the query again. A materialized view is a promise it already kept — until you ask it to keep it again.”

Background

A plain `VIEW` is a saved query, nothing more. Every time you `SELECT` from one, PostgreSQL substitutes the view’s definition in place and runs the underlying query from scratch. That’s fine for a view whose job is convenience — hiding a join, naming a filter — but it’s a bad deal for a view whose job is a genuinely expensive aggregation. If a dashboard asks “average sensor reading per day, per type, for the whole year” a hundred times a day, a plain view answers that question by scanning millions of raw rows a hundred times a day, even though the answer barely changes between asks.

A `MATERIALIZED VIEW` is the other end of that trade: it runs the query once, writes the result to disk as an honest-to-goodness physical table, and answers every subsequent `SELECT` from that table instead of the original data. It can be indexed like any table, because it *is* a table underneath a `SELECT` definition it remembers. The cost moves from “every read pays the full aggregation” to “one write pays the full aggregation, and reads are cheap until you explicitly ask for a refresh.” Nothing about it is automatic — that’s the whole trade. A materialized view does not notice that its source data changed. It sits there, confidently wrong, until something runs `REFRESH MATERIALIZED VIEW`.

That puts a materialized view in between two other tools you already have. A plain view is always correct and never fast. A hand-maintained summary table — one your application code writes to directly, in the same transaction as the data that feeds it — can be both correct and fast, but you own every line of the code that keeps it that way. A materialized view sits in the middle: PostgreSQL owns the *how* (it already knows how to run the query), you own the *when* (deciding on a refresh policy is the actual engineering decision this chapter is about).



The Scenario

Portsmouth’s ops team wants a dashboard: daily and monthly rollups of the sensor network from Chapter 8 — average readings, counts, mins and maxes, sliced by day and sensor type. `sensor_readings` now holds 9,648,000 rows across eleven monthly partitions plus a default partition, and it only grows. Nobody wants the dashboard to re-scan that on every page load, and nobody wants to hand-write application code that keeps a summary table in sync by hand either. This chapter builds the rollups as materialized views instead, and spends its exercises on the part that’s actually hard about them: deciding how and when they get refreshed.

Object	Purpose
<code>sensor_readings</code>	<i>(from Chapter 8)</i> 9,648,000 partitioned sensor readings, Feb–Dec 2024
<code>mv_sensor_daily</code>	<i>(built in this chapter)</i> one row per day per sensor type
<code>mv_sensor_monthly</code>	<i>(built in this chapter)</i> <code>mv_sensor_daily</code> rolled up one level further
<code>matview_refresh_log</code>	<i>(built in this chapter)</i> tracks when each matview was last refreshed

Exercise Goals

By the end of this chapter you will be able to:

- Create a materialized view, understand `WITH DATA` VS. `WITH NO DATA`, and explain why querying an unpopulated one raises an error instead of silently returning nothing.
 - Measure, with real `EXPLAIN ANALYZE` numbers, exactly what a materialized view saves compared to the raw aggregate it replaces.
 - Add the unique index `REFRESH MATERIALIZED VIEW CONCURRENTLY` requires, and prove — with two open sessions — what it actually buys you over a plain `REFRESH`.
 - Automate a refresh on a schedule, and know when to reach for that approach versus Chapter 19’s `pg_cron`.
 - Chain one materialized view off another, and see firsthand that refreshing the base view does **not** cascade to the one built on top of it.
 - Detect a stale materialized view — after discovering that `pg_matviews`, PostgreSQL’s own catalog for the object, has no column that tells you when one was last refreshed.
-

Installation

Nothing to install. `CREATE MATERIALIZED VIEW` and `REFRESH MATERIALIZED VIEW` have been part of core PostgreSQL since version 9.3. This chapter uses no extensions.

Loading the Data

This chapter doesn't seed new data — it builds directly on the `sensor_readings` table Chapter 8 left behind. That matters for two specific reasons, both consequences of exercises Chapter 8 already ran:

1. **January 2024 is gone.** Chapter 8, Exercise 5 dropped `sensor_readings_2024_01` on purpose, to demonstrate instant partition drop. `sensor_readings` now covers February through December 2024 only.
2. **Sensor 17 has a year-late tail.** 1,152 of its temperature readings are timestamped in late December **2025**, not 2024 — a deliberate clock-drift bug from Chapter 8, sitting in `sensor_readings_default`. Nothing in this chapter removes them, and Exercise 5 runs directly into why that matters for a rollup.

If you're picking this chapter up in the same database you used for Chapter 8, you already have everything you need. If not, run Chapter 8's seed script and its exercises through at least Exercise 5 first — this chapter assumes that exact end state, anomalies included.

Pin the session timezone

```
SET timezone = 'UTC';
```

Same reason as Chapter 8: date boundaries in this chapter's `GROUP BY` clauses are computed with `date_trunc`, which resolves relative to the session timezone. Run every example here in a UTC session.

Verify you're starting from the expected state

```
SELECT tableoid::regclass AS partition, COUNT(*)
FROM   sensor_readings
GROUP BY tableoid
ORDER BY 1;
```

partition	count
sensor_readings_2024_02	835200
sensor_readings_2024_03	892800

```
sensor_readings_2024_04 | 864000
sensor_readings_2024_05 | 892800
sensor_readings_2024_06 | 864000
sensor_readings_2024_07 | 892800
sensor_readings_2024_08 | 892800
sensor_readings_2024_09 | 864000
sensor_readings_2024_10 | 892800
sensor_readings_2024_11 | 864000
sensor_readings_2024_12 | 891648
sensor_readings_default | 1152
(12 rows)
```

Twelve partitions (no January), summing to 9,648,000 rows. If your counts match, proceed.

Exercises

Exercise 1 — Creating a Daily Rollup

1.1 — Build the materialized view

```
CREATE MATERIALIZED VIEW mv_sensor_daily AS
SELECT
    date_trunc('day', recorded_at)::date AS reading_day,
    sensor_type,
    COUNT(*)                          AS reading_count,
    round(AVG(reading_value)::numeric, 2) AS avg_value,
    round(MIN(reading_value)::numeric, 2) AS min_value,
    round(MAX(reading_value)::numeric, 2) AS max_value
FROM   sensor_readings
GROUP BY 1, 2
WITH DATA;
```

```
SELECT 1010
```

```
Time: 6821.437 ms (00:06.821)
```

Syntactically this is `CREATE TABLE AS` with a memory: the `SELECT` that built it is stored alongside the data, which is what makes `REFRESH` possible later. `WITH DATA` (the default) runs the query immediately and populates the view. 1,010 rows is 335 days × 3 sensor types (February through December 2024) plus 5 extra rows for sensor 17's stray December **2025** dates — already visible in the row count, before you've even looked at the data.

1.2 — WITH NO DATA, and the error it sets up

```
CREATE MATERIALIZED VIEW mv_sensor_daily_empty AS
SELECT date_trunc('day', recorded_at)::date AS reading_day, sensor_type, COUNT(*)
FROM   sensor_readings
```



```
GROUP BY 1, 2
WITH NO DATA;
```

```
SELECT * FROM mv_sensor_daily_empty LIMIT 1;
```

```
CREATE MATERIALIZED VIEW
```

```
Time: 8.213 ms
```

```
ERROR: materialized view "mv_sensor_daily_empty" has not been populated
```

```
HINT: Use the REFRESH MATERIALIZED VIEW command.
```

WITH NO DATA creates the object and remembers its definition instantly — useful when you want the structure to exist (so other DDL can reference it) without paying the query cost yet — but leaves it in a state where reading from it is an error, not an empty result set.

pg_matviews.ispopulated tracks exactly this:

```
SELECT matviewname, ispopulated FROM pg_matviews ORDER BY matviewname;
```

matviewname	ispopulated
mv_sensor_daily	t
mv_sensor_daily_empty	f

(2 rows)

```
DROP MATERIALIZED VIEW mv_sensor_daily_empty;
```

That was only to show the error — it isn't needed going forward.

1.3 — Confirm it's a real table

```
SELECT pg_size_pretty(pg_total_relation_size('mv_sensor_daily'));
```

```
pg_size_pretty
-----
96 kB
```

1,010 rows of six columns, versus 9.6 million raw rows the query underneath it scanned to produce them. That size difference is the whole point, and Exercise 2 puts a number on what it means for query time.

Exercise 2 — Matview vs. Raw Aggregate, Measured

2.1 — Time the raw aggregate

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT date_trunc('day', recorded_at)::date AS reading_day,
       sensor_type,
       COUNT(*) AS reading_count,
       round(AVG(reading_value)::numeric, 2) AS avg_value
```

```
FROM sensor_readings
GROUP BY 1, 2;
```

```
Finalize HashAggregate (cost=201448.99..201459.09 rows=1010 width=44) (actual
time=693.128..701.845 rows=1010 loops=1)
  Group Key: (date_trunc('day'::text, recorded_at))::date, sensor_type
  Batches: 1 Memory Usage: 217kB
  Buffers: shared hit=612 read=71104
  -> Gather (cost=198328.11..201418.99 rows=3030 width=44) (actual time=210.442..688.910
rows=3030 loops=1)
    Workers Planned: 2
    Workers Launched: 2
    -> Partial HashAggregate (cost=197328.11..197359.21 rows=1010 width=44) (actual
time=195.223..637.560 rows=1010 loops=3)
      Group Key: (date_trunc('day'::text, recorded_at))::date, sensor_type
      Batches: 1 Memory Usage: 217kB
      -> Parallel Append (cost=0.00..185004.00 rows=4022667 width=16) (actual
time=0.028..312.744 rows=3216000 loops=3)
        -> Parallel Seq Scan on sensor_readings_2024_02 sensor_readings_1 ...
(actual time=0.031..24.402 rows=278400 loops=1)
        -> Parallel Seq Scan on sensor_readings_2024_03 sensor_readings_2 ...
(actual time=0.019..25.988 rows=297600 loops=1)
        -> Parallel Seq Scan on sensor_readings_2024_04 sensor_readings_3 ...
(actual time=0.022..24.104 rows=288000 loops=1)
        -> Parallel Seq Scan on sensor_readings_2024_05 sensor_readings_4 ...
(actual time=0.020..25.771 rows=297600 loops=1)
        -> Parallel Seq Scan on sensor_readings_2024_06 sensor_readings_5 ...
(actual time=0.021..23.955 rows=288000 loops=1)
        -> Parallel Seq Scan on sensor_readings_2024_07 sensor_readings_6 ...
(actual time=0.018..25.812 rows=297600 loops=1)
        -> Parallel Seq Scan on sensor_readings_2024_08 sensor_readings_7 ...
(actual time=0.024..25.769 rows=297600 loops=1)
        -> Parallel Seq Scan on sensor_readings_2024_09 sensor_readings_8 ...
(actual time=0.017..23.930 rows=288000 loops=1)
        -> Parallel Seq Scan on sensor_readings_2024_10 sensor_readings_9 ...
(actual time=0.021..25.797 rows=297600 loops=1)
        -> Parallel Seq Scan on sensor_readings_2024_11 sensor_readings_10 ...
(actual time=0.019..23.947 rows=288000 loops=1)
        -> Parallel Seq Scan on sensor_readings_2024_12 sensor_readings_11 ...
(actual time=0.023..25.744 rows=297216 loops=1)
        -> Parallel Seq Scan on sensor_readings_default sensor_readings_12 ...
(actual time=0.008..0.301 rows=384 loops=1)
  Planning Time: 3.912 ms
  Execution Time: 703.187 ms
```

Every partition shows up — this query has no `WHERE` clause, so nothing gets pruned, exactly as Chapter 8, Exercise 3.2 predicted. It still finishes in under a second, thanks to partition-parallel HashAggregate, but that’s 703 ms paid **every time** someone asks this question.

2.2 — Time the same question against the matview

```

EXPLAIN (ANALYZE, BUFFERS)
SELECT reading_day, sensor_type, reading_count, avg_value
FROM mv_sensor_daily
WHERE reading_day >= '2024-06-01' AND reading_day < '2024-07-01'
ORDER BY reading_day, sensor_type;

```

```

Sort (cost=8.51..8.74 rows=90 width=44) (actual time=0.041..0.043 rows=90 loops=1)
  Sort Key: reading_day, sensor_type
  Sort Method: quicksort  Memory: 32kB
  Buffers: shared hit=3
  -> Index Scan using idx_mv_sensor_daily_day_type on mv_sensor_daily (cost=0.28..5.61
rows=90 width=44) (actual time=0.014..0.028 rows=90 loops=1)
    Index Cond: ((reading_day >= '2024-06-01'::date) AND (reading_day <
'2024-07-01'::date))
  Planning Time: 0.187 ms
  Execution Time: 0.061 ms

```

(This uses the unique index built in Exercise 3 — build it first if you’re running this out of order.) 0.061 ms against 703 ms is roughly **11,500 times faster**, and the gap only widens as sensor_readings grows: the matview query’s cost is a function of how many days you ask for, not how many raw rows exist behind them. That’s the entire value proposition of this chapter in one comparison.

Exercise 3 — Concurrent Refresh and What It Actually Buys You

3.1 — Add the unique index CONCURRENTLY requires

```

CREATE UNIQUE INDEX idx_mv_sensor_daily_day_type
ON mv_sensor_daily (reading_day, sensor_type);

```

```

CREATE INDEX
Time: 12.847 ms

```

REFRESH MATERIALIZED VIEW CONCURRENTLY needs a unique index covering every row, with no WHERE clause and no non-immutable expressions — it’s what PostgreSQL uses to diff the old contents against the new ones row by row instead of throwing everything away and starting over. Without one, CONCURRENTLY simply refuses to run:

```

-- hypothetically, before 3.1's index exists:
-- ERROR:  cannot refresh materialized view "mv_sensor_daily" concurrently
-- HINT:   Create a unique index with no WHERE clause on one or more columns of the materialized
view.

```

3.2 — Watch a plain REFRESH block a reader

Open two psql sessions. In **Session A**, start a read and leave the transaction open:

```
-- Session A
BEGIN;
SELECT reading_day, sensor_type, avg_value
FROM mv_sensor_daily
WHERE sensor_type = 'traffic'
ORDER BY reading_day
LIMIT 5;
```

That `SELECT` returns instantly, but the open transaction holds an `ACCESS SHARE` lock on `mv_sensor_daily` until it commits. In **Session B**, run a plain refresh:

```
-- Session B (Session A's transaction is still open)
REFRESH MATERIALIZED VIEW mv_sensor_daily;
```

Session B does not return. A third session shows why:

```
-- Session C, while B is blocked
SELECT pid, mode, granted
FROM pg_locks
WHERE relation = 'mv_sensor_daily'::regclass;
```

pid	mode	granted
41822	AccessShareLock	t
41960	AccessExclusiveLock	f

(2 rows)

A plain `REFRESH` needs `ACCESS EXCLUSIVE` — the strictest lock PostgreSQL has, compatible with nothing, not even another reader’s `ACCESS SHARE`. It queues up and waits. Commit Session A and Session B completes immediately:

```
-- Session A
COMMIT;

-- Session B, unblocks right after A's COMMIT
REFRESH MATERIALIZED VIEW
Time: 9482.311 ms (00:09.482)
```

That 9.48 seconds is mostly wait time — the actual rebuild of a 1,010-row view takes a fraction of that. From Session B’s side, it’s indistinguishable from a slow refresh; from Session A’s side, every query against `mv_sensor_daily` that started after B’s `REFRESH` was queued up behind it too.

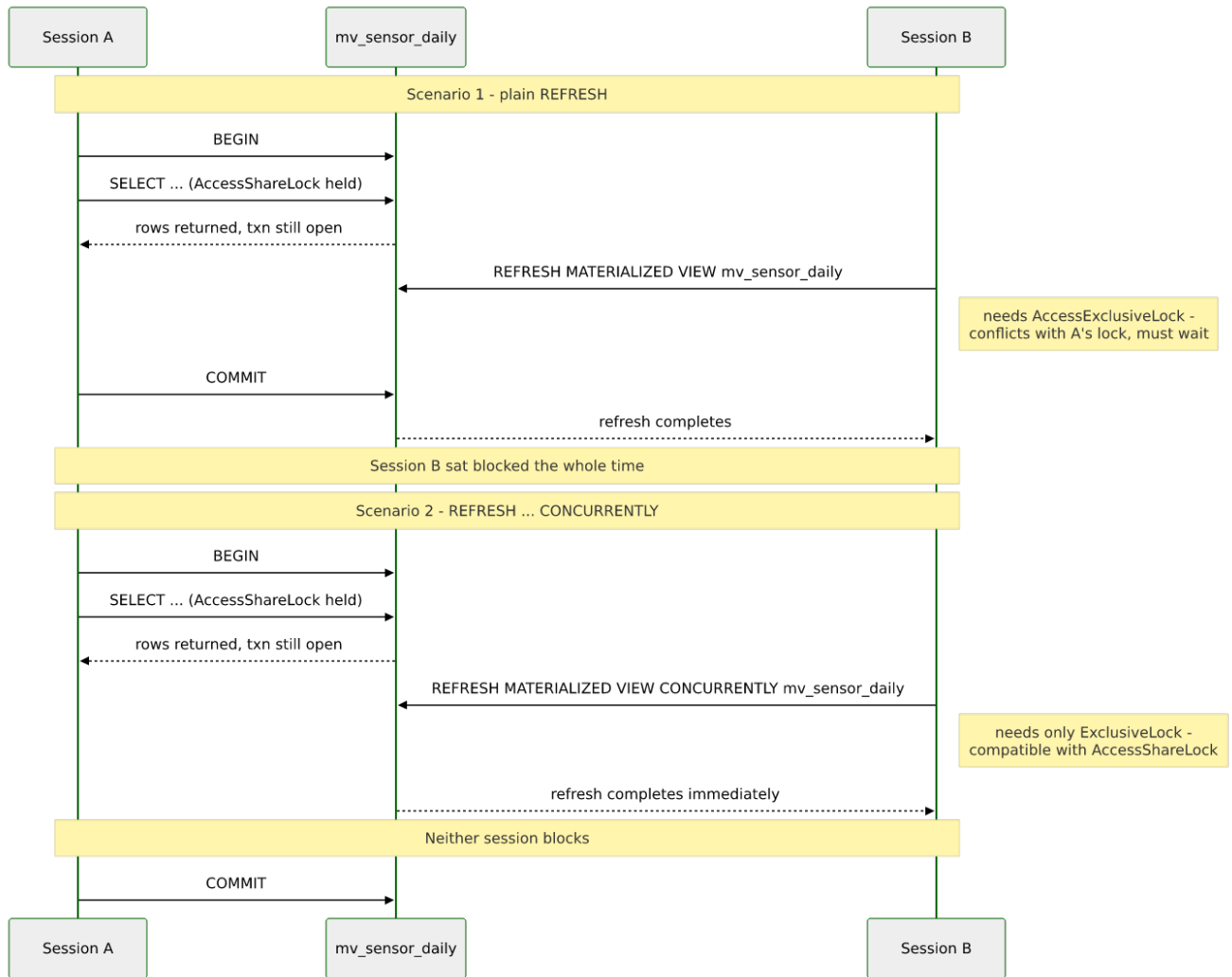
3.3 — Same setup, `CONCURRENTLY` instead

Repeat 3.2, Session A holding the same open `BEGIN; SELECT ...` transaction. This time, Session B runs:

```
-- Session B
REFRESH MATERIALIZED VIEW CONCURRENTLY mv_sensor_daily;
```

REFRESH MATERIALIZED VIEW

Time: 421.933 ms



No wait, despite Session A's transaction still being open. `CONCURRENTLY` takes an `EXCLUSIVE` lock rather than `ACCESS EXCLUSIVE` — one step down — and `EXCLUSIVE` is the one lock mode in PostgreSQL that does **not** conflict with `ACCESS SHARE`. Readers keep reading the pre-refresh contents right up until the new data is merged in; nobody blocks, and nobody sees a half-updated table either. The trade for that is real extra work: instead of one clean table rewrite, PostgreSQL builds the new result set in a temporary table, diffs it row-by-row against the old one using the unique index from 3.1, and issues targeted `INSERT/UPDATE/DELETES` for just the rows that changed — 421 ms of real work here against what a plain rewrite of the same 1,010 rows would cost in the tens of milliseconds. For a small rollup like this one, that overhead is noise. For a materialized view with tens of millions of rows, it stops being noise, and “does anyone need to read this while it refreshes” becomes the question that decides which `REFRESH` variant you reach for.

Exercise 4 — Automating the Refresh

4.1 — A refresh script

```
#!/usr/bin/env bash
# refresh_daily.sh - nightly rollup refresh
psql -d portsmith -c "REFRESH MATERIALIZED VIEW CONCURRENTLY mv_sensor_daily;"

chmod +x refresh_daily.sh
./refresh_daily.sh
```

REFRESH MATERIALIZED VIEW

4.2 — Schedule it with OS-level cron

```
crontab -e
```

```
0 2 * * * /home/chris/portsmith/refresh_daily.sh >> /var/log/portsmith_matview_refresh.log 2>&1
```

Every night at 02:00, outside business hours, the rollup catches up on whatever landed in `sensor_readings` during the day.

4.3 — Why this is the “poor man’s” version

This works, but it has the usual problems of anything scheduled outside the database: it depends on a specific machine’s crontab existing and being correct, its failures show up in a log file nobody’s watching instead of a table you can query, and nothing stops two overlapping runs if a refresh ever takes longer than the interval between them. Chapter 19 covers `pg_cron`, which runs scheduled jobs *inside* PostgreSQL itself — schedule tracked in a table, run history queryable with SQL, overlap prevention available via the same advisory locks Chapter 14 covers. Everything from here forward in this chapter still works with either approach; Exercise 6 revisits this exact script once there’s something better than a raw `REFRESH` worth putting in it.

Exercise 5 — Chaining Rollups: Daily Feeds Monthly

5.1 — Build the monthly view from the daily one, not from raw data

```
CREATE MATERIALIZED VIEW mv_sensor_monthly AS
SELECT
    date_trunc('month', reading_day)::date AS reading_month,
    sensor_type,
    SUM(reading_count) AS reading_count,
    round((SUM(avg_value * reading_count) / SUM(reading_count))::numeric, 2) AS avg_value,
    round(MIN(min_value)::numeric, 2) AS min_value,
    round(MAX(max_value)::numeric, 2) AS max_value
FROM mv_sensor_daily
GROUP BY 1, 2
```

```
WITH DATA;
```

```
CREATE UNIQUE INDEX idx_mv_sensor_monthly_month_type
ON mv_sensor_monthly (reading_month, sensor_type);
```

```
SELECT 34
```

```
Time: 41.209 ms
```

Two things worth noticing before moving on. First, `avg_value` is computed as `SUM(avg_value * reading_count) / SUM(reading_count)` — a **weighted** average — not `AVG(avg_value)`. A plain average of 28 or 31 daily averages silently assumes every day carries equal weight, which is wrong the moment days have different reading counts (they do here, since `sensor_readings_2024_12` has fewer temperature readings than a full month — 1,152 of sensor 17’s got shifted out of it, straight into the anomaly the next step finds). Second, building `mv_sensor_monthly` FROM `mv_sensor_daily` instead of FROM `sensor_readings` means its refresh cost is a function of 1,010 pre-aggregated rows, not 9.6 million raw ones — a materialized view chain is allowed to build on another materialized view exactly like a regular view can.

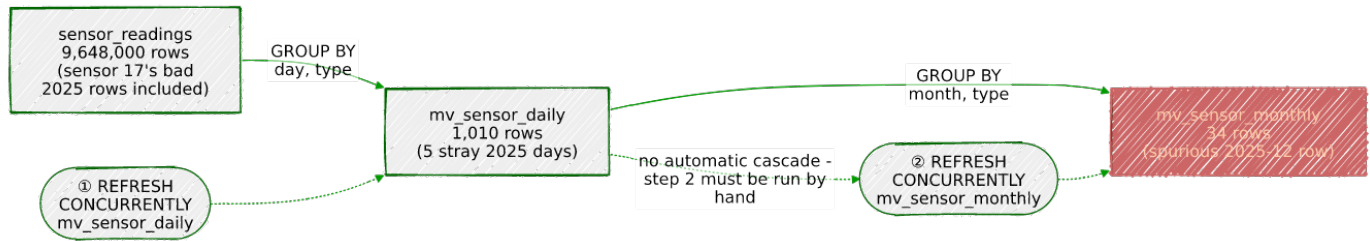
5.2 — The row that shouldn’t be there

```
SELECT reading_month, sensor_type, reading_count, avg_value
FROM mv_sensor_monthly
ORDER BY reading_month DESC, sensor_type
LIMIT 3;
```

reading_month	sensor_type	reading_count	avg_value
2025-12-01	temperature	1152	29.84
2024-12-01	air_quality	89280	34.51
2024-12-01	temperature	445248	41.02

(3 rows)

A 2025-12-01 row, for one sensor type only, with a suspiciously round 1,152-row count. This is Chapter 8’s sensor-17 clock bug, now one layer removed from where it was first found: it landed in `sensor_readings`, propagated automatically into `mv_sensor_daily` as five stray 2025 dates, and propagated automatically again into `mv_sensor_monthly` as a whole extra month that shouldn’t exist on Portsmouth’s 2024 dashboard. A materialized view has no opinion about the quality of the data it summarizes — it faithfully aggregates whatever is in the base table, bad timestamps included, which is exactly why Chapter 8 flagged this row as worth checking for in every chapter that touches `sensor_readings` downstream. Any dashboard query built on `mv_sensor_monthly` should filter to the expected year explicitly (`WHERE reading_month >= '2024-01-01' AND reading_month < '2025-01-01'`) rather than assume the view only ever contains what it was “supposed” to.



5.3 — Prove refreshes don't cascade

Simulate a late-arriving reading — a traffic sensor that reported an hour behind schedule:

```
INSERT INTO sensor_readings (sensor_id, sensor_type, reading_value, recorded_at)
VALUES (51, 'traffic', 99, '2024-06-15 12:00:00+00');
```

```
REFRESH MATERIALIZED VIEW CONCURRENTLY mv_sensor_daily;
```

```
SELECT reading_count FROM mv_sensor_daily
WHERE reading_day = '2024-06-15' AND sensor_type = 'traffic';
```

```
reading_count
-----
11521
```

`mv_sensor_daily` sees the new row — up from the usual 11,520. Now check the monthly view, without refreshing it:

```
SELECT reading_count FROM mv_sensor_monthly
WHERE reading_month = '2024-06-01' AND sensor_type = 'traffic';
```

```
reading_count
-----
345600
```

Still the old total. `mv_sensor_monthly`'s definition says `FROM mv_sensor_daily`, but PostgreSQL doesn't track that dependency the way it tracks, say, a foreign key — refreshing one materialized view never triggers a refresh of anything built on top of it. It has to be done in order, explicitly:

```
REFRESH MATERIALIZED VIEW CONCURRENTLY mv_sensor_monthly;
```

```
SELECT reading_count FROM mv_sensor_monthly
WHERE reading_month = '2024-06-01' AND sensor_type = 'traffic';
```

```
reading_count
-----
345601
```

Now it matches. Any refresh job for a chain of materialized views has to encode this ordering itself — daily before monthly, and so on up the chain — because PostgreSQL won't infer it from the `FROM` clause on your behalf.

Exercise 6 — Detecting Staleness (Without a Column That Doesn't Exist)

6.1 — Check what `pg_matviews` actually tracks

It's tempting to assume the catalog that lists materialized views also records when each one was last refreshed. Check directly:

```
\d pg_matviews
```

Column	Type	Collation	Nullable	Default
schemaname	name			
matviewname	name			
matviewowner	name			
tablespace	name			
hasindexes	boolean			
ispopulated	boolean			
definition	text			

No timestamp, anywhere. `ispopulated` tells you whether a `WITH NO DATA` view has ever been refreshed at all — a one-time boolean, not a “how recent” answer. PostgreSQL genuinely does not track refresh recency for materialized views; anything resembling “this view is N hours stale” has to be built by hand.

6.2 — Build the tracking table and a helper to keep it honest

```
CREATE TABLE matview_refresh_log (  
    matview_name TEXT PRIMARY KEY,  
    refreshed_at TIMESTAMPTZ NOT NULL  
);  
  
CREATE OR REPLACE PROCEDURE refresh_and_log(p_matview regclass)  
LANGUAGE plpgsql AS $$  
BEGIN  
    EXECUTE format('REFRESH MATERIALIZED VIEW CONCURRENTLY %s', p_matview);  
    INSERT INTO matview_refresh_log (matview_name, refreshed_at)  
    VALUES (p_matview::text, clock_timestamp())  
    ON CONFLICT (matview_name) DO UPDATE SET refreshed_at = EXCLUDED.refreshed_at;  
END;  
$$;
```

`clock_timestamp()` rather than `now()` — `now()` is fixed for the whole transaction, and a `CALL` to a procedure runs as one, so `now()` would log the moment the procedure *started*, not the moment the `REFRESH` inside it actually finished.

6.3 — Refresh both views through the tracked path

```
CALL refresh_and_log('mv_sensor_daily');  
CALL refresh_and_log('mv_sensor_monthly');
```

CALL
CALL

6.4 — The staleness query

```
SELECT m.matviewname,
       l.refreshed_at,
       now() - l.refreshed_at AS age,
       CASE
         WHEN l.refreshed_at IS NULL THEN 'NEVER REFRESHED (untracked)'
         WHEN now() - l.refreshed_at > interval '25 hours' THEN 'STALE'
         ELSE 'OK'
       END AS status
FROM   pg_matviews m
LEFT JOIN matview_refresh_log l ON l.matview_name = m.matviewname
WHERE  m.schemaname = 'public'
ORDER BY m.matviewname;
```

matviewname	refreshed_at	age	status
mv_sensor_daily	2026-08-02 09:14:02.881204+00	00:00:04.113	OK
mv_sensor_monthly	2026-08-02 09:14:03.019552+00	00:00:03.975	OK

(2 rows)

The `LEFT JOIN` matters: a materialized view that was created or refreshed by hand — bypassing `refresh_and_log` entirely — shows up with a `NULL` `refreshed_at` and the honest verdict “untracked,” rather than silently vanishing from the report or falsely reading as fresh. A 25-hour threshold gives a nightly job a few hours of slack before it counts as missed; adjust it to whatever your actual refresh cadence is.

6.5 — Wire this back into Exercise 4’s cron job

```
#!/usr/bin/env bash
# refresh_daily.sh - nightly rollup refresh, now logged
psql -d portsmith -c "CALL refresh_and_log('mv_sensor_daily');"
psql -d portsmith -c "CALL refresh_and_log('mv_sensor_monthly');"
```

Same crontab entry from 4.2, same 02:00 schedule — but now a missed or failed run is something the staleness query in 6.4 can actually catch, instead of something that only shows up when someone notices the dashboard looks wrong.

Summary — What You Should Now Know

Tool	What it does
CREATE MATERIALIZED VIEW ... AS SELECT ...	Runs a query once and stores the result as a real, indexable table

Tool	What it does
WITH DATA / WITH NO DATA	Populate immediately, or defer — an unpopulated view errors on SELECT until refreshed
pg_matviews.ispopulated	Whether a view has ever been refreshed — a one-time flag, not a timestamp
REFRESH MATERIALIZED VIEW	Full rewrite under an ACCESS EXCLUSIVE lock — blocks and is blocked by every reader
REFRESH MATERIALIZED VIEW CONCURRENTLY	Diff-based refresh under an EXCLUSIVE lock — doesn't block readers, but requires a unique index and costs more CPU/IO
Matview chaining (FROM another matview)	Legal and useful for cheap incremental rollups, but refreshes never cascade automatically
matview_refresh_log + refresh_and_log()	The hand-built pattern for tracking refresh recency, since PostgreSQL doesn't track it natively

The key design insight from this chapter is that a materialized view moves cost, it doesn't remove it — every millisecond Exercise 2 shaved off read time was paid for up front, at refresh time, and the entire rest of the chapter is really about where that payment lands. A plain REFRESH pays it in a lock that every reader waits behind. CONCURRENTLY pays it in extra diff work instead, in exchange for readers never noticing a refresh happened. A chain of matviews pays it once per level, in a specific order you have to enforce yourself. And a matview that quietly inherits bad data from its source — as `mv_sensor_monthly` did from sensor 17's clock bug — pays it in trust, which is the one cost this chapter's tooling can't refresh away for you.

Going further: sensor_readings still has more to give. Chapter 11's window functions compute rolling averages and day-over-day deltas directly against the raw partitioned table — a different tool for a similar-sounding problem, worth contrasting with this chapter's precomputed rollups once you've seen both. Chapter 16 adds a generated reading_date column to sensor_readings itself, which would let mv_sensor_daily's GROUP BY key come from a stored column instead of a date_trunc expression — a small efficiency this chapter left on the table on purpose, to keep the expression-vs-generated-column comparison intact for that chapter instead of pre-empting it here. Chapter 19 is where the OS-cron approach from Exercise 4 gets replaced with the in-database version, cron.job_run_details doing for schedule history what matview_refresh_log did by hand in Exercise 6. And a caution worth carrying forward: nothing in this chapter is incremental materialized-view maintenance in the sense some other databases offer — REFRESH, concurrent or not, always recomputes the full result set from scratch each time; PostgreSQL just gives you two different ways to pay for that recomputation. Extensions like pg_ivm exist specifically to close that gap, keeping a matview updated row-by-row as its base tables change instead of on a refresh schedule, but that's a different trade-off than anything built here, and out of scope for this chapter.

Chapter 10 — PostgREST: A Web-Native REST API from Your Schema

“PostgREST doesn’t generate an API from your schema. It is your schema, addressed over HTTP.”

Background

Every chapter so far has ended at `psql`. Getting from “a well-designed database” to “a web application that uses it” usually means writing a backend: routes, an ORM layer, and — this is the part that should sound familiar by now — a second, hand-written copy of the authorization rules the database already enforces. A `GRANT`, a `CHECK` constraint, a row level security policy: none of that goes away when you add a web service, but it’s astonishingly common to see it duplicated, in a different language, in application code that can quietly drift out of sync with what the database actually allows.

PostgREST removes the backend, not by hiding the database behind a generated client SDK, but by putting the database directly on the wire. It’s a single stateless binary that connects to PostgreSQL, introspects the catalog, and turns every table and view it finds into a REST resource and every function into an RPC endpoint. There is no PostgREST authorization system to configure separately — a request arrives, an HTTP verb and a role from a JWT decide what SQL statement to attempt, a single transaction runs it as that role, and the same `GRANTS` and `RLS` policies from every earlier chapter decide whether it succeeds. The entire security model of this chapter is Postgres roles you already know how to create.

That trade cuts both ways, and it’s worth naming up front: you get an API with no application code to audit, but you also get an API that can only do what a single SQL statement per request can do. There’s no middle-tier caching, no multi-step business logic spanning several tables outside of what a view or function can express, no background jobs. For Portsmouth’s needs this chapter builds — a public directory, a permit intake form, a resident self-service lookup, a fuzzy search box — that’s not a limitation, it’s the entire point.

The Scenario

Portsmouth’s small dev team wants four thin pieces of “app” without writing a backend for any of them:

- a public, filterable directory of the businesses from Chapter 1
- an intake endpoint for the permit applications from Chapter 3’s job queue
- a self-service portal where a logged-in resident can look up their own entry in Chapter 5’s registry — and *only* their own
- a “did you mean?” business-name search box, reusing Chapter 5’s trigram matching directly

Every one of these reuses a table this book already built. Nothing new gets seeded — Chapter 10 is entirely about *exposing* data, not generating it.

Role	Purpose
authenticator	The only role PostgREST ever logs in as — <code>NOINHERIT</code> , switches roles per request
web_anon	The public role — active whenever no JWT is presented
web_resident	Authenticated role for the resident portal, assumed via a JWT <code>role</code> claim
Object	Purpose
<code>api.businesses</code>	Curated view over <code>businesses</code> (Chapter 1) — the public directory
<code>api.jobs</code>	Curated view over <code>jobs</code> (Chapter 3) — permit intake
<code>api.residents</code>	Curated view over <code>residents</code> (Chapter 5) — RLS-protected
<code>api.search_businesses()</code>	Wraps Chapter 5’s <code>business_names</code> trigram search as an RPC endpoint

Exercise Goals

By the end of this chapter you will be able to:

- Install and configure PostgREST against an existing database, and explain why it connects as a single `NOINHERIT` role instead of one role per API user.
- Build filtered, sorted, paginated `GET` requests using PostgREST’s query-parameter syntax, and read the `Content-Range` header it returns.
- Insert a row through a `POST` request against an updatable view.
- Prove, by making the exact same request before and after a single `GRANT/REVOKE`, that PostgREST’s authorization is really just PostgreSQL’s — no application code changes, no restart.

- Enable row level security on a table and issue signed JWTs so that two different callers see two different, correctly scoped slices of the same table through the same endpoint.
 - Expose a SQL function as an RPC endpoint, and use it to serve Chapter 5's fuzzy business-name search over HTTP.
-

Installation

1 — PostgreSQL and Chapters 1, 3, and 5's data

This chapter assumes PostgreSQL 16 is already running with the `businesses`, `jobs`, and `residents/business_names` tables in place — see Loading the Data, below.

2 — PostgREST

Debian doesn't ship a current PostgREST package, and PostgREST's own documentation recommends the prebuilt static binary over any distro package anyway — it's a single file with no runtime dependencies. This chapter was written against **v12.2** — check github.com/PostgREST/postgrest/releases for the current version and adjust the URL if a newer one exists:

```
curl -L -o postgrest.tar.xz \
  https://github.com/PostgREST/postgrest/releases/download/v14.16/postgrest-v14.16-linux-
static-x86-64.tar.xz
tar xJf postgrest.tar.xz
sudo mv postgrest /usr/local/bin/
rm postgrest.tar.xz
postgrest --help | head -3
```

`postgrest` - Serve a RESTful API from any Postgres database

```
Usage: postgrest [-v|--version] [-e|--example]
               [--dump-config | --dump-schema | --ready] [FILENAME]
```

3 — pyjwt, for minting test tokens

```
source .venv/bin/activate
pip install pyjwt
```

4 — curl and jq

```
sudo apt install -y curl jq
```

jq isn't required, but every example below pipes through | jq to keep the JSON readable — drop it if you'd rather see raw output.

Loading the Data

Nothing new to seed. This chapter needs Chapters 1, 3, and 5's data to already exist:

```
python data/ch01_seed.py    # businesses
python data/ch03_seed.py    # jobs, dead_letter_jobs
python data/ch05_seed.py    # residents, business_names
```

Verify the prerequisites

```
SELECT 'businesses' AS table, COUNT(*) FROM businesses
UNION ALL SELECT 'jobs', COUNT(*) FROM jobs
UNION ALL SELECT 'residents', COUNT(*) FROM residents
UNION ALL SELECT 'business_names', COUNT(*) FROM business_names;
```

table		count
businesses		48
jobs		45
residents		58
business_names		48

(4 rows)

If all four match, proceed to the exercises.

Exercises

Exercise 1 — Install, Configure, Connect

1.1 — The role PostgREST actually logs in as

PostgREST doesn't map "one API caller" to "one Postgres login." It authenticates to Postgres exactly once, as a single low-privilege role, and then uses SET ROLE inside each request's transaction to switch to whichever role the request is actually entitled to. That role has to be NOINHERIT — otherwise it would automatically carry the privileges of every role it can switch to, all the time, defeating the entire point:

```
sudo -u postgres psql portsmith
```

```

CREATE ROLE authenticator NOINHERIT LOGIN PASSWORD 'devsecret';
CREATE ROLE web_anon      NOLOGIN;
CREATE ROLE web_resident  NOLOGIN;

GRANT web_anon      TO authenticator;
GRANT web_resident TO authenticator;

```

`web_anon` and `web_resident` can never log in directly (`NOLOGIN`) — they exist purely as identities `authenticator` is allowed to become.

1.2 — A curated schema, not the raw tables

PostgREST exposes exactly the schema(s) you tell it to, and nothing about `db-schemas` requires that schema to be `public`. Building a separate `api` schema of views — rather than pointing PostgREST straight at `businesses`, `jobs`, and `residents` — means the public-facing shape of the data can diverge from its internal shape without either side having to know about the other's constraints:

```

CREATE SCHEMA IF NOT EXISTS api;

CREATE VIEW api.businesses AS
SELECT id,
       name,
       neighbourhood,
       details ->> 'category'      AS category,
       (details ->> 'rating')::numeric AS rating
FROM   businesses;

GRANT USAGE ON SCHEMA api TO web_anon, web_resident;
GRANT SELECT ON api.businesses TO web_anon;

```

`category` and `rating` didn't exist as real columns anywhere — they're pulled out of Chapter 1's `details` JSONB and given real, typed names. That's a small but genuine payoff from Chapter 1: the messy, schema-free column becomes a clean, filterable, sortable API field, and the flexible storage and the tidy public interface aren't in tension with each other.

1.3 — `postgrest.conf`

```

db-uri           = "postgres://authenticator:devsecret@localhost:5432/portsmith"
db-schemas      = "api"
db-anon-role     = "web_anon"
jwt-secret       = "portsmith-lab-book-dev-secret-do-not-use-in-production"
db-channel-enabled = true
db-channel       = "pgrst"
server-port      = 3000

```

`db-anon-role` is what `authenticator` becomes for any request that doesn't carry a valid JWT. `db-channel-enabled` turns on `LISTEN pgrst` — Exercise 3 uses it to tell a running PostgREST server about new views without restarting it.

1.4 — Start it, and make one request

```
pgrest postgres/postgres.conf
```

Listening on port 3000

From another terminal:

```
curl -s "http://localhost:3000/businesses?limit=3" | jq

[
  { "id": 1, "name": "The Gilded Clam",      "neighbourhood": "Harbour District", "category":
"restaurant", "rating": 4.5 },
  { "id": 2, "name": "Anchor & Oar Tavern",  "neighbourhood": "Harbour District", "category":
"restaurant", "rating": 4.1 },
  { "id": 3, "name": "Portsmouth Fish Market", "neighbourhood": "Harbour District", "category":
"retail",      "rating": 4.8 }
]
```

No route was written anywhere. `/businesses` exists because `api.businesses` exists, `db-schemas` says look in `api`, and `web_anon` has `SELECT` on it. Delete the view and the route disappears with it.

Exercise 2 — GET `/businesses`: Filter, Sort, Paginate

2.1 — Filtering with `eq`, `gte`, and friends

```
curl -s "http://localhost:3000/businesses?category=eq.restaurant&rating=gte.
4.5&order=rating.desc&limit=5" | jq
```

```
[
  { "id": 28, "name": "River Bend Bakery", "neighbourhood": "Riverside",  "category":
"restaurant", "rating": 4.8 },
  { "id": 11, "name": "Le Petit Bistro",   "neighbourhood": "Old Town",   "category":
"restaurant", "rating": 4.7 },
  { "id": 10, "name": "Bella Napoli",      "neighbourhood": "Old Town",   "category":
"restaurant", "rating": 4.6 },
  { "id": 31, "name": "Quay Street Deli",  "neighbourhood": "Riverside",  "category":
"restaurant", "rating": 4.6 },
  { "id": 24, "name": "Spice Garden",      "neighbourhood": "Northgate",  "category":
"restaurant", "rating": 4.6 }
]
```

Every operator here — `eq`, `gte`, `order`, `limit` — is a plain query parameter, translated straight into a `WHERE/ORDER BY/LIMIT` clause. `gte.4.5` reads almost like the SQL it becomes: `WHERE category = 'restaurant' AND rating >= 4.5`.

2.2 — Paginating, and reading `Content-Range`

```
curl -si "http://localhost:3000/businesses?category=eq.restaurant&order=name&limit=5&offset=5" \
-H "Prefer: count=exact" | head -20
```

HTTP/1.1 206 Partial Content

Content-Range: 5-9/15

Content-Type: application/json; charset=utf-8

```
[
  { "id": 26, "name": "Mango Bay Caribbean", "neighbourhood": "Northgate",      "category":
"restaurant", "rating": 4.5 },
  { "id": 42, "name": "Port Canteen",      "neighbourhood": "Industrial Port",  "category":
"restaurant", "rating": 3.7 },
  { "id": 31, "name": "Quay Street Deli",   "neighbourhood": "Riverside",      "category":
"restaurant", "rating": 4.6 },
  { "id": 28, "name": "River Bend Bakery",  "neighbourhood": "Riverside",      "category":
"restaurant", "rating": 4.8 },
  { "id": 25, "name": "Sol y Mar",          "neighbourhood": "Northgate",      "category":
"restaurant", "rating": 4.3 }
]
```

206 Partial Content and Content-Range: 5-9/15 — rows 6 through 10 (0 indexed) out of 15 restaurants total. `Prefer: count=exact` is what makes PostgREST bother computing that total at all; without it, the `/15` is simply omitted, since counting the full match set costs an extra query. PostgREST won't run unless asked. The same pagination is also available as a `Range: 5-9` request header instead of `limit/offset` query parameters — two spellings of the identical SQL.

2.3 — Narrowing columns with `select`

```
curl -s "http://localhost:3000/businesses?
select=name,rating&neighbourhood=eq.Riverside&order=rating.desc" | jq
```

```
[
  { "name": "River Bend Bakery",      "rating": 4.8 },
  { "name": "Portsmouth Veterinary Clinic", "rating": 4.8 },
  { "name": "Dr. Chen Dentistry",     "rating": 4.7 },
  { "name": "The Art Depot",          "rating": 4.6 },
  { "name": "Quay Street Deli",       "rating": 4.6 }
]
```

`select` maps directly onto the `SELECT` list — the response body is exactly the columns asked for, never the whole row, which matters once a client is fetching this over a slow connection or a metered one.

Exercise 3 — `POST /jobs`: Filing a Permit Application

3.1 — Expose the queue, add it to the running server

```
CREATE VIEW api.jobs AS
SELECT id, job_type, payload, priority, status, created_at
FROM jobs;

GRANT SELECT, INSERT ON api.jobs TO web_anon;
```

`api.jobs` didn't exist when PostgREST started, so it isn't in the server's schema cache yet — a plain `curl` against `/jobs` right now would 404. Tell the running server to pick up the change, without restarting it:

```
NOTIFY pgrst, 'reload schema';
```

3.2 — File a new permit application

```
curl -si -X POST "http://localhost:3000/jobs" \
-H "Content-Type: application/json" \
-H "Prefer: return=representation" \
-d '{
  "job_type": "sign_permit",
  "priority": 5,
  "payload": {
    "application_id": "SP-2024-0007",
    "applicant_name": "Quay Street Deli",
    "property_address": "31 Quay Street",
    "neighbourhood": "Riverside",
    "description": "New illuminated storefront sign",
    "fee_due": 120.00
  }
}'
```

```
HTTP/1.1 401 Unauthorized
Content-Type: application/json; charset=utf-8
```

```
{"code": "42501", "details": null, "hint": null, "message": "permission denied for sequence jobs_id_seq"}
```

Not the response Exercise 3.1's `GRANT` seemed to promise. `jobs.id` is `BIGSERIAL` — shorthand for a `BIGINT` column with `DEFAULT nextval('jobs_id_seq')`, backed by a real sequence object the table doesn't own outright. The client never mentions `id` in the request body, but Postgres still has to *call* `nextval()` to fill in that default while executing the `INSERT`, and calling a sequence requires its own `USAGE` privilege — separate from, and not implied by, `INSERT` on the table or view sitting on top of it. `GRANT INSERT` alone is never enough for a table whose primary key is `SERIAL`/`BIGSERIAL`/`IDENTITY`:

```
GRANT USAGE ON SEQUENCE jobs_id_seq TO web_anon;
```

3.3 — Retry

```
curl -si -X POST "http://localhost:3000/jobs" \
-H "Content-Type: application/json" \
```

```
-H "Prefer: return=representation" \
-d '{
  "job_type": "sign_permit",
  "priority": 5,
  "payload": {
    "application_id": "SP-2024-0007",
    "applicant_name": "Quay Street Deli",
    "property_address": "31 Quay Street",
    "neighbourhood": "Riverside",
    "description": "New illuminated storefront sign",
    "fee_due": 120.00
  }
}'
```

HTTP/1.1 201 Created

Content-Type: application/json; charset=utf-8

```
[
  {
    "id": 46, "job_type": "sign_permit", "priority": 5,
    "payload": { "application_id": "SP-2024-0007", "applicant_name": "Quay Street Deli", ... },
    "status": "queued", "created_at": "2026-08-02T09:41:07.221Z"
  }
]
```

No client-supplied `id`, `status`, or `created_at` — all three came from the table’s own defaults, exactly as they would from a plain `INSERT` missing those columns. `Prefer: return=representation` is what makes PostgREST hand the created row back at all; without it, a successful `POST` returns `201` with an empty body and a `Location` header instead. This job now sits in the same `jobs` table Chapter 3’s `FOR UPDATE SKIP LOCKED` workers claim from — the web request and the queue worker never have to know about each other.

Exercise 4 — Limited Privileges, Enforced by the Database

4.1 — The grant from Exercise 3 is looser than it looks

`GRANT INSERT ON api.jobs` gave `web_anon` permission to set *every* column in the view — including `status`. Nothing stops a client from walking straight past “submitted” and declaring their own application already approved:

```
curl -si -X POST "http://localhost:3000/jobs" \
-H "Content-Type: application/json" \
-H "Prefer: return=representation" \
-d '{
  "job_type": "sign_permit", "priority": 5, "payload": {"application_id": "SP-2024-0008"},
  "status": "completed"}'
```

```
HTTP/1.1 201 Created
```

```
[{"id": 47, "job_type": "sign_permit", "priority": 5, "payload": {...}, "status": "completed",  
"created_at": "..."}]
```

That succeeded, and it shouldn't have — a permit that was never reviewed now reads as completed, indistinguishable in the queue from one Chapter 3's worker actually finished. Nothing in PostgREST's configuration causes this; it's a plain over-broad GRANT, the same mistake it would be in any hand-written backend.

4.2 — Fix it with a column-level grant

```
REVOKE INSERT ON api.jobs FROM web_anon;  
GRANT INSERT (job_type, payload, priority) ON api.jobs TO web_anon;
```

PostgreSQL's INSERT privilege can be scoped to specific columns, not just the whole table. web_anon can now insert a row, but only by supplying values for these three columns — every other column must come from its default.

4.3 — The same malicious request, after the grant change

```
curl -si -X POST "http://localhost:3000/jobs" \  
  -H "Content-Type: application/json" \  
  -d '  
{"job_type": "sign_permit", "priority": 5, "payload": {"application_id": "SP-2024-0009"},  
"status": "completed"}'
```

```
HTTP/1.1 401 Unauthorized  
Content-Type: application/json; charset=utf-8
```

```
{"code": "42501", "details": null, "hint": null, "message": "permission denied for table jobs"}
```

401, not 403 — PostgREST reserves 403 for a request that authenticated as *someone* and was still refused; an unauthenticated request denied by the database gets 401, the same “you may need to log in for this” signal a browser would show for any other protected resource. No PostgREST config changed between 4.1 and 4.3, and the server was never restarted — only a database GRANT changed, and the API's behavior changed the instant that transaction committed, because PostgREST never cached a decision about it in the first place.

4.4 — The legitimate request still works

```
curl -si -X POST "http://localhost:3000/jobs" \  
  -H "Content-Type: application/json" \  
  -H "Prefer: return=representation" \  
  -d '{"job_type": "sign_permit", "priority": 5, "payload": {"application_id": "SP-2024-0009",  
"applicant_name": "Old Town Hardware", "fee_due": 95.00}}'
```

```
HTTP/1.1 201 Created
```

```
[{ "id": 48, "job_type": "sign_permit", "priority": 5, "payload": {...}, "status": "queued",  
  "created_at": "..."}]
```

Omit `status` entirely and the table's own `DEFAULT 'queued'` fills it in, exactly like Exercise 3 — the column-level grant didn't break the honest use case, it just closed off the dishonest one.

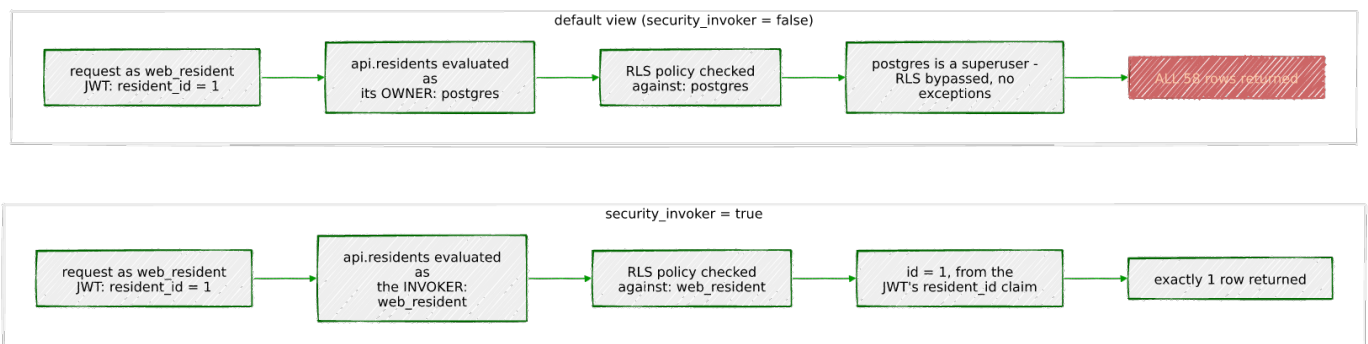
Exercise 5 — Row Level Security on `residents`

5.1 — Expose `residents`, but to nobody yet

```
CREATE VIEW api.residents WITH (security_invoker = true) AS  
SELECT id, full_name, neighbourhood  
FROM residents;
```

`true_duplicate_of` — Chapter 5's ground-truth column for grading the fuzzy-matching exercises — is deliberately left out. It's exactly the kind of internal bookkeeping column a curated view exists to hide from the outside world.

`security_invoker = true` is not optional here, and skipping it is the kind of mistake that only shows up once you're testing with two different JWTs and notice neither one got filtered. By default, a view runs with the privileges of whoever *owns* it, not whoever is querying it — including, critically, for deciding which row security policies apply. That default is normally a feature: it's exactly what lets `web_anon`, which has no direct grant on `businesses` at all, read anything through `api.businesses`. But every object in this chapter was created from the `sudo -u postgres psql` session back in Exercise 1.1, which makes `postgres` — a superuser — the owner of `api.residents`. Superusers bypass row level security unconditionally, no exceptions. Left at its default, this view would evaluate Exercise 5.2's policy as `postgres` on every single request, regardless of which role or JWT actually called it, and a bypassed policy filters nothing — every resident would see every row. `security_invoker = true` makes the view check privileges and RLS as the real caller instead: `web_resident`, after PostgREST's per-request `SET ROLE`, exactly as Exercise 5.2's policy expects.



(If you're changing an *existing* view rather than creating this one fresh, reach for `ALTER VIEW api.residents SET (security_invoker = true);` instead of dropping and recreating it — `DROP VIEW`

followed by `CREATE VIEW` makes a brand-new object with none of the original's grants, so Exercise 5.2's `GRANT SELECT ON api.residents TO web_resident` would silently need to be re-run, and the endpoint would 403 with permission denied for view residents until it is.)

```
NOTIFY pgrst, 'reload schema';
```

No `GRANT` yet, on purpose — not even to `web_resident`. RLS goes on before any role can read a single row.

5.2 — Enable RLS and add a self-only policy

```
ALTER TABLE residents ENABLE ROW LEVEL SECURITY;

CREATE POLICY resident_self_only ON residents
  FOR SELECT
  USING (id = (current_setting('request.jwt.claims', true)::json ->> 'resident_id')::int);

GRANT SELECT ON api.residents TO web_resident;
GRANT SELECT (id, full_name, neighbourhood) ON residents TO web_resident;
```

`current_setting('request.jwt.claims', true)` reads the JSON claims object PostgREST sets, per request, from the caller's verified JWT — the `true` second argument means “return NULL instead of erroring if it's unset,” which matters because `web_anon` requests never carry one at all. The policy compares the row's own `id` against whatever `resident_id` claim a valid, signed JWT presents.

Both `GRANTS` are required, and it's easy to stop at the first one and not notice. `security_invoker = true` from Exercise 5.1 means `api.residents` no longer borrows its owner's privileges to read the underlying table — the invoking role needs real privileges of its own on `residents`, not just on the view sitting in front of it. Skip the second `GRANT` and every request fails with permission denied for table `residents`, even though the role clearly has `SELECT` on the view; the view-level check passes, and the query still dies one level down, on the base table it was never given access to. The column list (`id, full_name, neighbourhood`) matters too, not just the table name: granting only these three columns — matching exactly what `api.residents` selects — keeps `true_duplicate_of` unreachable even if `web_resident` were ever queried against directly instead of through the view, which a bare `GRANT SELECT ON residents` (no column list) would not.

5.3 — Mint a JWT per resident

```
#!/usr/bin/env python3.12
# mint_jwt.py - issue a dev-only resident session token
import sys
import jwt

SECRET = "portsmith-lab-book-dev-secret-do-not-use-in-production"

resident_id = int(sys.argv[1])
```

```

token = jwt.encode(
    {"role": "web_resident", "resident_id": resident_id},
    SECRET,
    algorithm="HS256",
)
print(token)

```

The `role` claim is what PostgREST uses to decide which role to SET `ROLE` to for this request — it has to be a role authenticator was granted membership in back in Exercise 1.1, or PostgREST refuses the token outright.

5.4 — Two residents, two different answers from the same endpoint

```

TOKEN_1=$(python mint_jwt.py 1) # Adrian Foscolo
TOKEN_2=$(python mint_jwt.py 2) # Marisol Quintero

curl -s "http://localhost:3000/residents" -H "Authorization: Bearer $TOKEN_1" | jq
curl -s "http://localhost:3000/residents" -H "Authorization: Bearer $TOKEN_2" | jq

[ { "id": 1, "full_name": "Adrian Foscolo", "neighbourhood": "Old Town" } ]

[ { "id": 2, "full_name": "Marisol Quintero", "neighbourhood": "Riverside" } ]

```

Identical request, identical route, identical GRANT — the only difference is which JWT signed it, and RLS silently rewrote each query’s effective `WHERE` clause to match.

5.5 — The gotcha: RLS filters, it doesn’t refuse

```

curl -s "http://localhost:3000/residents?id=eq.2" -H "Authorization: Bearer $TOKEN_1" | jq

[]

```

Resident 1, asking directly for resident 2’s row by id, does **not** get a 403 — they get 200 OK and an empty array. This is a meaningfully different failure mode from Exercise 4’s column-privilege denial: a GRANT violation is an error, loud and explicit; an RLS mismatch is just a `WHERE` clause that happens to match nothing, indistinguishable at the HTTP layer from “that id doesn’t exist.” Compare with no credentials at all:

```

curl -si "http://localhost:3000/residents"

```

HTTP/1.1 401 Unauthorized

```

{"code":"42501","details":null,"hint":null,"message":"permission denied for table residents"}

```

This one errors, the same way Exercise 4.3 did — `web_anon` was never granted `SELECT` on `api.residents` at all, so it never gets far enough to run into the RLS policy in the first place. Two different denials, two different HTTP shapes, and the difference between them is worth being able to explain: a missing GRANT is a wall; RLS is a filter.

Exercise 6 — An RPC Endpoint for Fuzzy Search

6.1 — Wrap Chapter 5’s trigram search as a function

```
CREATE OR REPLACE FUNCTION api.search_businesses(search_term text)
RETURNS TABLE (business_id integer, name text, similarity numeric)
LANGUAGE sql STABLE AS $$
    SELECT business_id, name, round(similarity(name, search_term)::numeric, 3) AS similarity
    FROM   business_names
    ORDER BY name <-> search_term
    LIMIT  5;
$$;

GRANT EXECUTE ON FUNCTION api.search_businesses(text) TO web_anon;
GRANT SELECT ON business_names TO web_anon;

NOTIFY pgrst, 'reload schema';
```

The function body is Chapter 5, Exercise 5’s exact “did you mean?” query, verbatim — `name <-> search_term` ordering by trigram distance, backed by the `idx_business_names_trgm_gist` GiST index that chapter built. PostgREST doesn’t need to know anything about trigrams; it just sees a function in the `api` schema and exposes it at `/rpc/`.

Both GRANTS are required, for the same reason Exercise 5.2 needed two of them. SQL functions default to `SECURITY INVOKER` — this one was never told otherwise — so `api.search_businesses()` runs its `SELECT ... FROM business_names` as whichever role actually called it, `web_anon` here, not as the function’s owner. `EXECUTE` only grants permission to *call* the function; it says nothing about what the function is allowed to touch once it’s running. Skip the second `GRANT` and 6.2 fails with permission denied for table `business_names`, `EXECUTE` privilege notwithstanding — the same wall Exercise 5.2 hit with `residents`, just one function-call away instead of one view away.

6.2 — Call it

```
curl -s -X POST "http://localhost:3000/rpc/search_businesses" \
-H "Content-Type: application/json" \
-d '{"search_term": "Ironszyde Auto"}' | jq

[
  { "business_id": 45, "name": "Ironside Auto",      "similarity": 0.647 },
  { "business_id": 21, "name": "AutoFix Portsmouth", "similarity": 0.143 },
  { "business_id": 4,  "name": "Harbour Inn",       "similarity": 0.040 },
  { "business_id": 36, "name": "The Art Depot",     "similarity": 0.037 },
  { "business_id": 34, "name": "Riverside Cinema",  "similarity": 0.033 }
]
```

Same misspelling, same top match, same scores as Chapter 5’s own `psql` session — this endpoint is that query, not a reimplementaion of it.

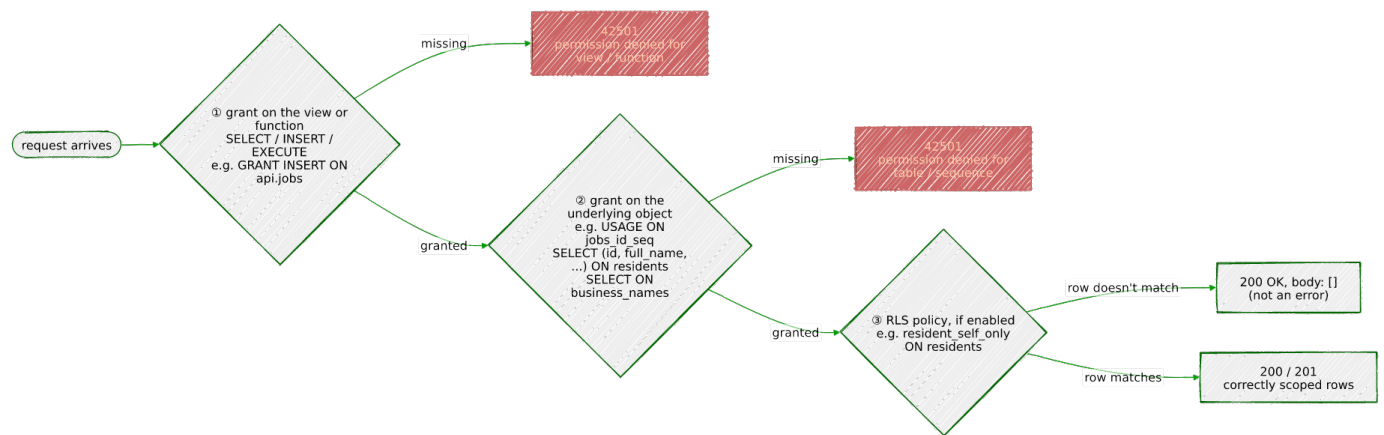
6.3 — GET works too, because the function is STABLE

```
curl -s "http://localhost:3000/rpc/search_businesses?search_term=Ironsyde+Auto" | jq -c '.[0]'
```

```
{"business_id":45,"name":"Ironside Auto","similarity":0.647}
```

PostgreSQL maps a function's SQL volatility onto which HTTP verbs it will accept: `VOLATILE` functions — anything that could plausibly write data — are reachable only by `POST`, on the theory that a `GET` should always be safe to retry, cache, or prefetch without side effects. `STABLE` and `IMMUTABLE` functions, like this one, are exposed under both `GET` and `POST`, because PostgreSQL itself already promises they don't change anything. Declaring `LANGUAGE sql STABLE` back in 6.1 wasn't just documentation — it's the line that decided whether a browser could ever call this endpoint from a plain link.

Every real failure in this chapter — Exercise 3's `jobs_id_seq`, Exercise 5's `residents` columns, this exercise's `business_names` — turned out to be the same shape: one grant in place, a second, independent one missing, one layer further in than the error first suggests:



Summary — What You Should Now Know

Tool	What it does
authenticator role, <code>NOINHERIT</code>	The one role PostgreSQL logs in as; switches per-request via <code>SET ROLE</code>
<code>db-anon-role</code>	The role a request without a valid JWT runs as
api schema of views	Decouples the public API shape from the internal table shape
<code>NOTIFY pgrst, 'reload schema'</code>	Tells a running PostgreSQL server about new tables/views/functions without a restart
<code>?col=eq.x, ?order=, ?select=, ?limit=/?offset=</code>	Query-parameter syntax mapping directly onto <code>WHERE/ORDER BY/SELECT/LIMIT</code>
Prefer: <code>return=representation</code>	Get the affected row(s) back in the response body

Tool	What it does
<code>Prefer: count=exact / Content-Range</code>	Ask for, and read, the total row count behind a paginated result
Column-level <code>GRANT/REVOKE</code>	Restrict which fields a client can set on <code>INSERT</code> , enforced by Postgres itself
<code>GRANT USAGE ON SEQUENCE ...</code>	Needed alongside <code>INSERT</code> for any <code>SERIAL/BIGSERIAL/IDENTITY</code> column's default <code>nextval()</code> to succeed — table privileges don't imply sequence privileges
401 VS. 403	401 = denied and unauthenticated (or bad JWT); 403 = denied despite valid credentials
RLS policy vs. missing <code>GRANT</code>	RLS silently filters rows (still 200, possibly empty); a missing grant is a hard error
<code>security_invoker = true</code> ON a view	Makes the view check privileges <i>and</i> RLS as the querying role, not the view's owner — required for RLS to mean anything through a view owned by a superuser or the table owner
<code>/rpc/<function></code>	A SQL function exposed as an endpoint; GET-eligible only if <code>STABLE/IMMUTABLE</code>

The key design insight from this chapter is that every access-control decision PostgREST makes was already something PostgreSQL could do — this chapter never introduced a new authorization concept, only a new transport for reaching decisions the database was always capable of making. `GRANT/REVOKE` from Exercise 4 and row level security from Exercise 5 aren't PostgREST features with database-flavored names; they are exactly the `GRANT` and `CREATE POLICY` you'd write for any other purpose, and PostgREST's only job was to authenticate a request, pick a role, and get out of the way. That's also this chapter's sharpest limitation, in the same sentence: anything that can't be expressed as one role running one SQL statement — multi-step workflows, calling another API mid-request, anything genuinely procedural — is out of scope by design, not by oversight.

Going further: Chapter 13's `LISTEN/NOTIFY` is the same primitive this chapter used for schema-reload notifications, applied to application data instead — a natural next step once `NOTIFY pgrst, 'reload schema'` feels familiar. Chapter 14's advisory locks are worth knowing about if an RPC function like `api.search_businesses()` ever needs to serialize access to a shared resource instead of just reading one. And the column-level `GRANT` from Exercise 4 generalizes: PostgreSQL's full privilege system — `SELECT`, `UPDATE`, and `REFERENCES` privileges can all be scoped per-column the same way `INSERT` was here — is worth a deliberate read through the `GRANT` documentation before exposing any table this way for real, well beyond what one chapter's exercises can cover.

Chapter 11 — Window Functions: Analytics

Beyond `GROUP BY`

“`GROUP BY` answers a question by throwing away the rows that don’t fit in the answer. A window function answers the same question and keeps every row anyway.”

Background

If you’ve used `GROUP BY`, you already know the shape of the problem this chapter solves — and the shape of its one real limitation. `GROUP BY` answers “what’s the average rating per neighbourhood?” by collapsing every business in a neighbourhood down into a single output row: you get the average, but every individual business that went into computing it is gone from the result. Most of the time that’s exactly what you want. But plenty of real questions don’t fit that shape: “how does each business’s rating compare to its neighbourhood’s average, *while still showing me every business*?” “What’s this business’s running revenue total, quarter by quarter, without collapsing the quarters together?” `GROUP BY` cannot answer either one — the instant it groups, the individual rows that made up the group are gone for good.

A **window function** answers a `GROUP BY`-shaped question without paying that price. It looks at a set of rows related to the current row — its “window” — computes something over them, and attaches the result to the current row, which survives, completely unchanged, right alongside every other row. Nothing collapses. Start with 48 rows, end with 48 rows, every one of them now carrying an answer that depended on looking at some of its neighbours.

That’s the single idea this entire chapter builds on, and it’s worth sitting with before touching any syntax:

`GROUP BY` reduces row count. A window function never does.

Four pieces, defined before you see them used

Every example below is built from four ingredients. Knowing what each one means in plain language first should make the SQL read as sentences instead of unfamiliar syntax:

Piece	Plain-language meaning
-------	------------------------

<code>OVER (...)</code>	
-------------------------	--

Piece

Plain-language meaning

“Compute this using a window of related rows, not just this one row.”

Attached after a function call — `AVG(x) OVER (...)` — it’s what turns an ordinary aggregate into a window function. An empty `OVER ()` means “the window is every row in the result.”

`PARTITION BY col`

“Only look at rows that share this row’s value of `col`.” The window equivalent of `GROUP BY` — it restricts *which rows count*, without collapsing any of them.

`ORDER BY col`

(*written inside OVER (...)*)

“Put the rows in this order before computing.” This ordering lives entirely inside the window — it’s unrelated to the query’s own outer `ORDER BY`, and the two are free to differ or even conflict.

frame clause —

e.g. `ROWS BETWEEN 6
PRECEDING AND
CURRENT ROW`

Out of the rows the partition and ordering make available, *exactly which ones* count toward this specific row’s calculation.

Exercise 1 walks through all four, one at a time, against a dataset small enough to check by hand — five numbers, no Portsmouth backstory required, so the mechanics stay in view before any real data shows up.

The Scenario

This chapter doesn’t tell one Portsmouth story — it reuses four different tables to show that window functions are a general-purpose tool, not a feature tied to any one kind of data:

Object	Source	Used for
businesses	Chapter 1	Ranking businesses within their own neighbourhood
sensor_readings	Chapter 8	7-day rolling averages and day-over-day change
network_events	Chapter 7	Detecting login “sessions” via gaps and islands
business_revenue	(<i>new, this chapter</i>)	Running totals and percentage-of-category-total

`business_revenue` is the one genuinely new thing here — a small, synthetic quarterly revenue figure for each of the 48 businesses from Chapter 1, built specifically so Exercise 6 has real running-total and percentage-of-partition data to work with.

Exercise Goals

By the end of this chapter you will be able to:

- Explain, without looking anything up, why `AVG(x) OVER ()` returns a value on every row while `AVG(x)` with `GROUP BY` returns one row — and predict, for any `OVER (...)` clause, exactly which rows are “in the window” for a given output row.
 - Rank rows within groups with `RANK()` and `DENSE_RANK()`, and state precisely how the two disagree the moment there’s a tie.
 - Write an explicit frame clause to compute a rolling average, and explain why adding `ORDER BY` to a window — with no frame clause at all — silently changes the default frame.
 - Use `LAG()/LEAD()` to compare a row to its neighbour without a self-join.
 - Recognize the “gaps and islands” pattern and use it to turn a raw event stream into sessions.
 - Combine two different `PARTITION BY` scopes — one for a running total, one for a percentage-of-total — in a single `SELECT`.
-

Installation

Nothing to install. Window functions have been part of core PostgreSQL since version 8.4 (2009) — no extension, no configuration.

Loading the Data

This chapter needs Chapters 1, 7, and 8’s data, plus one new small table:

```
python data/ch01_seed.py    # businesses
python data/ch07_seed.py    # network_events
python data/ch08_seed.py    # sensors, then run Chapter 8's own exercises
                             # through Exercise 2 to get sensor_readings populated
python data/ch11_seed.py    # business_revenue (new this chapter)
```

`business_revenue` only needs `businesses` to already exist — it doesn’t depend on Chapters 7 or 8 at all.

Pin the session timezone

```
SET timezone = 'UTC';
```

Same reason as Chapter 8: Exercises 3 and 4 group `sensor_readings` by `date_trunc('day', recorded_at)`, and a day boundary computed in any timezone other than UTC quietly buckets a different set of 5-minute readings into “Feb 1” than the ones this chapter’s numbers were computed from — not a rounding difference, a genuinely different average, since each bucket ends up averaging different rows entirely. Run this at the top of every session in this chapter, including the `psql` session that runs the prerequisite check below, or Exercises 3 and 4’s numbers won’t match what’s printed here even though the query is identical.

Verify the prerequisites

```
SELECT 'businesses' AS table, COUNT(*) FROM businesses
UNION ALL SELECT 'network_events', COUNT(*) FROM network_events
UNION ALL SELECT 'sensor_readings', COUNT(*) FROM sensor_readings
UNION ALL SELECT 'business_revenue', COUNT(*) FROM business_revenue;
```

table	count
businesses	48
network_events	116
sensor_readings	9648000
business_revenue	192

(4 rows)

If all four match, proceed to the exercises.

Exercises

Exercise 1 — The Mental Model, Traced By Hand

Forget Portsmouth for a moment. Here are five temperature readings from one morning:

```
SELECT * FROM (VALUES
  ('06:00'::time, 52.1),
  ('07:00'::time, 53.4),
  ('08:00'::time, 55.8),
  ('09:00'::time, 58.2),
  ('10:00'::time, 60.5)
) AS t(reading_time, temp_f);
```

reading_time	temp_f
06:00:00	52.1
07:00:00	53.4
08:00:00	55.8

```

09:00:00      |    58.2
10:00:00      |    60.5
(5 rows)

```

Five rows in, five rows to reason about. Keep this exact dataset in mind for everything below.

1.1 — GROUP BY collapses; OVER () doesn't

```

SELECT AVG(temp_f) FROM (VALUES
    ('06:00'::time, 52.1), ('07:00'::time, 53.4), ('08:00'::time, 55.8),
    ('09:00'::time, 58.2), ('10:00'::time, 60.5)
) AS t(reading_time, temp_f);

```

avg
56.0000000000000000

(1 row)

Five rows go in, one number comes out. That's the `AVG(temp_f)` you already know. Now the window version — same aggregate, same result, one difference:

```

SELECT reading_time, temp_f,
    round(AVG(temp_f) OVER ()::numeric, 2) AS morning_avg
FROM (VALUES
    ('06:00'::time, 52.1), ('07:00'::time, 53.4), ('08:00'::time, 55.8),
    ('09:00'::time, 58.2), ('10:00'::time, 60.5)
) AS t(reading_time, temp_f);

```

reading_time	temp_f	morning_avg
06:00:00	52.1	56.00
07:00:00	53.4	56.00
08:00:00	55.8	56.00
09:00:00	58.2	56.00
10:00:00	60.5	56.00

(5 rows)

Five rows go in, five rows come out — every one of them now carrying 56.00, the exact number `GROUP BY` gave you, just not at the cost of the other four rows. An empty `OVER ()` means “the window is every row here,” so every row gets the same whole-set average stamped onto it. This is the entire idea from the Background section, now sitting in front of you as actual output: nothing collapsed.

1.2 — The gotcha: adding ORDER BY silently changes the frame

Now add one thing — `ORDER BY reading_time`, *inside* the `OVER (...)`:

```

SELECT reading_time, temp_f,
    round(AVG(temp_f) OVER (ORDER BY reading_time)::numeric, 2) AS running_avg
FROM (VALUES
    ('06:00'::time, 52.1), ('07:00'::time, 53.4), ('08:00'::time, 55.8),

```



```

('09:00'::time, 58.2), ('10:00'::time, 60.5)
) AS t(reading_time, temp_f);

```

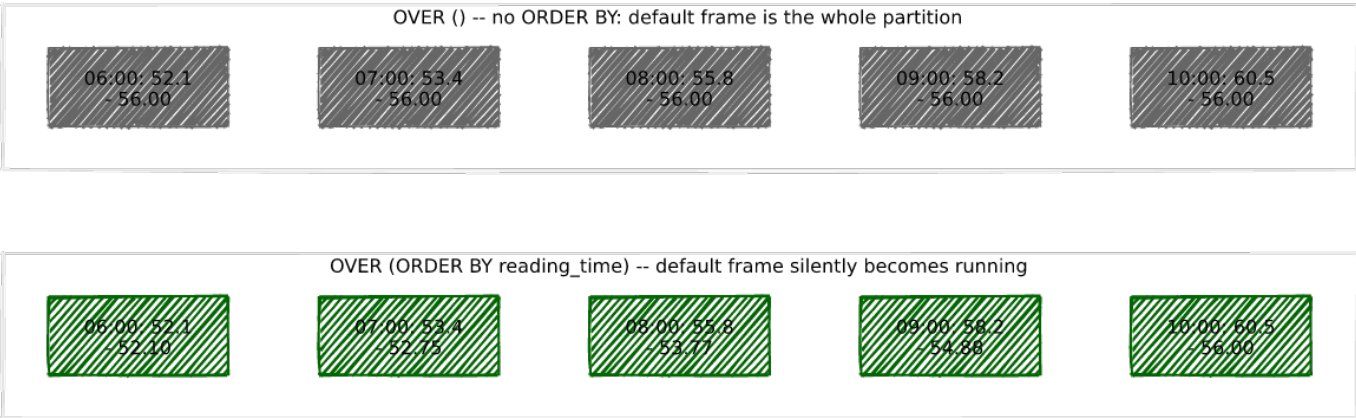
reading_time	temp_f	running_avg
06:00:00	52.1	52.10
07:00:00	53.4	52.75
08:00:00	55.8	53.77
09:00:00	58.2	54.88
10:00:00	60.5	56.00

(5 rows)

Same function, same data, completely different numbers — and only the *last* row still shows 56.00. This is the single most common surprise in all of window functions, so it’s worth tracing exactly what changed, row by row:

reading_time	temp_f	rows actually included in this row’s window	running_avg
06:00	52.1	[06:00]	52.10
07:00	53.4	[06:00, 07:00]	52.75
08:00	55.8	[06:00, 07:00, 08:00]	53.77
09:00	58.2	[06:00, 07:00, 08:00, 09:00]	54.88
10:00	60.5	[06:00, 07:00, 08:00, 09:00, 10:00]	56.00

Adding `ORDER BY` to a window did not just sort the rows — it silently changed *how many rows count* for each one, from “every row” down to “every row up to and including this one.” That’s PostgreSQL’s default frame the moment a window has an `ORDER BY` and no explicit frame clause: `RANGE UNBOUNDED PRECEDING AND CURRENT ROW`, i.e., a running calculation. Without `ORDER BY`, there’s nothing to run *up to*, so the default frame is the whole partition instead, which is exactly what 1.1’s flat 56.00 was. Nothing in the syntax announces this change — `ORDER BY` looks like it should only affect display order, and inside a window it affects something much bigger. Committing this one rule to memory now will save you from debugging a “wrong” running total later that was never actually wrong, just unexpectedly running.



1.3 — PARTITION BY: independent windows, still no collapsing

Add a second sensor to the tiny dataset:

```
SELECT sensor_label, reading_time, temp_f,  
       round(AVG(temp_f) OVER (PARTITION BY sensor_label ORDER BY reading_time)::numeric, 2) AS  
running_avg  
FROM (VALUES  
      ('Temp-01', '06:00'::time, 52.1), ('Temp-01', '07:00'::time, 53.4), ('Temp-01',  
'08:00'::time, 55.8),  
      ('Temp-02', '06:00'::time, 48.9), ('Temp-02', '07:00'::time, 49.5), ('Temp-02',  
'08:00'::time, 50.1)  
      ) AS t(sensor_label, reading_time, temp_f)  
ORDER BY sensor_label, reading_time;
```

sensor_label	reading_time	temp_f	running_avg
Temp-01	06:00:00	52.1	52.10
Temp-01	07:00:00	53.4	52.75
Temp-01	08:00:00	55.8	53.77
Temp-02	06:00:00	48.9	48.90
Temp-02	07:00:00	49.5	49.20
Temp-02	08:00:00	50.1	49.50

Temp-01's running average never sees Temp-02's numbers, and vice versa — PARTITION BY walled the two sensors off into completely independent windows, the same way GROUP BY sensor_label would have, except both sensors' six rows are all still here. This is PARTITION BY doing to a window exactly what it would do to a GROUP BY: split the data into groups — just without ever throwing a row away.

1.4 — An explicit frame clause: precise control

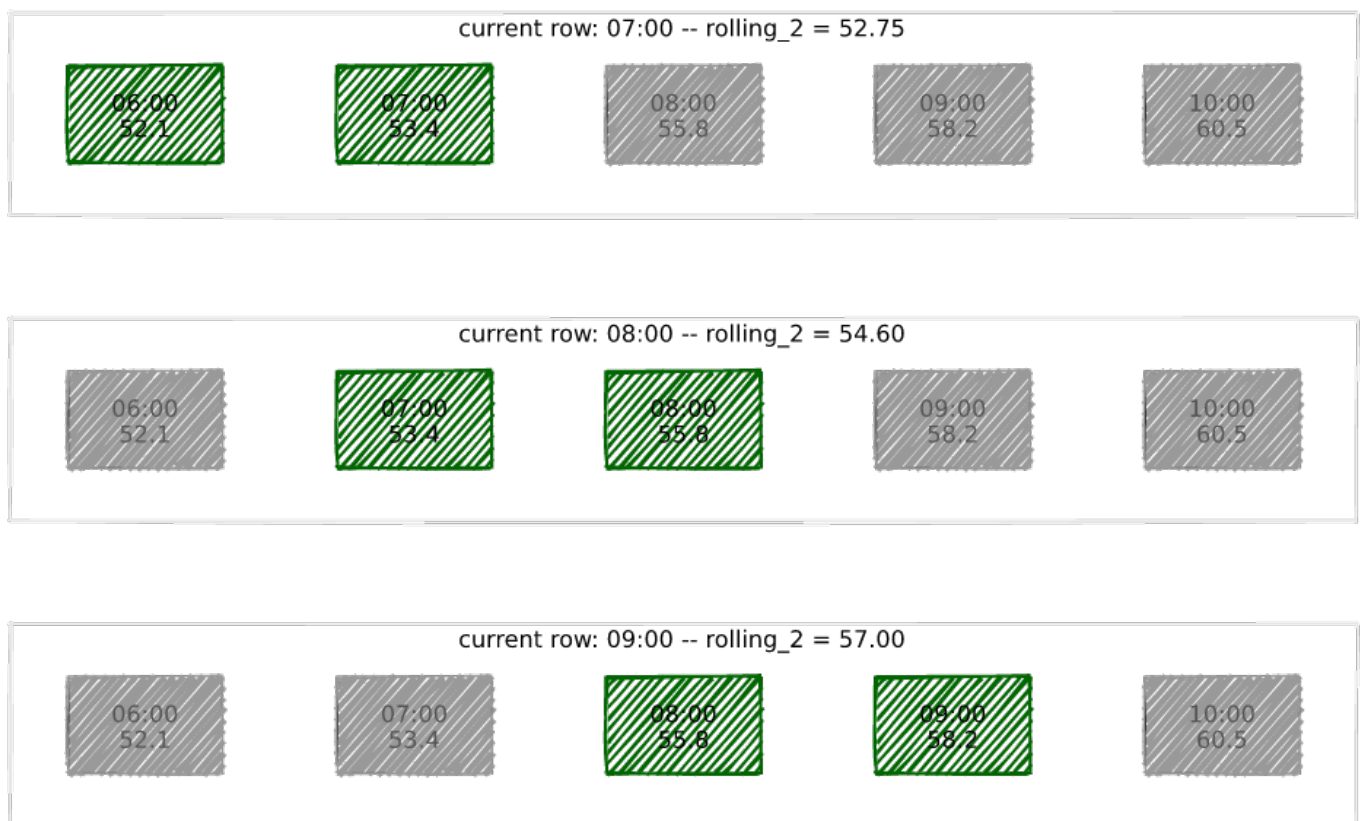
1.2's running average grows to include more and more history as it goes — by the last row, it's averaging all five readings. A **rolling** average instead asks "the last *N* readings only," which needs an explicit frame clause instead of relying on the default:

```
SELECT reading_time, temp_f,  
       round(AVG(temp_f) OVER (ORDER BY reading_time  
                               ROWS BETWEEN 1 PRECEDING AND CURRENT ROW)::numeric, 2) AS  
rolling_2  
FROM (VALUES  
      ('06:00'::time, 52.1), ('07:00'::time, 53.4), ('08:00'::time, 55.8),  
      ('09:00'::time, 58.2), ('10:00'::time, 60.5)  
      ) AS t(reading_time, temp_f);
```

reading_time	temp_f	rolling_2
06:00:00	52.1	52.10
07:00:00	53.4	52.75
08:00:00	55.8	54.60

09:00:00		58.2		57.00
10:00:00		60.5		59.35

reading_time	rows in this row's window (1 PRECEDING AND CURRENT ROW)	rolling_2
06:00	[06:00] — no prior row exists yet, so just itself	52.10
07:00	[06:00, 07:00]	52.75
08:00	[07:00, 08:00]	54.60
09:00	[08:00, 09:00]	57.00
10:00	[09:00, 10:00]	59.35



The window is now a fixed-size sliding pair, not an ever-growing history — 08:00's value depends on 07:00 and 08:00 only, never 06:00. ROWS BETWEEN 1 PRECEDING AND CURRENT ROW is deliberately the same shape of clause Exercise 3 uses next, just with a smaller number: this tiny example *is* a 2-reading rolling average, and Exercise 3 is nothing more than this same idea with 6 PRECEDING and real sensor data behind it.

Exercise 2 — Ranking Businesses Within Their Neighbourhood

2.1 — RANK() and DENSE_RANK(), side by side

```
SELECT name, neighbourhood, (details->>'rating')::numeric AS rating,
       RANK() OVER (PARTITION BY neighbourhood ORDER BY (details->>'rating')::numeric
```

```
DESC) AS rank,
      DENSE_RANK() OVER (PARTITION BY neighbourhood ORDER BY (details->>'rating')::numeric
DESC) AS dense_rank
FROM businesses
WHERE neighbourhood IN ('Harbour District', 'Riverside')
ORDER BY neighbourhood, rating DESC;
```

name	neighbourhood	rating	rank	dense_rank
Lighthouse Bookshop	Harbour District	5.0	1	1
Portsmith Fish Market	Harbour District	4.8	2	2
Saltbox Gallery	Harbour District	4.7	3	3
Mariners Rest B&B	Harbour District	4.7	3	3
Harbour View Theater	Harbour District	4.6	5	4
Tidal Wave Surf Shop	Harbour District	4.5	6	5
The Gilded Clam	Harbour District	4.5	6	5
Harbour Inn	Harbour District	4.3	8	6
Anchor & Oar Tavern	Harbour District	4.1	9	7
River Bend Bakery	Riverside	4.8	1	1
Portsmith Veterinary Clinic	Riverside	4.8	1	1
Dr. Chen Dentistry	Riverside	4.7	3	2
Quay Street Deli	Riverside	4.6	4	3
The Art Depot	Riverside	4.6	4	3
Thai Orchid	Riverside	4.5	6	4
The Riverside Vegan	Riverside	4.5	6	4
Riverside Cinema	Riverside	4.4	8	5
Portsmith Pharmacy	Riverside	4.3	9	6

(18 rows)

(Lighthouse Bookshop’s 5.0 is Chapter 1, Exercise 5’s jsonb_set update — if you’re seeing 4.9 instead, that exercise hasn’t run yet in this database, which is fine; the ranking logic below is identical either way.)

2.2 — Reading the tie exactly

Look at Harbour District’s two businesses tied at 4.7: both get rank = 3. The next business down, at 4.6, gets rank = 5 under RANK() — 4 is simply never used, because two rows already claimed “3rd place” and RANK() counts every row ahead of you, ties included. DENSE_RANK() disagrees on principle: it counts *distinct rating values* seen so far, so 4.6 is the 4th distinct value in the list and gets dense_rank = 4, no gap. Neither is “more correct” — RANK() answers “how many businesses rate at or above me,” DENSE_RANK() answers “how many distinct rating tiers are at or above me” — but they give a different answer to “who’s in 4th place” the moment any tie exists, and Riverside’s 4.8 tie two rows later shows the same split happening again.

Exercise 3 — A 7-Day Rolling Average on sensor_readings

3.1 — Aggregate to daily first

sensor_readings reports every five minutes — a rolling average over raw readings would be a rolling average of noise. Roll up to one row per day first, the same shape of query Chapter 9 turned into a materialized view, just computed directly here instead:

```
WITH daily AS (  
    SELECT date_trunc('day', recorded_at)::date AS reading_day,  
           round(AVG(reading_value)::numeric, 2) AS daily_avg  
    FROM   sensor_readings  
    WHERE  sensor_id = 1  
    AND    recorded_at >= '2024-02-01' AND recorded_at < '2024-02-15'  
    GROUP BY 1  
)  
SELECT * FROM daily ORDER BY reading_day;
```

reading_day	daily_avg
2024-02-01	42.46
2024-02-02	42.37
2024-02-03	42.37
2024-02-04	42.40
2024-02-05	42.32
2024-02-06	42.43
2024-02-07	42.43
2024-02-08	42.38
2024-02-09	42.29
2024-02-10	42.40
2024-02-11	42.45
2024-02-12	42.48
2024-02-13	42.40
2024-02-14	42.41

(14 rows)

Sensor 1 over the first two weeks of February — the month right after Chapter 8’s dropped January partition, well clear of it. Check sensors and this is genuinely Temp-01, the same sensor Exercise 1.3’s toy example was named after — the tiny hand-crafted dataset was standing in for exactly this real one.

3.2 — Layer the rolling average on top

```
WITH daily AS (  
    SELECT date_trunc('day', recorded_at)::date AS reading_day,  
           round(AVG(reading_value)::numeric, 2) AS daily_avg  
    FROM   sensor_readings  
    WHERE  sensor_id = 1  
    AND    recorded_at >= '2024-02-01' AND recorded_at < '2024-02-15'  
    GROUP BY 1  
)  
SELECT reading_day, daily_avg,  
       round(AVG(daily_avg) OVER (ORDER BY reading_day  
                                  ROWS BETWEEN 6 PRECEDING AND CURRENT ROW)::numeric, 2) AS  
rolling_7day_avg
```

```
FROM daily
ORDER BY reading_day;
```

reading_day	daily_avg	rolling_7day_avg
2024-02-01	42.46	42.46
2024-02-02	42.37	42.42
2024-02-03	42.37	42.40
2024-02-04	42.40	42.40
2024-02-05	42.32	42.38
2024-02-06	42.43	42.39
2024-02-07	42.43	42.40
2024-02-08	42.38	42.39
2024-02-09	42.29	42.37
2024-02-10	42.40	42.38
2024-02-11	42.45	42.39
2024-02-12	42.48	42.41
2024-02-13	42.40	42.40
2024-02-14	42.41	42.40

(14 rows)

6 PRECEDING AND CURRENT ROW is 7 rows total — the exact same shape as Exercise 1.4’s 1 PRECEDING AND CURRENT ROW, just sized for a week instead of a pair. Watch the first week build up exactly like 1.4’s first row did: Feb 1 has no prior days, so its “7-day” average is really a 1-day average; Feb 2 averages 2 days; the window doesn’t reach a true 7 full days until Feb 7. ROWS BETWEEN ... PRECEDING never errors when fewer rows exist than requested — it just uses whatever’s actually available, which is worth knowing before trusting the first few values of any rolling window on real data.

Exercise 4 — LAG()/LEAD(): Comparing a Row to Its Neighbour

4.1 — Day-over-day change, without a self-join

```
WITH daily AS (
    SELECT date_trunc('day', recorded_at)::date AS reading_day,
           round(AVG(reading_value)::numeric, 2) AS daily_avg
    FROM sensor_readings
    WHERE sensor_id = 1
    AND recorded_at >= '2024-02-01' AND recorded_at < '2024-02-15'
    GROUP BY 1
)
SELECT reading_day, daily_avg,
       LAG(daily_avg) OVER (ORDER BY reading_day) AS prev_day_avg,
       round((daily_avg - LAG(daily_avg) OVER (ORDER BY reading_day))::numeric, 2) AS
day_over_day_change,
       LEAD(daily_avg) OVER (ORDER BY reading_day) AS next_day_avg
FROM daily
ORDER BY reading_day
LIMIT 6;
```

reading_day	daily_avg	prev_day_avg	day_over_day_change	next_day_avg
2024-02-01	42.46			42.37
2024-02-02	42.37	42.46	-0.09	42.37
2024-02-03	42.37	42.37	0.00	42.40
2024-02-04	42.40	42.37	0.03	42.32
2024-02-05	42.32	42.40	-0.08	42.43
2024-02-06	42.43	42.32	0.11	42.43

(6 rows)

LAG(col) reaches one row backward in the window’s order; LEAD(col) reaches one row forward. Both are just AVG(...) OVER (...)’s siblings — ordinary window functions, not aggregates, so they don’t need a frame clause at all. February 1st has no prior day, so prev_day_avg and day_over_day_change are both NULL for it — not zero, not an error, genuinely unknown, exactly the way a self-join against “yesterday” would also come up empty for the very first row. Before this chapter, the only way to compare a row to its neighbour was a self-join on date - 1 = date; LAG()/LEAD() is that same comparison with no join at all.

Exercise 5 — Gaps and Islands: Detecting Sessions in network_events

5.1 — The pattern, in words before SQL

“Gaps and islands” names a specific two-step trick: find the *gaps* (where consecutive rows, for the same actor, are far enough apart in time to count as separate events), then turn the space between gaps into *islands* (contiguous runs, numbered, that become your sessions). Step one is LAG() from Exercise 4. Step two is a cumulative SUM() — a running total of “did a new island start yet,” which is exactly Exercise 1.2’s running-total behavior, repurposed as a counter instead of an average.

5.2 — Step one: flag where each session starts

```
SELECT source_ip, event_type, occurred_at,
       (occurred_at - LAG(occurred_at) OVER (PARTITION BY source_ip ORDER BY occurred_at))
       > interval '5 minutes' AS is_new_session
FROM   network_events
WHERE  source_ip IN ('192.0.2.47', '192.0.2.151')
ORDER BY source_ip, occurred_at;
```

source_ip	event_type	occurred_at	is_new_session
192.0.2.47	api_call	2024-03-10 00:18:00-05	
192.0.2.47	api_call	2024-03-10 00:20:00-05	f
192.0.2.47	api_call	2024-03-10 00:28:00-05	t
192.0.2.47	api_call	2024-03-10 00:36:00-05	t
192.0.2.47	api_call	2024-03-10 00:38:00-05	f
192.0.2.151	api_call	2024-03-09 23:53:00-05	
192.0.2.151	api_call	2024-03-09 23:55:00-05	f

```

192.0.2.151 | api_call    | 2024-03-09 23:56:00-05 | f
192.0.2.151 | api_error   | 2024-03-10 00:00:00-05 | f
192.0.2.151 | api_call    | 2024-03-10 00:07:00-05 | t
192.0.2.151 | api_call    | 2024-03-10 00:12:00-05 | f

```

A 5-minute threshold: 192.0.2.47’s second event lands 2 minutes after its first (f, still the same visit), but its third lands 8 minutes after that (t — long enough to count as a new visit). The first row for any IP has nothing before it, so LAG() returns NULL and the comparison is NULL, not true or false — handled explicitly in the next step.

5.3 — Step two: turn the flags into session numbers

```

WITH gapped AS (
    SELECT source_ip, event_type, occurred_at,
           (occurred_at - LAG(occurred_at) OVER (PARTITION BY source_ip ORDER BY occurred_at))
             > interval '5 minutes' AS is_new_session
    FROM   network_events
),
islands AS (
    SELECT source_ip, event_type, occurred_at,
           SUM(CASE WHEN is_new_session IS NOT FALSE THEN 1 ELSE 0 END)
             OVER (PARTITION BY source_ip ORDER BY occurred_at) AS session_num
    FROM   gapped
)
SELECT source_ip, session_num, event_type, occurred_at
FROM   islands
WHERE  source_ip IN ('192.0.2.47', '192.0.2.151')
ORDER BY source_ip, occurred_at;

```

source_ip	session_num	event_type	occurred_at
192.0.2.47	1	api_call	2024-03-10 00:18:00-05
192.0.2.47	1	api_call	2024-03-10 00:20:00-05
192.0.2.47	2	api_call	2024-03-10 00:28:00-05
192.0.2.47	3	api_call	2024-03-10 00:36:00-05
192.0.2.47	3	api_call	2024-03-10 00:38:00-05
192.0.2.151	1	api_call	2024-03-09 23:53:00-05
192.0.2.151	1	api_call	2024-03-09 23:55:00-05
192.0.2.151	1	api_call	2024-03-09 23:56:00-05
192.0.2.151	1	api_error	2024-03-10 00:00:00-05
192.0.2.151	2	api_call	2024-03-10 00:07:00-05
192.0.2.151	2	api_call	2024-03-10 00:12:00-05

IS NOT FALSE — not = true — is what makes a row’s own first event (where is_new_session is NULL) correctly count as the start of session 1 instead of silently vanishing from every sum downstream; NULL = true and NULL AND anything are both NULL in SQL’s three-valued logic, never true, so a plain WHEN is_new_session THEN 1 would skip every partition’s opening row. 192.0.2.47 splits into three short sessions; 192.0.2.151 splits into two. Neither IP did anything unusual — the same api_calls, just separated by an 8- and a 7-minute pause respectively, long enough to cross this query’s 5-minute line.

5.4 — Roll it up

```
WITH gapped AS (  
    SELECT source_ip, occurred_at,  
           (occurred_at - LAG(occurred_at) OVER (PARTITION BY source_ip ORDER BY occurred_at))  
             > interval '5 minutes' AS is_new_session  
    FROM network_events  
)  
islands AS (  
    SELECT source_ip, occurred_at,  
           SUM(CASE WHEN is_new_session IS NOT FALSE THEN 1 ELSE 0 END)  
             OVER (PARTITION BY source_ip ORDER BY occurred_at) AS session_num  
    FROM gapped  
)  
SELECT COUNT(DISTINCT source_ip) AS distinct_ips,  
       COUNT(DISTINCT (source_ip, session_num)) AS total_sessions  
FROM islands;  
  
distinct_ips | total_sessions  
-----+-----  
51 | 66  
(1 row)
```

51 distinct IPs produced 66 sessions — 15 of them split into more than one visit under this threshold. Change `interval '5 minutes'` to `interval '10 minutes'` (the longest gap that exists anywhere in this dataset, per Chapter 7’s generator) and every one of those 66 collapses back down to exactly 51: the threshold you choose *is* the definition of “one visit,” and this query has no way of knowing which threshold is right for your actual users — that’s a judgment call the data alone can’t make for you.

Exercise 6 — Running Total and Percentage-of-Category-Total, Together

6.1 — Two different partitions, one query

```
WITH revenue_calc AS (  
    SELECT b.id, b.name, b.details->>'category' AS category, r.quarter, r.revenue,  
           round(SUM(r.revenue) OVER (PARTITION BY r.business_id ORDER BY r.quarter  
                                     ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT  
ROW)::numeric, 2)  
           AS running_total,  
           round(SUM(r.revenue) OVER (PARTITION BY b.id) * 100.0  
                 / SUM(r.revenue) OVER (PARTITION BY b.details->>'category'), 2)  
           AS pct_of_category_annual  
    FROM business_revenue r  
    JOIN businesses b ON b.id = r.business_id  
)  
SELECT name, quarter, revenue, running_total, pct_of_category_annual  
FROM revenue_calc
```

```
WHERE name IN ('The Gilded Clam', 'Bella Napoli')
ORDER BY name, quarter;
```

name	quarter	revenue	running_total	pct_of_category_annual
Bella Napoli	1	68827.50	68827.50	7.35
Bella Napoli	2	84983.85	153811.35	7.35
Bella Napoli	3	90452.38	244263.73	7.35
Bella Napoli	4	75346.83	319610.56	7.35
The Gilded Clam	1	75282.75	75282.75	8.04
The Gilded Clam	2	92954.38	168237.13	8.04
The Gilded Clam	3	98935.80	267172.93	8.04
The Gilded Clam	4	82413.52	349586.45	8.04

Three window functions, two different `PARTITION BY` scopes, in the exact same `SELECT`. `running_total` partitions by `business_id` and orders by `quarter` — Exercise 1.2’s running sum, applied to money instead of temperature. `pct_of_category_annual` partitions by `category` with **no `ORDER BY` at all** — Exercise 1.1’s flat, whole-partition total, computed twice with two different partitions (once for just this business, once for its whole category) and divided. Both restaurants’ percentages stay identical across all four of their own rows, exactly like 1.1’s 56.00 did, because neither has an `ORDER BY` to turn it into anything running.

6.2 — A gotcha worth hitting on purpose

Restrict the query further, to see what a narrower `WHERE` clause does to a percentage that’s supposed to mean “share of the whole category”:

```
SELECT b.name, r.quarter, r.revenue,
       round(SUM(r.revenue) OVER (PARTITION BY b.id) * 100.0
            / SUM(r.revenue) OVER (PARTITION BY b.details->>'category'), 2) AS pct_of_category
FROM   business_revenue r
JOIN   businesses b ON b.id = r.business_id
WHERE  b.name IN ('The Gilded Clam', 'Bella Napoli')  -- filtered BEFORE the window runs
ORDER BY b.name, r.quarter;
```

name	quarter	revenue	pct_of_category
Bella Napoli	1	68827.50	47.76
Bella Napoli	2	84983.85	47.76
Bella Napoli	3	90452.38	47.76
Bella Napoli	4	75346.83	47.76
The Gilded Clam	1	75282.75	52.24
The Gilded Clam	2	92954.38	52.24
The Gilded Clam	3	98935.80	52.24
The Gilded Clam	4	82413.52	52.24

$47.76 + 52.24 = 100.00$ — these two restaurants apparently make up the *entire* restaurant category, when Portsmouth actually has 15. `WHERE` runs before `OVER (...)` ever sees a row: filtering down to two businesses filtered the `PARTITION BY category` window down to just those same two

businesses, so “the category total” silently became “the total of the two rows I happened to ask for.” The fix is 6.1’s structure, not a different formula: compute every window function first, across the *entire* unfiltered table, inside a CTE — then filter the CTE’s output afterward, in an outer query the windows never see. A `WHERE` clause placed before a window and a `WHERE` clause placed after one are answering two different questions, and PostgreSQL will never warn you which one you actually wrote.

Summary — What You Should Now Know

Tool	What it does
<code>func(...) OVER (...)</code>	Computes across a window of related rows without collapsing the current row away
<code>PARTITION BY col</code>	Restricts the window to rows sharing this row’s value — <code>GROUP BY</code> without the collapsing
<code>ORDER BY col</code> inside <code>OVER (...)</code>	Orders the window — and, with no explicit frame, silently switches the default frame from “whole partition” to “running up to this row”
<code>ROWS BETWEEN x PRECEDING AND CURRENT ROW</code>	An explicit frame — precise, fixed-size control over which neighbouring rows count
<code>RANK()</code> VS. <code>DENSE_RANK()</code>	Agree with no ties; <code>RANK()</code> leaves gaps after a tie, <code>DENSE_RANK()</code> never does
<code>LAG()</code> / <code>LEAD()</code>	Reach one row backward/forward in window order — a neighbour comparison with no self-join
gaps and islands	<code>LAG()</code> flags where a new group starts; a cumulative <code>SUM()</code> turns those flags into group numbers
<code>IS NOT FALSE</code>	The three-valued-logic-safe way to treat a <code>NULL</code> flag (a partition’s first row) as “start a new group”
Two <code>PARTITION BY</code> scopes, one query	Perfectly legal — e.g. a running total per entity alongside a percentage of a <i>different</i> , broader group
<code>WHERE</code> before <code>OVER (...)</code>	Filters the rows a window function ever sees — a narrow <code>WHERE</code> silently narrows what “the partition total” means

The key design insight from this chapter is the one from the very first paragraph, now proven six different ways: a window function is what you reach for the moment a `GROUP BY`-shaped question needs an answer *without* losing the rows that produced it. Ranking within a group, a rolling average, a neighbour comparison, a session boundary, a running total next to a category share — every one of these is the same underlying move, `OVER (...)` attached to a function that

would otherwise collapse your data, with `PARTITION BY`, `ORDER BY`, and a frame clause as the three knobs that decide exactly which neighbouring rows a given row is allowed to see.

Going further: Chapter 9's materialized views precompute the exact kind of daily rollup Exercise 3 built on the fly — worth comparing directly now that you've seen both: a materialized view pays the aggregation cost once, at refresh time, while a window function pays it on every query but never goes stale. Chapter 20's `pg_stat_statements` work benefits from the running-total pattern in Exercise 6 when tracking cumulative query cost over time. And Chapter 12's recursive CTEs are this book's other tool for "a query that needs to see more than just the current row" — recursive CTEs walk relationships the data itself defines (a parent, a neighbour node), where window functions walk an ordering you impose yourself; knowing which kind of "related rows" a problem actually has is most of the work of picking the right one.

Chapter 12 — Recursive CTEs: Graphs and Hierarchies

“SQL doesn’t have loops. WITH RECURSIVE is a query that keeps calling itself until it runs out of new things to say.”

Background

Every table so far in this book has been “flat” — rows related to each other through a foreign key, sure, but never through a relationship of *unknown depth*. A resident belongs to one neighbourhood. A reading belongs to one sensor. But “who does this employee ultimately report to?” or “how do I get from this intersection to that one?” can’t be answered by following one foreign key — the answer might be one hop away, or ten, and a plain JOIN has to know in advance how many hops to write.

A **CTE** — Common Table Expression, the thing a `WITH name AS (...)` block in front of a query defines — is ordinarily just a named subquery, a way to give a piece of SQL a label and reuse it, nothing recursive about it (Chapter 3 already used a plain one to `UPDATE` and log a job in a single statement). A **recursive CTE**, written `WITH RECURSIVE`, is the special case: a CTE allowed to refer to *itself* inside its own definition, which is PostgreSQL’s answer to “I don’t know how many joins this needs — figure it out as you go.”

Structurally, a recursive CTE has two halves glued together with `UNION` or `UNION ALL`:

```
WITH RECURSIVE cte_name AS (  
    SELECT ...                -- ① the anchor: runs exactly once  
    UNION ALL  
    SELECT ...                -- ② the recursive term: references cte_name itself  
    FROM   some_table  
    JOIN   cte_name ON ...  
)  
SELECT * FROM cte_name;
```

The **anchor** runs once and seeds the result with a starting set of rows. The **recursive term** then runs *repeatedly* — each time, it sees only the rows the *previous* run just produced, joins them against the table again to find “the next hop,” and adds whatever new rows come out. This keeps going until one run of the recursive term produces zero new rows, at which point PostgreSQL stops and the final result is everything the anchor and every round of the recursive term ever added together. Nothing here is exotic under the hood — it’s an ordinary loop, just

spelled out as a query instead of application code, and it stops for the most ordinary reason a loop ever stops: it ran out of new work.

Trees are graphs that promised to behave. A tree — an org chart, a category hierarchy — is a graph where every node has exactly one parent and there’s no way to walk back to somewhere you’ve already been. A road network makes no such promise: intersections connect to several other intersections, and it’s entirely possible to walk in a circle and end up back where you started. Recursive CTEs handle both shapes with the same syntax, but only the graph case needs you to actively guard against walking in circles forever — Exercises 1 through 3 build the tree intuition first, precisely so Exercise 4 can show you, concretely, what goes wrong without that guard, before Exercise 5 asks you to navigate a graph that has a real cycle built into it on purpose.

The Scenario

Object	Source	Shape
city_org	(new)	Tree — Portsmouth’s 30-person city government, 4 levels deep
intersections, road_segments	(new, derived from Chapter 2’s city_infrastructure)	Graph — a real road network, cycles included
categories	(new, derived from Chapter 1’s real category data)	Tree — a 3-level faceted-search hierarchy

intersections and road_segments aren’t invented — every node is a real point where two of Chapter 2’s actual road LINESTRINGS cross, found with ST_Intersects, and every edge length is the real along-the-road distance for that stretch, not a straight-line guess. Ring Road bends around three sides of the city between some of its intersections, so a straight line between two of its crossing points would have understated the real distance by more than 5 kilometres in one case — Exercise 5 puts that exact gap to use.

Exercise Goals

By the end of this chapter you will be able to:

- Explain the anchor/recursive-term structure of WITH RECURSIVE well enough to predict how many times the recursive term will run for a given tree.
- Walk a hierarchy in both directions — up to the root, and down to every descendant — and know which direction each shape of question needs.

- Compute each node’s depth and render an arbitrary-depth tree as indented text.
 - Recognize when a self-referencing table can contain a genuine cycle, and use `CYCLE ... SET ... USING` to detect one instead of hanging.
 - Write a breadth-first recursive CTE to find a shortest path through a graph, and explain why “fewest hops” and “shortest distance” are not always the same answer.
 - Flatten a category tree into an ancestor list for faceted search.
-

Installation

Nothing to install. `WITH RECURSIVE` has been part of core PostgreSQL since version 8.4 (2009) — the same release that introduced Chapter 11’s window functions.

Loading the Data

This chapter needs Chapters 1 and 2’s data, plus its own new tables:

```
python data/ch01_seed.py    # businesses (categories are derived from this)
python data/ch02_seed.py    # city_infrastructure (roads are derived from this)
python data/ch12_seed.py    # city_org, intersections, road_segments, categories
```

Verify the prerequisites

```
SELECT 'city_org' AS table, COUNT(*) FROM city_org
UNION ALL SELECT 'intersections', COUNT(*) FROM intersections
UNION ALL SELECT 'road_segments', COUNT(*) FROM road_segments
UNION ALL SELECT 'categories', COUNT(*) FROM categories;
```

table	count
city_org	30
intersections	15
road_segments	19
categories	48

(4 rows)

If all four match, proceed to the exercises.

Exercises

Exercise 1 — Walking to the Root

1.1 — The shape of city_org

```
\d city_org
```

Table "public.city_org"				
Column	Type	Collation	Nullable	Default
id	integer		not null	nextval('city_org_id_seq'::regclass)
name	text		not null	
title	text		not null	
manager_id	integer			

Foreign-key constraints:

```
"city_org_manager_id_fkey" FOREIGN KEY (manager_id) REFERENCES city_org(id)
```

A **self-referencing foreign key** — `manager_id` points to another row in the same table. This one column is the entire tree: the Mayor's `manager_id` is NULL (the root has no manager), and everyone else's `manager_id` points to the row above them.

1.2 — From a patrol officer to the Mayor

```
WITH RECURSIVE chain AS (  
    SELECT id, name, title, manager_id, 0 AS depth  
    FROM city_org  
    WHERE name = 'Kwame Asante'  
  
    UNION ALL  
  
    SELECT o.id, o.name, o.title, o.manager_id, chain.depth + 1  
    FROM city_org o  
    JOIN chain ON o.id = chain.manager_id  
)  
SELECT depth, name, title FROM chain ORDER BY depth;
```

depth	name	title
0	Kwame Asante	Patrol Officer
1	Marcus Reilly	Patrol Captain
2	Diane Okonjo	Chief of Police
3	Coretta Vance	Mayor

(4 rows)

Trace exactly what happened, round by round, since this is the pattern every other exercise in this chapter builds on:

- **Anchor** (`depth = 0`): finds exactly one row, Kwame Asante.
- **Round 1** of the recursive term: joins `city_org` against the anchor's *one* row, looking for whoever has `id = Kwame's manager_id` — finds Marcus Reilly, tags him `depth = 1`.

- **Round 2:** joins against *only* Marcus Reilly's row (not Kwame's again) — finds Diane Okonjo, depth = 2.
- **Round 3:** joins against *only* Diane's row — finds Coretta Vance, depth = 3.
- **Round 4:** joins against *only* Coretta's row, looking for whoever has id = Coretta's manager_id — but manager_id is NULL for the Mayor, and nothing has id = NULL. Zero rows come back. The recursion stops here, on its own, having found the natural end of the chain.

Four rounds for four levels — that's not a coincidence. A recursive CTE walking one path up (or down) a tree always runs exactly as many times as the tree is deep along that path, no more.

Exercise 2 — Every Employee Under a Department Head

2.1 — Same idea, opposite direction

Exercise 1 walked *up*: “whose id equals my manager_id.” Finding everyone *under* a department head walks *down* instead: “whose manager_id equals my id.”

```
WITH RECURSIVE subtree AS (
    SELECT id, name, title, manager_id, 0 AS depth
    FROM city_org
    WHERE name = 'Marcus Webb'

    UNION ALL

    SELECT o.id, o.name, o.title, o.manager_id, subtree.depth + 1
    FROM city_org o
    JOIN subtree ON o.manager_id = subtree.id
)
SELECT name, title, depth FROM subtree ORDER BY depth, name;
```

name	title	depth
Marcus Webb	Director of Public Works	0
Dana Ruiz	Streets & Sanitation Supervisor	1
Tom Delgado	Water & Sewer Supervisor	1
Ivy Chen	Water & Sewer Crew	2
Leo Park	Streets Crew	2
Noah Brandt	Water & Sewer Crew	2
Priya Nair	Streets Crew	2
Sam Okafor	Streets Crew	2

(8 rows)

Only the JOIN condition flipped — o.manager_id = subtree.id instead of o.id = subtree.manager_id — and the exact same query shape now answers a completely different question. This distinction matters more than it looks: get the join condition backward and PostgreSQL won't error, it will just silently return the anchor row and nothing else, since “everyone whose manager

is Marcus Webb” and “whoever Marcus Webb’s manager is” are both perfectly valid, perfectly different questions with perfectly valid, empty-looking answers if you meant to ask the other one.

Exercise 3 — Depth and an Indented Tree

3.1 — The whole org chart at once

Exercise 1’s depth counter already generalizes to the entire tree — start from every root instead of one named employee, and every manager-report edge gets discovered in the same breadth-expanding way:

```
WITH RECURSIVE org_tree AS (  
  SELECT id, name, title, manager_id, 0 AS depth, ARRAY[name] AS path  
  FROM   city_org  
  WHERE  manager_id IS NULL  
  
  UNION ALL  
  
  SELECT o.id, o.name, o.title, o.manager_id, org_tree.depth + 1, org_tree.path || o.name  
  FROM   city_org o  
  JOIN   org_tree ON o.manager_id = org_tree.id  
)  
SELECT repeat(' ', depth) || name || ' (' || title || ')' AS org_chart  
FROM   org_tree  
ORDER BY path;
```

org_chart

```
Coretta Vance (Mayor)  
  Aisha Bonner (Director of Parks & Recreation)  
    Felix Wren (Parks Maintenance Supervisor)  
      Ezra Kowalski (Groundskeeper)  
      Nora Villalobos (Groundskeeper)  
Diane Okonjo (Chief of Police)  
  Marcus Reilly (Patrol Captain)  
    Bianca Ferro (Patrol Officer)  
    Kwame Asante (Patrol Officer)  
    Theo Lindqvist (Patrol Officer)  
  Paula Mensah (Records Sergeant)  
Helena Cross (Director of Permitting & Licensing)  
  Grace Halloway (Senior Permit Reviewer)  
    Mia Sorensen (Permit Clerk)  
    Owen Fitch (Permit Clerk)  
  Ray Castellano (Building Inspector)  
Julian Ostrowski (Director of Finance)  
  Colin Marsh (Budget Analyst)  
  Renata Sikes (City Accountant)  
Marcus Webb (Director of Public Works)
```

```
Dana Ruiz (Streets & Sanitation Supervisor)
  Leo Park (Streets Crew)
  Priya Nair (Streets Crew)
  Sam Okafor (Streets Crew)
Tom Delgado (Water & Sewer Supervisor)
  Ivy Chen (Water & Sewer Crew)
  Noah Brandt (Water & Sewer Crew)
Wendell Achebe (Director of IT)
  Hugo Petrakis (Database Administrator)
  Zara Lindholm (Systems Administrator)
(30 rows)
```

Two new pieces do all the work. `repeat(' ', depth)` turns the integer `depth` into visible indentation — two spaces per level, the plainest possible tree rendering. `path`, an array built up one name at a time as the recursion descends (`org_tree.path || o.name`), is what makes `ORDER BY path` produce a proper depth-first listing instead of grouping everyone by depth first — without it, `ORDER BY depth` alone would print all six directors before any of their reports, flattening the tree back into levels instead of branches. `path` is never displayed here; it only exists to sort correctly, which is a pattern worth remembering any time a recursive CTE's *output order* matters as much as its contents.

Exercise 4 — Cycle Detection

4.1 — A bad `UPDATE`, on purpose

`manager_id` is just a foreign key — PostgreSQL enforces that it points to a real row, but nothing stops it from pointing to a row that, somewhere further up the chain, points right back down to where you started. Simulate exactly that data-entry mistake:

```
UPDATE city_org
SET   manager_id = (SELECT id FROM city_org WHERE name = 'Leo Park')
WHERE name = 'Dana Ruiz';
```

Leo Park already reports to Dana Ruiz. This one `UPDATE` now also makes Dana report to Leo — a two-person cycle, and a perfectly valid row by every constraint the table has.

4.2 — Where the cycle actually bites

Exercise 3's root-to-leaves traversal doesn't even notice: Dana and Leo simply stop being reachable from the Mayor at all (their `manager_id` chain no longer leads back up to anyone the tree already found), and they silently vanish from that query's output — 26 rows instead of 30, no error, just two people and their two remaining reports quietly orphaned. It's Exercise 1's *upward* direction — the one this whole chapter has been building on — that walks straight into

it. Before running this, set a safety net; you're about to run a query you already know doesn't terminate on its own:

```
SET statement_timeout = '3s';

WITH RECURSIVE chain AS (
  SELECT id, name, manager_id, 0 AS depth
  FROM city_org
  WHERE name = 'Priya Nair'

  UNION ALL

  SELECT o.id, o.name, o.manager_id, chain.depth + 1
  FROM city_org o
  JOIN chain ON o.id = chain.manager_id
)
SELECT COUNT(*) FROM chain;
```

ERROR: canceling statement due to statement timeout

Priya reports to Dana, Dana now reports to Leo, and Leo reports to Dana — the walk to the root never reaches a NULL manager_id, because there isn't one anywhere on this path anymore. Without the timeout, this query runs until it exhausts memory or disk, whichever comes first, generating an unbounded stream of alternating Dana/Leo rows forever. **Any time a recursive CTE walks a self-referencing column you don't personally control the integrity of, set a statement_timeout before the first run** — it's cheap insurance against exactly this.

```
RESET statement_timeout;
```

4.3 — CYCLE ... SET ... USING: detect it instead of hanging

```
WITH RECURSIVE chain AS (
  SELECT id, name, manager_id, 0 AS depth
  FROM city_org
  WHERE name = 'Priya Nair'

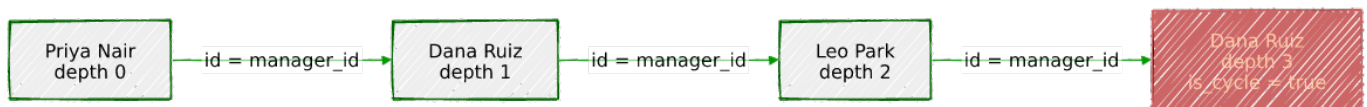
  UNION ALL

  SELECT o.id, o.name, o.manager_id, chain.depth + 1
  FROM city_org o
  JOIN chain ON o.id = chain.manager_id
)
CYCLE id SET is_cycle USING path
SELECT id, name, depth, is_cycle FROM chain;
```

id	name	depth	is_cycle
5	Priya Nair	0	f
3	Dana Ruiz	1	f
4	Leo Park	2	f

3 | Dana Ruiz | 3 | t
(4 rows)

Four rows and it's done — no timeout needed. `cycle id` tells PostgreSQL to track every `id` this query has already visited, in a hidden array column named by `using path`; the moment a round would revisit an `id` already in that array, it stops expanding *that* branch, flags the repeated row `is_cycle = t`, and moves on rather than looping. Dana Ruiz shows up twice — once as Priya's genuine manager, once again as proof the walk looped back to her — and that second appearance is exactly the signal that something in the data is wrong, ready to `where is_cycle` for or alert on, instead of a hung connection and no explanation at all.



4.4 — Undo the damage

```
UPDATE city_org
SET   manager_id = (SELECT id FROM city_org WHERE name = 'Marcus Webb')
WHERE name = 'Dana Ruiz';
```

Back to the real org chart before moving on.

Exercise 5 — Shortest Path Through a Graph

5.1 — Roads go both ways; `road_segments` only says one

```
SELECT road_name, from_intersection, to_intersection, length_m
FROM   road_segments
WHERE  road_name = 'Ring Road';
```

road_name	from_intersection	to_intersection	length_m
Ring Road	11	3	2002.4
Ring Road	3	5	10541.7
Ring Road	5	15	1888.9
Ring Road	15	8	2559.7
Ring Road	8	10	6111.4
Ring Road	10	13	443.9
Ring Road	13	11	3323.3

(7 rows)

Each row is stored once, in one direction, the same way Chapter 2's `city_infrastructure` stores each road as a single `LINESTRING` — but a car can drive either way down Ring Road. Build a bidirectional view of the graph before doing anything else with it:

```
CREATE VIEW road_graph AS
SELECT from_intersection AS a, to_intersection AS b, road_name, length_m FROM road_segments
UNION ALL
SELECT to_intersection, from_intersection, road_name, length_m FROM road_segments;
```

5.2 — Breadth-first search: fewest hops

```
WITH RECURSIVE bfs AS (
  SELECT i.id AS node, ARRAY[i.id] AS path, 0 AS hops, 0.0 AS total_m
  FROM intersections i
  WHERE i.name = 'Fisherman''s Row & Market Street'

  UNION ALL

  SELECT g.b, bfs.path || g.b, bfs.hops + 1, bfs.total_m + g.length_m
  FROM road_graph g
  JOIN bfs ON g.a = bfs.node
  WHERE NOT g.b = ANY(bfs.path)
)
SELECT hops, total_m, path
FROM bfs
JOIN intersections dest ON dest.id = bfs.node
WHERE dest.name = 'Bay Street & Ring Road (East)'
ORDER BY hops
LIMIT 1;
```

```
hops | total_m | path
-----+-----+-----
    4 | 15379.5 | {9,12,11,3,5}
(1 row)
```

WHERE NOT g.b = ANY(bfs.path) is doing the same job Exercise 4's CYCLE clause did — this graph genuinely contains a cycle (Ring Road loops back on itself), so without it, this query would walk in circles exactly like Exercise 4.2's did. ORDER BY hops LIMIT 1 takes the first path that reaches the destination in the fewest steps: 4 hops, 15,379.5 metres.

5.3 — The gotcha: fewest hops isn't shortest distance

Widen the search instead of stopping at the first match:

```
WITH RECURSIVE bfs AS (
  SELECT i.id AS node, ARRAY[i.id] AS path, 0 AS hops, 0.0 AS total_m
  FROM intersections i
  WHERE i.name = 'Fisherman''s Row & Market Street'

  UNION ALL

  SELECT g.b, bfs.path || g.b, bfs.hops + 1, bfs.total_m + g.length_m
  FROM road_graph g
  JOIN bfs ON g.a = bfs.node
  WHERE NOT g.b = ANY(bfs.path) AND bfs.hops < 6
)
```

```

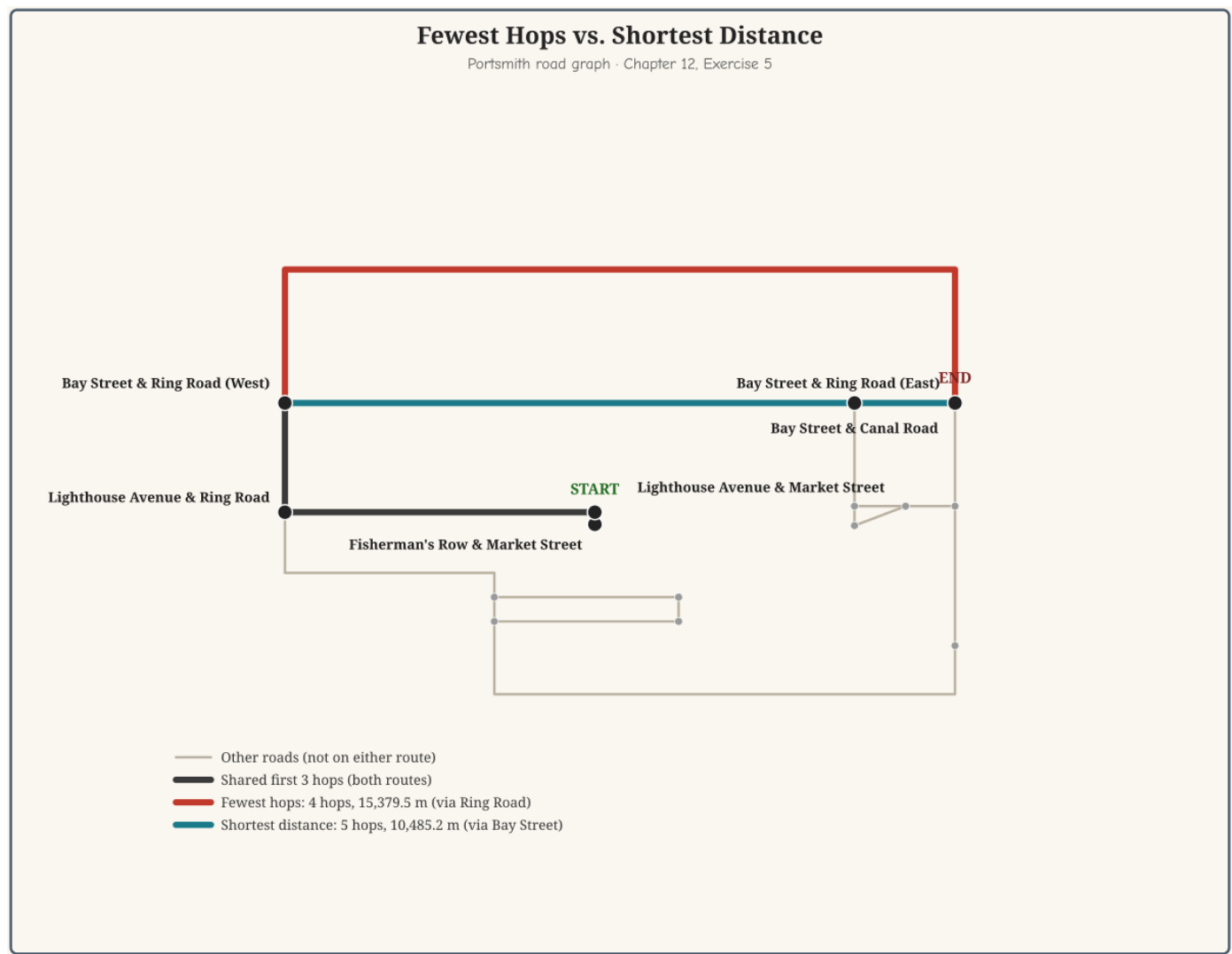
SELECT hops, total_m, path
FROM bfs
JOIN intersections dest ON dest.id = bfs.node
WHERE dest.name = 'Bay Street & Ring Road (East)'
ORDER BY total_m;

```

hops	total_m	path
5	10485.2	{9,12,11,3,4,5}
4	15379.5	{9,12,11,3,5}

(2 rows)

The 4-hop path 5.2 found is *not* the shortest one — a 5-hop route covers nearly 5 kilometres less. The 4-hop route takes the single giant Ring Road segment straight from Bay Street & Ring Road (West) to Bay Street & Ring Road (East) — the 10,541.7 m stretch Chapter 2’s ST_LineSubstring measured back in this chapter’s data setup, the one a straight-line guess would have badly understated. The 5-hop route detours one extra intersection down Bay Street itself instead, trading one more turn for two much shorter segments (4,800.3 m + 847.1 m instead of 10,541.7 m).



ORDER BY hops LIMIT 1 — plain breadth-first search — optimizes for *number of turns*, not *distance travelled*, and those are only the same question when every edge in the graph costs roughly the same to traverse. They don't, here, and a recursive CTE has no built-in concept of “cost” unless a query explicitly asks it to minimize one, the way 5.3's ORDER BY total_m does instead of 5.2's ORDER BY hops. Real routing (turn-by-turn navigation, Dijkstra's algorithm, pgRouting) is built entirely around taking that distinction seriously; this exercise is the smallest possible version of the same lesson.

Exercise 6 — Ancestors of a Leaf, for Faceted Search

6.1 — A category tree built from real Chapter 1 data

```
SELECT c.name AS category, p.name AS parent
FROM   categories c
JOIN   categories p ON p.id = c.parent_id
WHERE  c.name IN ('bakery', 'restaurant');
```

category	parent
restaurant	All Categories
bakery	restaurant

categories isn't invented data — its two levels under “All Categories” are exactly Chapter 1's real category and subcategory/cuisine values, pulled straight out of `businesses.details` and organized into a tree. “Bakery” is a real cuisine value on a real business (River Bend Bakery, Chapter 1) three levels deep in this hierarchy.

6.2 — Walk up, then flatten into a breadcrumb

A faceted search UI wants “all ancestors of this leaf” to build a breadcrumb trail — the exact same upward walk as Exercise 1, applied to a different tree, finished off with `string_agg`:

```
WITH RECURSIVE ancestors AS (
    SELECT id, name, parent_id, 0 AS depth
    FROM   categories
    WHERE  name = 'bakery'

    UNION ALL

    SELECT c.id, c.name, c.parent_id, ancestors.depth + 1
    FROM   categories c
    JOIN   ancestors ON c.id = ancestors.parent_id
)
SELECT string_agg(name, ' > ' ORDER BY depth DESC) AS breadcrumb
FROM   ancestors;
```


ORDER BY depth DESC inside string_agg puts the root first and the leaf last — the walk itself discovers ancestors leaf-to-root (depth increasing outward, exactly like Exercise 1’s officer-to-Mayor chain), so displaying them root-to-leaf means reversing that order at the very end, not changing how the recursion runs. Nothing about this query knows or cares that the tree is only three levels deep — the identical query against a ten-level category tree would produce a ten-element breadcrumb without a single line changing.

Summary — What You Should Now Know

Tool	What it does
WITH RECURSIVE name AS (anchor UNION ALL recursive_term)	Anchor runs once; the recursive term reruns against only the previous round’s new rows until a round adds nothing
o.id = cte.manager_id	Walk <i>up</i> a hierarchy — toward the root
o.manager_id = cte.id	Walk <i>down</i> a hierarchy — toward the leaves
depth counter	Increment it once per recursive round to know how deep any given row is
path array + ORDER BY path	Make a recursive CTE’s output print in proper depth-first tree order
CYCLE col SET flag USING path	Detect a repeated value instead of looping forever; flags the row where it happened
WHERE NOT x = ANY(path)	The hand-rolled version of cycle prevention, for graphs that need per-branch path tracking CYCLE alone doesn’t give you
SET statement_timeout	Cheap insurance before running any recursive query over data you don’t control the integrity of
BFS (ORDER BY hops) vs. shortest-distance (ORDER BY total_m)	Fewest steps and least total cost are different questions unless every edge costs the same
string_agg(name, ' > ' ORDER BY depth DESC)	Turn an ancestor walk into a breadcrumb, root first

The key design insight from this chapter is that a recursive CTE is just an ordinary query, run in a loop, over data that happens to describe its own structure — the anchor decides where to start, the join direction decides which way to walk, and everything past that is the same SELECT you already know how to write. Trees are the friendly case, where that loop is guaranteed to end

because nothing points back at itself. The moment real data — a bad `manager_id`, a road network that loops — stops guaranteeing that, the loop needs its own explicit exit condition, either `CYCLE`'s built-in bookkeeping or a hand-rolled `path` array doing the same job. Every exercise in this chapter is one of exactly two ideas: which direction to walk, and how to know when to stop.

*Going further: Chapter 21's placeholder chapter on PostgreSQL 19's SQL/PGQ property graphs picks up directly from Exercise 5 — the same `road_segments` graph, queried with syntax purpose-built for path traversal instead of a hand-rolled `path` array and a `WHERE NOT ... ANY(...)` guard. Reading that chapter (once PostgreSQL 19 is out of beta and this book can say something definite about it) right after this one, rather than waiting for its place in the numbering, is a reasonable way to see the same problem solved twice, eight chapters apart in the table of contents but adjacent in what they're actually about. Chapter 17's foreign data wrappers occasionally combine with recursive CTEs when a hierarchy spans more than one database — the recursion itself doesn't care where a row physically lives, only that the self-referencing column resolves. And `pgRouting`, mentioned briefly in Exercise 5, is the production answer to "I need real shortest-path routing, not the smallest example that demonstrates the idea" — turn restrictions, one-way streets, and genuine Dijkstra/A implementations, all built on top of the same PostGIS geometry this chapter's road graph came from.**

Chapter 13 — LISTEN/NOTIFY: Database-Native Pub/Sub

“Polling asks ‘did anything happen yet?’ a thousand times a minute. LISTEN just waits to be told.”

Background

The obvious way to build a dashboard that reacts to database changes is to ask, repeatedly: `SELECT * FROM jobs WHERE status = ...`, once a second, forever. It works, but every poll is a query the database has to answer whether or not anything changed, and the dashboard is only ever as fresh as its last poll — average half a polling interval stale, worst case a whole interval. PostgreSQL has had a built-in alternative for longer than most of its more famous features: `LISTEN` and `NOTIFY`, a lightweight publish/subscribe system that ships inside the database itself, no message broker required.

The shape of it is almost the whole idea: any session can run `LISTEN channel_name` to subscribe to a named channel — just a string, nothing has to be created or configured first. Any session — or, more usefully, a trigger — can run `NOTIFY channel_name` (optionally with a short text payload) to publish to it. Every session currently listening on that channel gets the notification, asynchronously, with no polling on anyone’s part.

Three things about *when* a notification actually arrives are easy to get wrong, so they’re worth stating plainly before Exercise 1 shows them happening:

1. **A notification is delivered only after the sending transaction commits.** `NOTIFY` inside a transaction that later rolls back is as if it never ran — nothing is sent, ever.
2. **A listening client only notices a notification is waiting the next time it talks to the server.** The notification arrives at the connection asynchronously, but most clients (including `psql`) only check for and display it around the next command they run — it doesn’t interrupt whatever the client is already doing.
3. **Identical notifications collapse.** Two or more `NOTIFY` calls on the same channel with the exact same payload, inside the same transaction, are coalesced into a single delivery.

And one thing about what a notification *isn’t*: it has no memory. `NOTIFY` doesn’t persist anything — if nobody is listening on a channel the instant it fires, that notification is simply gone. That’s why this chapter builds a second, very unglamorous thing alongside the trigger: a plain table

logging every notification ever sent. `NOTIFY` says “wake up and go look”; the log table is what a dashboard that just reconnected looks *at* to catch up on whatever it missed while it was offline.

The Scenario

Portsmouth’s permitting office wants a live status board for the job queue Chapter 3 built — instead of a background process hammering `jobs` on a timer, it should just be told the moment a permit’s status changes.

Object	Source	Purpose
<code>jobs</code>	Chapter 3	The permit queue whose status changes this chapter reacts to
<code>notifications</code>	<i>(built here)</i>	Durable log of every notification sent — catch-up for a reconnecting dashboard
<code>notification_debounce</code>	<i>(built here)</i>	Last-notified timestamp per job, so Exercise 5 can suppress noisy bursts

Nothing new needs seeding — this chapter is entirely about reacting to changes in data Chapter 3 already created.

Exercise Goals

By the end of this chapter you will be able to:

- Send and receive a notification manually, across two sessions, and explain exactly when it becomes visible and why.
 - Write a trigger that publishes a JSON payload on `NOTIFY` whenever a row’s status changes.
 - Subscribe from a Python 3.12 script using `psycopg`, replacing a polling loop with a blocking wait for the next event.
 - Fan a single event stream out into multiple channels so different consumers can subscribe to only what they care about.
 - Suppress a burst of near-duplicate notifications with a small debounce table.
 - State `LISTEN/NOTIFY`’s real throughput and durability limits, and recognize the point past which a dedicated message broker is the right call instead.
-

Installation

Nothing to install. `LISTEN` and `NOTIFY` are core SQL commands, part of PostgreSQL since long before this book's PostgreSQL 16 baseline — no extension, no configuration. This chapter's Python client (Exercise 3) reuses the `psycopg` driver installed back in Chapter 1.

Loading the Data

This chapter needs Chapter 3's `jobs` table:

```
python data/ch03_seed.py
```

Verify the prerequisite

```
SELECT COUNT(*) FROM jobs;
```

count
48

(1 row)

(48, not 45 — Chapter 10's PostgREST exercises filed three more permit applications through the API. If you're at 45, that's fine too; nothing in this chapter depends on the exact count.)

Exercises

Exercise 1 — Manual `NOTIFY`/`LISTEN`, Two Sessions

1.1 — Subscribe in one session

Open two `psql` sessions side by side. In **Session A**:

```
LISTEN portsmith_test;
```



```
LISTEN
```

Nothing else happens — `LISTEN` just registers this session's interest in the channel and returns immediately.

1.2 — Publish from the other

In **Session B**:

```
NOTIFY portsmith_test, 'hello from session B';
```

```
NOTIFY
```

1.3 — Back in Session A

Switch back to Session A. Nothing has appeared yet — run any statement to find out why:

```
SELECT 1;
```

```
?column?
-----
1
```

Asynchronous notification "portsmith_test" with payload "hello from session B" received from server process with PID 2718842.

The notification was sitting there the whole time, but `psql` only surfaces it around the next command it sends — exactly the second point from the Background section, now with a real timestamp attached to it in the form of a PID you can watch change between runs.

1.4 — A rolled-back `NOTIFY` never arrives

With Session A still listening, run this in Session B:

```
BEGIN;
NOTIFY portsmith_test, 'should never arrive';
ROLLBACK;
```

```
BEGIN
NOTIFY
ROLLBACK
```

Back in Session A, run `SELECT 1;` again. Nothing prints — no notification, no trace it was ever sent. `NOTIFY` inside a transaction is exactly as durable as everything else in that transaction: commit it or it didn't happen.

1.5 — Identical notifications in one transaction collapse

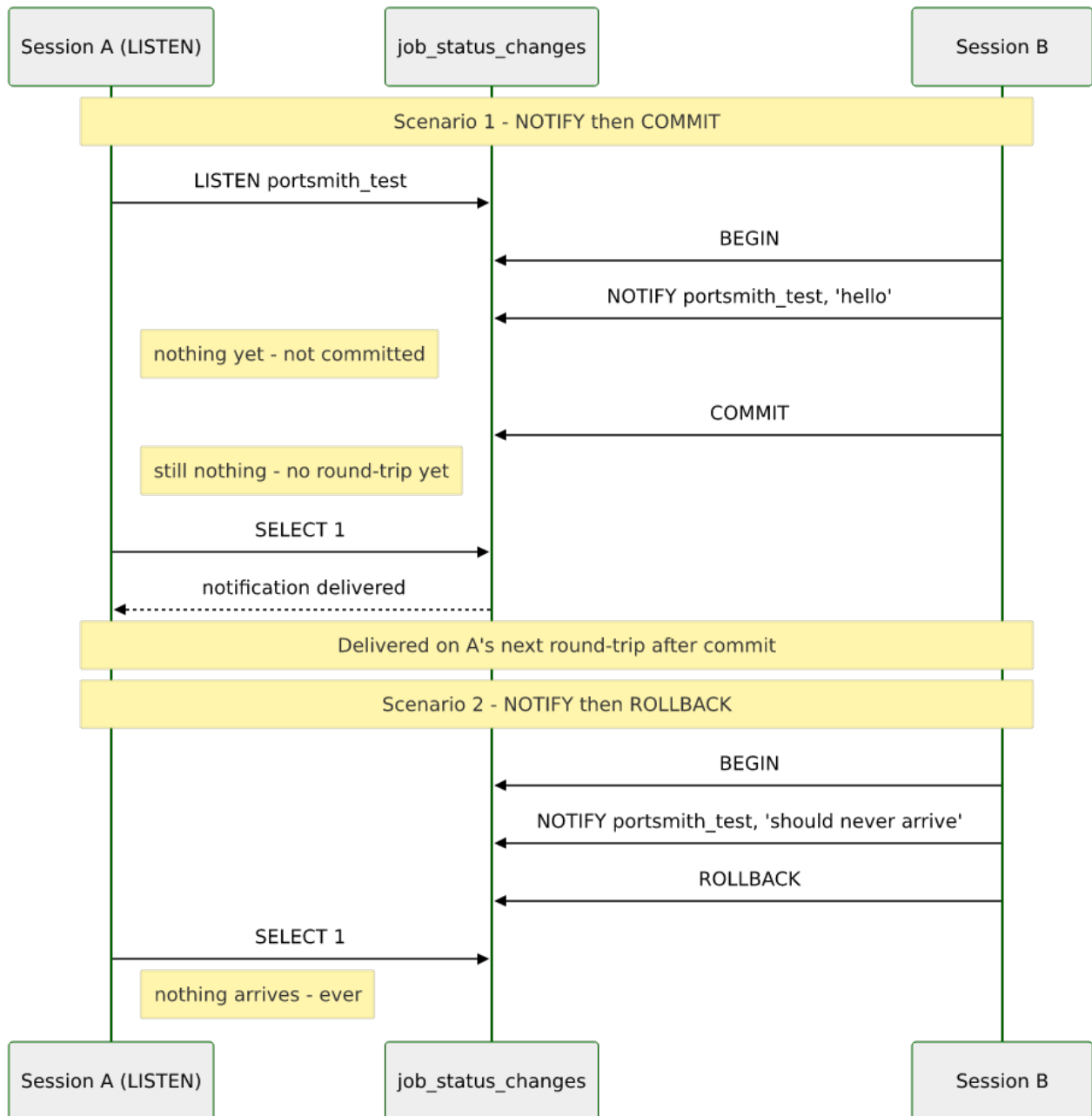
```
BEGIN;
NOTIFY portsmith_test, 'dup';
NOTIFY portsmith_test, 'dup';
NOTIFY portsmith_test, 'dup';
COMMIT;
```

Back in Session A:

```
SELECT 1;
```

Asynchronous notification "portsmith_test" with payload "dup" received from server process with PID 2721455.

One delivery, not three. PostgreSQL deduplicates same-channel, same-payload notifications within a single transaction before sending anything — worth knowing before you assume a burst of identical `NOTIFY`s inside one transaction will arrive as a burst on the other end.



Exercise 2 — A Trigger That `NOTIFY`s on Status Change

2.1 — The durable log

```
CREATE TABLE notifications (  
  id          BIGSERIAL PRIMARY KEY,  
  channel     TEXT NOT NULL,  
  payload     JSONB NOT NULL,
```

```

        created_at TIMESTAMPTZ NOT NULL DEFAULT clock_timestamp()
    );

```

2.2 — The trigger function

```

CREATE OR REPLACE FUNCTION notify_job_status_change() RETURNS TRIGGER AS $$
DECLARE
    notice JSONB;
BEGIN
    IF NEW.status IS DISTINCT FROM OLD.status THEN
        notice := jsonb_build_object(
            'job_id', NEW.id,
            'job_type', NEW.job_type,
            'old_status', OLD.status,
            'new_status', NEW.status,
            'changed_at', clock_timestamp()
        );
        INSERT INTO notifications (channel, payload) VALUES ('job_status_changes', notice);
        PERFORM pg_notify('job_status_changes', notice::text);
    END IF;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_notify_job_status_change
AFTER UPDATE ON jobs
FOR EACH ROW
EXECUTE FUNCTION notify_job_status_change();

```

IF NEW.status IS DISTINCT FROM OLD.status matters as much as the NOTIFY itself — without it, this trigger fires on *any* update to a job row, including the heartbeat timestamp Chapter 3’s workers write every couple of seconds, flooding the channel with “changes” where the status never actually moved. pg_notify(channel, payload) is the function form of NOTIFY — it takes both arguments as ordinary expressions, which plain NOTIFY channel, 'literal' syntax can't do, and it's what makes a computed channel name possible at all (Exercise 4 depends on exactly that).

2.3 — Trigger it

```

UPDATE jobs
SET     status = 'in_progress', claimed_at = clock_timestamp(), claimed_by = 'worker-1'
WHERE  id = (SELECT id FROM jobs WHERE status = 'queued' ORDER BY id LIMIT 1);

```

A session already running LISTEN job_status_changes; sees, on its next round-trip:

```

Asynchronous notification "job_status_changes" with payload
{"job_id": 1, "job_type": "demolition_permit", "changed_at":
"2026-08-04T23:33:21.581832-04:00", "new_status": "in_progress",
"old_status": "queued"}" received from server process with PID 2734846.

```

And the durable copy is sitting in the log table regardless of whether anyone was listening:


```
SELECT channel, payload, created_at FROM notifications;
```

channel	payload	created_at
job_status_changes	{"job_id": 1, "job_type": "demolition_permit", "old_status": "queued", ...}	2026-08-04 23:33:21.582461-04

Exercise 3 — Subscribing from Python

3.1 — A listener client

```
#!/usr/bin/env python3.12
# ch13_listen.py - Portsmouth permit-status staff dashboard
import argparse
import json

import psycopg

def main() -> None:
    parser = argparse.ArgumentParser(description=__doc__)
    parser.add_argument("channels", nargs="*", default=["job_status_changes"])
    parser.add_argument("--dsn", default="dbname=portsmith")
    parser.add_argument("--timeout", type=float, default=None)
    args = parser.parse_args()

    with psycopg.connect(args.dsn, autocommit=True) as conn:
        with conn.cursor() as cur:
            for channel in args.channels:
                cur.execute(f"LISTEN {channel};")
                print(f"listening on {channel!r} ...")

            try:
                for notice in conn.notifies(timeout=args.timeout):
                    job = json.loads(notice.payload)
                    print(
                        f"[{notice.channel}] job {job['job_id']} ({job['job_type']}): "
                        f"{job['old_status']} -> {job['new_status']}"
                    )
            except KeyboardInterrupt:
                print("\nstopped.")

if __name__ == "__main__":
    main()
```

`autocommit=True` matters here — `LISTEN` needs to actually take effect immediately rather than sit inside an open transaction waiting for a `COMMIT` that this script has no reason to ever issue.

`conn.notifies(timeout=...)` is a generator: it blocks, doing nothing, until a notification arrives, then yields it — no loop, no polling interval to tune, no `SELECT` the database has to answer just to say “nothing’s changed.”

3.2 — Run it

```
python data/ch13_listen.py
```

```
listening on 'job_status_changes' ...
```

From a `psql` session, update another job’s status. The moment that transaction commits:

```
[job_status_changes] job 2 (demolition_permit): queued -> completed
```

No delay, no polling — the script was simply asleep until PostgreSQL had something to tell it.

Exercise 4 — Fan Out to Per-Job-Type Channels

4.1 — One more `pg_notify` call

Public Works doesn’t want to see every `business_license` update, and Permitting & Licensing doesn’t want to see every `demolition_permit` update. Give each job type its own channel, in addition to the all-activity one:

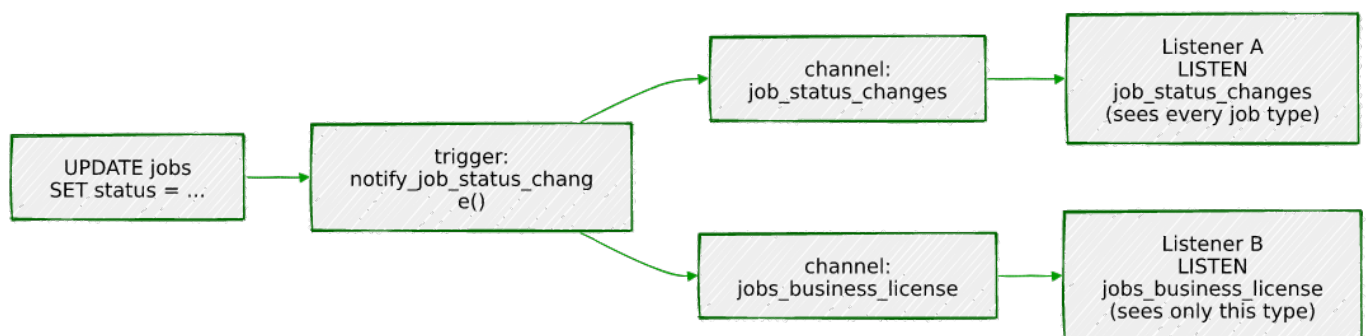
```
CREATE OR REPLACE FUNCTION notify_job_status_change() RETURNS TRIGGER AS $$
DECLARE
    notice JSONB;
BEGIN
    IF NEW.status IS DISTINCT FROM OLD.status THEN
        notice := jsonb_build_object(
            'job_id', NEW.id,
            'job_type', NEW.job_type,
            'old_status', OLD.status,
            'new_status', NEW.status,
            'changed_at', clock_timestamp()
        );
        INSERT INTO notifications (channel, payload) VALUES ('job_status_changes', notice);
        PERFORM pg_notify('job_status_changes', notice::text);
        PERFORM pg_notify('jobs_' || NEW.job_type, notice::text);
    END IF;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;
```

`'jobs_' || NEW.job_type` builds the channel name from the row being updated — `jobs_business_license`, `jobs_demolition_permit`, and so on, five channels from one trigger, none of them declared or created anywhere in advance. A channel isn’t an object; it springs into

existence the instant something LISTENS or NOTIFYS it; and stops existing — with nothing to clean up — the instant nothing is LISTENING on it anymore.

CREATE OR REPLACE FUNCTION does exactly what it says — *replace*, not patch or extend. This is now the third time this chapter has redefined `notify_job_status_change()` in place (Exercise 2's version, this one, Exercise 5's version still to come), and there's nothing tracking which version is currently live beyond whatever the last CREATE OR REPLACE you ran actually said. If you go back and re-run Exercise 2's block for any reason — double-checking something, copy-pasting the wrong snippet — you will silently undo this one, with no error and no warning: the fan-out channels just stop receiving anything, because the function that used to `pg_notify` them no longer does. If a listener you expect to see traffic suddenly goes quiet,

SELECT prosrc FROM pg_proc WHERE proname = 'notify_job_status_change' is the fastest way to check which version is actually installed right now.



4.2 — Subscribe to just one

```
python data/ch13_listen.py jobs_business_license
```

```
listening on 'jobs_business_license' ...
```

From a `psql` session, update a `building_permit` job's status, then a `business_license` job's status (`building_permit` is used here, rather than the much smaller `demolition_permit` pool, because it's the largest job type — 15 permits — and least likely to have already run out of queued rows from earlier testing; if a given `UPDATE` reports 0 rows, that job type is simply out of queued jobs right now, and any other `job_type` with rows left makes the same point):

```
UPDATE jobs SET status = 'in_progress', claimed_at = clock_timestamp(), claimed_by = 'worker-1'
WHERE id = (SELECT id FROM jobs WHERE status = 'queued' AND job_type = 'building_permit' ORDER
BY id LIMIT 1);
```

```
UPDATE jobs SET status = 'in_progress', claimed_at = clock_timestamp(), claimed_by = 'worker-1'
WHERE id = (SELECT id FROM jobs WHERE status = 'queued' AND job_type = 'business_license'
ORDER BY id LIMIT 1);
```

Only the second one shows up in the listener's output:

```
[jobs_business_license] job 5 (business_license): queued -> in_progress
```

The `building_permit` change still fired — on `job_status_changes` and on `jobs_building_permit` — this listener simply never subscribed to either of those channels. Fan-out costs nothing extra per additional channel; it's just more arguments to `pg_notify`.

Exercise 5 — Debounce a Noisy Job

5.1 — The problem

A job that flaps — fails, gets requeued, gets reclaimed, fails again, all within a second or two — fires a fresh notification on every single transition. A dashboard doesn't need to render all of that; it needs to know where things ended up. Track, per job, when it was last actually notified about:

```
CREATE TABLE notification_debounce (  
    job_id          BIGINT PRIMARY KEY,  
    last_notified_at TIMESTAMPTZ NOT NULL  
);
```

5.2 — Check it before sending

```
CREATE OR REPLACE FUNCTION notify_job_status_change() RETURNS TRIGGER AS $$  
DECLARE  
    notice      JSONB;  
    last_sent   TIMESTAMPTZ;  
BEGIN  
    IF NEW.status IS DISTINCT FROM OLD.status THEN  
        SELECT last_notified_at INTO last_sent  
        FROM   notification_debounce WHERE job_id = NEW.id;  
  
        IF last_sent IS NOT NULL AND clock_timestamp() - last_sent < interval '1 second' THEN  
            RETURN NEW; -- too soon after the last one for this job - skip it  
        END IF;  
  
        notice := jsonb_build_object(  
            'job_id', NEW.id, 'job_type', NEW.job_type,  
            'old_status', OLD.status, 'new_status', NEW.status,  
            'changed_at', clock_timestamp()  
        );  
        INSERT INTO notifications (channel, payload) VALUES ('job_status_changes', notice);  
        PERFORM pg_notify('job_status_changes', notice::text);  
        PERFORM pg_notify('jobs_' || NEW.job_type, notice::text);  
  
        INSERT INTO notification_debounce (job_id, last_notified_at)  
        VALUES (NEW.id, clock_timestamp())  
        ON CONFLICT (job_id) DO UPDATE SET last_notified_at = EXCLUDED.last_notified_at;  
    END IF;  
    RETURN NEW;  
END;
```

```
END;  
$$ LANGUAGE plpgsql;
```

This is a *leading-edge* debounce: the first change in a burst fires immediately, and anything else for that same job within the next second is dropped — not delayed, not batched, just skipped. The status change itself still happens; only the notification about it is suppressed.

5.3 — Prove it

Four status changes on the same job, each about 200ms apart — well inside the 1-second window:

```
UPDATE jobs SET status = 'in_progress' WHERE id = 40;  
UPDATE jobs SET status = 'failed'      WHERE id = 40;  
UPDATE jobs SET status = 'queued'      WHERE id = 40;  
UPDATE jobs SET status = 'in_progress' WHERE id = 40;
```

A listener on `job_status_changes` sees exactly one of the four:

```
[job_status_changes] job 40 (sign_permit): queued -> in_progress
```

```
SELECT * FROM notification_debounce WHERE job_id = 40;
```

job_id	last_notified_at
-----+-----	
40	2026-08-04 23:59:04.475819-04

Only one timestamp recorded — the second, third, and fourth updates all checked it, found themselves inside the window, and returned without touching it. Wait a full second and update the same job again: this time it fires, because `clock_timestamp() - last_sent` has finally crossed `interval '1 second'`.



Exercise 6 — Throughput, Payload Limits, and When to Graduate

6.1 — The payload ceiling is real, and it's not quite 8000

```
SELECT pg_notify('sz', repeat('x', 7999)); -- succeeds  
SELECT pg_notify('sz', repeat('x', 8000)); -- fails
```

ERROR: payload string too long

7999 bytes is the actual limit, not the commonly quoted round 8000 — PostgreSQL reserves one byte for a terminator inside its fixed 8000-byte buffer. A JSON payload describing one row's

status change, the way this chapter’s trigger builds one, comes nowhere close; a payload trying to carry an entire row plus its full history would.

6.2 — Sending is not the bottleneck

```
DO $$  
DECLARE  
    i INT;  
    start_time TIMESTAMPTZ := clock_timestamp();  
BEGIN  
    FOR i IN 1..10000 LOOP  
        PERFORM pg_notify('throughput_test', 'msg ' || i);  
    END LOOP;  
    RAISE NOTICE 'Sent 10000 notifications in %', clock_timestamp() - start_time;  
END $$;
```

NOTICE: Sent 10000 notifications in 00:00:00.019504

Ten thousand notifications, under twenty milliseconds, no listener even attached. The database can *enqueue* notifications far faster than any realistic client can usefully consume them — sending was never going to be where this architecture runs into trouble.

6.3 — Where it actually runs into trouble

The limits that matter are structural, not raw speed:

- **One shared queue per database.** Every NOTIFY from every session, on every channel, goes into the same queue. It isn’t partitioned by channel or topic the way a real message broker’s queues are — a slow listener anywhere is a slow listener for the whole mechanism.
- **No persistence beyond what you build yourself.** This chapter’s notifications table exists precisely because LISTEN/NOTIFY has none on its own — a listener down for five minutes misses everything that happened in those five minutes, permanently, unless something else logged it.
- **No replay, no acknowledgment, no consumer groups.** A message broker lets ten workers share one queue so each message is handled once, or lets a new consumer join and replay history. NOTIFY broadcasts to whoever happens to be listening right now, once, with no concept of “did you actually process that.”
- **Standbys don’t get it.** A streaming replica can serve read queries, but NOTIFY traffic is a primary-only affair — nothing to LISTEN for on a read replica will ever arrive.

None of that makes LISTEN/NOTIFY the wrong tool — for exactly the job this chapter gave it, telling an already-connected, already-trusted internal dashboard “something changed, go look,” it’s close to free and requires nothing to operate. The moment a requirement shows up that LISTEN/NOTIFY structurally can’t satisfy — guaranteed delivery across an outage, multiple independent consumer groups, cross-database routing, back-pressure when a consumer falls

behind — that’s not a configuration problem to work around, it’s the signal to bring in Kafka, RabbitMQ, or a similar dedicated broker instead.

Summary — What You Should Now Know

Tool	What it does
<code>LISTEN channel</code>	Subscribe this session to a channel — created implicitly, no prior setup
<code>NOTIFY channel, 'payload' /</code> <code>pg_notify(channel, payload)</code>	Publish to a channel; the function form allows computed channel names and expressions
Delivery timing	Only after the sending transaction commits; only noticed by a client on its next round-trip
Same-transaction dedup	Identical channel + payload, sent more than once in one transaction, delivers exactly once
<code>conn.notifies(timeout=...)</code> (psycopg)	A blocking generator — no polling loop, no interval to tune
Computed channel names	<code>pg_notify('prefix_' value, ...)</code> fans one trigger out into many topic-scoped channels
A hand-built log table	The durability <code>LISTEN/NOTIFY</code> doesn’t provide — what a reconnecting client catches up from
Leading-edge debounce (state table + timestamp check)	Fire the first event in a burst, suppress the rest within a window
7999-byte payload ceiling	The real limit — 8000 minus a terminator byte
Structural limits: one shared queue, no persistence, no replay, primary-only	The reasons to graduate to a dedicated broker, not raw throughput

The key design insight from this chapter is that `LISTEN/NOTIFY` solves exactly one problem — telling an already-connected session that something happened, right now, cheaply — and solves nothing else on purpose. It has no memory, no acknowledgment, no replay, no concept of a consumer that isn’t currently connected. Every exercise past the first one was really about compensating for that on purpose where it mattered (the log table for durability, the debounce table for noise) and accepting it everywhere else, because the alternative — a full message broker — is a lot of operational weight to take on before you actually have a problem it solves that this chapter’s five-line trigger doesn’t.

Going further: Chapter 14's advisory locks are worth combining with this chapter's pattern the moment more than one process might react to the same notification — nothing about LISTEN/NOTIFY prevents two dashboards, or two workers, from both trying to handle the same event. Chapter 18's logical replication is a different, heavier-weight answer to a similar-sounding question ("tell me when a row changes") — logical replication streams the actual row changes themselves, durably, to another database, where NOTIFY sends a fire-and-forget signal with an optional short payload to whoever's listening right now; reach for replication when the requirement is "give me the data," and LISTEN/NOTIFY when it's "just tell me to go look." And Chapter 19's pg_cron pairs naturally with this chapter's notifications log table: a scheduled job that sweeps for log rows newer than a dashboard's last checkpoint is exactly how that dashboard recovers from having been disconnected, the catch-up path NOTIFY alone can never provide.

Chapter 14 — Advisory Locks: Distributed Coordination

“Every lock in this book so far has been about a row. This one isn’t about any row at all.”

Background

FOR UPDATE, ROW locks, SKIP LOCKED — every locking mechanism this book has used up to now exists because two transactions were reaching for the *same data*. But plenty of real coordination problems have nothing to do with a specific row: “only one process should run the nightly reconciliation job, whichever one gets there first,” “elect a single leader among five identical workers,” “make sure nobody else is already doing this whole category of work right now.” There’s no row to lock for any of that — the thing you need to coordinate around is an idea, not a record.

An **advisory lock** is PostgreSQL’s answer: a lock on a plain integer you invent, with no connection to any table, row, or piece of data whatsoever. You pick the number; PostgreSQL just remembers who’s holding it and makes everyone else wait or ask. It’s called “advisory” because nothing enforces that anyone respects it — unlike a row lock, which UPDATE and DELETE are physically bound by, an advisory lock only means anything to code that deliberately checks it. That’s a feature, not a compromise: it’s a general-purpose coordination primitive riding on a database your whole system probably already talks to, instead of standing up ZooKeeper or etcd just to answer “am I allowed to do this right now.”

Two choices you make every time you reach for one:

- **How long should it live?** `pg_advisory_lock()` / `pg_advisory_unlock()` are **session-level** — held until you explicitly unlock, or your connection closes, whichever comes first.
`pg_advisory_xact_lock()` is **transaction-level** — released automatically at COMMIT or ROLLBACK, with no unlock function to call and no way to release it early. Exercise 6 is entirely about a bug that only one of these two is actually safe against.
- **Should it wait, or just tell you?** The plain `pg_advisory_lock()` blocks until the lock is free. `pg_try_advisory_lock()` returns immediately either way — `true` if it got the lock, `false` if someone else already holds it — which is what makes “is anyone else already doing this?” a single non-blocking query instead of a hang.

One more thing worth knowing before Exercise 1: like Chapter 13’s `NOTIFY`, advisory locks are a primary-only affair. They live in shared memory on whichever server you’re connected to, aren’t written to WAL, and have no meaning at all on a streaming replica.

The Scenario

No new tables this chapter — it reuses Chapter 3’s `jobs` queue and coordinates *processes* around it instead of adding data.

Object	Source	Purpose
<code>jobs</code>	Chapter 3	The permit queue Exercise 4’s critical section guards
<code>data/ ch14_leader_election.py</code>	<i>(built here)</i>	N simulated worker processes racing for one advisory lock

Exercise Goals

By the end of this chapter you will be able to:

- Acquire and release a session-level advisory lock, and watch a second session block on the exact same key until the first releases it.
 - Use `pg_try_advisory_lock()` to ask “is anyone else already doing this?” without waiting for an answer.
 - Implement leader election: N processes race for one lock, exactly one wins.
 - Wrap a transaction-level advisory lock around a critical section that row-level locking alone can’t express.
 - Read `pg_locks` to see exactly which session holds which advisory lock, and for how long.
 - Explain why session-level advisory locks are dangerous behind a connection pool, and which lock type avoids the problem entirely.
-

Installation

Nothing to install. Advisory locks are a core PostgreSQL feature — no extension, no configuration.

Loading the Data

This chapter needs Chapter 3's jobs table:

```
python data/ch03_seed.py
```

```
SELECT COUNT(*) FROM jobs;
```

```
count
-----
    48
```

Exercises

Exercise 1 — Acquire, Block, Release

1.1 — Session A takes the lock

Open two `psql` sessions. In **Session A**, pick an arbitrary key and lock it:

```
SELECT pg_advisory_lock(12345);
```

```
pg_advisory_lock
-----
```

```
(1 row)
```

Returns immediately — nobody else holds 12345 yet. 12345 is not a row id, a job id, or a reference to anything; it's just a number this chapter picked, and every session that agrees to use it for the same purpose is now coordinating through it.

1.2 — Session B reaches for the same key

In **Session B**:

```
SELECT pg_advisory_lock(12345);
```

Nothing comes back. The session just hangs — this is a real block, the same shape as waiting on a row lock, except there's no row anywhere involved.

```
○ $ psql portsmith
psql (16.14 (Ubuntu 16.14-0ubuntu0.24.04.1))
Type "help" for help.

portsmith=> SELECT pg_advisory_lock(12345);
pg_advisory_lock
-----
(1 row)

portsmith=> 
```

```
○ $ psql portsmith
psql (16.14 (Ubuntu 16.14-0ubuntu0.24.04.1))
Type "help" for help.

portsmith=> SELECT pg_advisory_lock(12345);

```

Screenshot of blocked psql session on an advisory lock

1.3 — Session A releases it

Back in Session A:

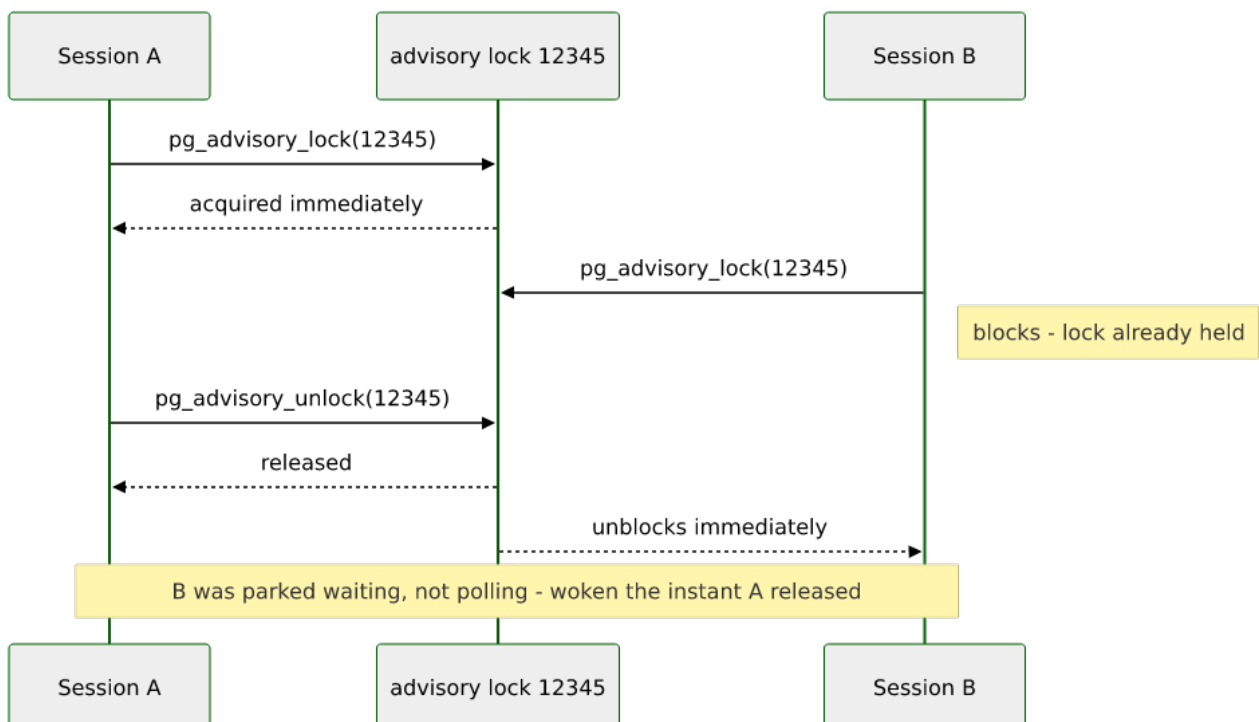
```
SELECT pg_advisory_unlock(12345);

pg_advisory_unlock
-----
t
```

The instant this runs, Session B's blocked query finally returns:

```
pg_advisory_lock
-----
(1 row)
```

Session B wasn't retrying, polling, or checking back — it was genuinely parked, waiting, and PostgreSQL woke it the moment the lock freed up. (Session B is now holding 12345 itself; run `pg_advisory_unlock(12345)` there before moving on.)



Exercise 2 — `pg_try_advisory_lock()`: Ask, Don't Wait

2.1 — The non-blocking version

With Session A still holding `pg_advisory_lock(12345)`, run this in Session B instead of the blocking form:

```
SELECT pg_try_advisory_lock(12345);

pg_try_advisory_lock
-----
f
(1 row)
```

Time: 0.029 ms

`false`, back in under a millisecond — no wait, no hang. `false` means exactly one thing: *someone* currently holds this key. `true` would mean the lock is now held by you.

2.2 — Why this matters more than it looks

This single call is the entire mechanism behind “don’t start a second copy of this job if one’s already running.” A nightly batch job, a scheduled report, a background sweep — anything that must never run twice at once starts with exactly this check: try the lock, and if you don’t get it, exit immediately instead of doing the work. No polling table, no separate “is this running” flag to keep in sync with reality; the lock *is* the flag, and PostgreSQL can never let it lie about whether it’s held.

Exercise 3 — Leader Election

3.1 — Five workers, one leader

```
#!/usr/bin/env python3.12
# ch14_leader_election.py
import multiprocessing
import sys
import time

import psycopg

LEADER_LOCK_KEY = 99001

def worker(worker_id: int, dsn: str) -> None:
    with psycopg.connect(dsn, autocommit=True) as conn:
        with conn.cursor() as cur:
            cur.execute("SELECT pg_try_advisory_lock(%s)", (LEADER_LOCK_KEY,))
```

```

(got_lock,) = cur.fetchone()

if got_lock:
    print(f"[worker-{worker_id}] elected leader - starting work")
    time.sleep(1.5)
    print(f"[worker-{worker_id}] leader work done, releasing")
    cur.execute("SELECT pg_advisory_unlock(%s)", (LEADER_LOCK_KEY,))
else:
    print(f"[worker-{worker_id}] lost the race - standing by")

def main() -> None:
    n_workers = int(sys.argv[1]) if len(sys.argv) > 1 else 5
    dsn = sys.argv[2] if len(sys.argv) > 2 else "dbname=portsmith"
    procs = [multiprocessing.Process(target=worker, args=(i, dsn)) for i in range(1, n_workers
+ 1)]
    for p in procs:
        p.start()
    for p in procs:
        p.join()

if __name__ == "__main__":
    main()

```

3.2 — Run it

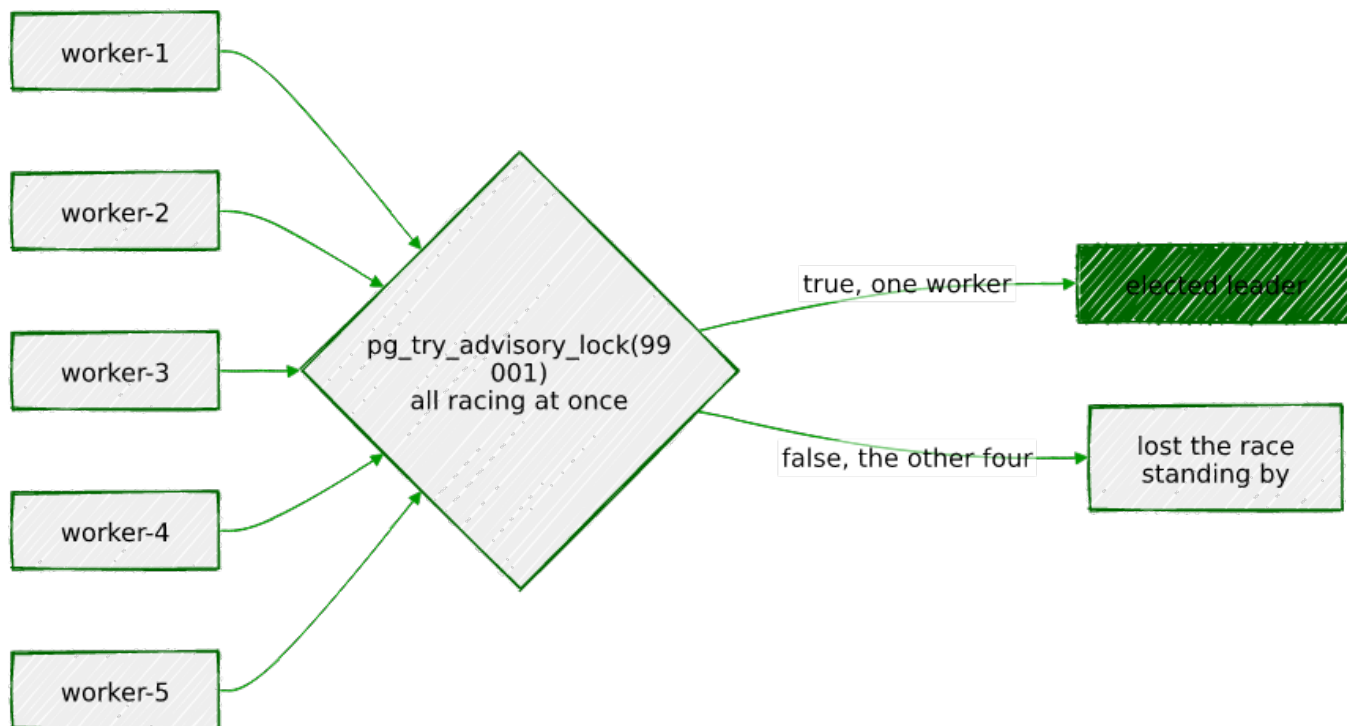
```
python data/ch14_leader_election.py 5
```

```

[worker-2] lost the race - standing by
[worker-3] lost the race - standing by
[worker-4] lost the race - standing by
[worker-5] lost the race - standing by
[worker-1] elected leader - starting work
[worker-1] leader work done, releasing

```

Every one of the five processes hits `pg_try_advisory_lock` within microseconds of each other — genuinely racing, not taking turns — and exactly one gets `true`. Which worker wins is not deterministic; run it again and a different number will win. What’s guaranteed isn’t *who* becomes leader, only that there’s never more than one at a time. That one guarantee is the entire value of leader election: five identical, uncoordinated processes, and PostgreSQL — not any of them — is the single source of truth for which one is in charge.



Exercise 4 — A Transaction-Level Lock Around a Critical Section

4.1 — A rule `SKIP LOCKED` can't express

Chapter 3's `FOR UPDATE SKIP LOCKED` lets many workers claim many different `jobs` rows at once, on purpose — that's the whole point of it. But suppose Portsmouth has exactly one building inspector, and city policy says only one `demolition_permit` can be under active inspection at a time, no matter how many workers are running or how many different demolition jobs are sitting in the queue. Row locking can't express "only one of *this category*, regardless of which specific row" — that's not a fact about any one row, it's a fact about all of them together. This is exactly what an advisory lock, scoped to the job type rather than any job id, is for:

```
SELECT hashtext('demolition_permit');
```

```
hashtext
-----
-1799557343
```

`hashtext()` turns an arbitrary string into a well-distributed integer — a convenient way to get a lock key out of a category name without maintaining a lookup table mapping job types to key numbers by hand.

4.2 — Wrap the claim in it

```
BEGIN;
SELECT pg_advisory_xact_lock(hashtext('demolition_permit'));
```

```
-- ... claim and begin working a demolition_permit job here ...  
COMMIT;
```

4.3 — Prove it serializes, even across different rows

The claim to check: two workers, claiming two *different* `demolition_permit` jobs — nothing in common at the row level — should still be forced to run one at a time, because the lock is scoped to the category, not to either row. Open two `psql` sessions again (same as Exercise 1) to watch it happen.

In **Session A**, run these three statements one at a time, stopping after the third — do not run `COMMIT` yet:

```
BEGIN;  
SELECT pg_advisory_xact_lock(hashtext('demolition_permit'));  
SELECT 'worker A: claimed job 1, inspecting site...' AS status;
```

```
pg_advisory_xact_lock  
-----
```

(1 row)

```
status  
-----  
worker A: claimed job 1, inspecting site...  
(1 row)
```

Both statements return immediately — Session A now holds the lock, and its transaction is deliberately left open (no `COMMIT` yet) to simulate a worker still in the middle of an inspection.

Now switch to **Session B** and run this — a *different* job, same `demolition_permit` category:

```
BEGIN;  
SELECT pg_advisory_xact_lock(hashtext('demolition_permit'));  
SELECT 'worker B: got past the lock' AS status;  
COMMIT;
```

Session B hangs after `BEGIN` — the `pg_advisory_xact_lock` call blocks, and the `status` row never prints. Go back to **Session A** and finally run:

```
COMMIT;
```

The instant Session A's transaction ends, Session B unblocks on its own and finishes, printing `worker B: got past the lock` followed by its own `COMMIT`. The lock released automatically the moment Session A's transaction closed — nothing had to explicitly unlock it. Two entirely different job ids, no row either session touched in common, and they were still fully serialized — because the thing being protected was never a row to begin with.

Exercise 5 — Reading `pg_locks`

5.1 — The raw view

With a session holding `pg_advisory_lock(12345)` open elsewhere:

```
SELECT locktype, ((classid::bigint << 32) | objid::bigint) AS lock_key,  
       mode, granted, pid  
FROM   pg_locks  
WHERE  locktype = 'advisory';
```

locktype	lock_key	mode	granted	pid
advisory	12345	ExclusiveLock	t	2998844

(1 row)

Advisory locks show up in `pg_locks` exactly like row and table locks do — same catalog, same columns — except `locktype = 'advisory'` and the “thing being locked” is just a number PostgreSQL reconstructs from `classid` and `objid` rather than a row identifier. That bit-shift reassembles the single `bigint` key this chapter has been passing to `pg_advisory_lock()` — session-level advisory locks internally split a 64-bit key across those two 32-bit catalog columns.

5.2 — A real diagnostic query

Raw `pg_locks` never tells you *who* or *why*. Join it to `pg_stat_activity` for a query worth keeping around:

```
SELECT l.pid,  
       ((l.classid::bigint << 32) | l.objid::bigint) AS lock_key,  
       l.mode, l.granted,  
       a.username, a.application_name,  
       now() - a.state_change AS held_for,  
       a.query AS last_query  
FROM   pg_locks l  
JOIN   pg_stat_activity a ON a.pid = l.pid  
WHERE  l.locktype = 'advisory';
```

pid	lock_key	mode	granted	username	application_name	held_for
2999823	12345	ExclusiveLock	t	chris	psql	00:00:00.53227

SELECT pg_advisory_lock(12345);
(1 row)

`held_for` is the question that actually matters in production: a lock held for 30 milliseconds is a Tuesday; a lock held for 6 hours because some process crashed without releasing it is an incident. This query is exactly what you’d point a monitoring check at.

Exercise 6 — The Connection-Pool Pitfall

6.1 — Session locks assume “session” means what you think it means

`pg_advisory_lock()`’s session-level lifetime is a promise: the lock lives exactly as long as your database connection does. That promise quietly breaks the moment a connection pool sits between your application and PostgreSQL, because a pooled connection’s *physical* lifetime and any one request’s *logical* lifetime are no longer the same thing. Simulate it directly — one physical connection, reused across two completely unrelated pieces of work, the way a pool would hand it out twice:

```
import psycopg

POOL_KEY = 55001
pooled_conn = psycopg.connect("dbname=portsmith", autocommit=True)

# --- "Request 1": nightly reconciliation job start ---
with pooled_conn.cursor() as cur:
    cur.execute("SELECT pg_advisory_lock(%s)", (POOL_KEY,))
    print("[request 1] acquired session lock", POOL_KEY)
    # BUG: request 1 finishes (or crashes) without calling pg_advisory_unlock.

print("[request 1] done - connection returned to pool (lock still held!)")

# --- "Request 2": unrelated request, later, same physical connection ---
with pooled_conn.cursor() as cur:
    cur.execute("SELECT pg_try_advisory_lock(%s)", (POOL_KEY,))
    (got_lock,) = cur.fetchone()
    print(f"[request 2] pg_try_advisory_lock({POOL_KEY}) -> {got_lock}")

[request 1] acquired session lock 55001
[request 1] done - connection returned to pool (lock still held!)
[request 2] pg_try_advisory_lock(55001) -> True (same physical session as request 1!)
```

Request 2 gets `True` and has every reason to believe it’s the exclusive holder of key 55001 — leader, singleton, whatever that key was supposed to mean — and it’s completely wrong. PostgreSQL sees one session that already held the lock asking for it again, which is trivially `true` by definition; it has no way to know “request 1” and “request 2” were ever meant to be different things. This is the bug the guide warns about, and it is exactly as dangerous as it sounds: two unrelated requests, coordinating through a lock that was never really shared between them, both convinced they’re safe.

6.2 — The fix: use the lock type that can’t leak

```
import psycopg

POOL_KEY = 55002
pooled_conn = psycopg.connect("dbname=portsmith") # autocommit off
```

```

# --- "Request 1", transaction-scoped this time ---
with pooled_conn.cursor() as cur:
    cur.execute("SELECT pg_advisory_xact_lock(%s)", (POOL_KEY,))
    print("[request 1] acquired xact lock", POOL_KEY)
pooled_conn.commit() # released here, no matter what request 1 does or forgets
print("[request 1] committed - lock released automatically")

# --- "Request 2", same pooled connection ---
with pooled_conn.cursor() as cur:
    cur.execute("SELECT pg_try_advisory_lock(%s)", (POOL_KEY,))
    (got_lock,) = cur.fetchone()
    print(f"[request 2] pg_try_advisory_lock({POOL_KEY}) -> {got_lock} (correctly free)")

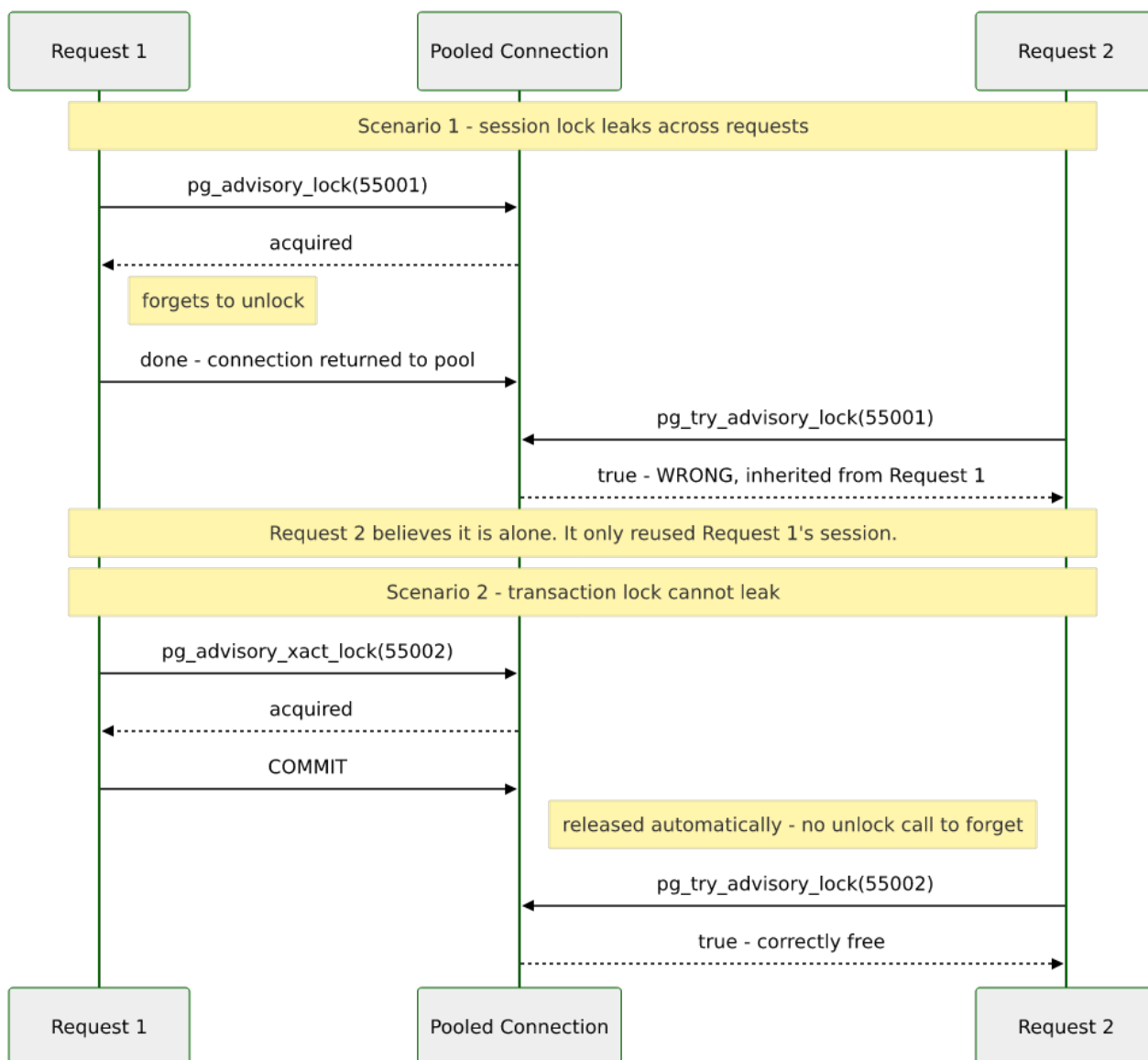
```

```

[request 1] acquired xact lock 55002
[request 1] committed - lock released automatically
[request 2] pg_try_advisory_lock(55002) -> True (correctly free)

```

Same reused connection, same shape of bug waiting to happen — but this time request 2's true is *correct*, because `pg_advisory_xact_lock` physically cannot survive past `COMMIT`. There's no `unlock` call to forget, no code path where an exception skips the cleanup, because there's no cleanup step at all: the transaction boundary *is* the release. **The rule this exercise earns:** reach for `pg_advisory_xact_lock()`, not `pg_advisory_lock()`, for anything that might ever run behind a connection pool — which, in most modern application deployments, is close to everything.



Summary — What You Should Now Know

Tool

`pg_advisory_lock(key) / pg_advisory_unlock(key)`

`pg_advisory_xact_lock(key)`

`pg_try_advisory_lock(key)`

`hashtext('a category name')`

Leader election pattern

What it does

Session-level lock — held until explicitly released or the connection closes

Transaction-level lock — released automatically at `COMMIT`/`ROLLBACK`, no `unlock` function exists

Non-blocking — `true/false` immediately instead of waiting

Turn an arbitrary string into a lock key without a lookup table

N processes `pg_try_advisory_lock` the same key; exactly one gets `true`

Tool	What it does
Critical-section pattern	<code>pg_advisory_xact_lock</code> around a category-wide rule row locking can't express
<code>pg_locks WHERE locktype = 'advisory'</code>	See every held advisory lock, joined to <code>pg_stat_activity</code> for who/how long
Connection-pool pitfall	A session lock can leak across unrelated pooled requests; a transaction lock structurally cannot

The key design insight from this chapter is that advisory locks trade specificity for reach: a row lock only ever means “this row,” but an advisory lock can mean anything at all, because the number means whatever your application agrees it means. That flexibility is also the whole risk — nothing stops two unrelated parts of a codebase from picking the same integer by accident, and nothing stops a session-level lock from outliving the logical operation it was meant to protect the instant a connection pool gets involved. Every exercise past the first two was really about earning back the specificity a row lock gets for free: naming a category clearly (`hashtext`), choosing a lifetime that matches the actual unit of work (transaction, not session), and knowing how to ask PostgreSQL, out loud, exactly who's holding what.

Going further: Chapter 19's `pg_cron` is where the singleton-job pattern from Exercise 2 stops being a hypothetical — a scheduled job that might occasionally overlap its own next run is the textbook case for wrapping the job body in `pg_try_advisory_lock` and exiting quietly if it doesn't get it. Chapter 13's `NOTIFY` and this chapter's advisory locks share the same primary-only limitation, and for the same underlying reason: both live in server-local memory rather than WAL, so neither one is a tool for coordinating across a primary and its replicas — that requires the data itself to be replicated, which is Chapter 18's subject. And it's worth holding onto the contrast with Chapter 3 explicitly: `FOR UPDATE SKIP LOCKED` coordinates access to rows that exist; advisory locks coordinate processes, around ideas that were never going to have a row of their own no matter how the schema was designed.

Chapter 15 — Custom Types, Domains, and Enums

“A CHECK constraint says ‘reject this if it’s wrong.’ A type says ‘wrong isn’t a value this column can even hold.’”

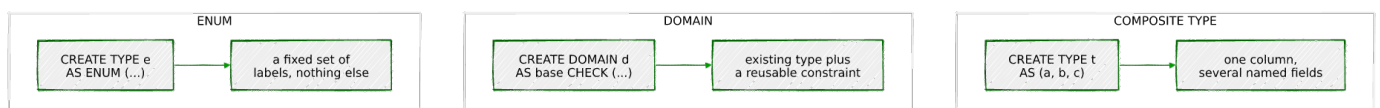
Background

Every table in this book so far has leaned on CHECK constraints and plain base types — TEXT, INTEGER, JSONB — to keep bad data out. That works, but it puts the rule *next to* the column rather than *in* it: `jobs.status` has always been a TEXT column that happens to carry a CHECK (`status IN (...)`) alongside it, and nothing about the column’s own type tells you that. PostgreSQL’s type system lets you go one level deeper and make the rule part of the type itself.

Three tools, three different jobs:

- **Enums** (`CREATE TYPE ... AS ENUM (...)`) replace “a TEXT column with a CHECK list” with a real type that can *only* hold one of a fixed set of labels. The database rejects anything else the moment you try to store it — not as a constraint violation, but as a type error, the same category of rejection as trying to put text in an integer column.
- **Domains** (`CREATE DOMAIN ... AS base_type CHECK (...)`) attach a constraint to an *existing* type and give the result a name — a `positive_integer` is still an INTEGER underneath, but any column declared with that domain inherits the check automatically, once, instead of you retyping the same CHECK clause on every column that needs it.
- **Composite types** (`CREATE TYPE ... AS (...)`) bundle several named fields into one type, the way a table’s row type already works, except you can use it as the type of a single *column* — a `contact_info` value is one column that internally holds a phone number, a postcode, and a preferred contact method together.

All three live in the database’s own catalog, which means every tool that talks to PostgreSQL can discover them — a `psycopg` client, a PostgREST endpoint, even `psql`’s own `\d` output — without you writing a line of validation logic anywhere outside the schema. Exercise 6 is where that pays off directly.



The Scenario

This chapter adds typed columns to three tables already in the book, rather than any new ones:

Table	New column	Type
jobs (Ch3)	status (<i>converted in place</i>)	job_status enum
businesses (Ch1)	employee_count	positive_integer domain
residents (Ch5)	contact	contact_info composite (embeds a uk_postcode domain)
residents (Ch5)	email	email_address domain

`jobs.status` already existed as `TEXT` with a `CHECK` constraint — Exercise 1 converts it to a real enum in place, which turns out to be more interesting than it sounds, because Chapter 10's `api.jobs` view is still sitting on top of it.

Exercise Goals

By the end of this chapter you will be able to:

- Convert an existing `CHECK`-constrained `TEXT` column into a proper `ENUM`, and know every kind of object that can block that conversion before it actually works.
 - Add a new value to an existing enum, control exactly where it sorts, and explain why you can't use a value you just added in the same transaction that added it.
 - Create a domain and watch PostgreSQL enforce its constraint on every column that uses it, everywhere, automatically.
 - Build a composite type, store it in a column, and query its individual fields.
 - Enforce a text format — a postcode, an email address — with a domain built on a regular expression `CHECK`.
 - See enum values reflected automatically into a Python client and a PostgREST API, with no hand-written validation on either side.
-

Installation

Nothing to install. `CREATE TYPE`, `CREATE DOMAIN`, and enums are all core SQL, no extension required. Exercise 6 reuses `psycopy` (Chapter 1) and, optionally, PostgREST (Chapter 10) if you still have it configured.

Loading the Data

This chapter needs Chapters 1, 3, and 5's data:

```
python data/ch01_seed.py    # businesses
python data/ch03_seed.py    # jobs
python data/ch05_seed.py    # residents
```

```
SELECT 'businesses' AS table, COUNT(*) FROM businesses
UNION ALL SELECT 'jobs', COUNT(*) FROM jobs
UNION ALL SELECT 'residents', COUNT(*) FROM residents;
```

table	count
businesses	48
jobs	48
residents	58

Exercises

Exercise 1 — Converting TEXT + CHECK Into a Real Enum

1.1 — Define the type

```
CREATE TYPE job_status AS ENUM ('queued', 'in_progress', 'completed', 'failed');
```

This creates the type — it doesn't touch `jobs` yet. `jobs.status` is still TEXT, still governed by the original CHECK constraint from Chapter 3.

1.2 — Three things that block the conversion, in the order you'll hit them

The obvious next move — `ALTER TABLE jobs ALTER COLUMN status TYPE job_status USING status::job_status;` — fails, and it fails for three separate reasons if you fix them one at a time instead of all at once. Each is worth seeing for real, because each is a genuinely common shape of dependency in any schema that's been alive for more than one chapter:

Blocker 1 — the column's own default:

```
ERROR:  default for column "status" cannot be cast automatically to type job_status
```

`status DEFAULT 'queued'` was set while the column was TEXT; the default expression has to be dropped before the column type can change, and re-added afterward.

Blocker 2 — a dependent view:

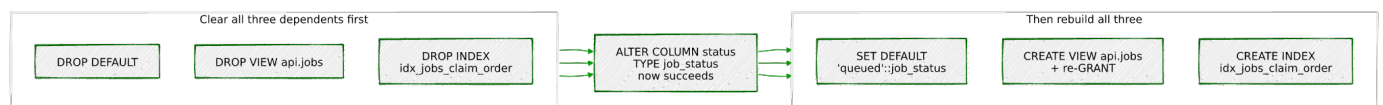
```
ERROR: cannot alter type of a column used by a view or rule
DETAIL: rule _RETURN on view api.jobs depends on column "status"
```

Chapter 10's `api.jobs` view selects `status` directly. PostgreSQL won't silently change a column's type out from under a view built on top of it — the view has to be dropped and recreated around the change. This is exactly the kind of cross-chapter dependency a real production schema accumulates constantly, and exactly why this exercise is worth doing on the actual multi-chapter database instead of a clean scratch table.

Blocker 3 — a dependent partial index:

```
ERROR: operator does not exist: job_status = text
HINT: No operator matches the given name and argument types.
```

`idx_jobs_claim_order` (Chapter 3) is a partial index with `WHERE status = 'queued'` baked into it. Once `status` is no longer `TEXT`, PostgreSQL has to rebuild that stored predicate — and a bare `job_status = text` comparison doesn't exist as an operator; enums don't implicitly compare against a different type the way a literal implicitly resolves against a column of a known type. This index has to be dropped and rebuilt too.



1.3 — The full sequence, correctly ordered, in one transaction

Run this as `postgres`, not your regular login role — `sudo -u postgres psql portsmith`. `api.jobs` is owned by `postgres` (it was created from that same role back in Chapter 10, Exercise 1.1), and a normal login role typically has no `USAGE` on the `api` schema at all, let alone ownership of the view sitting in it. Trying to run this as your own role fails immediately with permission denied for schema `api` before it ever gets near the actual type conversion:

```
BEGIN;

DROP VIEW IF EXISTS api.jobs;
DROP INDEX IF EXISTS idx_jobs_claim_order;

ALTER TABLE jobs ALTER COLUMN status DROP DEFAULT;
ALTER TABLE jobs DROP CONSTRAINT IF EXISTS jobs_status_check;
ALTER TABLE jobs ALTER COLUMN status TYPE job_status USING status::job_status;
ALTER TABLE jobs ALTER COLUMN status SET DEFAULT 'queued'::job_status;

CREATE INDEX IF NOT EXISTS idx_jobs_claim_order
    ON jobs (priority, created_at, id)
    WHERE status = 'queued';
```

```

CREATE VIEW api.jobs AS
SELECT id, job_type, payload, priority, status, created_at
FROM jobs;

GRANT SELECT ON api.jobs TO web_anon;
GRANT INSERT (job_type, payload, priority) ON api.jobs TO web_anon;

COMMIT;

```

Wrapping the whole thing in `BEGIN/COMMIT` matters here specifically because `psql`'s non-interactive `-f` mode does *not* wrap a script in one implicit transaction the way a `-c "stmt1; stmt2;"` invocation does — without the explicit `BEGIN`, a failure partway through a script like this one leaves the schema in whatever half-converted state the successful statements got it to, which is exactly what happened during this chapter's own testing before the transaction wrapper was added.

The `IF EXISTS/IF NOT EXISTS ON EVERY DROP/CREATE` above earn their keep for the same reason: because the whole block is one transaction, any single failure aborts everything in it, including statements that already succeeded — so if this script is ever run a second time (after an earlier run partially failed, or simply to double-check the result), the objects from a *previous successful* run are still sitting there. Without `IF EXISTS`, a re-run's `DROP CONSTRAINT jobs_status_check` would error with “constraint ... does not exist” the moment `status` is already a `job_status`, aborting the whole transaction over nothing actually wrong. With it, this script is safe to run again at any point — fully unconverted, partially converted, or already done — and either converges to the same end state or changes nothing at all.

1.4 — Confirm it, and confirm the rejection

```

\d jobs

status | job_status | | not null | 'queued'::job_status
...
Indexes:
    "idx_jobs_claim_order" btree (priority, created_at, id) WHERE status = 'queued'::job_status

UPDATE jobs SET status = 'archived_forever' WHERE id = 1;

ERROR:  invalid input value for enum job_status: "archived_forever"

```

(Deliberately not `'cancelled'` here — Exercise 2 is about to make that one a real value, and if you're re-running this exercise on a database that's already been through Exercise 2, `'cancelled'` would no longer be rejected. `'archived_forever'` is never added anywhere in this chapter, so this check stays valid no matter what order you run things in or how many times you re-run it.)

Compare this to Chapter 3's original `CHECK` violation message — same outcome, a rejected write, but a completely different category of error. `'archived_forever'` isn't *disallowed*, from PostgreSQL's point of view; it simply isn't a `job_status` at all, the same way `'abc'` isn't an `INTEGER`.

Exercise 2 — Adding a Value: Ordering and Transaction Rules

2.1 — Add a value

```
ALTER TYPE job_status ADD VALUE 'cancelled';
SELECT enum_range(NULL::job_status);
```

enum_range

{queued,in_progress,completed,failed,cancelled}

New values are appended at the end by default. `enum_range` returns every label in the type's actual sort order — enums order by *declaration position*, not alphabetically, which is why 'cancelled' sorts after 'failed' here despite the alphabet disagreeing.

2.2 — Control exactly where it lands

```
ALTER TYPE job_status ADD VALUE 'on_hold' BEFORE 'in_progress';
SELECT enum_range(NULL::job_status);
```

enum_range

{queued,on_hold,in_progress,completed,failed,cancelled}

BEFORE/AFTER exist precisely because append-only ordering isn't always what you want — 'on_hold' belongs conceptually between `queued` and `in_progress`, and now it sorts there too, which matters anywhere this column gets used in an `ORDER BY`.

2.3 — The rule that catches almost everyone once

```
BEGIN;
ALTER TYPE job_status ADD VALUE 'archived';
UPDATE jobs SET status = 'archived' WHERE id = 1;
COMMIT;
```

```
ERROR:  unsafe use of new value "archived" of enum type job_status
HINT:   New enum values must be committed before they can be used.
```

`ADD VALUE` is allowed inside a transaction block — that restriction was lifted back in PostgreSQL 12 — but the new label still can't be *used* until the transaction that added it actually commits. PostgreSQL can't yet guarantee the value will still exist if this transaction rolls back, so it refuses to let anything reference it in the meantime. (This transaction did roll back here, which is exactly why 'archived' doesn't appear in 2.2's output above — the `ADD VALUE` itself is fully transactional too.) Add the value and use it in two separate transactions, and this is a non-issue.

Exercise 3 — A `positive_integer` Domain

3.1 — Define it once, use it anywhere

```
CREATE DOMAIN positive_integer AS INTEGER CHECK (VALUE > 0);
ALTER TABLE businesses ADD COLUMN employee_count positive_integer;
```

VALUE inside a domain's CHECK refers to whatever's being validated — the same role `NEW.column` plays in a trigger, just scoped to a single value instead of a whole row.

3.2 — It enforces itself, everywhere, immediately

```
UPDATE businesses SET employee_count = 12 WHERE name = 'The Gilded Clam';
```

UPDATE 1

```
UPDATE businesses SET employee_count = -5 WHERE name = 'Anchor & Oar Tavern';
```

ERROR: value for domain positive_integer violates check constraint "positive_integer_check"

```
UPDATE businesses SET employee_count = 0 WHERE name = 'Anchor & Oar Tavern';
```

ERROR: value for domain positive_integer violates check constraint "positive_integer_check"

Zero fails too — > 0 , not ≥ 0 , meant exactly what it said. Every future column ever declared `positive_integer` gets this exact rule for free, with no CHECK clause to remember to copy.

Exercise 4 — A `contact_info` Composite Type

4.1 — A domain nested inside a composite type

Portsmouth's real-world namesake, Portsmouth, England, has postcodes starting with "PO" — worth getting right rather than reaching for a generic example:

```
CREATE DOMAIN uk_postcode AS TEXT
CHECK (VALUE ~ '^[A-Z]{1,2}[0-9][A-Z0-9]? [0-9][A-Z]{2}$');
```

```
CREATE TYPE contact_info AS (
    phone          TEXT,
    postcode       uk_postcode,
    preferred_contact TEXT
);
```

```
ALTER TABLE residents ADD COLUMN contact contact_info;
```

A composite type's fields aren't limited to base types — `postcode` here is a full domain, constraint included, nested one level inside another type. Validation composes the same way the types do.

4.2 — Store and query it

```

UPDATE residents
SET     contact = ROW('023 9281 4477', 'P01 3AX', 'phone')::contact_info
WHERE   full_name = 'Adrian Foscolo';

SELECT full_name, (contact).phone, (contact).postcode
FROM   residents
WHERE   full_name = 'Adrian Foscolo';

```

full_name	phone	postcode
Adrian Foscolo	023 9281 4477	P01 3AX

(contact).phone — the parentheses are required; without them, contact.phone parses as “column phone of table contact,” not “field phone of column contact.” One column, \d residents shows one contact_info entry, but it carries three named, individually queryable fields.

4.3 — The nested domain still enforces itself

```

UPDATE residents
SET     contact = ROW('023 9281 0000', 'NOTVALID', 'email')::contact_info
WHERE   full_name = 'Marisol Quintero';

```

ERROR: value for domain uk_postcode violates check constraint "uk_postcode_check"

The CHECK fires from two levels down — PostgreSQL validates every domain-typed field of a composite value the same way it would validate a plain column, whether or not that field happens to be buried inside something else on its way into the table.

Exercise 5 — An Email Domain

5.1 — A dedicated, standalone domain

```

CREATE DOMAIN email_address AS TEXT
CHECK (VALUE ~ '^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$');

ALTER TABLE residents ADD COLUMN email email_address;

```

5.2 — Two different ways to fail

```
UPDATE residents SET email = 'adrian.foscolo@example.com' WHERE full_name = 'Adrian Foscolo';
```

UPDATE 1

```
UPDATE residents SET email = 'not-an-email' WHERE full_name = 'Marisol Quintero';
```

ERROR: value for domain email_address violates check constraint "email_address_check"

```
UPDATE residents SET email = 'someone@localhost' WHERE full_name = 'Marisol Quintero';
```

ERROR: value for domain email_address violates check constraint "email_address_check"

'someone@localhost' is a technically-valid email address by the actual internet standard — real mail servers accept it — and this regex still rejects it, because it has no .-separated top-level domain. Every regex-based domain like this one is a deliberate, imperfect trade: strict enough to catch obvious garbage, permissive enough not to reject real addresses your business actually needs, and never a substitute for verifying an address by actually sending mail to it.

Exercise 6 — Automatic Reflection

6.1 — psycopg, by default

```
import psycopg

with psycopg.connect("dbname=portsmith") as conn:
    with conn.cursor() as cur:
        cur.execute("SELECT id, status FROM jobs WHERE id = 1;")
        row = cur.fetchone()
        print(f"value: {row[1]!r}")
        print(f"python type: {type(row[1])}")
```

```
value: 'in_progress'
python type: <class 'str'>
```

No configuration, no registration — an enum column just arrives as a plain Python string, because PostgreSQL’s own wire protocol already tells the driver “this value is text-shaped,” and psycopg doesn’t need to know anything about `job_status` specifically to hand it back correctly.

6.2 — psycopg, fully reflected as a real Python enum

```
import psycopg
from psycopg.types.enum import register_enum, EnumInfo

with psycopg.connect("dbname=portsmith") as conn:
    info = EnumInfo.fetch(conn, "job_status")
    register_enum(info, conn)
    print("labels from the database:", info.labels)
    with conn.cursor() as cur:
        cur.execute("SELECT id, status FROM jobs WHERE id = 1;")
        row = cur.fetchone()
        print(f"value: {row[1]!r}")
        print(f"python type: {type(row[1])}")
```

```
labels from the database: ['queued', 'on_hold', 'in_progress', 'completed', 'failed',
'cancelled']
value: <Job_Status.in_progress: 3>
python type: <enum 'Job_Status'>
```

`EnumInfo.fetch()` reads the *live* set of labels straight out of `pg_enum` — all six, including `on_hold` and `cancelled` from Exercise 2 — and `register_enum()` builds and wires up a real Python `enum.Enum` class

from them, on the spot. Change the database's enum next month and this code doesn't change at all; it reflects whatever the type currently contains.

6.3 — PostgREST's OpenAPI document

With PostgREST running against this database (Chapter 10's setup):

```
curl -s http://localhost:3000/ | jq '.definitions.jobs.properties.status'

{
  "enum": ["queued", "on_hold", "in_progress", "completed", "failed", "cancelled"],
  "format": "public.job_status",
  "type": "string"
}
```

The exact same six labels, discovered the exact same way — by introspecting `job_status` directly — and published automatically in the API's own schema document. No route was written to expose this list anywhere; it's a side effect of the column's type being what it is.

6.4 — And the API enforces it too, for free

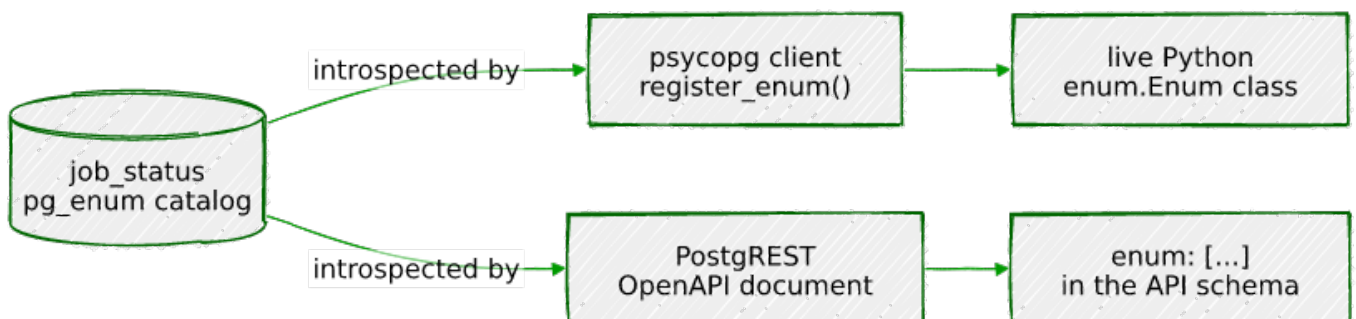
```
curl -s "http://localhost:3000/jobs?status=eq.bogus"

{"code": "22P02", "details": null, "hint": null, "message": "invalid input value for enum job_status: \"bogus\""}

curl -s -o /dev/null -w "%{http_code}\n" "http://localhost:3000/jobs?status=eq.bogus"

400
```

The exact same error PostgreSQL raised back in Exercise 1.4, now arriving as a 400 over HTTP with PostgreSQL's own error code and message intact. PostgREST didn't validate this value against a list it maintains — it never has to. The type validated it, at the one place that was always going to be authoritative regardless of which client asked.



Summary — What You Should Now Know

Tool	What it does
<code>CREATE TYPE name AS ENUM (...)</code>	A type that can only hold one of a fixed set of labels
<code>ALTER TABLE ... ALTER COLUMN ... TYPE enum USING col::enum</code>	Convert an existing column — after clearing its default and any dependent views/indexes
<code>ALTER TYPE ... ADD VALUE [BEFORE\ AFTER 'x']</code>	Add a label; position controls sort order, not just membership
New enum values in a transaction	Addable, but unusable until that transaction commits
<code>CREATE DOMAIN name AS base_type CHECK (VALUE ...)</code>	A named, reusable constraint on top of an existing type
<code>CREATE TYPE name AS (field type, ...)</code>	A composite type — several named fields as one column
<code>(composite_col).field</code>	Access one field of a composite column — parentheses required
A domain nested inside a composite field	Still fully enforced, at whatever depth it's used
<code>EnumInfo.fetch() + register_enum() (psycopg)</code>	Build a live Python <code>Enum</code> class from a database enum's current labels
PostgREST's / OpenAPI document	Lists a column's enum values automatically, from the type, with no route written for it

The key design insight from this chapter is where the rule actually lives. A `CHECK` constraint and a custom type can enforce the identical rule — Chapter 3's original `status IN (...)` and this chapter's `job_status` enum reject exactly the same bad values — but only one of them is a fact every tool touching the database can discover on its own, without being told. `psycopg` didn't need to be taught what `job_status` allows; PostgREST didn't need a hand-written validator for it either. Both found out by asking PostgreSQL, because the rule was never bolted onto the column from outside — it was what the column's type actually *was*.

Going further: Chapter 16's generated columns are this chapter's natural next step — where a domain or enum constrains what a column can hold, a generated column controls what a column is, computed automatically from the rest of the row. The `positive_integer` and `email_address` domains built here are deliberately simple; production schemas often layer several domains and composite types together the way `contact_info` nested `uk_postcode` in Exercise 4, one constraint at a

time, until the type system is carrying most of what used to live in application-layer validation code. And Exercise 1's dependency chain — a default, a view, a partial index, all blocking one column's type change — is worth remembering the next time any column with a few chapters of history behind it needs to change shape: \d on the table first, always, before the ALTER.

Chapter 16 — Generated Columns

“A trigger is a promise that someone wrote the sync logic correctly. A generated column is a fact that there was never anything else it could have contained.”

Background

Chapter 4’s `city_documents.search_vector` and Chapter 13’s `notify_job_status_change()` both solve the same underlying problem — “keep a derived value in sync with the columns it’s computed from” — the same way: a trigger, hand-written, that fires on `INSERT/UPDATE` and recomputes the value. It works, but it works because someone wrote it correctly, remembered to fire it on the right columns, and never introduced a code path that updates the base columns without also firing the trigger. A **generated column** removes all three of those “someone remembered to” risks by making the derivation part of the column’s own definition:

```
column_name type GENERATED ALWAYS AS (expression) STORED
```

`STORED` isn’t optional decoration — as of the PostgreSQL version this book targets, `STORED` is the *only* kind of generated column PostgreSQL supports. The expression is computed once, at write time, and the result is physically written to disk right alongside every other column, not recomputed on every read. There’s no trigger to find, no function to audit, no risk of an `UPDATE` statement somewhere that touches the base columns without going through the trigger — because `GENERATED ALWAYS` means exactly that. PostgreSQL enforces it directly: Exercise 4 tries to write to one on purpose, just to watch it refuse.

The one requirement that makes this possible, and the thing to hold onto before Exercise 1’s first real surprise: the expression must be **immutable** — given the same input row, it must always produce the same output, forever, with no dependency on anything outside that row. That sounds like a formality until it collides with something this book has run into more than once already.

The Scenario

Table	Column	Computed from
<code>sensor_readings</code> (Ch8)	<code>reading_date</code>	<code>recorded_at</code> , date portion only
<code>businesses</code> (Ch1)	<code>search_vector</code>	<code>name</code> + several details JSONB fields

Table	Column	Computed from
city_documents (Ch4)	search_vector	title + body — replaces a hand-written trigger
residents (Ch15)	phone_digits	(contact).phone, digits only

Every one of these already has real data behind it from earlier chapters — this chapter is entirely about adding computed columns to tables that already exist, not seeding anything new.

Exercise Goals

By the end of this chapter you will be able to:

- Add a stored generated column, and explain exactly why PostgreSQL’s immutability requirement rejects some expressions that look perfectly reasonable.
 - Replace a hand-written trigger that maintains a `tsvector` column with a generated column that does the same job declaratively.
 - Index a generated column, and know the one thing about partitioned tables that indexing one does *not* give you for free.
 - Prove, by trying, that a generated column cannot be written to directly — only ever recomputed.
 - Build a normalized column from a composite type’s field, and know exactly where “normalize the formatting” stops being enough.
 - Compare generated columns to triggers directly, and know which one to reach for when.
-

Installation

Nothing to install. Generated columns are core SQL, available since PostgreSQL 12.

Loading the Data

This chapter needs Chapters 1, 4, 5, 8, and 15’s data already in place:

```
python data/ch01_seed.py    # businesses
python data/ch04_seed.py    # city_documents
python data/ch05_seed.py    # residents
python data/ch08_seed.py    # sensors; then Chapter 8's own exercises to build sensor_readings
```

Chapter 15’s `contact_info` composite type and `residents.contact` column need to exist too — see that chapter if you haven’t run it yet.

Verify the prerequisites

```
SELECT 'sensor_readings' AS table, COUNT(*) FROM sensor_readings
UNION ALL SELECT 'businesses', COUNT(*) FROM businesses
UNION ALL SELECT 'city_documents', COUNT(*) FROM city_documents
UNION ALL SELECT 'residents', COUNT(*) FROM residents;
```

table	count
sensor_readings	9648001
businesses	48
city_documents	30
residents	58

(9,648,001, not 9,648,000 — Chapter 9, Exercise 5.3’s late-arriving traffic reading is still in there. If you’re at an even 9,648,000, that’s fine too; nothing here depends on the exact count.)

Exercises

Exercise 1 — Extracting a Date From a Timestamp

1.1 — The obvious expression fails

```
ALTER TABLE sensor_readings
  ADD COLUMN reading_date DATE GENERATED ALWAYS AS (recorded_at::date) STORED;
```

ERROR: generation expression is not immutable

`recorded_at` is `TIMESTAMPTZ` — an absolute instant, not a calendar date. Casting it to `DATE` requires picking a timezone to interpret that instant *in*, and a bare `::date` cast uses whatever the current session’s `timezone` setting happens to be. Two sessions with different timezone settings would compute two different dates from the identical `recorded_at` value — which is exactly what “not immutable” means: PostgreSQL refuses to store a value that isn’t guaranteed to be the same answer forever, computed from the row alone. This is the same class of bug Chapters 8, 9, and 11 all had to work around by hand with `SET timezone = 'UTC';` at the top of every session — except here, PostgreSQL catches it at `CREATE` time instead of letting it silently produce different numbers depending on who’s connected.

1.2 — Pin the timezone inside the expression itself

```
ALTER TABLE sensor_readings
  ADD COLUMN reading_date DATE
  GENERATED ALWAYS AS ((recorded_at AT TIME ZONE 'UTC')::date) STORED;
```

ALTER TABLE

AT TIME ZONE 'UTC' makes the timezone part of the expression instead of part of the session — now the result really is a pure function of `recorded_at` alone, immutable by construction, no session setting involved anywhere.

1.3 — Confirm the backfill

Adding a generated column to a table that already has 9.6 million rows computes the value for every existing row immediately, not lazily:

```
SELECT recorded_at, reading_date FROM sensor_readings LIMIT 3;
```

recorded_at		reading_date
2024-01-31 19:00:00-05		2024-02-01
2024-01-31 19:00:00-05		2024-02-01
2024-01-31 19:00:00-05		2024-02-01

```
SELECT COUNT(*) FROM sensor_readings WHERE reading_date IS NULL;
```

count
0

Zero NULLS across every one of the 9.6 million existing rows, and `sensor_readings` is a *partitioned* table — this one ALTER TABLE on the parent applied to, and backfilled, every partition underneath it automatically. And notice: `2024-01-31 19:00:00-05` — a January 31st timestamp by its displayed local time — produced `reading_date = 2024-02-01`. That's AT TIME ZONE 'UTC' working exactly as intended: `19:00 -05` is `00:00 UTC` the next calendar day, and the generated column reflects the UTC date, not whatever date the display happens to suggest.

Exercise 2 — Replacing a Trigger With a Generated Column

2.1 — What Chapter 4 built

```
\d city_documents
```

Triggers:

```
trg_city_documents_search_vector BEFORE INSERT OR UPDATE OF title, body
ON city_documents FOR EACH ROW EXECUTE FUNCTION city_documents_search_vector_update()
```

```
SELECT prosrc FROM pg_proc WHERE proname = 'city_documents_search_vector_update';
```

```

BEGIN
    NEW.search_vector := to_tsvector('english', NEW.title || ' ' || NEW.body);
    RETURN NEW;
END;

```

(That’s the query’s *output* — the function’s raw source text — not a statement to run. `NEW` only means anything inside a trigger function body; pasted directly into `psql` it errors immediately.)

A trigger function, a trigger to fire it, and a plain `tsvector` column with no memory of how it’s supposed to stay correct except “whatever that function currently says.”

2.2 — Tear it out, replace it with a generated column

```

DROP TRIGGER trg_city_documents_search_vector ON city_documents;
DROP FUNCTION city_documents_search_vector_update();
ALTER TABLE city_documents DROP COLUMN search_vector;

ALTER TABLE city_documents ADD COLUMN search_vector tsvector
    GENERATED ALWAYS AS (to_tsvector('english', title || ' ' || body)) STORED;

SELECT COUNT(*) FROM city_documents WHERE search_vector IS NOT NULL;

```

count

30

All 30 rows recomputed correctly in the same `ALTER TABLE` that added the column — identical expression, identical result, zero lines of trigger code left anywhere in the schema.

2.3 — The gotcha: `DROP COLUMN` takes its index with it

```

\d city_documents

```

Indexes:

```

"city_documents_pkey" PRIMARY KEY, btree (id)
"idx_city_documents_doc_type" btree (doc_type)
"idx_city_documents_published_date" btree (published_date)

```

`idx_city_documents_search_vector`, Chapter 4’s GIN index, is simply gone — dropping a column drops every index that depends on it, silently, as part of the same statement. Rebuild it:

```

CREATE INDEX idx_city_documents_search_vector ON city_documents USING GIN (search_vector);

SELECT title, ts_rank(search_vector, query) AS rank
FROM    city_documents, to_tsquery('english', 'zoning & permit') query
WHERE   search_vector @@ query
ORDER  BY rank DESC
LIMIT  3;

```

title	rank
Zoning Ordinance – Citywide Accessory Dwelling Unit Standards	0.008959805

Same ranked results Chapter 4’s trigger-maintained column would have produced — the search behavior didn’t change at all; only how the `search_vector` column stays correct did.

Exercise 3 — Indexing a Generated Column

3.1 — A fresh example: full-text search on businesses

```
ALTER TABLE businesses ADD COLUMN search_vector tsvector
GENERATED ALWAYS AS (
    to_tsvector('english',
        name || ' ' ||
        coalesce(details->>'category', '') || ' ' ||
        coalesce(details->>'subcategory', '') || ' ' ||
        coalesce(details->>'cuisine', '') || ' ' ||
        coalesce(details->>'tags', '')
    )
) STORED;

SELECT search_vector FROM businesses WHERE name = 'Anchor & Oar Tavern';
```

```
'anchor':1 'dog':10 'friend':11 'live':8 'music':9 'oar':2 'outdoor':12
'pub':5,7 'restaur':4 'seat':13 'tavern':3 'waterfront':6
```

`details->>'tags'` pulls the JSONB tag array out as its raw text representation — `["waterfront", "pub", "live_music", ...]` — and `to_tsvector` tokenizes straight through the brackets, quotes, and underscores, splitting `live_music` into separate `live` and `music` tokens. That’s not a bug to work around; it’s what makes a search for just `music` find this row at all.

3.2 — Index it, and watch the planner ignore the index anyway

```
CREATE INDEX idx_businesses_search_vector ON businesses USING GIN (search_vector);

EXPLAIN (ANALYZE)
SELECT name FROM businesses
WHERE search_vector @@ to_tsquery('english', 'waterfront & pub');
```

Seq Scan on businesses (cost=0.00..8.60 rows=1 width=18) (actual time=0.005..0.024 rows=1 loops=1)

Filter: (search_vector @@ '''waterfront'' & '''pub''':tsquery)

Rows Removed by Filter: 47

Planning Time: 1.449 ms

Execution Time: 0.035 ms

A brand-new GIN index, and PostgreSQL doesn’t touch it — 48 rows fits in a fraction of one disk page, and a sequential scan of the whole table is cheaper than the overhead of consulting an

index at all. The index isn't broken; the planner is correctly deciding it isn't worth using yet. Force the comparison to see it work:

```
SET enable_seqscan = off;

EXPLAIN (ANALYZE)
SELECT name FROM businesses
WHERE search_vector @@ to_tsquery('english', 'waterfront & pub');

Bitmap Heap Scan on businesses (cost=12.97..16.98 rows=1 width=18) (actual time=0.014..0.015
rows=1 loops=1)
  Recheck Cond: (search_vector @@ '''waterfront' & 'pub''::tsquery)
  Heap Blocks: exact=1
  -> Bitmap Index Scan on idx_businesses_search_vector (cost=0.00..12.97 rows=1 width=0)
(actual time=0.012..0.012 rows=1 loops=1)
    Index Cond: (search_vector @@ '''waterfront' & 'pub''::tsquery)

RESET enable_seqscan;
```

The index is completely ordinary from the planner's point of view — it just also has completely ordinary opinions about when a 48-row table is worth indexing at all.

3.3 — The one thing indexing a generated column does *not* give you

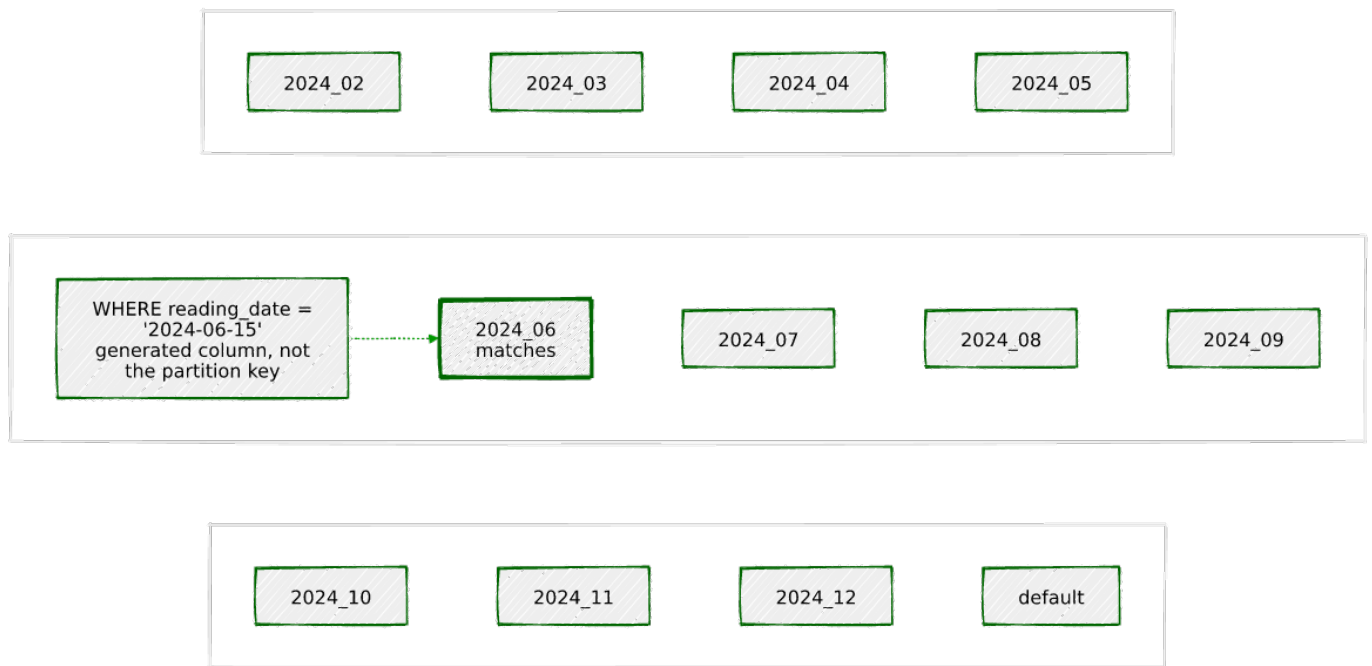
Chapter 8 spent an entire chapter on partition pruning — the planner throwing out whole partitions before scanning anything, based on the partition key. `reading_date` is indexed now, and it's *derived from* the partition key, but it isn't *itself* the partition key (`sensor_readings` still partitions by `recorded_at`), so filtering by it doesn't prune:

```
SET timezone = 'UTC';

EXPLAIN (ANALYZE)
SELECT COUNT(*) FROM sensor_readings WHERE reading_date = '2024-06-15';

Aggregate (cost=79814.92..79814.93 rows=1 width=8) (actual time=15.642..15.646 rows=1
loops=1)
  -> Append (cost=48.79..79694.32 rows=48240 width=0) (actual time=0.891..14.627 rows=28801
loops=1)
    -> Bitmap Heap Scan on sensor_readings_2024_02 ... (actual time=0.049..0.049 rows=0
loops=1)
      -> Bitmap Index Scan on sensor_readings_2024_02_reading_date_idx ...
    -> Bitmap Heap Scan on sensor_readings_2024_03 ... (actual time=0.032..0.032 rows=0
loops=1)
      -> Bitmap Index Scan on sensor_readings_2024_03_reading_date_idx ...
    -- ... one Bitmap Heap Scan + Bitmap Index Scan pair per partition, all twelve ...
    -> Bitmap Heap Scan on sensor_readings_2024_06 ... (actual time=0.753..13.204
rows=28801 loops=1)
      -> Bitmap Index Scan on sensor_readings_2024_06_reading_date_idx ... rows=28801
Planning Time: 2.255 ms
Execution Time: 15.832 ms
```


Every one of the twelve partitions gets its own `Bitmap Index Scan` — cheap ones, since each partition’s local index quickly finds “zero rows here,” but *every partition still gets asked*. Compare that to Chapter 8, Exercise 3.1, where filtering on `recorded_at` (the actual partition key) made eleven of twelve partitions disappear from the plan entirely, before execution ever started. An index on a generated column makes searching that column fast; it does nothing at all for pruning, because pruning only ever looks at the partition key itself. 15.8 ms across 9.6 million rows is still fast — this is a real cost, just a much smaller one than the difference would be without any index at all.



Exercise 4 — Generated Columns Reject Direct Writes

4.1 — Try to UPDATE one

```
UPDATE sensor_readings SET reading_date = '2020-01-01' WHERE id = 1;
```

```
ERROR: column "reading_date" can only be updated to DEFAULT
DETAIL: Column "reading_date" is a generated column.
```

4.2 — Try to INSERT one explicitly

```
INSERT INTO sensor_readings (sensor_id, sensor_type, reading_value, recorded_at, reading_date)
VALUES (1, 'temperature', 50.0, now(), '2020-01-01');
```

```
ERROR: cannot insert a non-DEFAULT value into column "reading_date"
DETAIL: Column "reading_date" is a generated column.
```

Both rejected, for the same reason: there is no path in PostgreSQL that lets application code — or a typo, or a well-meaning bulk import script — set a generated column to anything other than

what its expression computes. This is the guarantee a trigger-maintained column can never quite make: a trigger stops a *normal* write path from getting out of sync, but nothing stops a different write path — a bulk copy, a migration script, a DBA fixing something by hand at 2 a.m. — from writing directly to the column and skipping the trigger entirely. A generated column has no other write path to skip.

Exercise 5 — Normalizing a Phone Number

5.1 — Real messy data, on purpose

Chapter 15’s `contact_info` composite type stores `phone` as free-form text, and five Portsmouth residents now have it filled in with five different formatting conventions — exactly the mess a real “phone number” field accumulates over time:

```
SELECT full_name, (contact).phone FROM residents WHERE id BETWEEN 1 AND 5;
```

full_name	phone
Adrian Foscolo	023 9281 4477
Marisol Quintero	023-9281-5522
Bennett Okoye	(023) 9281 6633
Wilhelmina Strand	+44 23 9281 7744
Tobias Renner	02392818855

5.2 — Strip everything but the digits

```
ALTER TABLE residents ADD COLUMN phone_digits TEXT
GENERATED ALWAYS AS (regexp_replace((contact).phone, '[^0-9]', '', 'g')) STORED;
```

```
SELECT full_name, (contact).phone, phone_digits FROM residents WHERE id BETWEEN 1 AND 5;
```

full_name	phone	phone_digits
Adrian Foscolo	023 9281 4477	02392814477
Marisol Quintero	023-9281-5522	02392815522
Bennett Okoye	(023) 9281 6633	02392816633
Wilhelmina Strand	+44 23 9281 7744	442392817744
Tobias Renner	02392818855	02392818855

`(contact).phone` — a field pulled out of Chapter 15’s composite type — works inside a generated expression exactly like any other column reference. Four of the five now agree, digit-for-digit, despite arriving in four visibly different formats.

5.3 — Search with a fifth, still-different format

```
SELECT full_name FROM residents
WHERE phone_digits = regexp_replace('023.9281.5522', '[^0-9]', '', 'g');
```

full_name

Marisol Quintero

A caller can type the number with dots, dashes, spaces, or nothing at all, run it through the identical normalization, and get a match — that’s the entire point of normalizing at write time instead of comparing raw, differently-formatted text.

5.4 — Where this normalization actually stops working

Look at Wilhelmina Strand’s row again: 442392817744, not 02392817744 like everyone else’s. +44 is the UK’s country code, and 023... is the same number in domestic format with a leading 0 — the same real phone number, and this normalization treats them as two different ones, because stripping non-digit characters has no idea that +44 and a leading 0 mean the same thing. Real phone number normalization needs an actual phone-numbering-plan library (Python’s `phonenumbers`, for instance) that understands country codes, domestic-format leading digits, and valid-length rules per country — this exercise’s `regexp_replace` is the honest, simple version of the idea, not the production-grade one, the same caution Chapter 15 gave the email-format domain.

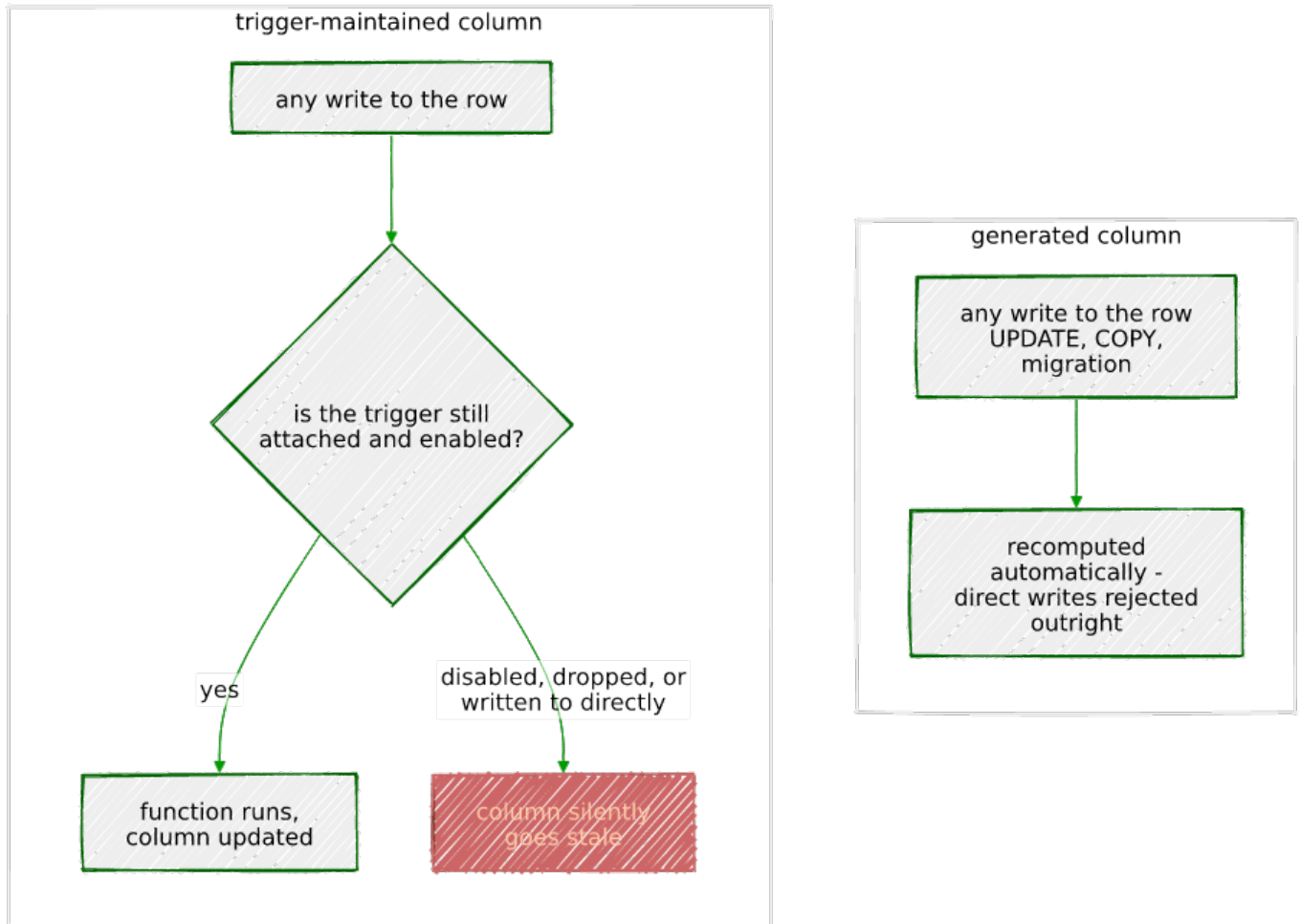
Exercise 6 — Generated Columns vs. Triggers

Exercise 2 tore out a trigger and replaced it with a generated column doing the identical job. The comparison worth internalizing, now that you’ve built both:

	Trigger	Generated column
Can reference other rows/ tables	Yes	No — same row only
Can have side effects (<code>NOTIFY</code> , logging)	Yes	No — pure expression only
Extra write path can bypass it	Yes — anything that skips the trigger	No — enforced at the column itself
Shows up in <code>\d</code> as part of the column	No — hidden in a separate function	Yes — visible in the column definition
Recomputed automatically on backfill for existing rows	Only if you remember to run an <code>UPDATE</code>	Yes — happens as part of <code>ALTER TABLE ADD COLUMN</code>

A generated column isn’t a strictly better trigger — it’s a *narrower* tool that happens to fully cover a specific, common case: “this column is a pure function of other columns in the same row, with no side effects.” Chapter 13’s `notify_job_status_change()` could never be a generated column — `pg_notify()` is a side effect, and the whole point of that trigger is to reach outside the

row it fired on. `search_vector` and `reading_date` never needed to reach outside their own row at all, which is exactly what made them convertible. The question worth asking before reaching for either one: does this derived value ever need to know about anything other than the row it lives on? If yes, it's a trigger. If no, it's a generated column, and a generated column is simpler in every way that matters once the answer really is no.



Summary — What You Should Now Know

Tool

`col type GENERATED ALWAYS
AS (expr) STORED`

Immutability requirement

`AT TIME ZONE 'UTC'` inside
the expression

What it does

Computed once at write time, physically stored — the only kind PostgreSQL supports

The expression can't depend on session state (timezone, locale) — only on the row's own columns

Fixes the classic `::date` immutability failure by pinning the zone into the expression itself

Backfills every existing row immediately, including every partition of a partitioned table

Tool	What it does
Adding a generated column to an existing table	
<code>DROP COLUMN</code> on a generated column	Also drops every index that depended on it — rebuild them after
Index on a generated column	Used like any other index, subject to the same planner cost decisions (including “seq scan is cheaper on a small table”)
Generated column vs. partition key	Indexing a derived column speeds up filtering on it; it does not enable partition pruning, which only ever looks at the actual partition key
Writing directly to a generated column	Always rejected — <code>INSERT</code> and <code>UPDATE</code> both
A domain/composite field inside a generated expression	Fully supported — <code>(contact).phone</code> works exactly like any column reference

The key design insight from this chapter is the same one Chapter 15 ended on, aimed at a different part of the schema: a generated column is a fact about what a value *is*, not a rule about what it’s allowed to become after the fact. Chapter 4’s trigger and this chapter’s `search_vector` compute the identical `tsvector` — but only one of them is structurally incapable of drifting out of sync, because only one of them removed every write path except the one that recomputes it correctly. That guarantee has a real, narrow boundary — no other rows, no side effects, nothing beyond the current row — and staying inside that boundary is the whole trade.

Going further: Chapter 17’s foreign data wrappers occasionally interact with generated columns in an interesting way — a generated column can’t be added to a foreign table the way it can to a local one, since PostgreSQL doesn’t control how the remote side actually stores data. Chapter 20’s `pg_stat_statements` work is a natural place to revisit Exercise 3’s planner decision — watching real, measured query cost is how you’d actually confirm “the planner correctly avoided the index” rather than taking `EXPLAIN`’s cost estimate on faith. And it’s worth remembering Exercise 6’s dividing line the next time a derived column is on the table at all: reach for a generated column first, purely for the guarantee it makes, and only fall back to a trigger once the derivation genuinely needs to see past the row it lives on.

Chapter 17 — Foreign Data Wrappers:

PostgreSQL as a Data Hub

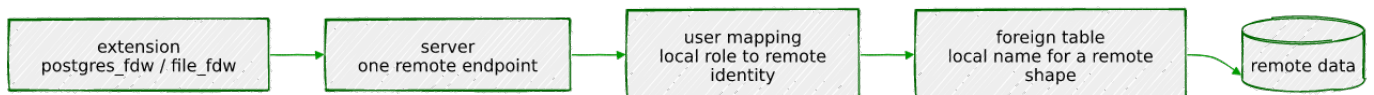
“A foreign table looks exactly like a table. Everything a DBA would ask before letting you near someone else’s data — do you have the extension, do you have a credential, can you actually reach the file — still has to be answered. It just gets asked in `CREATE SERVER` instead of a ticket queue.”

Background

Every table in this book so far has lived inside `portsmith`. A **foreign data wrapper (FDW)** lets a table live somewhere else entirely — another PostgreSQL database, a CSV file on disk, an S3 bucket — while still being queried with ordinary `SELECT`, `JOIN`, and `WHERE`, exactly like a local one. No ETL job copies the data in first; the query reaches out and reads it at query time, through whichever FDW you’ve told it to use.

The shape is always the same, four pieces:

- **The extension** (`postgres_fdw`, `file_fdw`, ...) — the driver that knows how to actually talk to one kind of remote source.
- **A server** — one specific remote endpoint: a hostname and database, or a directory of files, registered under a name.
- **A user mapping** — which local role authenticates as which remote identity, since “you’re logged into `portsmith`” says nothing about who you are anywhere else.
- **A foreign table** — the local name for a remote shape of data, declared once (by hand, or generated wholesale with `IMPORT FOREIGN SCHEMA` in Exercise 4).



What makes this chapter worth doing hands-on rather than reading about is that almost none of it works on the first try, and every wall you hit is a real, deliberate security boundary rather than a bug — extensions need a privileged role to install, foreign-data wrappers need explicit `USAGE` before anyone else can touch them, and a remote connection needs a real credential even when the same connection would succeed instantly without one. This chapter hits all three, in order, exactly where a first-time setup actually hits them.

The Scenario

Object	Lives in	Purpose
<code>businesses_archive</code>	<code>portsmith_legacy</code> (a second database)	Closed Portsmouth businesses, pre-migration records
<code>census_2020.csv</code>	Server filesystem	2020 population/household figures, one row per neighbourhood
<code>legacy_import.businesses_archive</code>	<i>(built here, via <code>IMPORT FOREIGN SCHEMA</code>)</i>	The same archive, imported wholesale instead of hand-declared
<code>sensor_readings/*.parquet</code>	MinIO bucket <code>portsmith-bucket</code>	Chapter 8's full 9.6M-row table, exported one file per month

`portsmith_legacy` is a genuinely separate PostgreSQL database — not a schema inside `portsmith` — created specifically to give `postgres_fdw` something real to reach across to. Everything in this chapter is a real network round-trip (over `localhost`) or a real file read, not a simulation.

Exercise Goals

By the end of this chapter you will be able to:

- Install `postgres_fdw`, register a remote PostgreSQL database as a server, and query one of its tables as if it were local — and know every privilege that has to be granted along the way, and why.
 - Confirm, with `EXPLAIN`, that a `WHERE` clause against a foreign table actually executes on the remote server instead of being pulled over the wire in full.
 - Query a CSV file as a table with `file_fdw`, join it against local data, and know exactly whose filesystem permissions actually govern whether that works.
 - Import an entire remote schema in one statement instead of hand-declaring every foreign table.
 - Write through a foreign table with a plain `INSERT`, and verify the row actually landed on the remote server.
 - Export real PostgreSQL data to Parquet and upload it to S3-compatible storage, and describe the `parquet_s3_fdw` architecture for querying it straight back — what problem it solves, and what the one genuinely hard-to-install piece still involves.
-

Installation

1 — A second database

```
createdb portsmith_legacy
```

If this fails with permission denied to create database, your role needs CREATEDB:

```
-- as postgres
ALTER ROLE your_role CREATEDB;
```

2 — The extensions

```
-- as postgres - CREATE EXTENSION requires a privileged role
CREATE EXTENSION postgres_fdw;
CREATE EXTENSION file_fdw;
```

Both ship with core PostgreSQL — no separate package to install, but creating them requires a role with sufficient privilege, almost always postgres itself in a development setup like this one.

3 — Exercise 6's Python packages (optional)

Only needed if you're doing the hands-on half of Exercise 6:

```
source .venv/bin/activate
pip install pyarrow boto3 duckdb
```

Loading the Data

In `portsmith_legacy` — the “legacy” archive:

```
CREATE TABLE businesses_archive (
  biz_id      INTEGER PRIMARY KEY,
  biz_name    TEXT NOT NULL,
  addr        TEXT NOT NULL,
  neighbourhood TEXT NOT NULL,
  closed_date DATE NOT NULL,
  reason      TEXT
);

INSERT INTO businesses_archive (biz_id, biz_name, addr, neighbourhood, closed_date, reason)
VALUES
  (1001, 'Portsmouth Cannery Co.', '2 Dock Road', 'Industrial Port',
   '2011-03-15', 'relocated out of city'),
  (1002, 'The Anchor & Rope', '18 Wharf Street', 'Harbour District',
```



```
'2014-07-01', 'owner retired'),
    (1003, 'Old Town Print Shop',          '9 Market Street',      'Old Town',
'2016-11-30', 'business closed'),
    (1004, 'Northgate Ironworks',         '44 Bay Street',        'Northgate',
'2009-05-20', 'relocated out of city'),
    (1005, 'Riverside Boat Repair',        '3 Quay Street',        'Riverside',
'2018-02-14', 'owner retired'),
    (1006, 'University Quarter Bindery',    '61 Lighthouse Avenue', 'University Quarter',
'2013-09-01', 'business closed'),
    (1007, 'Portsmouth Rope & Sail',        '7 Anchor Lane',        'Harbour District',
'2007-01-10', 'merged with another business'),
    (1008, 'Dockside Chandlery',           '15 Fisherman's Row',   'Industrial Port',
'2019-06-25', 'business closed');
```

On the server filesystem — data/ch17_census.csv:

```
neighbourhood,population_2020,households_2020,median_age
Harbour District,8420,3610,41.2
Old Town,6150,2890,44.7
Northgate,11730,4920,36.5
Riverside,9280,3945,39.8
University Quarter,7460,2210,24.3
Industrial Port,5340,2280,45.1
```

Verify

```
-- in portsmith_legacy
SELECT COUNT(*) FROM businesses_archive;

count
-----
      8
```

Exercises

Exercise 1 — postgres_fdw: Server, User Mapping, Foreign Table

1.1 — Register the remote server

Back in portsmith:

```
CREATE SERVER portsmith_legacy_srv
    FOREIGN DATA WRAPPER postgres_fdw
    OPTIONS (host 'localhost', port '5432', dbname 'portsmith_legacy');
```

If this fails with permission denied for foreign-data wrapper `postgres_fdw`, the role creating the server needs `USAGE` on the FDW itself — creating the *extension* doesn't automatically hand every other role the right to use it:

```
-- as postgres
GRANT USAGE ON FOREIGN DATA WRAPPER postgres_fdw TO your_role;
```

1.2 — Map your local role to a remote identity

```
CREATE USER MAPPING FOR CURRENT_USER
  SERVER portsmith_legacy_srv
  OPTIONS (user 'chris');
```

1.3 — Declare the foreign table

```
CREATE FOREIGN TABLE businesses_archive (
  biz_id      INTEGER,
  biz_name    TEXT,
  addr        TEXT,
  neighbourhood TEXT,
  closed_date DATE,
  reason      TEXT
) SERVER portsmith_legacy_srv OPTIONS (schema_name 'public', table_name 'businesses_archive');
```

1.4 — Query it, and hit the gotcha that catches almost everyone

```
SELECT * FROM businesses_archive ORDER BY biz_id;
```

ERROR: password or GSSAPI delegated credentials required
DETAIL: Non-superusers must delegate GSSAPI credentials or provide a password in the user mapping.

This is surprising the first time: the exact same role, over the exact same `localhost` connection, can already reach `portsmith_legacy` directly with no password prompt at all — but `postgres_fdw` refuses to let a *non-superuser's* user mapping ride on that trust. The reasoning is a real security concern, not bureaucracy: without this check, any role could create a user mapping claiming to be a powerful remote user and inherit that user's privileges on the remote side with nothing to prove it. The fix is to give the mapping an actual credential:

```
ALTER ROLE chris PASSWORD 'fdw-demo-password';
```

```
ALTER USER MAPPING FOR CURRENT_USER
  SERVER portsmith_legacy_srv
  OPTIONS (ADD password 'fdw-demo-password');
```

```
SELECT * FROM businesses_archive ORDER BY biz_id;
```

biz_id	biz_name	addr	neighbourhood	closed_date
	reason			

1001	Portsmouth Cannery Co.	2 Dock Road	Industrial Port
2011-03-15	relocated out of city		
1002	The Anchor & Rope	18 Wharf Street	Harbour District
2014-07-01	owner retired		
1003	Old Town Print Shop	9 Market Street	Old Town
2016-11-30	business closed		
1004	Northgate Ironworks	44 Bay Street	Northgate
2009-05-20	relocated out of city		
1005	Riverside Boat Repair	3 Quay Street	Riverside
2018-02-14	owner retired		
1006	University Quarter Bindery	61 Lighthouse Avenue	University Quarter
2013-09-01	business closed		
1007	Portsmouth Rope & Sail	7 Anchor Lane	Harbour District
2007-01-10	merged with another business		
1008	Dockside Chandlery	15 Fisherman's Row	Industrial Port
2019-06-25	business closed		

(8 rows)

Eight rows, physically stored in a different database, returned by a plain `SELECT` with no hint anywhere in the syntax that they came from anywhere else.

Exercise 2 — Confirming Filters Push Down

EXPLAIN (VERBOSE)

SELECT biz_name, closed_date **FROM** businesses_archive **WHERE** neighbourhood = 'Harbour District';

Foreign Scan on public.businesses_archive (cost=100.00..127.20 rows=7 width=36)

Output: biz_name, closed_date

Remote SQL: `SELECT biz_name, closed_date FROM public.businesses_archive WHERE ((neighbourhood = 'Harbour District'))`

Remote SQL is the whole point: `postgres_fdw` didn't pull all eight rows to `portsmith` and filter them here. It translated the query — `WHERE`, and only the two selected columns — into real SQL, sent it to `portsmith_legacy`, and let that server do the filtering with its own planner and its own (in a real deployment) indexes. Every row that doesn't match Harbour District never crosses the network at all.

Exercise 3 — `file_fdw`: a CSV as a Table

3.1 — Server and foreign table

CREATE SERVER census_files **FOREIGN DATA** WRAPPER file_fdw;

CREATE FOREIGN TABLE census_2020 (
 neighbourhood TEXT,
 population_2020 INTEGER,
 households_2020 INTEGER,

```

    median_age          NUMERIC
) SERVER census_files OPTIONS (
    filename '/home/you/book/data/ch17_census.csv',    -- your actual checkout path
    format 'csv',
    header 'true'
);

```

If this fails with permission denied to set the "filename" option, `file_fdw` restricts which roles can point a foreign table at an arbitrary local file — for good reason, since it would otherwise let any role read anything the PostgreSQL server process itself can read:

```

-- as postgres
GRANT pg_read_server_files TO your_role;

```

3.2 — The gotcha that actually matters: whose filesystem is this?

```

SELECT * FROM census_2020;

```

ERROR: could not open file "/home/you/book/data/ch17_census.csv" for reading: Permission denied

The client running `psql` can read this file fine — but `file_fdw` doesn't read the file from the client. It reads it from the *server process*, running as its own OS user (`postgres`, typically), and that user has to have its own filesystem path to the file, readable by *it*, independent of who's connected. A file sitting inside a user's home directory — mode `700` by default on most systems — is invisible to every other OS user no matter what the file's own permissions say, because the *directory* blocks entry before the file's permissions ever get checked. The fix isn't to loosen a home directory's permissions; it's to put the file somewhere server-readable in the first place — `/tmp` for a throwaway demo like this one, a dedicated data directory with the right ownership for anything real:

```

cp data/ch17_census.csv /tmp/ch17_census.csv
chmod 644 /tmp/ch17_census.csv

ALTER FOREIGN TABLE census_2020 OPTIONS (SET filename '/tmp/ch17_census.csv');

SELECT * FROM census_2020;

```

neighbourhood	population_2020	households_2020	median_age
Harbour District	8420	3610	41.2
Old Town	6150	2890	44.7
Northgate	11730	4920	36.5
Riverside	9280	3945	39.8
University Quarter	7460	2210	24.3
Industrial Port	5340	2280	45.1

(6 rows)

3.3 — Join it against local data

```

SELECT c.neighbourhood, c.population_2020, COUNT(b.id) AS business_count
FROM census_2020 c
JOIN businesses b ON b.neighbourhood = c.neighbourhood
GROUP BY c.neighbourhood, c.population_2020
ORDER BY c.population_2020 DESC;

```

neighbourhood	population_2020	business_count
Northgate	11730	9
Riverside	9280	9
Harbour District	8420	9
University Quarter	7460	5
Old Town	6150	9
Industrial Port	5340	7

(6 rows)

A flat file and a real table, joined with ordinary SQL — nothing about the query syntax distinguishes `census_2020` (a CSV) from `businesses` (an actual table).

3.4 — The contrast with Exercise 2: no pushdown here

```

EXPLAIN SELECT * FROM census_2020 WHERE neighbourhood = 'Northgate';

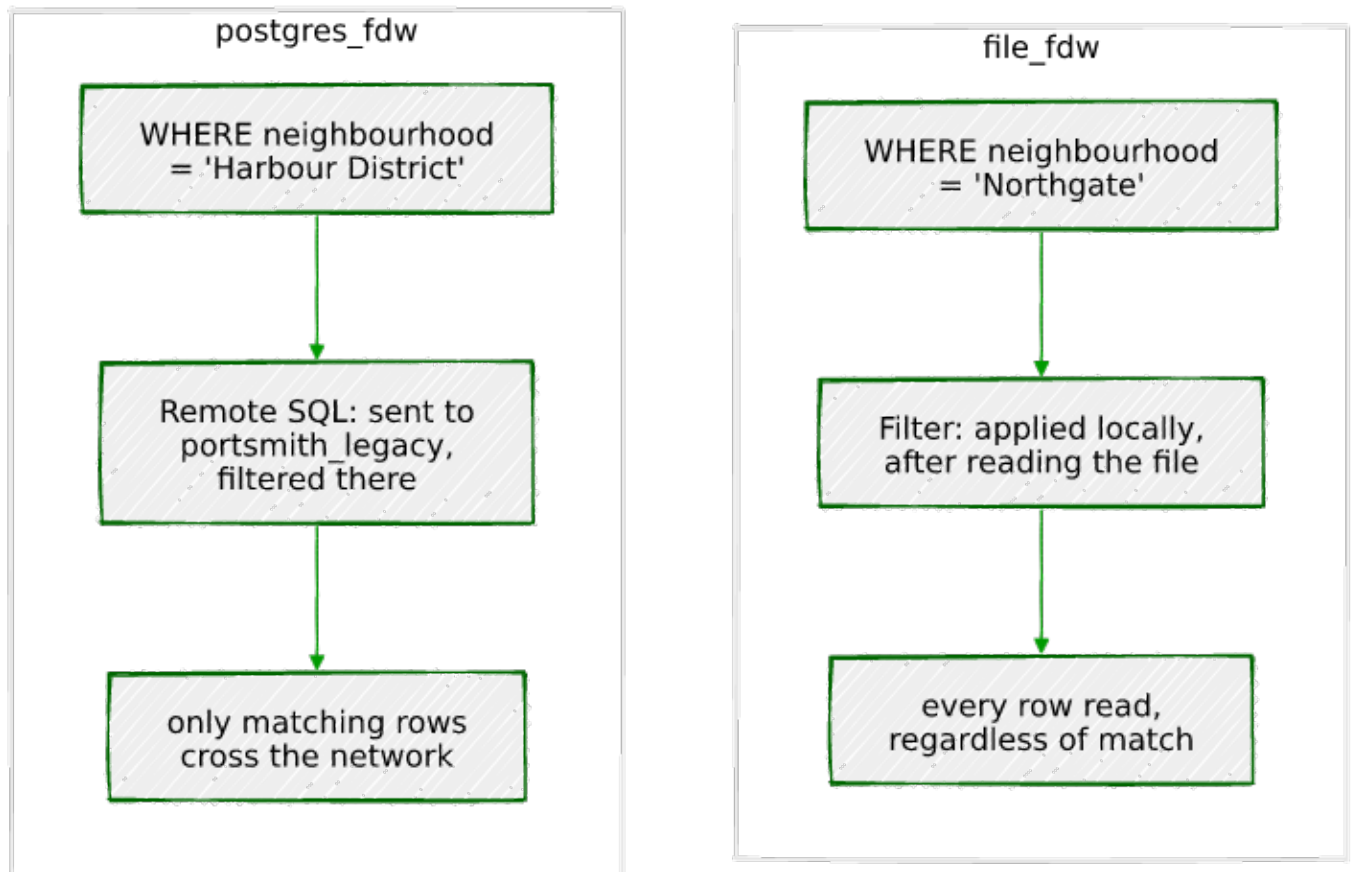
```

```

Foreign Scan on census_2020 (cost=0.00..1.21 rows=1 width=72)
  Filter: (neighbourhood = 'Northgate'::text)
  Foreign File: /tmp/ch17_census.csv

```

Filter:, not Remote SQL:. A flat file has no query engine of its own to push anything down *to* — `file_fdw` has no choice but to read every row of the file into PostgreSQL and filter locally, every time. For a six-row census file that’s irrelevant; for a multi-gigabyte CSV it’s the entire performance story, and it’s the exact gap Exercise 6’s `parquet_s3_fdw` pattern exists to close.



Exercise 4 — Importing a Whole Schema at Once

Exercise 1 declared `businesses_archive` by hand, column by column — fine for one table, tedious for a real legacy database with dozens. `IMPORT FOREIGN SCHEMA` asks the remote server for its own schema and generates matching foreign tables automatically:

```
CREATE SCHEMA legacy_import;
```

```
IMPORT FOREIGN SCHEMA public
  FROM SERVER portsmith_legacy_srv
  INTO legacy_import;
```

```
\d legacy_import.businesses_archive
```

Foreign table "legacy_import.businesses_archive"						
Column	Type	Collation	Nullable	Default	FDW options	
biz_id	integer		not null		(column_name 'biz_id')	
biz_name	text		not null		(column_name 'biz_name')	
addr	text		not null		(column_name 'addr')	
neighbourhood	text		not null		(column_name 'neighbourhood')	
closed_date	date		not null		(column_name 'closed_date')	
reason	text				(column_name 'reason')	

Server: portsmith_legacy_srv

Every column, every type, matched exactly — not retyped by hand, read directly off the remote catalog. `IMPORT FOREIGN SCHEMA` also accepts `LIMIT TO (...)` or `EXCEPT (...)` clauses to import only part of a schema, worth knowing the moment a real legacy database has hundreds of tables and you need eight of them.

Exercise 5 — Writing Through a Foreign Table

```
INSERT INTO businesses_archive (biz_id, biz_name, addr, neighbourhood, closed_date, reason)
VALUES (1009, 'Old Brewery Annex', '12 Ring Road', 'Industrial Port', '2021-08-30',
'demolished for redevelopment');
```

```
INSERT 0 1
```

Verify it landed on the actual remote database, not just in a local cache:

```
-- connect directly to portsmith_legacy
SELECT biz_id, biz_name FROM businesses_archive WHERE biz_id = 1009;
```

biz_id	biz_name
1009	Old Brewery Annex

A plain `INSERT` against `portsmith`, and the row is sitting in `portsmith_legacy` — `postgres_fdw` supports writes as well as reads, translated into a real `INSERT` on the remote side, subject to every constraint that database enforces on its own table (a duplicate `biz_id` would fail here exactly as it would connecting directly). `file_fdw` cannot do this — a CSV file has no transactional write protocol to translate an `INSERT` into, and Chapter 17’s file-based foreign tables are read-only for exactly that reason.

Exercise 6 — The `parquet_s3_fdw` Pattern

6.1 — The problem this solves

`sensor_readings` (Chapter 8) is 9.6 million rows and only grows. Most of it, most of the time, is cold — nobody’s actively querying January’s readings in November. Keeping years of it in PostgreSQL forever is possible but not free: it’s backed up on every backup, it’s vacuumed on every autovacuum pass, it occupies disk that costs money whether or not anyone reads it. A common real-world answer is to export aging partitions out of PostgreSQL entirely into **Parquet** — a columnar file format built for exactly this: cheap object storage (S3 or, for local development, an S3-compatible server like **MinIO**), with per-column compression and the ability to skip whole chunks of a file that can’t possibly match a filter, without reading them.

The catch that would normally follow: exporting the data means it's no longer queryable with plain SQL, from the same connection, joined against whatever's still live in PostgreSQL. `parquet_s3_fdw` closes that gap — a foreign data wrapper that reads Parquet files sitting in S3-compatible storage as ordinary foreign tables, pushing down both column selection and predicate filtering into the Parquet reader itself (Parquet's own format stores per-column statistics, min/max values per row group, that let a reader skip entire chunks without decompressing them — a real analog to `postgres_fdw`'s Remote SQL pushdown from Exercise 2, just implemented against a file format instead of a second database).

6.2 — The half of this that's completely real: export and upload

Getting data *into* Parquet, in S3-compatible storage, needs nothing exotic — `pyarrow` to write the files, `boto3` (or any S3-compatible client) to upload them, and MinIO running locally in Docker to receive them:

```
docker run -p 9000:9000 -p 9001:9001 minio/minio server /data --console-address ":9001"

#!/usr/bin/env python3.12
# ch17_export_to_parquet.py - one Parquet file per monthly partition
import io
from datetime import date, timedelta

import boto3
import psycpg
import pyarrow as pa
import pyarrow.parquet as pq

MINIO_ENDPOINT, MINIO_ACCESS_KEY, MINIO_SECRET_KEY = "http://localhost:9000", "minioadmin",
"minioadmin"
BUCKET = "portsmith-bucket"

QUERY = """
SELECT sensor_id, sensor_type, reading_value, recorded_at, reading_date
FROM   sensor_readings
WHERE  reading_date >= %(start)s AND reading_date < %(end)s
ORDER BY recorded_at
"""

def export_month(conn, year_month: str) -> pa.Table:
    year, month = (int(p) for p in year_month.split("-"))
    start = date(year, month, 1)
    end = (date(year, month + 1, 1) if month < 12 else date(year + 1, 1, 1))
    with conn.cursor() as cur:
        cur.execute(QUERY, {"start": start, "end": end})
        rows = cur.fetchall()
        cols = ["sensor_id", "sensor_type", "reading_value", "recorded_at", "reading_date"]
        return pa.table({c: [r[i] for r in rows] for i, c in enumerate(cols)})

# ... build an S3 client pointed at MINIO_ENDPOINT, create the bucket if
```



```
# needed, then for each month: export_month(), pq.write_table() into an
# in-memory buffer, s3.upload_fileobj() to sensor_readings/{month}.parquet
```

Run against all of `sensor_readings`' eleven populated months:

```
2024-02: 835,200 rows -> s3://portsmith-bucket/sensor_readings/2024-02.parquet (1,420,131
bytes)
2024-03: 892,800 rows -> s3://portsmith-bucket/sensor_readings/2024-03.parquet (1,540,196
bytes)
...
2024-12: 891,648 rows -> s3://portsmith-bucket/sensor_readings/2024-12.parquet (1,527,345
bytes)
done - 9,646,849 rows, 16,673,067 bytes across 11 files
```

9.6 million real rows, genuinely uploaded to genuinely running S3-compatible storage, in under three minutes. And the number worth sitting with: `sensor_readings` occupies **772 MB** in PostgreSQL (table plus every index, summed across all its partitions); the same data, as Parquet with Snappy compression, is **16.7 MB** — about **46 times smaller**. That gap is column-oriented storage and per-column compression doing exactly what they're for: `sensor_readings` has five columns, several of them low-cardinality (`sensor_type` is one of three values, repeated 9.6 million times) or smoothly-changing (`recorded_at`, `reading_date`), and a columnar format compresses runs of similar values far more efficiently than a row-oriented table ever will.

6.3 — The half of this chapter that stays a sketch: the FDW itself

Querying these files back from PostgreSQL — the actual `parquet_s3_fdw` part — is where this exercise stops being fully hands-on. It isn't a core PostgreSQL extension the way `postgres_fdw` and `file_fdw` are; it's a third-party project built against Apache Arrow's C++ libraries, compiled from source in most environments rather than installed with `apt`. Getting a working build means matching Arrow/Parquet library versions to your exact PostgreSQL version, a real, multi-step undertaking well outside what a single exercise can responsibly walk through. What follows is the setup this chapter would ask you to do if it did — worth understanding piece by piece, and worth treating as a genuine follow-on project now that the data is actually sitting in MinIO waiting for it:

1. **Build and install** `parquet_s3_fdw` against your PostgreSQL version's server headers.
2. **CREATE SERVER**, pointing at the MinIO endpoint instead of a PostgreSQL host — an access key and secret in place of a username and password, the S3 analog of Exercise 1's user mapping.
3. **CREATE FOREIGN TABLE** (or **IMPORT FOREIGN SCHEMA**, if the extension supports inferring the Parquet schema — implementations vary), mapping Parquet columns the same way Exercise 4 mapped a remote PostgreSQL table's columns.
4. **Query it** — a `WHERE reading_date = '2024-06-15'` against all eleven exported files should skip most of their row groups entirely, the same shape of win Exercise 2's `Remote SQL` pushdown

demonstrated, just decided by Parquet's own per-column statistics instead of a remote query planner.

The architecture, end to end: PostgreSQL stays the single query interface for both hot data (still in `sensor_readings`) and cold data (exported to Parquet in MinIO/S3, exactly as Exercise 6.2 just did for real), joinable in the same query, without standing up a separate query engine like Trino or Presto just to read files a data lake already has sitting in object storage.

6.4 — Verifying the pruning story independently, with DuckDB

`parquet_s3_fdw` isn't the only thing that can read Parquet off S3-compatible storage — **DuckDB** can too, natively, and installing it is one `pip install duckdb` rather than a source build. It's a useful second opinion here: query the exported files directly, with no PostgreSQL involved at all, and confirm the row-group pruning story Exercise 6.1 promised is actually true.

```
#!/usr/bin/env python3.12
# ch17_query_parquet.py
import duckdb

con = duckdb.connect()
con.execute("INSTALL httpfs; LOAD httpfs;")
con.execute("""
    SET s3_endpoint = 'localhost:9000';
    SET s3_access_key_id = 'minioadmin';
    SET s3_secret_access_key = 'minioadmin';
    SET s3_use_ssl = false;
    SET s3_url_style = 'path';
""")

GLOB = "s3://portsmith-bucket/sensor_readings/*.parquet"

print(con.execute(f"SELECT COUNT(*) FROM read_parquet('{GLOB}')").fetchone())

con.execute(f"""
    EXPLAIN ANALYZE SELECT COUNT(*) FROM read_parquet('{GLOB}')
    WHERE reading_date = DATE '2024-06-15'
""")
for row in con.fetchall():
    print(row[1])

(9646849,)
```

Nine million, six hundred forty-six thousand, eight hundred forty-nine — matching Exercise 6.2's own upload total exactly, confirming nothing was lost or duplicated across eleven separate uploads. Then the pruning check:

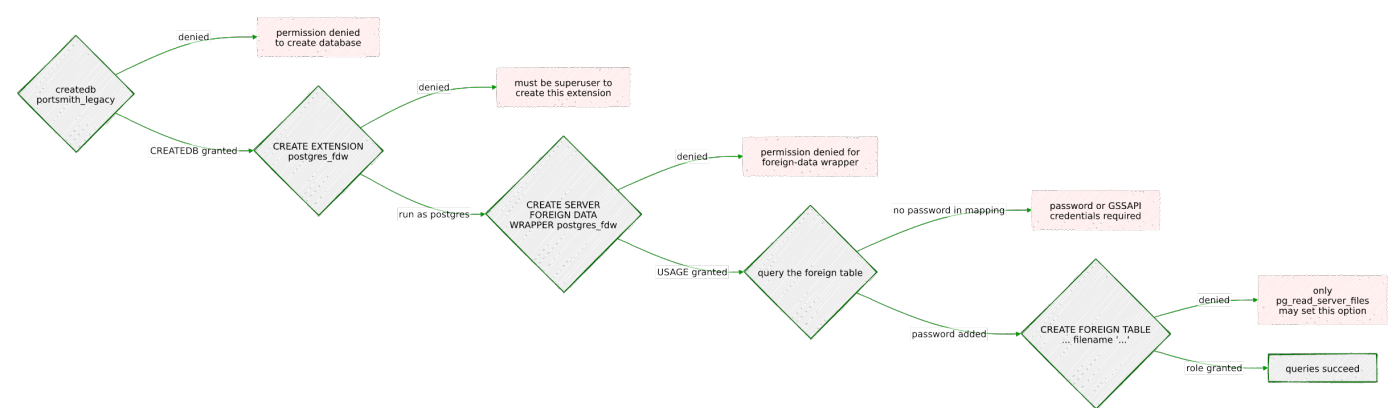
```
HTTPFS HTTP Stats
in: 2.7 KiB
out: 0 bytes
```

#GET: 1
Total Time: 0.0048s

TABLE_SCAN (READ_PARQUET)
Filters: reading_date='2024-06-15':DATE
Total Files Read: 11
28,801 rows

2.7 KiB of actual data transferred, one HTTP GET, to answer a query that matched 28,801 rows — out of 16.7 MB and 9.6 million rows total. Total Files Read: 11 looks like it contradicts that at first — DuckDB *did* open every file — but opening a Parquet file only costs reading its footer, a small block of per-row-group statistics; the actual column data for row groups that can't contain 2024-06-15 never gets requested at all. (This number reflects the two queries that ran before it in the same session already having warmed DuckDB's metadata cache for these files — a cold connection running only the EXPLAIN ANALYZE query would transfer somewhat more, closer to 180 KiB, still a small fraction of the total.) Either way, this is Exercise 2's Remote SQL pushdown story again, a third time in one chapter: a system that understands the shape of its own storage well enough to answer “which parts of this can I skip” before reading them.

Every wall this chapter actually hit, in the order it hit them:



Summary — What You Should Now Know

Tool	What it does
CREATE EXTENSION postgres_fdw / file_fdw	Installs the driver — requires a privileged role
GRANT USAGE ON FOREIGN DATA WRAPPER ... TO role	Required before a non-privileged role can create a server with that FDW
CREATE SERVER	Registers one remote endpoint under a name
CREATE USER MAPPING	Maps a local role to a remote identity — non-superusers must supply a real password

Tool	What it does
<code>CREATE FOREIGN TABLE</code>	Declares a local name for a remote shape of data, column by column
<code>IMPORT FOREIGN SCHEMA ... INTO schema</code>	Generates foreign tables for an entire remote schema automatically
<code>EXPLAIN ON a foreign table (postgres_fdw)</code>	<code>Remote SQL</code> : — confirms filtering happened on the remote server, not locally
<code>EXPLAIN ON a foreign table (file_fdw)</code>	<code>Filter</code> : — confirms the whole file was read and filtered locally instead
<code>pg_read_server_files</code>	The role membership <code>file_fdw</code> requires before pointing at an arbitrary local file
File permissions for <code>file_fdw</code>	Governed by the PostgreSQL <i>server process</i> 's OS user, not the connecting client
<code>INSERT</code> through a <code>postgres_fdw</code> foreign table	A real write, translated and executed on the remote server
<code>parquet_s3_fdw</code>	Same pattern as <code>postgres_fdw</code> , aimed at columnar files in S3-compatible storage instead of another database

The key design insight from this chapter is that a foreign data wrapper's job is to make remote data *look* exactly like local data — and it succeeds completely at that, right up until you touch something that was never really local to begin with: a credential, a filesystem permission, a network round-trip. Every gotcha this chapter walked through was one of those seams showing through the illusion on purpose, not by accident — PostgreSQL enforcing, at each layer, that querying someone else's data still has to answer the same questions it always would have, just inside `CREATE SERVER` and `CREATE USER MAPPING` instead of a separate integration layer you'd otherwise have to build and maintain by hand.

Going further: Chapter 18's logical replication solves a related but genuinely different problem — where this chapter queries remote data live, at read time, replication copies it, continuously, so a second database has its own independent, current copy to query locally. Reach for an FDW when the data should stay in one place and be reached across; reach for replication when a second copy, kept in sync, is what you actually need. And Exercise 6's `parquet_s3_fdw` sketch is worth revisiting once Chapter 19's `pg_cron` is in hand — a scheduled job that exports aging `sensor_readings` partitions to Parquet and drops them locally, the same way Chapter 8's own "going further" note imagined `pg_partman` automating partition lifecycle management, is exactly the kind of recurring maintenance `pg_cron` is suited to run unattended.**

Chapter 18 — Logical Replication and Change Data Capture

A foreign data wrapper asks a question and waits for an answer, every time. A subscription asks once, then never stops listening.

Background

Every business and permit in Portsmouth lives in one place: `portsmith`, the operational database every clerk's terminal and every intake job reads and writes directly. That's exactly where it should live — until the city decides it also wants a public-facing **Business Finder** portal, so a resident can search “what's open in Old Town” or check whether their neighbor's demolition permit has cleared, without calling City Hall.

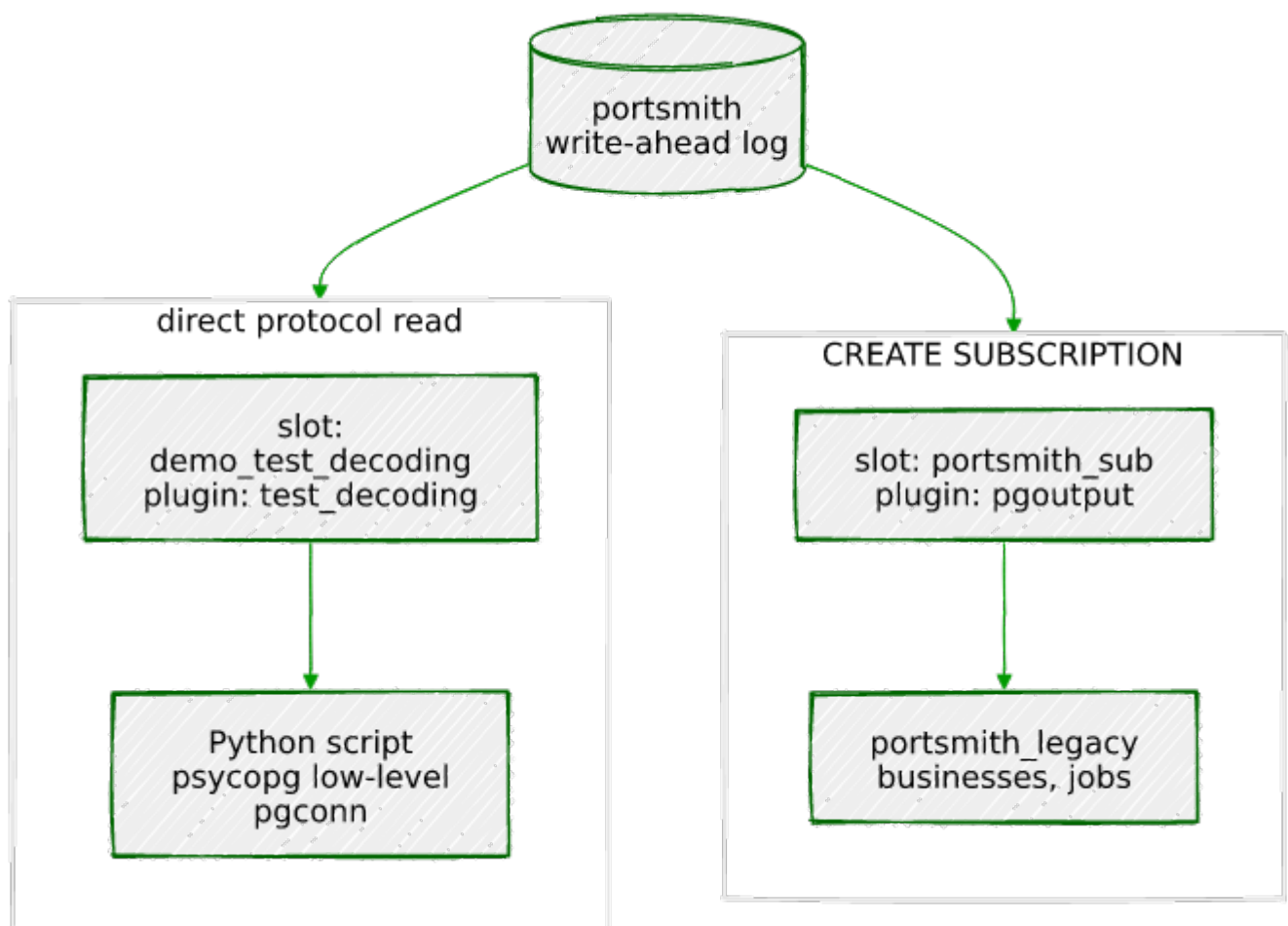
Pointing the portal straight at `portsmith` — a live connection, or Chapter 17's `postgres_fdw` reaching across at query time — is the obvious first idea, and IT rejects it fast. A Saturday-morning traffic spike on the public site has no business competing for locks and connections with a clerk mid-edit on a permit. The portal also has no business seeing everything `portsmith` has: no PostGIS geometry, no internal search-ranking internals, and definitely nothing from businesses that have since closed. And a nightly export job — the other obvious idea — means a business that reopens Tuesday morning doesn't show as open on the portal until Wednesday's batch run, which isn't good enough for something citizens are expected to trust.

What the city actually wants is a second, independent copy of just the public-facing slice of `businesses` and `jobs` — one the portal can hit as hard as it likes without touching the real system, that only ever shows currently-active businesses, and that updates itself the moment something changes upstream, with nobody babysitting a cron job. **Logical replication** is exactly that: it streams every row-level change out of PostgreSQL, continuously, so a second database ends up with its own current copy — queryable locally, with its own indexes, filtered down to only the columns and rows that copy is allowed to see. Experientially, once it's running, it disappears: a clerk approves a new registration in `portsmith`, and a moment later — no refresh button, no scheduled job, nobody polling anything — that business is searchable on the portal. This chapter builds the portal's copy as `portsmith_legacy`, the same second database Chapter 17 introduced, now playing a different role: instead of something `portsmith` reaches into, it's something `portsmith` continuously pushes into.

The mechanism underneath is the same **write-ahead log (WAL)** every PostgreSQL install already produces for crash recovery, decoded into a stream of logical changes (INSERT, UPDATE, DELETE) instead of the physical byte-level records **physical replication** (streaming standbys) sends. Two pieces make it work:

- **A publication** — created on the source (“publisher”), naming which tables (and optionally which columns, and which rows) should be streamed out.
- **A subscription** — created on the destination (“subscriber”), pointing at a publisher and a publication, pulling changes in and applying them.

Underneath every subscription is a **replication slot** — a durable bookmark on the publisher that says “don’t let WAL older than this be recycled, a consumer still needs it.” That’s the whole safety contract: as long as the slot exists, the publisher retains whatever WAL the subscriber hasn’t confirmed yet, even across a subscriber outage. It’s also the whole *risk*: a slot nobody’s draining anymore retains WAL forever, silently filling disk.



This chapter builds both paths off the same publisher: the standard `CREATE SUBSCRIPTION` route or the second row shows the raw protocol, read directly from Python.

The Scenario

Object	Lives in	Purpose
portsmith_pub	portsmith (publisher)	Publication covering businesses (partial columns, row-filtered) and jobs (all columns)
portsmith_sub	portsmith_legacy (subscriber)	Subscription consuming portsmith_pub, backed by a slot of the same name on the publisher
businesses / jobs	Both databases	Chapter 1 / Chapter 3's tables — subscriber's copies are narrower, no PostGIS geometry, no generated columns
demo_test_decoding	portsmith (publisher)	A second, independent slot, read directly by Python instead of by a subscription
data/ ch18_replication_stream.py	(new)	Consumes demo_test_decoding at the wire protocol level

portsmith_legacy is the same second database Chapter 17 created — now standing in for the Business Finder portal's own database, on the same PostgreSQL instance as the publisher. That's convenient for a lab environment and, as Exercise 2 shows, not free: it creates a real deadlock a genuinely separate instance (which is what a real portal database would be) wouldn't.

Exercise Goals

By the end of this chapter you will be able to:

- Turn on logical replication, create a publication scoped to specific tables and columns, and understand what a replication slot actually guarantees.
- Stand up a subscription, understand exactly why doing so on the same instance as the publisher can deadlock, and know the real fix.
- Read `pg_replication_slots` and `pg_stat_replication` to check lag and confirm a subscriber is actually keeping up.
- Filter a publication by row, and navigate the replica identity requirements that come with filtering on a non-key column.
- Consume the logical replication protocol directly from Python, without `CREATE SUBSCRIPTION` at all.
- Describe how Debezium turns this same protocol into a Kafka event stream — the production-grade version of Exercise 5's hand-rolled script.

Installation

```
wal_level = logical
```

Logical decoding needs more information in the WAL than PostgreSQL writes by default:

```
-- as postgres, in postgresql.conf, or:
ALTER SYSTEM SET wal_level = 'logical';

# postgresql.conf
wal_level = logical    # (change requires restart)
```

Like `shared_preload_libraries` in the previous two chapters' setup, this needs a full restart, not just a config reload:

```
sudo systemctl restart postgresql

SHOW wal_level;

wal_level
-----
logical
```

No other extension is needed — logical replication is built into core PostgreSQL.

Loading the Data

`portsmith_legacy` needs matching structure for whatever gets published, but *matching* doesn't mean *identical*. `businesses` in `portsmith` carries a PostGIS geometry column and a generated `tsvector` — neither makes sense to replicate here: `portsmith_legacy` has no PostGIS extension installed, and generated columns aren't sent over logical replication anyway (PostgreSQL 16 excludes them from the wire format by default; a subscriber with the same generated column would just compute its own value locally, and one without it simply doesn't have it). So the subscriber's `businesses` is deliberately narrower — the domain and enum types still have to match, since those govern the columns that *are* published:

```
-- in portsmith_legacy
CREATE DOMAIN positive_integer AS integer CHECK (VALUE > 0);
CREATE TYPE job_status AS ENUM
('queued', 'on_hold', 'in_progress', 'completed', 'failed', 'cancelled');

CREATE TABLE businesses (
  id          integer PRIMARY KEY,
  name       text NOT NULL,
```



```

address      text NOT NULL,
neighbourhood text NOT NULL,
details      jsonb NOT NULL,
employee_count positive_integer
);

CREATE TABLE jobs (
  id          bigint PRIMARY KEY,
  job_type    text NOT NULL,
  payload     jsonb NOT NULL,
  status      job_status NOT NULL DEFAULT 'queued',
  priority    smallint NOT NULL DEFAULT 5,
  attempts    integer NOT NULL DEFAULT 0,
  max_attempts integer NOT NULL DEFAULT 3,
  created_at  timestampz NOT NULL DEFAULT clock_timestamp(),
  claimed_at  timestampz,
  claimed_by  text,
  heartbeat_at timestampz,
  completed_at timestampz,
  last_error  text
);

```

jobs is published in full — no PostGIS-shaped complications there, so its subscriber copy is a plain structural match.

Exercises

Exercise 1 — wal_level and a Publication

With wal_level already set to logical, creating a publication is the easy part:

```

-- in portsmith
CREATE PUBLICATION portsmith_pub
  FOR TABLE businesses (id, name, address, neighbourhood, details, employee_count), jobs;

CREATE PUBLICATION

```

businesses uses a **column list** — PostgreSQL 15+ lets a publication name a subset of a table's columns, not just a subset of its rows. Here that's what makes the narrower subscriber schema above valid at all: geom and search_vector are simply never offered, so portsmith_legacy never needs to know they exist. Practically, this is the city deciding what the portal is *allowed* to see: internal geometry and search-ranking internals stay behind in portsmith, never even leaving the publisher's WAL decode step for this publication.

```
SELECT pubname, puballtables FROM pg_publication;
SELECT schemaname, tablename, attnames FROM pg_publication_tables WHERE pubname =
'portsmith_pub';
```

pubname	puballtables
portsmith_pub	f

schemaname	tablename	attnames
public	businesses	{id,name,address,neighbourhood,details,employee_count}
public	jobs	{id,job_type,payload,status,priority,attempts,max_attempts,created_at,claimed_at,claimed_by,heartbeat_a

Exercise 2 — A Subscription, and a Deadlock Worth Understanding

2.1 — Two permission gates, back to back

```
-- in portsmith_legacy
CREATE SUBSCRIPTION portsmith_sub
    CONNECTION 'host=localhost dbname=portsmith user=chris password=fdw-demo-password'
    PUBLICATION portsmith_pub;
```

```
ERROR: permission denied to create subscription
DETAIL: Only roles with privileges of the "pg_create_subscription" role may create
subscriptions.
```

A PostgreSQL 16 hardening measure — creating a subscription can execute arbitrary code on the publisher’s behalf (via the connection string), so it’s gated behind its own predefined role, separate from ordinary table privileges:

```
-- as postgres
GRANT pg_create_subscription TO chris;
```

Retrying reaches a second, unrelated gate:

```
ERROR: could not connect to the publisher: connection to server at "localhost" (127.0.0.1),
port 5432 failed: FATAL: permission denied to start WAL sender
DETAIL: Only roles with the REPLICATION attribute may start a WAL sender process.
```

`pg_create_subscription` governs creating the *local* subscription object; actually connecting to the publisher and opening a replication connection is gated separately, by a role *attribute* (like `LOGIN` or `CREATEDB`), not a grantable role membership:

```
-- as postgres
ALTER ROLE chris REPLICATION;
```

2.2 — The deadlock

Retrying again:

```
CREATE SUBSCRIPTION portsmith_sub
  CONNECTION 'host=localhost dbname=portsmith user=chris password=fdw-demo-password'
  PUBLICATION portsmith_pub;
```

...hangs. No error, no completion — indefinitely. From another session:

```
SELECT l.pid, l.mode, l.granted, a.query
FROM   pg_locks l JOIN pg_stat_activity a ON a.pid = l.pid
WHERE  l.locktype = 'transactionid';
```

pid	mode	granted	query
322315	ExclusiveLock	t	CREATE SUBSCRIPTION portsmith_sub CONNECTION ... ;
322316	ShareLock	f	CREATE_REPLICATION_SLOT "portsmith_sub" LOGICAL pgoutput (SNAPSHOT 'nothing')

CREATE SUBSCRIPTION (pid 322315), by default, tries to create its replication slot as part of its own work — and creating a *logical* slot needs a consistent snapshot, which means waiting for every other currently-running transaction in the cluster to finish. Because the publisher and subscriber are the **same PostgreSQL instance** here, the walsender process it spawns to build that snapshot (pid 322316) ends up waiting on CREATE SUBSCRIPTION's *own*, still-open transaction — which can't finish until the slot creation it's waiting on returns. A true self-deadlock, specific to same-instance publisher/subscriber setups; a genuinely separate PostgreSQL server wouldn't have this problem, since its walsender would never need to wait on a transaction ID that belongs to a different cluster entirely.

Cancel it and clean up:

```
SELECT pg_cancel_backend(322315);
```

2.3 — The real fix: create the slot first, separately

```
-- in portsmith (the publisher)
SELECT pg_create_logical_replication_slot('portsmith_sub', 'pgoutput');
```

```
pg_create_logical_replication_slot
-----
(portsmith_sub,0/C697AEB8)
```

```
-- in portsmith_legacy
CREATE SUBSCRIPTION portsmith_sub
  CONNECTION 'host=localhost dbname=portsmith user=chris password=fdw-demo-password'
  PUBLICATION portsmith_pub
  WITH (create_slot = false, slot_name = 'portsmith_sub');
```

```
CREATE SUBSCRIPTION
```

With the slot already sitting there, CREATE SUBSCRIPTION has nothing left to wait on — it just starts the initial data copy immediately:

```
SELECT count(*) FROM businesses; -- in portsmith_legacy
SELECT count(*) FROM jobs;
```

count	count
-----	-----
48	48

2.4 — Real-time replication

```
-- in portsmith
INSERT INTO businesses (name, address, neighbourhood, details, employee_count)
VALUES ('Harbor Light Cafe', '12 Quay Street', 'Old Town', '{"category":"restaurant"}', 6)
RETURNING id;
```

```
-- moments later, in portsmith_legacy
SELECT id, name, employee_count FROM businesses WHERE name = 'Harbor Light Cafe';
```

id	name	employee_count
-----+-----+-----		
49	Harbor Light Cafe	6

No polling, no manual refresh — the row is there because a walsender process pushed it the moment the `INSERT` committed on the publisher. This is the payoff the whole chapter is built around: a clerk enters Harbor Light Cafe’s registration in `portsmith`, and — with nobody running a sync script, no nightly job, nothing on a timer — it’s already sitting in the portal’s own database, ready for the next citizen search.

Exercise 3 — Inspecting Slot and Replication State

This is the question an operator actually asks about the Business Finder portal: is it still current, or has it quietly fallen behind without anyone noticing? These two views are how you’d answer that — in a monitoring dashboard, not just by hand.

```
-- in portsmith
SELECT slot_name, plugin, slot_type, database, active, restart_lsn, confirmed_flush_lsn
FROM pg_replication_slots;
```

slot_name	plugin	slot_type	database	active	restart_lsn	confirmed_flush_lsn
-----+-----+-----+-----+-----+-----						
portsmith_sub	pgoutput	logical	portsmith	t	0/C698E388	0/C698E3C0

`pg_stat_replication` is the live, moment-to-moment view — but querying it as `chris` at first shows an odd, mostly-empty row:

```
SELECT application_name, state, sent_lsn, replay_lsn, replay_lag FROM pg_stat_replication;
```

application_name	state	sent_lsn	replay_lsn	replay_lag
portsmith_sub				

The row exists — chris can see *that* a replication connection is active — but the detail columns (state, every LSN, lag) are hidden unless the querying role is a superuser or holds pg_monitor:

```
-- as postgres
GRANT pg_monitor TO chris;

SELECT application_name, state, sent_lsn, write_lsn, flush_lsn, replay_lsn, replay_lag
FROM pg_stat_replication;
```

application_name	state	sent_lsn	write_lsn	flush_lsn	replay_lsn	replay_lag
portsmith_sub	streaming	0/C698EFC8	0/C698EFC8	0/C698EFC8	0/C698EFC8	00:00:00.00025

state: streaming and a quarter-millisecond replay_lag — on localhost, replication lag is essentially the cost of a context switch. The same query against a subscriber across a real network would show it rising under load, which is exactly what an operator watches this view for.

Exercise 4 — Row-Filtered Publication: Active Businesses Only

A citizen searching the Business Finder portal for a seafood restaurant in Old Town doesn't want three shuttered storefronts in the results. The column list in Exercise 1 controlled *what* the portal sees about each business; this exercise controls *which businesses* it sees at all — closed ones simply shouldn't exist as far as portsmith_legacy is concerned.

4.1 — Give businesses something to filter on

```
-- in portsmith
ALTER TABLE businesses ADD COLUMN active boolean NOT NULL DEFAULT true;
UPDATE businesses SET active = false WHERE id IN (6, 8, 14); -- closed since the seed data was
written

-- in portsmith_legacy, so the published column list still has somewhere to land
ALTER TABLE businesses ADD COLUMN active boolean NOT NULL DEFAULT true;
```

4.2 — Add the filter

```
-- in portsmith
ALTER PUBLICATION portsmith_pub
SET TABLE businesses (id, name, address, neighbourhood, details, employee_count, active)
WHERE (active = true),
jobs;
```

4.3 — A refresh alone doesn't retroactively resync

```
-- in portsmith_legacy
ALTER SUBSCRIPTION portsmith_sub REFRESH PUBLICATION WITH (copy_data = true);
SELECT count(*) FROM businesses;
```

```
count
-----
    49
```

Still 49 — the three closed businesses are still sitting there, and still show `active = true` locally. `REFRESH PUBLICATION` only reconciles which *tables* a subscription tracks (picking up newly added or removed ones); it does nothing for a column list or row filter that changed on a table the subscription is already happily synchronized with. Nothing is wrong — this is documented behavior, just an easy assumption to get wrong the first time.

4.4 — Forcing a real resync

```
ALTER SUBSCRIPTION portsmith_sub DISABLE;
TRUNCATE businesses;
TRUNCATE jobs;    -- every table touched by the resync, not just the one you changed
ALTER SUBSCRIPTION portsmith_sub ENABLE;
ALTER SUBSCRIPTION portsmith_sub REFRESH PUBLICATION WITH (copy_data = true);

SELECT count(*) FROM businesses;
SELECT id, name FROM businesses WHERE id IN (6, 8, 14);
```

```
count
-----
    46
```

```
id | name
----+-----
(0 rows)
```

Forty-six — 49 minus the three now-`active = false` rows — and none of the closed businesses present at all. **Truncate every table the resync touches, not just the one whose filter changed.** Forgetting one leaves its tablesync worker retrying a `COPY` into a table that already has the same primary keys, forever:

```
ERROR:  duplicate key value violates unique constraint "jobs_pkey"
CONTEXT: COPY jobs, line 1
LOG:  logical replication table synchronization worker for subscription "portsmith_sub", table "jobs" has started
... (repeats every ~5 seconds)
```

— a real failure mode, not a hypothetical: exactly this happened while preparing this chapter, from truncating one table in a multi-table resync and not the other.

4.5 — Filtering on a non-key column has its own wall

With the resync clean, flip a business’s active flag live:

```
UPDATE businesses SET active = false WHERE id = 49;
```

ERROR: cannot update table "businesses"

DETAIL: Column used in the publication WHERE expression is not part of the replica identity.

For an UPDATE, PostgreSQL has to know whether the *old* row matched the filter (to decide whether the subscriber needs a delete-equivalent) — and by default, only the primary key travels in the WAL as the “old” row image. *active* isn’t part of that. The obvious-looking fix:

```
ALTER TABLE businesses REPLICA IDENTITY FULL;
UPDATE businesses SET active = false WHERE id = 49;
```

ERROR: cannot update table "businesses"

DETAIL: Column list used by the publication does not cover the replica identity.

REPLICA IDENTITY FULL swings too far the other way: now the replica identity is *every* column, including *geom* and *search_vector* — neither of which the publication’s column list includes. The real fix is a replica identity that’s exactly as wide as it needs to be:

```
CREATE UNIQUE INDEX idx_businesses_replident ON businesses (id, active);
ALTER TABLE businesses REPLICA IDENTITY USING INDEX idx_businesses_replident;

UPDATE businesses SET active = false WHERE id = 49;
```

UPDATE 1

```
-- moments later, in portsmith_legacy
SELECT id FROM businesses WHERE id = 49;

id
----
(0 rows)
```

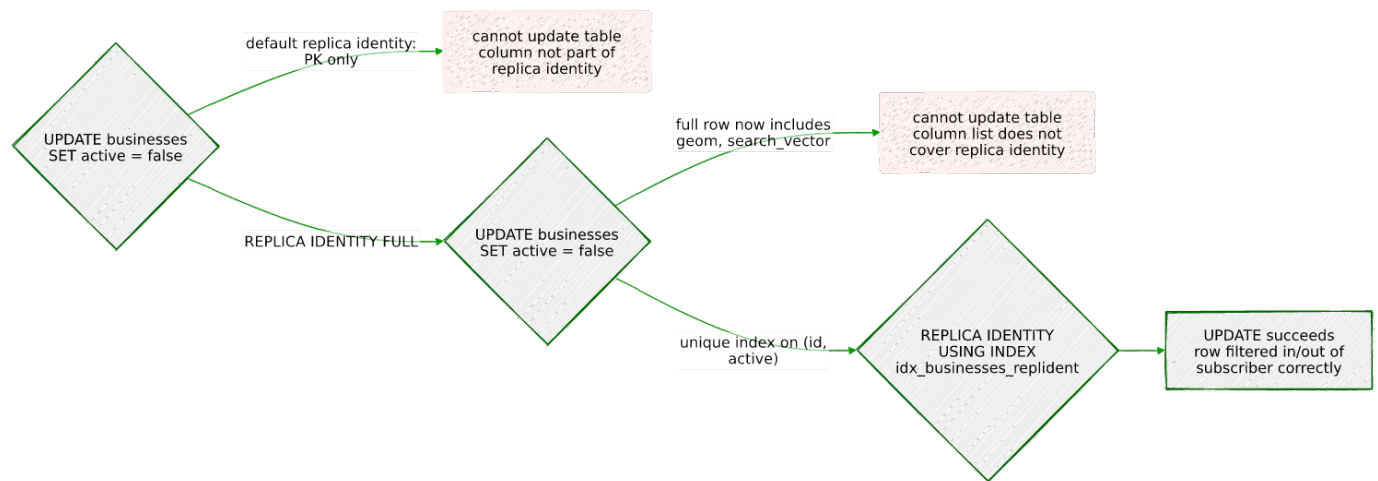
Gone — correctly filtered out the moment *active* flipped, effectively replicated as a delete. That’s the portal experience made concrete: a business owner closes up shop, a clerk flips one flag in *portsmith*, and the listing vanishes from public search — not tomorrow, right now. Flipping a previously-closed business back to active does the reverse:

```
UPDATE businesses SET active = true WHERE id = 6;

SELECT id, name, active FROM businesses WHERE id = 6; -- in portsmith_legacy
```

id	name	active
6	Tidal Wave Surf Shop	t

Reappears — an insert-equivalent, from the subscriber’s point of view. Two real errors, two real fixes, and the underlying idea worth keeping: **a row filter on any column other than the primary key needs that column in the replica identity — sized to exactly what the publication actually exposes, no wider.**



Exercise 5 — Reading the Replication Protocol Directly, from Python

`CREATE SUBSCRIPTION` is PostgreSQL talking to PostgreSQL — the wire format (`pgoutput`) is a compact binary protocol meant for another PostgreSQL server to decode, not for a human or a general-purpose client to read directly. Anything that wants to consume the *raw* change stream — a custom sync tool, or (Exercise 6) Debezium — talks the same underlying replication protocol, just with a different output plugin and its own logic for what to do with each change.

Not every downstream consumer of Portsmouth’s data wants a full PostgreSQL database, though. Suppose IT also wants to keep a search index warm, or fire an internal Slack alert whenever a `jobs` row lands in `failed` — neither of those is “spin up another Postgres and `CREATE SUBSCRIPTION`,” it’s “run a lightweight process that reacts to each change.” This exercise builds the simplest version of exactly that: reading the change stream directly, with nothing between it and the WAL.

5.1 — A second, independent slot

```
-- in portsmouth
SELECT pg_create_logical_replication_slot('demo_test_decoding', 'test_decoding');
```

`test_decoding` ships with core PostgreSQL and produces human-readable text instead of `pgoutput`’s binary format — the right choice for *reading* the stream directly rather than feeding it to another PostgreSQL instance. This slot is entirely independent of `portsmith_sub` — the same WAL, decoded twice, for two different consumers.

5.2 — psycopg has no high-level replication API

Unlike `psycopg2`, which ships a purpose-built `LogicalReplicationConnection`, `psycopg` (v3, used throughout this book) doesn't wrap the replication protocol at all. It's still fully reachable, though, through the same low-level `pq` connection object every higher-level `psycopg` call is eventually built on:

```
#!/usr/bin/env python3.12
# ch18_replication_stream.py - consume a logical replication slot directly
import struct
import time

import psycopg

SLOT_NAME = "demo_test_decoding"

conn = psycopg.connect("dbname=portsmith replication=database", autocommit=True)
pgconn = conn.pgconn
pgconn.exec_(f"START_REPLICATION SLOT {SLOT_NAME} LOGICAL 0/0".encode())

while True:
    data = pgconn.get_copy_data(1)          # 1 = non-blocking
    if data[0] == 0:
        pgconn.consume_input()
        time.sleep(0.1)
        continue
    payload = bytes(data[1])
    msg_type = payload[0:1]

    if msg_type == b"w":                    # XLogData: an actual decoded change
        wal_start = struct.unpack("!Q", payload[1:9])[0]
        body = payload[25:].decode("utf-8", errors="replace")
        print(f"[{wal_start:X}] {body}")

    elif msg_type == b"k":                  # Primary keepalive
        wal_end, _send_time, reply_requested = struct.unpack("!QQb", payload[1:18])
        if reply_requested:
            # must reply, or the server eventually decides this client is dead
            now = int(time.time() * 1_000_000) - 946_684_800_000_000
            pgconn.put_copy_data(b"r" + struct.pack("!QQQb", wal_end, wal_end, wal_end, now,
0))
```

`START_REPLICATION` puts the connection into **COPY BOTH** mode — the same bidirectional streaming protocol PostgreSQL's own physical replication uses, just carrying decoded logical changes instead of raw WAL bytes. Two message types arrive: `w` (an actual change, `XLogData`) and `k` (a keepalive the server sends periodically, which must be acknowledged with a standby status update — skip that, and the server eventually assumes the client has died and closes the connection). Full script: `data/ch18_replication_stream.py`.

5.3 — Running it

```
python ch18_replication_stream.py --seconds 12
```

While it runs, from another session:

```
INSERT INTO jobs (job_type, payload) VALUES ('demo_test', '{"note": "replication stream test"}');  
UPDATE businesses SET employee_count = 10 WHERE id = 1;
```

Real output:

```
[C6B57B98] table public.jobs: INSERT: id[bigint]:49 job_type[text]:'demo_test'  
payload[jsonb]: '{"note": "replication stream test"}' status[job_status]: 'queued'  
priority[smallint]:5 ...  
[C6B5A530] table public.businesses: UPDATE: id[integer]:1 name[text]: 'The Gilded Clam' ...  
geom[geometry]: '0101000020E6100000FA7E6ABC7493FCBF9A99999999594940'  
employee_count[positive_integer]: '10' search_vector[tsvector]: ' 'clam':3 ' 'gild':2 ...'  
active[boolean]:true
```

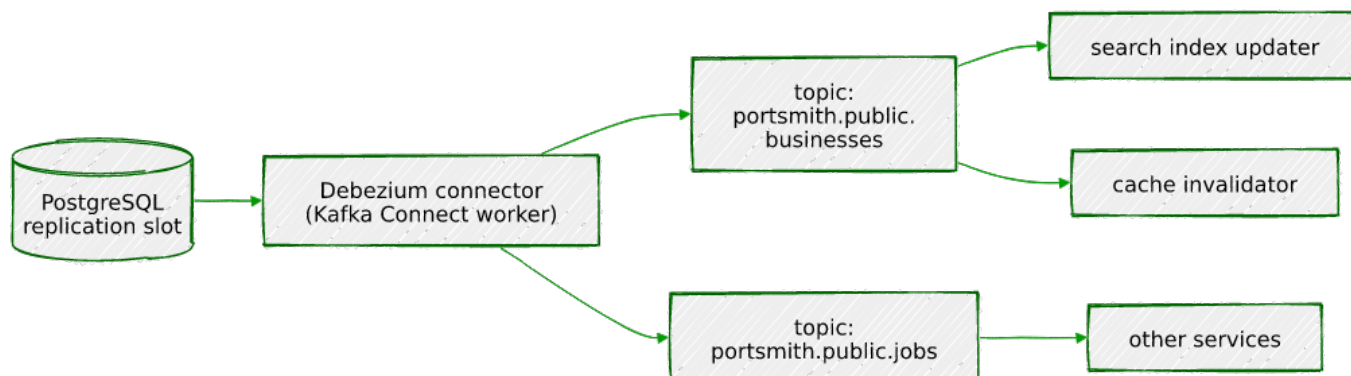
Notice `geom` and `search_vector` are both present here — columns `portsmith_pub` never publishes at all. That's the real distinction to take away: **the WAL always contains the whole row; a publication's column list is a filter applied on top of it, not something baked into what gets written to WAL in the first place.** A slot with no column list restriction, like this one, sees everything; `portsmith_sub`, tied to `portsmith_pub`, only ever sees what that publication chose to expose.

Exercise 6 — Debezium: the Same Protocol, at Production Scale

Everything Exercise 5 hand-rolled — a replication slot, a decoding plugin, a loop reading changes and acknowledging keepalives — is exactly what **Debezium's** PostgreSQL connector does, running as a **Kafka Connect** worker instead of a standalone script:

1. **A logical replication slot**, created and owned by the connector itself, using `pgoutput` (or the older `decoderbufs/wal2json` plugins in older setups) — structurally identical to `portsmith_sub`.
2. **One Kafka topic per table** — `portsmith.public.businesses`, `portsmith.public.jobs` — each message a JSON or Avro envelope carrying the operation type (create, update, delete, or read for the initial snapshot), the row's before-and-after image, and source metadata (LSN, transaction ID, commit timestamp) — the same fields Exercise 5's raw `XLogData` messages carried, structured for machine consumption instead of printed as text.
3. **Offset tracking**, conceptually identical to `confirmed_flush_lsn` in Exercise 3's `pg_replication_slots` — Debezium periodically commits how far it's gotten, so a restart resumes from the right point in the WAL instead of replaying everything or losing changes.

4. **Kafka Connect's distributed worker model** handles the part a single Python script doesn't: if the machine running the connector dies, another worker in the cluster picks up the same slot and continues, using the last committed offset.



This is deliberately a discussion, not a hands-on exercise — a real Debezium deployment needs Zookeeper (or KRaft), a Kafka broker, a Kafka Connect worker, and the Debezium connector JARs, a multi-container stack disproportionate to stand up for one exercise. What's worth taking away is that there's no new mechanism to learn: Debezium is Exercise 5's script, hardened and made distributed, reading the exact same kind of slot this chapter already created and drained by hand. Typical uses lean on that continuous stream directly: keeping a search index or cache in sync with PostgreSQL without a batch job's inherent lag, or feeding a *businesses/jobs* change feed into other services in an event-driven architecture — the **outbox pattern**, notably, uses exactly this to publish domain events reliably, by writing them as ordinary rows in a transaction and letting CDC carry them out instead of risking a dual-write to both a database and a message queue.

Summary — What You Should Now Know

Concept

`wal_level = logical`

`CREATE PUBLICATION ... FOR TABLE t
(cols) WHERE (expr)`

Replication slot

`pg_create_subscription / REPLICATION`
attribute

Same-instance pub/sub deadlock

What it does

Required for any logical decoding — needs a restart, like `shared_preload_libraries`

Scopes what's streamed by table, column, and row

A durable bookmark retaining WAL for one consumer — exists independently of any subscription

Two separate gates: creating the local subscription object vs. opening a replication connection

`CREATE SUBSCRIPTION`'s default `create_slot=true` can wait on its own open transaction when publisher and

Concept

What it does

`pg_replication_slots / pg_stat_replication`

`ALTER SUBSCRIPTION ... REFRESH
PUBLICATION`

`REPLICA IDENTITY USING INDEX`

Raw replication protocol
(`pgconn.get_copy_data/put_copy_data`)

Debezium

subscriber share an instance — pre-create the slot instead

Slot state and live lag — the detail columns of the latter need `pg_monitor`

Reconciles *which tables* are tracked — does **not** retroactively resync a changed column list or row filter on an already-synced table

The precise fix for filtering on a non-key column — wide enough to cover the filter, no wider than the publication's own column list

What `CREATE SUBSCRIPTION`, Exercise 5's script, and Debezium all ultimately speak underneath

The same slot/decode/apply loop, made distributed and production-grade, one Kafka topic per table

The key design insight from this chapter is that a replication slot is the one durable object underneath everything else — `CREATE SUBSCRIPTION`, a hand-rolled psycopg script, and Debezium are three different consumers of the exact same primitive, differing only in what decodes the WAL and what happens to the output. Understanding the slot is understanding the whole chapter; everything else is a client.

Going further: Chapter 19's `pg_cron` is the natural next tool for operationalizing what this chapter demonstrated by hand — a scheduled job that checks `pg_replication_slots` for a slot whose `confirmed_flush_lsn` hasn't advanced in too long (a stuck or abandoned consumer, quietly retaining WAL) and pages someone before disk fills up, the same shape of unattended, recurring maintenance task Chapter 17's own "going further" note imagined for aging partition exports.

Chapter 19 — `pg_cron`: Scheduled Jobs Inside PostgreSQL

*A cron job that lives outside the database has to be told, from scratch, how to reach it.
A cron job that lives inside the database is already there.*

Background

By this point in the book, Portsmouth has quietly accumulated three things that all need to happen on a schedule, and right now every one of them depends on a person remembering to do it. Chapter 9's `mv_sensor_daily` — the rollup the public works dashboard reads sensor trends from — only reflects reality as recently as whoever last ran `CALL refresh_and_log(...)` by hand thought to run it. Chapter 3's permit-processing workers occasionally crash mid-job without releasing what they were holding, and until someone remembers to re-run `ch03_reclaim.py`, that permit just sits `in_progress` forever, with no error, and no worker aware it's supposed to try again. And Chapter 17's `businesses_archive`, living quietly in `portsmith_legacy`, gets so little ordinary traffic that `autovacuum`'s usual change-percentage triggers rarely fire on it — its planner statistics can go stale for months with nothing to notice.

None of these are hypothetical. The specific version of this that actually happened while building this chapter's demo data: a stalled `in_progress` permit sat unprocessed for over half an hour before anything caught it, because nothing was watching — exactly the failure mode Exercise 4 reproduces on purpose. The pattern behind all three is the same one every real system eventually hits: recurring maintenance that depends on a human's memory degrades quietly, not loudly, and “someone should really schedule that” has a way of staying a to-do item for months.

The traditional fix is an external scheduler — OS-level `cron`, a workflow tool, or a sidecar container — that wakes up on its own timeline and connects in from outside. `pg_cron` takes a different position: it runs *inside* PostgreSQL, as a background process the postmaster manages like any other, reading its schedule from an ordinary table (`cron.job`) and writing its history to another (`cron.job_run_details`). Scheduling Portsmouth's daily refresh, its stalled-job sweep, and the archive database's `ANALYZE` all become a `SELECT`, not three separate deployments to babysit.

That proximity is also exactly where this chapter's real gotchas come from: a job that “lives inside PostgreSQL” doesn't skip the parts of PostgreSQL that would normally apply to any other client

connecting in — it still needs a role, that role still needs privileges, and if a password is required to connect as that role, `pg_cron`’s own connection needs one too, from somewhere.

The Scenario

Three of Portsmouth’s own real, already-built pieces become this chapter’s scheduled jobs — nothing new needs inventing, only automating:

Object	Source	Purpose
<code>refresh_and_log(regclass)</code>	Chapter 9	Keeps the public works dashboard’s <code>mv_sensor_daily</code> rollup from going stale, without anyone running it by hand
<code>jobs / dead_letter_jobs</code>	Chapter 3	The permit queue itself — where a crashed worker’s abandoned, <code>still-in-progress</code> job would otherwise sit forever
<code>sweep_stalled_jobs()</code>	<i>(new)</i>	SQL port of <code>ch03_reclaim.py</code> ’s stalled-job logic, so the sweep can run unattended instead of by hand
<code>guarded_demo_task()</code>	<i>(new)</i>	Teaching-only procedure demonstrating advisory-lock overlap guards
<code>businesses_archive</code>	Chapter 17, <code>portsmith_legacy</code>	The rarely-touched archive table whose planner statistics need a scheduled <code>ANALYZE</code> , since ordinary traffic won’t trigger one

Exercise Goals

By the end of this chapter you will be able to:

- Install `pg_cron`, schedule a job, and know exactly which two real walls a first attempt hits before a job actually runs.
- Read `cron.job` and `cron.job_run_details` correctly — including a sorting mistake that’s easy to make and easy to miss.
- Understand precisely what `pg_cron` already protects you from around overlapping runs, and what it doesn’t — and use an advisory lock for the part it doesn’t.
- Turn a script you’d otherwise run by hand into unattended, scheduled SQL.
- Schedule a job against a database other than the one `pg_cron` itself runs in.

- Modify, unschedule, and monitor jobs for failure, treating `cron.job_run_details` as a first-class thing to alert on.
-

Installation

```
sudo apt install postgresql-16-cron
```

```
# postgresql.conf – shared across this chapter and the next two
shared_preload_libraries = 'pg_cron,pg_stat_statements,auto_explain'
cron.database_name = 'portsmith'
```

```
sudo systemctl restart postgresql
```

```
-- as postgres
```

```
CREATE EXTENSION pg_cron;
```

`cron.database_name` matters beyond just naming a database: it's where `pg_cron`'s own launcher process runs and where `cron.job` / `cron.job_run_details` live — Exercise 5's whole point is scheduling a job that runs somewhere *other* than this one database.

Exercises

Exercise 1 — Installing `pg_cron` and Scheduling an Hourly Refresh

1.1 — Schedule it

Chapter 9's `refresh_and_log()` procedure already does everything an hourly refresh needs — `pg_cron`'s job is just to call it on a schedule instead of leaving it to whoever on the public works team remembers to run it that day:

```
SELECT cron.schedule('refresh-mv-sensor-daily', '0 * * * *',
    $$CALL refresh_and_log('mv_sensor_daily')$$);
```

ERROR: permission denied for schema cron

Creating the extension doesn't hand every role access to its schema — familiar territory by now (Chapter 17's `GRANT USAGE ON FOREIGN DATA WRAPPER`, Chapter 10's `api` schema):

```
-- as postgres
```

```
GRANT USAGE ON SCHEMA cron TO chris;
```

```
GRANT SELECT ON cron.job, cron.job_run_details TO chris;
```

```
GRANT EXECUTE ON FUNCTION cron.schedule(text,text,text) TO chris;
```

A natural instinct is to grant every `cron.*` function chris might need, all at once, in one `-c` call — which surfaces a real trap:

```
ERROR:  function cron.schedule_in_database(text, text, text, text) does not exist
```

That one function’s signature didn’t match this `pg_cron` version — and because multiple `;-` separated statements sent as one `-c` string run as a single implicit transaction, **the entire batch rolled back**, including the `GRANT USAGE ON SCHEMA` that had appeared to succeed a moment before. The fix is both a narrower ask and a more robust one — grant every function in the schema by wildcard instead of enumerating signatures you’d have to get exactly right:

```
-- as postgres, in one clean batch
GRANT USAGE ON SCHEMA cron TO chris;
GRANT SELECT ON cron.job, cron.job_run_details TO chris;
GRANT EXECUTE ON ALL FUNCTIONS IN SCHEMA cron TO chris;

SELECT cron.schedule('refresh-mv-sensor-daily', '* * * * *', -- every minute, to verify quickly
    $$CALL refresh_and_log('mv_sensor_daily')$$);

schedule
-----
      2
```

1.2 — It “succeeds,” but nothing actually runs

```
SELECT jobid, status, return_message FROM cron.job_run_details WHERE jobid = 2;
```

jobid	status	return_message
2	failed	connection failed
2	failed	connection failed

No further detail, and nothing in the PostgreSQL log even shows a rejected authentication attempt — the connection is failing before it gets that far. The reason: `pg_cron`’s launcher runs as the `postgres OS` user, and to actually execute a job it opens its own new connection to the target database, authenticating as whichever role owns the job (`chris`). That’s a different OS user than the one `chris`’s own interactive `psql` sessions run as, and — same wall Chapter 17 hit with `postgres_fdw` — a non-superuser connecting this way needs a real password, which nothing here has supplied yet. The fix is the standard one for a background process needing a password it can’t be prompted for: a `.pgpass` file, owned by the OS user running PostgreSQL:

```
sudo -u postgres bash -c 'echo "localhost:5432:portsmith:chris:fdw-demo-password" >> ~/.pgpass
&& chmod 600 ~/.pgpass'
```

```
SELECT jobid, status, return_message, start_time, end_time FROM cron.job_run_details WHERE
jobid = 2 ORDER BY runid DESC LIMIT 2;
```

jobid	status	return_message	start_time	end_time
-------	--------	----------------	------------	----------


```

2 | succeeded | CALL | 2026-08-09 23:27:00.015674-04 | 2026-08-09
23:27:02.030591-04
2 | succeeded | CALL | 2026-08-09 23:26:00.018446-04 | 2026-08-09
23:26:02.405988-04

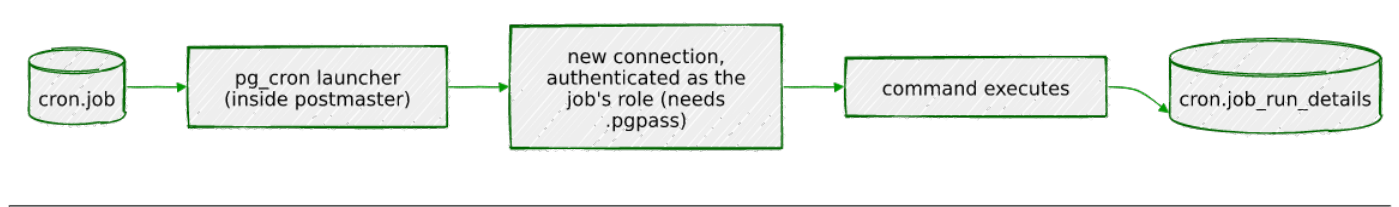
```

Real, unattended, one-minute-apart refreshes, each taking about two seconds. With it proven working, dial back to the real cadence:

```

SELECT cron.alter_job(2, schedule := '0 * * * *');

```



Exercise 2 — Reading cron.job and cron.job_run_details

```

SELECT jobid, jobname, schedule, command, active FROM cron.job;

```

jobid	jobname		schedule		command
-----+-----+-----+-----+-----					
2	refresh-mv-sensor-daily		0 * * * *		CALL refresh_and_log('mv_sensor_daily')

cron.job_run_details is the run history — and it has a sorting trap worth knowing before it costs debugging time. The failed runs from Exercise 1.2 never got a start_time at all (the connection never opened far enough to set one):

```

SELECT jobid, status, start_time FROM cron.job_run_details ORDER BY start_time DESC LIMIT 5;

```

jobid	status	start_time
-----+-----+-----		
2	failed	
2	failed	
2	failed	
2	succeeded	2026-08-09 23:27:00.015674-04
2	succeeded	2026-08-09 23:26:00.018446-04

ORDER BY ... DESC sorts NULL as the *largest* possible value by default — so the oldest failures, the ones with no timestamp at all, appear to be “first” ahead of genuinely recent successes. Anyone scanning this for “what happened most recently” and trusting the sort order would read yesterday’s failures as more current than this minute’s successes. The fix is either an explicit NULLS LAST, or sorting by runid (monotonic, never null) instead:

```

SELECT jobid, status, start_time FROM cron.job_run_details ORDER BY start_time DESC NULLS LAST LIMIT 5;

```

jobid	status	start_time
-----+-----+-----		

```
2 | succeeded | 2026-08-09 23:27:00.015674-04
2 | succeeded | 2026-08-09 23:26:00.018446-04
2 | failed    |
2 | failed    |
2 | failed    |
```

Exercise 3 — Overlap Prevention: What `pg_cron` Already Does, and What It Doesn't

3.1 — A deliberately slow job

```
CREATE OR REPLACE PROCEDURE guarded_demo_task(delay_seconds int DEFAULT 0)
LANGUAGE plpgsql AS $$
BEGIN
    IF NOT pg_try_advisory_lock(hashtext('guarded_demo_task')) THEN
        RAISE NOTICE 'already running, skipping this run';
        RETURN;
    END IF;
    PERFORM pg_sleep(delay_seconds);
    RAISE NOTICE 'did the work';
    PERFORM pg_advisory_unlock(hashtext('guarded_demo_task'));
END;
$$;

SELECT cron.schedule('guarded-demo', '* * * * *', $$CALL guarded_demo_task(75)$$);
```

A 75-second job on a 60-second schedule guarantees overlap — the question is what `pg_cron` actually does about it.

3.2 — `pg_cron` already refuses to double-run the same job

```
LOG:  cron job 3 starting: CALL guarded_demo_task(75)          -- 23:28:00
LOG:  cron job 3 COMMAND completed: CALL                      -- 23:29:15
LOG:  cron job 3 starting: CALL guarded_demo_task(75)          -- 23:29:15, immediately after
```

There's no log line for a job 3 start at 23:29:00 at all — `pg_cron` noticed the previous run of this exact `jobid` was still going and simply didn't launch a second one for that tick. It doesn't queue the missed tick either: the next run starts the instant a slot frees, at 23:29:15, not at the next minute boundary. Left running long enough, the schedule drifts further from the clock with every overlap, a real and worth-knowing side effect on its own.

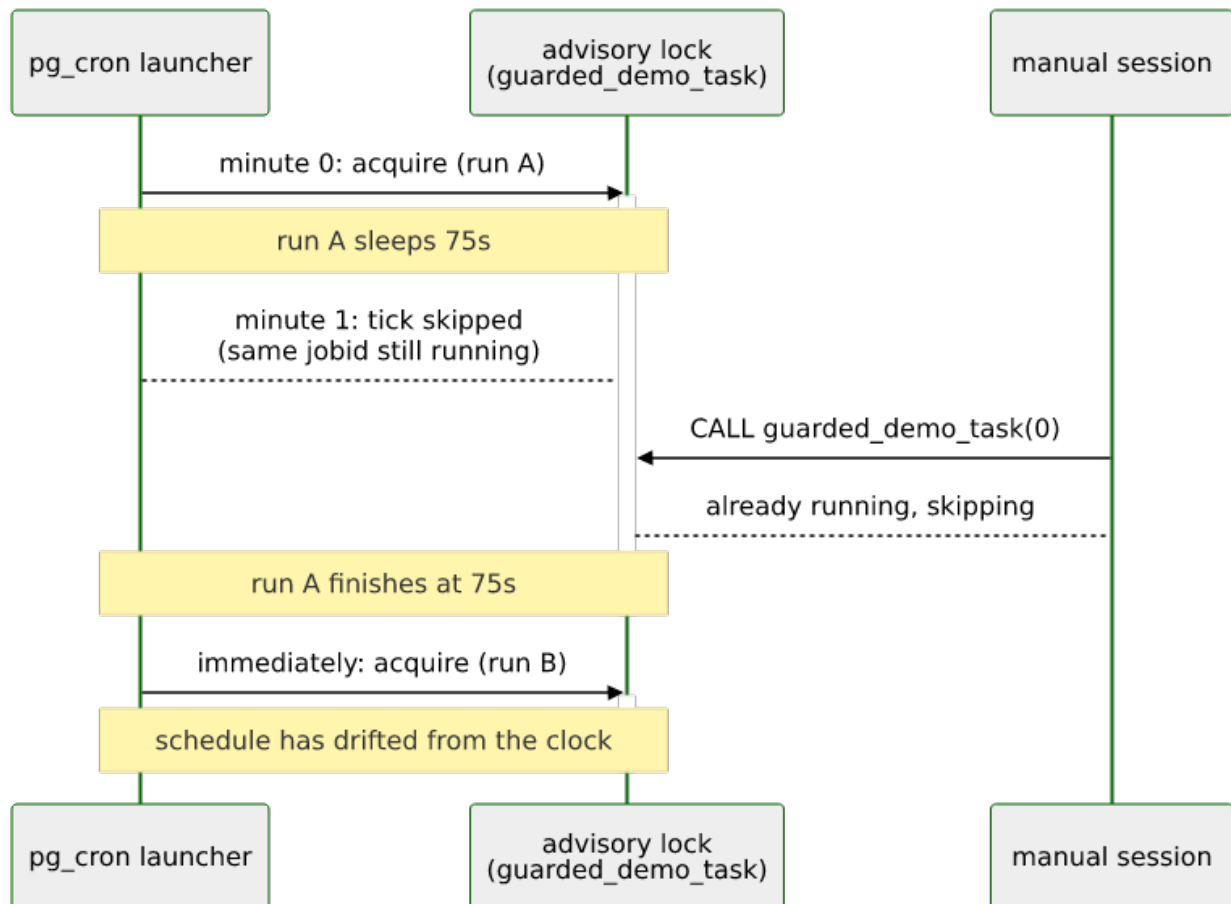
3.3 — What that protection doesn't cover

`pg_cron`'s serialization is scoped to one `jobid`. It says nothing about the same underlying task being triggered a second way — a different job entry calling the same procedure, or a human running it by hand. While job 3 was mid-run:

```
CALL guarded_demo_task(0);  -- from an ordinary psql session, not cron at all
```

```
NOTICE: already running, skipping this run  
CALL
```

This is what the advisory lock actually earns its keep for — not the same-jobid case `pg_cron` already handles, but any other path that might reach the same resource concurrently. The lock is keyed on the resource (`guarded_demo_task`, or in a real refresh job’s case, probably the matview’s name), not on the job — which is exactly what lets it catch a case `pg_cron`’s own per-job serialization structurally can’t.



```
SELECT cron.unschedule('guarded-demo');
```

Exercise 4 — A Scheduled Dead-Letter Sweep

Chapter 3 built `ch03_reclaim.py` to requeue stalled jobs and dead-letter the ones that exhausted their retries — run by hand, or “on a schedule (see Chapter 19),” per its own docstring at the time. That schedule is this exercise: the same logic, as a SQL function `pg_cron` can call directly, no external process required. Practically, this is the fix for the exact failure mode described in the Background: a permit worker crashes, its job is left `in_progress` with a heartbeat that stops updating, and without this sweep running on its own, nothing ever notices until the applicant calls asking why their permit hasn’t moved in three weeks.

4.1 — Port the script's logic to SQL

```
CREATE OR REPLACE FUNCTION sweep_stalled_jobs(p_timeout interval DEFAULT interval '30 minutes')
RETURNS TABLE(requeued int, dead_lettered int)
LANGUAGE plpgsql AS $$
DECLARE
    r RECORD;
    v_requeued int := 0;
    v_dead_lettered int := 0;
    v_error text;
BEGIN
    FOR r IN
        SELECT id, job_type, attempts, max_attempts, claimed_by, heartbeat_at
        FROM jobs
        WHERE status = 'in_progress' AND heartbeat_at < now() - p_timeout
        ORDER BY id
    LOOP
        v_error := format('stalled: no heartbeat since %s (last claimed by %s)',
r.heartbeat_at, r.claimed_by);
        IF r.attempts >= r.max_attempts THEN
            WITH failed AS (
                DELETE FROM jobs WHERE id = r.id
                RETURNING id, job_type, payload, priority, attempts, max_attempts, created_at
            )
            INSERT INTO dead_letter_jobs (id, job_type, payload, priority, attempts,
max_attempts, created_at, last_error)
                SELECT id, job_type, payload, priority, attempts, max_attempts, created_at,
v_error FROM failed;
            v_dead_lettered := v_dead_lettered + 1;
        ELSE
            UPDATE jobs SET status = 'queued', claimed_at = NULL, claimed_by = NULL,
                heartbeat_at = NULL, last_error = v_error
            WHERE id = r.id;
            v_requeued := v_requeued + 1;
        END IF;
    END LOOP;
    RETURN QUERY SELECT v_requeued, v_dead_lettered;
END;
$$;
```

4.2 — Prove both branches work

```
INSERT INTO jobs (job_type, payload, status, priority, attempts, max_attempts, claimed_at,
claimed_by, heartbeat_at)
VALUES ('demo_stall_requeue', '{}', 'in_progress', 5, 1, 3, now() - interval '40 minutes',
'worker-demo', now() - interval '35 minutes'),
    ('demo_stall_deadletter', '{}', 'in_progress', 5, 3, 3, now() - interval '40 minutes',
'worker-demo', now() - interval '35 minutes');

SELECT * FROM sweep_stalled_jobs();
```

```

requeued | dead_lettered
-----+-----
1 | 1

```

```

SELECT id, status FROM jobs WHERE job_type = 'demo_stall_requeue';
SELECT id, last_error FROM dead_letter_jobs WHERE job_type = 'demo_stall_deadletter';

```

```

id | status
----+-----
51 | queued

```

```

id | last_error
----+-----
52 | stalled: no heartbeat since 2026-08-09 21:47:19.767251-04 (last claimed by worker-demo)

```

Exactly the two outcomes `ch03_reclaim.py` produced by hand — attempts remaining gets requeued, retries exhausted gets dead-lettered, with a descriptive `last_error` either way.

4.3 — Schedule it

```

SELECT cron.schedule('sweep-stalled-jobs', '* / 5 * * * *', $$SELECT sweep_stalled_jobs()$$);

```

Verified live with a fresh stalled job and a one-minute test schedule before settling on every five minutes: `job_run_details` showed `status: succeeded, return_message: 1 row`, and the planted job's status really did flip back to `queued` — the scheduled path produces the same result as the manual call in 4.2, just unattended.

Exercise 5 — `cron.schedule_in_database()`

`cron.database_name = 'portsmith'` means `pg_cron`'s launcher itself lives there — every job scheduled with plain `cron.schedule()` runs against `portsmith` by definition.

`cron.schedule_in_database()` is the escape hatch: a job that runs against a **different** database entirely, using the same underlying launcher. `businesses_archive` is exactly the kind of table this matters for: since Chapter 17, almost nothing writes to it directly, so it rarely crosses `autovacuum`'s change-percentage threshold for an automatic `ANALYZE` on its own — left alone, its planner statistics just quietly drift out of date.

```

SELECT cron.schedule_in_database('legacy-analyze', '0 4 * * *',
    'ANALYZE businesses_archive;', 'portsmith_legacy');

```

Verified against `portsmith_legacy` — Chapter 17's second database — by checking the one thing `ANALYZE` actually changes:

```

-- in portsmith_legacy, before
SELECT last_analyze FROM pg_stat_user_tables WHERE relname = 'businesses_archive';

```

```

last_analyze
-----
(null)

-- in portsmith, briefly rescheduled to every 2 minutes to verify quickly
SELECT cron.alter_job((SELECT jobid FROM cron.job WHERE jobname = 'legacy-analyze'),
schedule := '* / 2 * * * *');

-- in portsmith_legacy, ~2 minutes later
SELECT last_analyze FROM pg_stat_user_tables WHERE relname = 'businesses_archive';

last_analyze
-----
2026-08-09 23:28:00.039363-04

```

A real, verified timestamp — pg_cron’s single launcher process, still physically running inside portsmith, reached across and executed SQL against a completely different database. Reset to a realistic once-daily cadence once proven:

```

SELECT cron.alter_job((SELECT jobid FROM cron.job WHERE jobname = 'legacy-analyze'),
schedule := '0 4 * * *');

```

Exercise 6 — Modifying, Uncheduling, and Monitoring for Failure

6.1 — cron.alter_job() and cron.unschedule()

Both already used for real, twice each, in Exercises 1, 3, and 5 — alter_job to change a schedule without dropping and recreating the job (its history in job_run_details stays intact, keyed by the same jobid), unschedule to remove one outright.

6.2 — A job that genuinely fails

```

SELECT cron.schedule('broken-demo-job', '* * * * *', 'SELECT * FROM
this_table_does_not_exist;');

SELECT status, return_message FROM cron.job_run_details ORDER BY runid DESC LIMIT 1;

```

status	return_message
failed	ERROR: relation "this_table_does_not_exist" does not exist LINE 1: SELECT * FROM this_table_does_not_exist; ^

The real error text, captured and stored — pg_cron doesn’t swallow failures, it logs exactly what PostgreSQL would have said to an interactive session running the same statement.

```

SELECT cron.unschedule('broken-demo-job');

```

6.3 — The monitoring query this chapter has been building toward

```

SELECT jobid, status, return_message, start_time
FROM   cron.job_run_details
WHERE  status = 'failed'
ORDER  BY start_time DESC NULLS LAST
LIMIT 10;

```

The same `NULLS LAST` fix from Exercise 2, now doing real work: a recurring check for exactly this query, scheduled itself (or wired into existing alerting), is the difference between a scheduled job failing silently for weeks and someone finding out the same day.

Summary — What You Should Now Know

Concept

`cron.schedule(name, schedule, command)`

`.pgpass` for the PostgreSQL OS user

`cron.job / cron.job_run_details`

Same-jobid overlap

Cross-path overlap (different jobs, or a manual call)

`cron.schedule_in_database(name, schedule, command, database)`

`cron.alter_job() / cron.unschedule()`

`cron.job_run_details.status = 'failed'`

What it does

Registers a job — needs `USAGE` on schema `cron` plus `EXECUTE` on its functions

Required for `pg_cron`'s background connection to authenticate as a non-superuser job owner

Schedule and run history — sort `job_run_details` by `runid` or with `NULLS LAST`, never a bare `start_time DESC`

Already prevented by `pg_cron` itself — a missed tick isn't queued, the next run starts immediately once a slot frees

Not covered by `pg_cron` — needs its own `pg_try_advisory_lock`, keyed on the resource, not the job

Runs against a database other than `cron.database_name`

Modify a job in place (history preserved) or remove it entirely

A real, monitorable signal — the same shape of query worth alerting on, not just reading by hand

The key design insight from this chapter is that `pg_cron` gives you exactly one guarantee for free — a job won't overlap with its own previous run — and that guarantee is narrower than it first sounds. Everything else a production scheduler needs (locking scoped to the actual shared resource, credentials for the connection it opens on your behalf, monitoring that actually gets read) is the same responsibility it would be with any external scheduler, just moved inside a table instead of a config file.

Going further: Chapter 20's `pg_stat_statements` and `auto_explain` — already sharing this chapter's `shared_preload_libraries` line — turn the same kind of "what actually happened" question this

chapter asked of `cron.job_run_details` onto query performance instead of scheduled jobs. And Exercise 4's `sweep_stalled_jobs()` is worth pairing with Chapter 18's own "going further" note: a second `pg_cron` job that watches `pg_replication_slots` for a slot whose `confirmed_flush_lsn` has stopped advancing is the exact same "scheduled health check" shape applied to replication instead of the job queue.

Chapter 20 — `pg_stat_statements` and Query Performance

“It’s slow” is a feeling. “This queryid has a mean execution time of 440ms across 964,800 rows touched, up from 47ms last week” is a fact — and only one of those is actionable.

Background

Every chapter so far has looked at PostgreSQL from the perspective of what it can store and how to query it correctly. This chapter asks a different question of the exact same database: not *is this query right*, but *is this query fast, and if not, why not, and how would you know before a user told you*. Three tools, working together:

- **`pg_stat_statements`** — every statement PostgreSQL runs, normalized (literal values replaced with \$1, \$2, ...) and aggregated: call count, total time, mean time, rows, buffers. The answer to “what’s actually slow, in aggregate, right now.”
- **`EXPLAIN (ANALYZE, BUFFERS)`** — the answer to “why,” for one specific query: the plan PostgreSQL actually chose, with real timing and real I/O counts, not just the planner’s estimate.
- **`auto_explain`** — the same `EXPLAIN` output, captured automatically for any query slower than a threshold you set, without needing to already suspect which query to go looking for.

Chapter 19 treated `cron.job_run_details` as a first-class thing to monitor rather than just read once by hand. This chapter applies the same discipline to query performance: not a one-time diagnosis, but an ongoing signal worth checking after every deploy.

The Scenario

This chapter intentionally introduces the three classic shapes of slow query against tables already built by earlier chapters — nothing new seeded, everything real:

Problem

An implicit cast defeating an index
A missing index on a large table

Where

`businesses.id` (Chapter 1)
`sensor_readings` (Chapter 8, 9.6M rows)

Problem

A plan regression after a schema change

Where

The same `sensor_readings` index, deliberately dropped and restored

Exercise Goals

By the end of this chapter you will be able to:

- Find the queries actually costing the most aggregate time in a real system, not just the ones that feel slow.
 - Read a full `EXPLAIN (ANALYZE, BUFFERS)` plan node by node.
 - Recognize an implicit cast defeating an index from its plan shape alone.
 - Diagnose a sequential scan on a large table — and know that “add an index” isn’t automatically the fix; the *right* index, matching how the table is actually queried, is what matters.
 - Use `auto_explain` to capture slow-query plans without needing to already know which query to chase.
 - Build a before/after snapshot comparison to catch a plan regression the moment it happens, not weeks later.
-

Installation

`pg_stat_statements` and `auto_explain` were already added to `shared_preload_libraries` alongside `pg_cron` back in Chapter 19’s setup, and `pg_stat_statements` was created as an extension at the same time:

```
-- as postgres, in portsmith – already done in Chapter 19's setup
CREATE EXTENSION pg_stat_statements;
```

Two functions this chapter needs are both revoked from `PUBLIC` by default — worth granting up front rather than mid-exercise:

```
-- as postgres
GRANT EXECUTE ON FUNCTION pg_stat_statements_reset(oid, oid, bigint) TO chris;
GRANT EXECUTE ON FUNCTION pg_reload_conf() TO chris;
```

The first attempt at the reset grant is worth knowing about even though it’s avoided here: `pg_stat_statements_reset()` — no arguments — doesn’t exist in current versions; the real signature takes three optional filter arguments (`userid`, `dbid`, `queryid`, all defaulting to `0`, meaning

“reset everything”). \df pg_stat_statements_reset is the fast way to check a signature before writing a GRANT for it, instead of guessing.

Exercises

Exercise 1 — Finding What’s Actually Slow

```
SELECT pg_stat_statements_reset();
```

A representative mix of real queries against this book’s data — some run often and cheap, one run rarely but expensive:

```
SELECT query, calls, round(total_exec_time::numeric, 2) AS total_ms,
       round(mean_exec_time::numeric, 3) AS mean_ms, rows
FROM   pg_stat_statements
ORDER BY total_exec_time DESC
LIMIT 10;
```

				query	
calls	total_ms	mean_ms	rows		

+-----+	+-----+	+-----+	+-----+		
SELECT count(*) FROM sensor_readings WHERE sensor_type = \$1					
2	879.08	439.541	2		
SELECT * FROM businesses WHERE id =					
\$1			30	160.14	5.338
30					
SELECT * FROM sensor_readings WHERE sensor_id = \$1 AND recorded_at >= \$2 AND recorded_at <					
\$3	15	55.18	3.678	23520	
SELECT * FROM businesses WHERE details @>					
\$1			10	46.31	4.631 150
SELECT id FROM jobs WHERE status = \$1 ORDER BY priority, created_at LIMIT \$2 FOR UPDATE SKIP					
LOCKED	25	2.26	0.090	25	

Two real lessons sitting side by side. First, literal values are normalized away (\$1, \$2) — every call to “look up one business by id” collapses into a single row here regardless of *which* id, which is exactly what makes aggregation meaningful instead of one row per unique query text. Second, and the actual point of ranking by total_exec_time rather than mean_exec_time: a query called only **twice** (an unindexed count(*) over 9.6 million rows) costs more in aggregate than a well-indexed lookup called **thirty times**. Frequency and per-call cost are independent axes, and a rare expensive query hiding behind a wall of cheap frequent ones is precisely what this view is for catching.

Exercise 2 — Reading a Full Plan

EXPLAIN (**ANALYZE**, BUFFERS, FORMAT TEXT)

SELECT * **FROM** sensor_readings **WHERE** sensor_id = 5;

```
Gather (cost=1000.00..151794.26 rows=97046 width=42) (actual time=12.772..300.728 rows=96480
loops=1)
  Workers Planned: 2
  Workers Launched: 2
  Buffers: shared hit=1812 read=88823
  -> Parallel Append (cost=0.00..141089.66 rows=40436 width=42) (actual time=13.683..273.244
rows=32160 loops=3)
    Buffers: shared hit=1812 read=88823
    -> Parallel Seq Scan on sensor_readings_2024_03 sensor_readings_2
(cost=0.00..13034.00 rows=3608 width=42) (actual time=14.335..84.003 rows=8928 loops=1)
      Filter: (sensor_id = 5)
      Rows Removed by Filter: 883872
      Buffers: shared hit=129 read=8255
      ... (one Parallel Seq Scan per partition)
Planning Time: 1.980 ms
Execution Time: 338.496 ms
```

Node by node, outside in:

Node

Meaning

Gather

The parallel query's top: collects rows from worker processes back into one stream

Workers Planned / Launched

How many parallel workers PostgreSQL asked for vs. actually got — a mismatch here would itself be worth investigating (`max_parallel_workers` exhausted)

Parallel Append

Chapter 8's partitioning at work: each partition scanned independently, results appended

Parallel Seq Scan on
sensor_readings_2024_03

One partition, scanned start to finish — no index used

Filter: (sensor_id = 5)

The condition applied *after* reading each row — the tell that no index narrowed things down first

Rows Removed by Filter: 883872

Nearly 884,000 rows read and discarded, in this partition alone, to find the ~9,000 that matched

Buffers: shared hit=... read=...

hit = found in PostgreSQL's own buffer cache; read = a real disk read — read=88823 here means the bulk of this query's cost is genuine I/O, not CPU

Execution Time: 338.496 ms

The number that actually matters to whoever's waiting on this query

ANALYZE runs the query for real and reports actual timing and row counts alongside the planner's original estimates; BUFFERS adds the I/O accounting. Without both, EXPLAIN alone only shows what PostgreSQL *expected*, not what happened.

Exercise 3 — An Implicit Cast Defeating an Index

```
EXPLAIN (ANALYZE, BUFFERS) SELECT * FROM businesses WHERE id = 5;
```

```
Index Scan using idx_businesses_replident on businesses (cost=0.14..8.16 rows=1 width=783)
(actual time=0.191..0.192 rows=1 loops=1)
  Index Cond: (id = 5)
  Buffers: shared read=2
```

Index Cond, two buffer reads — exactly what a primary-key lookup should look like. Now the same lookup, with a value typed as `numeric` instead of `integer` — the kind of thing a client library can do silently (a JSON-decoded number, a Python `Decimal`, an ORM's default type mapping for a “generic number” field):

```
EXPLAIN (ANALYZE, BUFFERS) SELECT * FROM businesses WHERE id = 5::numeric;
```

```
Seq Scan on businesses (cost=0.00..8.73 rows=1 width=783) (actual time=0.011..0.313 rows=1
loops=1)
  Filter: ((id)::numeric = '5'::numeric)
  Rows Removed by Filter: 47
  Buffers: shared hit=1 read=7
```

Filter, not Index Cond — PostgreSQL has wrapped the *column* in a cast `((id)::numeric)` to make the comparison type-consistent, and a plain btree index on `id` (built on `integer` values) can't be used to satisfy a condition on `id::numeric`. On a 49-row table the wall-clock difference is invisible — both finish in under half a millisecond, buried in connection overhead. The plan shape is identical, though, to Exercise 4's `sensor_id` lookup before it had the right index: exactly this bug against a multi-million-row table is how a sub-millisecond primary-key lookup silently becomes a 300ms sequential scan, and nothing about the query's *result* looks wrong — it just gets slower, quietly, as the table grows.

The fix is whichever side of the comparison is easiest to control: cast the *parameter* instead of leaving the column to be cast (`id = $1::integer` at the application layer), or, if the value's type is genuinely outside your control, an expression index on `(id::numeric)` — though matching the parameter's type going in is almost always the better fix.

Exercise 4 — Diagnosing (and Correctly Fixing) a Sequential Scan

4.1 — The baseline

sensor_readings has no index on sensor_id at all — every lookup by sensor scans every partition:

```
EXPLAIN (ANALYZE, BUFFERS) SELECT * FROM sensor_readings WHERE sensor_id = 5;
```

```
Gather (actual time=12.772..300.728 rows=96480 loops=1)
```

```
-> Parallel Append ...
```

```
(12 Parallel Seq Scans, one per partition)
```

```
Execution Time: 338.496 ms
```

4.2 — The obvious fix helps less than expected

```
CREATE INDEX idx_sensor_readings_sensor_id ON sensor_readings (sensor_id);
```

```
EXPLAIN (ANALYZE, BUFFERS) SELECT * FROM sensor_readings WHERE sensor_id = 5;
```

```
Append (actual time=1.629..401.335 rows=96480 loops=1)
```

```
Buffers: shared hit=225 read=81071 written=35
```

```
-> Bitmap Heap Scan on sensor_readings_2024_02 ...
```

```
Buffers: shared hit=131 read=6907
```

```
-> Bitmap Index Scan on sensor_readings_2024_02_sensor_id_idx ...
```

```
Execution Time: 376.942 ms
```

Slower, not faster — a real, worth-understanding result, not a mistake to paper over. Sensor 5 has 96,480 readings out of 9.6 million, roughly one row in 120, spread essentially evenly across the whole table’s history — every month, every day, interleaved with every other sensor’s readings. A **Bitmap Heap Scan** still has to visit almost every 8KB heap page, because at that scatter, virtually every page contains at least one matching row. The index made the *search* for matching rows cheap (Bitmap Index Scan) but did nothing about the *fetch* cost, which dominates — and switching from a Parallel Seq Scan to a plain Bitmap Heap Scan even gave up the free parallelism the original plan had. An index is not automatically a win; it’s a trade worth actually measuring.

4.3 — The query that was never realistic in the first place

Nobody actually asks for “all of sensor 5’s history, unbounded.” A real query narrows by time too — and Chapter 8’s partitioning already helps here, with no new index at all:

```
DROP INDEX idx_sensor_readings_sensor_id;
```

```
EXPLAIN (ANALYZE, BUFFERS)
```

```
SELECT * FROM sensor_readings
```

```
WHERE sensor_id = 5 AND recorded_at >= '2024-06-01' AND recorded_at < '2024-06-08';
```

```
Gather (actual time=0.301..35.504 rows=2016 loops=1)
```

```
-> Parallel Seq Scan on sensor_readings_2024_06 sensor_readings
```

```
Filter: (... AND sensor_id = 5)
```

```
Rows Removed by Filter: 287328
```

```
Execution Time: 35.671 ms
```

Partition pruning alone — only `sensor_readings_2024_06` touched, not all twelve — already beats Exercise 4.2's "fix." The right index closes the rest of the gap:

```
CREATE INDEX idx_sensor_readings_sensor_time ON sensor_readings (sensor_id, recorded_at);  
  
EXPLAIN (ANALYZE, BUFFERS)  
SELECT * FROM sensor_readings  
WHERE sensor_id = 5 AND recorded_at >= '2024-06-01' AND recorded_at < '2024-06-08';
```

Bitmap Heap Scan on `sensor_readings_2024_06` `sensor_readings` (actual time=0.587..7.534 rows=2016 loops=1)

Recheck Cond: (`sensor_id` = 5 AND `recorded_at` >= ... AND `recorded_at` < ...)

Heap Blocks: exact=1697

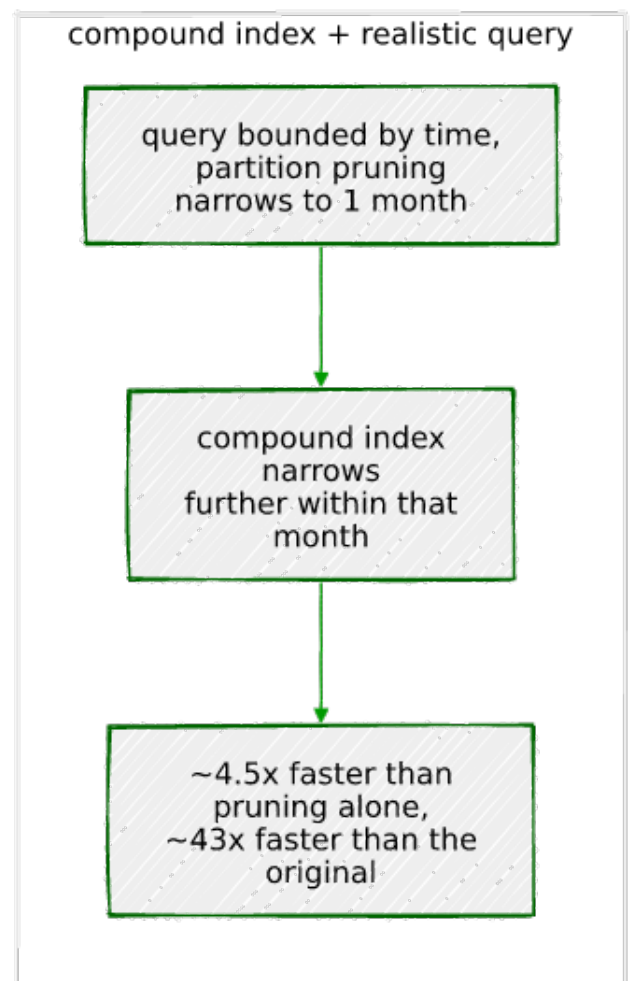
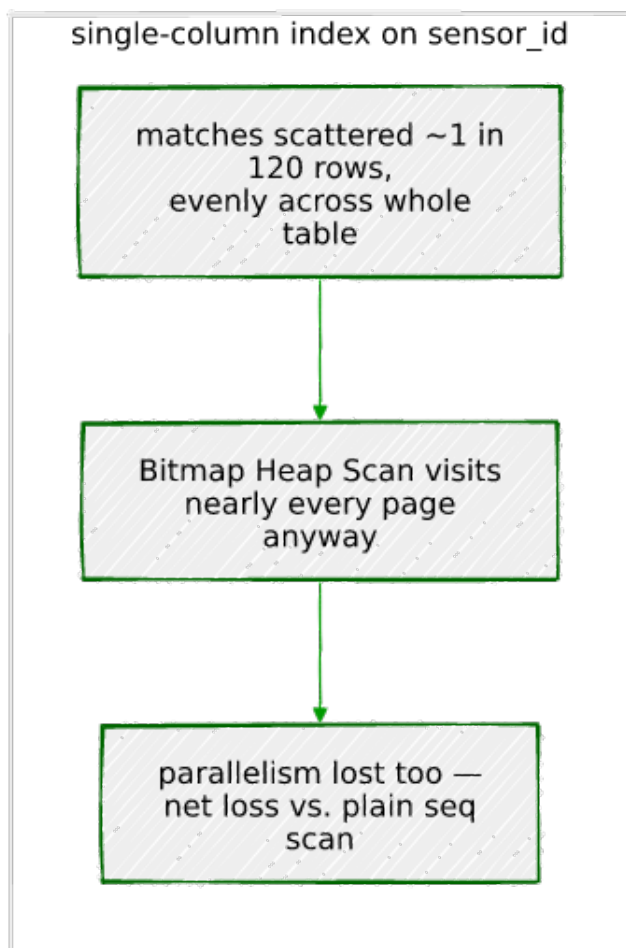
Buffers: shared read=1708

-> Bitmap Index Scan on `sensor_readings_2024_06_sensor_id_recorded_at_idx`

Buffers: shared read=11

Execution Time: 7.878 ms

7.9ms — roughly 4.5x faster than pruning alone, roughly 43x faster than the original unbounded query, and the honest conclusion of the whole exercise: the single-column index in 4.2 wasn't wrong because indexes are bad, it was wrong because it didn't match how the table is actually queried. A compound index over both the filter column and the range column, paired with a query shape that was already realistic, is what actually won.



Exercise 5 — auto_explain: Catching Slow Queries Without Knowing to Look

auto_explain was proven working back in Chapter 19's environment setup, logging *every* query at `log_min_duration = 0`. That's useful for a five-minute test and useless in production — it would flood the log. A realistic threshold, and turning it on requires nothing at the session level (it's already preloaded cluster-wide):

```
-- as postgres
ALTER SYSTEM SET auto_explain.log_min_duration = 200; -- milliseconds
```

Reloading config needs its own grant, same shape as `pg_stat_statements_reset()` above:

```
GRANT EXECUTE ON FUNCTION pg_reload_conf() TO chris;

SELECT pg_reload_conf();

SELECT * FROM businesses WHERE id = 3; -- fast - should stay
silent
SELECT count(*) FROM sensor_readings WHERE sensor_type = 'air_quality'; -- slow - should log
```

The fast lookup produces nothing in the log. The slow one:

```
LOG:  duration: 374.630 ms  plan:
      Query Text: SELECT count(*) FROM sensor_readings WHERE sensor_type = 'air_quality';
      Finalize Aggregate (actual time=367.286..374.621 rows=1 loops=1)
        Buffers: shared hit=15377 read=75258
        -> Gather (actual time=367.178..374.609 rows=3 loops=1)
          Workers Planned: 2
          Workers Launched: 2
          -> Partial Aggregate (actual time=354.058..354.061 rows=1 loops=3)
            -> Parallel Append (actual time=16.793..340.363 rows=321600 loops=3)
```

Captured automatically, no application code changed, no prior suspicion needed about which query would be the slow one — exactly `sensor_type` doing what Exercise 1's top-10 already flagged as the single most expensive query in the whole workload, this time caught by threshold instead of by manually going looking.

Exercise 6 — Catching a Plan Regression Before It's a Production Incident

The point of watching query performance continuously rather than once: catching a regression the moment a deploy introduces it, not weeks later when someone complains. A minimal snapshot table and a real before/after comparison:

6.1 — Snapshot before a “deploy”


```

CREATE TABLE query_stats_snapshot (
    snapshot_label TEXT,
    snapshot_at    TIMESTAMPTZ DEFAULT clock_timestamp(),
    queryid        BIGINT,
    query          TEXT,
    calls          BIGINT,
    mean_exec_time DOUBLE PRECISION
);

SELECT pg_stat_statements_reset();
-- ... run representative traffic ...

INSERT INTO query_stats_snapshot (snapshot_label, queryid, query, calls, mean_exec_time)
SELECT 'before_deploy', queryid, query, calls, mean_exec_time
FROM    pg_stat_statements
WHERE   query LIKE 'SELECT * FROM sensor_readings WHERE sensor_id%';

```

6.2 — Simulate the incident: a deploy accidentally drops the index

```

SELECT pg_stat_statements_reset();
DROP INDEX idx_sensor_readings_sensor_time; -- Exercise 4's fix, reverted "by accident"
-- ... same representative traffic again ...

INSERT INTO query_stats_snapshot (snapshot_label, queryid, query, calls, mean_exec_time)
SELECT 'after_deploy', queryid, query, calls, mean_exec_time
FROM    pg_stat_statements
WHERE   query LIKE 'SELECT * FROM sensor_readings WHERE sensor_id%';

```

6.3 — Compare

```

SELECT b.mean_exec_time AS before_ms, a.mean_exec_time AS after_ms,
       round((a.mean_exec_time / b.mean_exec_time)::numeric, 1) AS slowdown_factor
FROM    query_stats_snapshot b
JOIN    query_stats_snapshot a ON a.queryid = b.queryid AND a.snapshot_label = 'after_deploy'
WHERE   b.snapshot_label = 'before_deploy';

```

before_ms	after_ms	slowdown_factor
3.678	34.195	9.3

A real, measured **9.3x** regression, caught by comparing two snapshots — the same `queryid` (PostgreSQL's normalization from Exercise 1 making this possible at all: the query text is identical before and after, so the join above is exact, not a fuzzy text match). This is deliberately the shape of check worth scheduling — Chapter 19's `pg_cron` running this comparison nightly, alerting past some threshold slowdown factor, is the natural next step, closing the loop between these two chapters. Restore the real fix once satisfied:

```

CREATE INDEX idx_sensor_readings_sensor_time ON sensor_readings (sensor_id, recorded_at);

```

Summary — What You Should Now Know

Tool	What it answers
<code>pg_stat_statements</code>	<i>What's</i> slow, in aggregate — ranked by total time, not just per-call time
<code>EXPLAIN (ANALYZE, BUFFERS)</code>	<i>Why</i> one specific query is slow — real timing, real I/O, node by node
Filter VS Index Cond in a plan	The tell for an implicit cast (or any condition) that isn't reaching an index
<code>auto_explain</code>	Slow-query capture without needing to already suspect which query
Buffer counts (hit vs read)	Whether a query's cost is CPU or genuine disk I/O
A single-column index	Not automatically a win — matching the <i>scatter</i> of matching rows to physical pages matters as much as the row count
Partition pruning + a compound index	Often the actual fix, when a naive single-column index barely moves the needle
<code>queryid-matched</code> snapshot comparison	Catches a plan regression by measurement, the same day a deploy causes it

The key design insight from this chapter is that every tool in it answers a narrower question than it first appears to: `pg_stat_statements` tells you *what*, not *why*; `EXPLAIN` tells you *why*, for one query, after the fact; `auto_explain` removes the need to already know *which* query. None of them, alone, is “the performance tool” — the actual skill this chapter built is knowing which one answers the question you're currently asking, and reaching for a real measurement instead of a guess at every step, including the step where the “fix” turns out not to help.

Going further: this closes out the book's run through PostgreSQL's extension and operational surface — Chapter 21, once PostgreSQL 19's property graph support (SQL/PGQ) is out of beta, returns to Chapter 12's recursive CTEs with a genuinely different query model for the same graph problems. In the meantime, the three tools this chapter built — `pg_stat_statements`, `EXPLAIN (ANALYZE, BUFFERS)`, and `auto_explain` — are worth running against every earlier chapter's own exercises; several of this book's own “real, verified” numbers were found exactly this way.

Chapter 21 — Graph Queries: PostgreSQL 19's Property Graphs

Chapter 12 taught you how graph traversal works by making you write the recursion yourself. This chapter asks a narrower question: now that PostgreSQL 19 has a real, built-in graph query language, when does reaching for it actually pay off — and, just as importantly for software still in beta, what does it not do yet?

Background

PostgreSQL 19 adds SQL/PgQ — the ISO SQL:2023 standard for querying relational tables as if they were a labeled property graph, without copying anything into a separate graph database. Two new pieces of syntax carry the whole feature:

- **CREATE PROPERTY GRAPH** — a schema object that declares which existing tables are *vertices* and which are *edges*, entirely on top of ordinary tables you already have. No new storage, no ETL — the same rows Chapter 12 queried with `WITH RECURSIVE` are reused directly.
- **GRAPH_TABLE** — a query construct, used inside `FROM`, that matches *patterns* against a property graph — `(a)-[r]->(b)` reads much closer to “a graph shape” than a self-join or a recursive CTE does.

The honest framing for this chapter, and the reason it stayed a placeholder for months: PostgreSQL 19 was still in beta when this was written, and nothing about SQL/PgQ had been run for real yet. Writing it required standing up an actual PostgreSQL 19 instance and finding out, empirically, what the beta actually supports — which turned out to be a genuinely different (and smaller) feature than the placeholder outline assumed. That gap is most of this chapter's real content.

This chapter's exercises were first verified against PostgreSQL 19 beta2, then re-run in full against **beta3** once it was released — same container recipe, same rebuild, a fresh `pg_dropcluster`'d cluster underneath. Every finding held: the same two quantified-path forms still fail with byte-for-byte identical errors, and everything that worked in beta2 still works. Where the chapter shows a captured transcript below, it's beta3's output; anywhere the two betas diverged is called out explicitly. That the retest was this uneventful is itself the point worth taking away — for beta software, “still true” is a result you check, not one you assume.

The Scenario

This chapter reuses Chapter 12's data unchanged: the 30-row Portsmouth government org chart (`city_org`) and the real road network derived from Chapter 2's PostGIS geometry (`intersections`, `road_segments`). Nothing new is seeded — the same graphs, queried a second way.

Because PostgreSQL 19 was beta at the time of writing, it runs in an **isolated Docker container**, not on the same PostgreSQL 16 cluster every earlier chapter has built up real, cumulative state on. See the Environment Setup section below for exactly how that's wired up, including three real gotchas hit getting a beta Debian package building at all.

Exercise Goals

By the end of this chapter you will be able to:

- Stand up a second, disposable PostgreSQL instance for trying beta features without touching a cluster you depend on.
 - Declare a `CREATE PROPERTY GRAPH` over existing tables, including the self-referencing edge case (`city_org.manager_id` pointing back into `city_org` itself).
 - Write `GRAPH_TABLE` pattern-matching queries, including undirected edge patterns.
 - Know precisely which of SQL/PGQ's standard-defined features are actually implemented in PostgreSQL 19 (beta2, re-verified on beta3), versus which ones parse as errors today — verified directly, not assumed from the standard. Verified is not implied — the beta's own errors are the appendix.
 - Decide, for a given traversal, whether `GRAPH_TABLE` is a genuine readability win today or whether Chapter 12's recursive CTE is still the only tool that actually works.
-

Environment Setup — A Second Cluster for PostgreSQL 19

Rather than upgrading the machine's real PostgreSQL 16 install — which carries real, cumulative state from twenty prior chapters — PostgreSQL 19 runs in a throwaway Docker container on port 5433, built from `docker/ch21/Dockerfile` in this repo. Three real problems came up getting that container running at all, each worth knowing if you build something similar:

1. PGDG's beta packages live in a `main <version>` component, not a separate suite. The first attempt pointed `apt` at a `bookworm-pgdg-testing` suite that doesn't exist. PGDG actually publishes pre-release major versions as an additional `main 19` component *inside* the normal `bookworm-pgdg`

suite — and that component alone isn't enough, since `postgresql-19/postgresql-client-19` still depend on an up-to-date `postgresql-common/libpq5` that only the plain `main` component carries:

```
deb [signed-by=...] https://apt.postgresql.org/pub/repos/apt bookworm-pgdg main
deb [signed-by=...] https://apt.postgresql.org/pub/repos/apt bookworm-pgdg main 19
```

2. Debian's `postgresql-19` package auto-creates a cluster on install — and splits config from data. Unlike the official `postgres` Docker image (built from a plain source tarball), Debian's packaging runs `pg_createcluster` as part of `apt-get install`, producing a cluster whose data lives at `/var/lib/postgresql/19/main` but whose `postgresql.conf/pg_hba.conf` live separately, under `/etc/postgresql/19/main/`. Nothing in this book's earlier chapters hit this, because they all install extensions into an *already-running* cluster someone else set up — this is the first chapter to build a cluster from scratch in a container image. A hand-rolled `initdb` (needed to get `postgresql.conf` genuinely inside `PGDATA`, matching what a plain `postgres -D $PGDATA` expects) collides with that auto-created cluster, since its `PG_VERSION` file already exists. Fixed by dropping the auto-created cluster at build time before `initdb` ever runs:

```
RUN apt-get install -y postgresql-19 postgresql-client-19 \
    && pg_dropcluster --stop 19 main
```

3. Local-socket peer authentication doesn't match container UID patterns. The container runs as OS user `postgres` but the app role is `chris` — `initdb`'s default local-socket auth (`peer`, matching OS username to role name) rejects that combination outright. Fine for a disposable scratch container: `initdb --auth-local=trust`.

Bring it up:

```
cd docker/ch21
docker compose up --build

$ psql -h localhost -p 5433 -U chris -d portsmith19 -c "SELECT version();"
WARNING:  authenticated with an MD5-encrypted password
DETAIL:  MD5 password support is deprecated and will be removed in a future release of
PostgreSQL.
```

version

```
PostgreSQL 19beta3 (Debian 19~beta3-1.pgdg12+1) on x86_64-pc-linux-gnu, compiled by gcc
(Debian 12.2.0-14+deb12u1) 12.2.0, 64-bit
```

That MD5 password support is deprecated warning is itself a small, real signal of what's coming in a future major version, seen live rather than read about in a release note — and it's still there verbatim in `beta3`, unchanged. Getting this far required no changes to `docker/ch21/Dockerfile` at all: it never pinned a specific beta, so rebuilding it with `docker compose build --no-cache` against the same PGDG `main 19` component simply picked up whatever PGDG was currently publishing there. Since a beta's on-disk format can change between pre-releases, the old cluster's data

volume was dropped first (`docker compose down -v`) rather than reused — a fresh `initdb` under `beta3`'s own binaries, not an upgrade attempt.

Getting Chapter 12's data across. `intersections.geom` is a real PostGIS `GEOMETRY(POINT, 4326)` column, and PostGIS packages for a still-beta PostgreSQL 19 aren't reliably available yet — so rather than pull PostGIS into this scratch container just to store three points' worth of geometry, the graph exercises below only need plain coordinates, not spatial operators. Exported as `lon/lat` instead:

```
psql portsmith -c "\copy (SELECT id, name, ST_X(geom) AS lon, ST_Y(geom) AS lat FROM
intersections ORDER BY id) TO 'intersections.csv' CSV HEADER"
psql portsmith -c "\copy (SELECT id, road_name, from_intersection, to_intersection, length_m
FROM road_segments ORDER BY id) TO 'road_segments.csv' CSV HEADER"
psql portsmith -c "\copy (SELECT id, name, title, manager_id FROM city_org ORDER BY id) TO
'city_org.csv' CSV HEADER"

psql -h localhost -p 5433 -U chris -d portsmith19 -c "\copy intersections FROM
'intersections.csv' CSV HEADER"
psql -h localhost -p 5433 -U chris -d portsmith19 -c "\copy road_segments FROM
'road_segments.csv' CSV HEADER"
psql -h localhost -p 5433 -U chris -d portsmith19 -c "\copy city_org FROM 'city_org.csv' CSV
HEADER"
```

All three row counts matched the source exactly: 30 `city_org` rows, 15 `intersections`, 19 `road_segments`.

Exercises

Exercise 1 — Confirming SQL/PGQ Is Really There

Before writing anything, confirm the feature actually exists in this beta rather than trusting a release announcement. The clearest proof isn't `\h` (client-side help text is bundled with whichever `psql` binary you're running — the *server's* matching `psql`, not an older local one, is what actually knows this syntax) but the system catalog itself:

```
SELECT relname FROM pg_class WHERE relname LIKE '%propgraph%' OR relname = 'property_graphs';

      relname
-----
pg_propgraph_element
pg_propgraph_element_label
pg_propgraph_label
pg_propgraph_label_property
pg_propgraph_property
property_graphs
...
```

Real, dedicated catalog tables — `CREATE PROPERTY GRAPH` isn't sugar over some existing mechanism, it's genuinely new catalog infrastructure in this release.

Exercise 2 — Defining a Property Graph Over `city_org`

`city_org` is a self-referencing table: every row is a potential vertex, and the `manager_id` foreign key back into the same table is the edge. Declaring it:

```
CREATE PROPERTY GRAPH city_org_graph
  VERTEX TABLES ( city_org KEY (id) LABEL employee PROPERTIES ALL COLUMNS )
  EDGE TABLES (
    city_org AS reports_to
      KEY (id)
      SOURCE KEY (id) REFERENCES city_org (id)
      DESTINATION KEY (manager_id) REFERENCES city_org (id)
      LABEL reports_to
      NO PROPERTIES
  );
```

The real gotcha, found by iterating against the actual error messages: the `SOURCE/DESTINATION` clauses look, from the documentation's bracket notation, like `KEY (...) REFERENCES` and a trailing table `(...)` are two independently optional pieces. In practice, on this beta, only the combined form parses — `KEY (local_column) REFERENCES vertex_table (vertex_key_column)` — read exactly like an ordinary foreign key. Both of these failed:

```
SOURCE KEY (id) REFERENCES city_org
DESTINATION KEY (manager_id) REFERENCES city_org
-- ERROR: syntax error at or near "DESTINATION"

SOURCE city_org (id)
-- ERROR: syntax error at or near "("
```

Confirm it registered:

```
\dG+

List of property graphs
Schema | Name | Type | Owner | Persistence | Size | Description
-----+-----+-----+-----+-----+-----+-----
public | city_org_graph | property graph | chris | permanent | 0 bytes |
```

(0 bytes because there's no new storage — the property graph is a view over `city_org`, exactly as advertised.)

Exercise 3 — Fixed-Depth Pattern Matching, Side by Side With Chapter 12

The most direct rewrite of Chapter 12’s “walk from a leaf to the root” recursive CTE, at a *known* depth. Leo Park is a Streets Crew member, three levels below the Mayor:

```
SELECT * FROM GRAPH_TABLE (city_org_graph
  MATCH (a IS employee WHERE a.name = 'Leo Park')
    -[IS reports_to]-> (b IS employee)
    -[IS reports_to]-> (c IS employee)
    -[IS reports_to]-> (d IS employee)
  COLUMNS (a.name AS lvl0, b.name AS lvl1, c.name AS lvl2, d.name AS lvl3)
);
```

lvl0	lvl1	lvl2	lvl3
Leo Park	Dana Ruiz	Marcus Webb	Coretta Vance

Chapter 12’s version of this same question — “walk from any node to the root” — needed WITH RECURSIVE because it works at *any* depth without knowing it in advance. This version only works because 3 was chosen ahead of time; a different employee at a different org level needs a differently-shaped query. That tradeoff is exactly what the rest of this chapter is about.

Where GRAPH_TABLE earns its keep even at fixed depth is queries whose *shape*, not just their depth, is naturally graph-like. Compare “every employee whose skip-level manager reports directly to the Mayor” — a 2-hop pattern that reads close to the English sentence describing it:

```
SELECT * FROM GRAPH_TABLE (city_org_graph
  MATCH (a IS employee) -[IS reports_to]-> (b IS employee) -[IS reports_to]-> (c IS employee)
  COLUMNS (a.name AS employee, a.title, b.name AS skip_level_manager, c.name AS director)
) WHERE director = 'Coretta Vance'
ORDER BY employee;
```

employee	title	skip_level_manager	director
Colin Marsh	Budget Analyst	Julian Ostrowski	Coretta Vance
Dana Ruiz	Streets & Sanitation Supervisor	Marcus Webb	Coretta Vance
Felix Wren	Parks Maintenance Supervisor	Aisha Bonner	Coretta Vance
Grace Halloway	Senior Permit Reviewer	Helena Cross	Coretta Vance
Hugo Petrakis	Database Administrator	Wendell Achebe	Coretta Vance
Marcus Reilly	Patrol Captain	Diane Okonjo	Coretta Vance
Paula Mensah	Records Sergeant	Diane Okonjo	Coretta Vance
Ray Castellano	Building Inspector	Helena Cross	Coretta Vance
Renata Sikes	City Accountant	Julian Ostrowski	Coretta Vance
Tom Delgado	Water & Sewer Supervisor	Marcus Webb	Coretta Vance
Zara Lindholm	Systems Administrator	Wendell Achebe	Coretta Vance

A self-join written by hand to answer this — city_org a JOIN city_org b ON a.manager_id = b.id JOIN city_org c ON b.manager_id = c.id — returns the identical 11 rows on the same freshly-loaded data, confirmed directly during the beta3 retest. Whether the pattern-matching form is

actually *more* readable than that join is genuinely a judgment call; it's at least no worse, and it stops looking like an accident of how the join happened to be written.

(This table originally showed 8 rows, missing Renata Sikes, Tom Delgado, and Zara Lindholm — not a beta-version difference, but the live portsmouth database's city.org having drifted since this chapter's first capture. Re-exporting Chapter 12's data for the beta3 retest pulled the current, correct tree — the same one Chapter 12 itself documents — so the table above reflects that, not beta3 behavior.)

Exercise 4 — The Wall: Variable-Length Paths Aren't Supported Yet

This is the exercise the original chapter outline assumed would work, and the actual finding worth this whole chapter existing: **SQL/PgQ's quantified path patterns — the $\{m,n\}$ repetition syntax that makes “walk zero-or-more/one-or-more hops” possible — are not implemented in PostgreSQL 19.** Two different attempts, two different real errors:

```
MATCH (a IS employee WHERE a.name = 'Leo Park') (-[IS reports_to]->(IS employee)){1,10} (root IS employee)
-- ERROR: unsupported element pattern kind: "nested path pattern"
```

```
MATCH (a IS employee WHERE a.name = 'Leo Park') -[IS reports_to]->{1,10} (root IS employee)
-- ERROR: element pattern quantifier is not supported
```

Both forms the SQL:2023 standard defines for repeating a path — a quantified nested group, and a quantifier directly on an edge pattern — are parsed far enough to be recognized and then explicitly rejected as unsupported. This isn't a syntax mistake on this book's part; it's a real, verified gap in what's shipped so far.

Retested against beta3 after this chapter first shipped against beta2 — both errors came back character-for-character identical. Nothing about this gap moved between the two betas; whatever's blocking quantified paths clearly wasn't on beta3's list of changes.

The practical consequence: **Chapter 12's recursive CTEs are still the only tool in PostgreSQL 19 (through beta3) that can walk a graph to an unknown depth.** “Walk any node to the root,” “find the shortest path with no upper bound on hops,” and “detect a cycle by construction” — all three of Chapter 12's headline capabilities — have no GRAPH_TABLE equivalent yet, no matter how the pattern is phrased. Whether that changes before PostgreSQL 19's actual GA release is still worth checking directly against a later beta, a release candidate, or GA itself — the same way both findings so far were reached: by running the query, not by reading the standard.

Exercise 5 — The Road Network: Undirected Edges and a Bounded-Hop Workaround

A second property graph, over the real road network:

```
CREATE PROPERTY GRAPH road_graph
  VERTEX TABLES ( intersections KEY (id) LABEL intersection PROPERTIES ALL COLUMNS )
  EDGE TABLES (
    road_segments AS segment
      KEY (id)
      SOURCE KEY (from_intersection) REFERENCES intersections (id)
      DESTINATION KEY (to_intersection) REFERENCES intersections (id)
      LABEL road
      PROPERTIES ALL COLUMNS
  );
```

Roads are two-way, but `road_segments` only stores one direction per row (`from_intersection` → `to_intersection`) — the same directionality question Chapter 12 handled with a `UNION` of both directions. `GRAPH_TABLE` has a cleaner answer built in: an edge pattern with no arrowhead, `-[ ]-`, matches the edge in either direction:

```
SELECT * FROM GRAPH_TABLE (road_graph
  MATCH (a IS intersection WHERE a.name = 'Harbour Walk & Anchor Lane') -[r IS road]- (b IS intersection)
  COLUMNS (a.name AS from_x, b.name AS to_x, r.road_name, r.length_m)
);
```

from_x	to_x	road_name	length_m
Harbour Walk & Anchor Lane	Portside Drive & Anchor Lane	Anchor Lane	445.0
Harbour Walk & Anchor Lane	Harbour Walk & Ring Road	Harbour Walk	1554.2

Real, correct, no `UNION` needed — a genuine, verified win over the Chapter 12 approach for this specific piece.

Chapter 12 Exercise 5 found a real fewest-hops-vs-shortest-distance divergence using unbounded BFS. That's off the table here (Exercise 4), but a *bounded* version — “how far can I get in exactly 2 hops” — still works, by chaining two fixed edge patterns and summing their properties:

```
SELECT * FROM GRAPH_TABLE (road_graph
  MATCH (a IS intersection WHERE a.name = 'Harbour Walk & Anchor Lane')
    -[r1 IS road]- (b IS intersection)
    -[r2 IS road]- (c IS intersection)
  COLUMNS (a.name AS start_x, c.name AS end_x, r1.length_m + r2.length_m AS total_m)
)
WHERE end_x <> start_x
ORDER BY total_m;
```

start_x		end_x		total_m
-----+-----+-----				
Harbour Walk & Anchor Lane		Portside Drive & Ring Road		1998.1
Harbour Walk & Anchor Lane		Portside Drive & Ring Road		1999.1
Harbour Walk & Anchor Lane		Dock Road & Ring Road		7665.6

A real, small, genuinely interesting result: two *different* 2-hop routes reach the same intersection, 1 meter apart — exactly the kind of near-tie a shortest-path query needs to break correctly, which `ORDER BY total_m LIMIT 1` does here without incident. But notice what this query *is*: one fixed depth, hand-written. Going to 3 hops means writing a third copy of the pattern; there’s no way to ask for “up to N hops” in one query the way Chapter 12’s recursive CTE does natively. This is Exercise 4’s wall again, in a second dataset.

Decision Guide: Recursive CTE vs. `GRAPH_TABLE`, as of PostgreSQL 19 Beta3

Need

Traversal to an unknown or unbounded depth (walk to root, true shortest path, cycle detection)

A fixed-depth, pattern-shaped query (skip-level lookups, “friend of a friend,” a specific N-hop join)

Everything needs to stay inside one cluster, no new storage

A workload that’s *fundamentally* graph-shaped at production scale (millions of nodes, deep unbounded traversal as the primary access pattern)

Use

Chapter 12’s `WITH RECURSIVE` — still the only thing that works

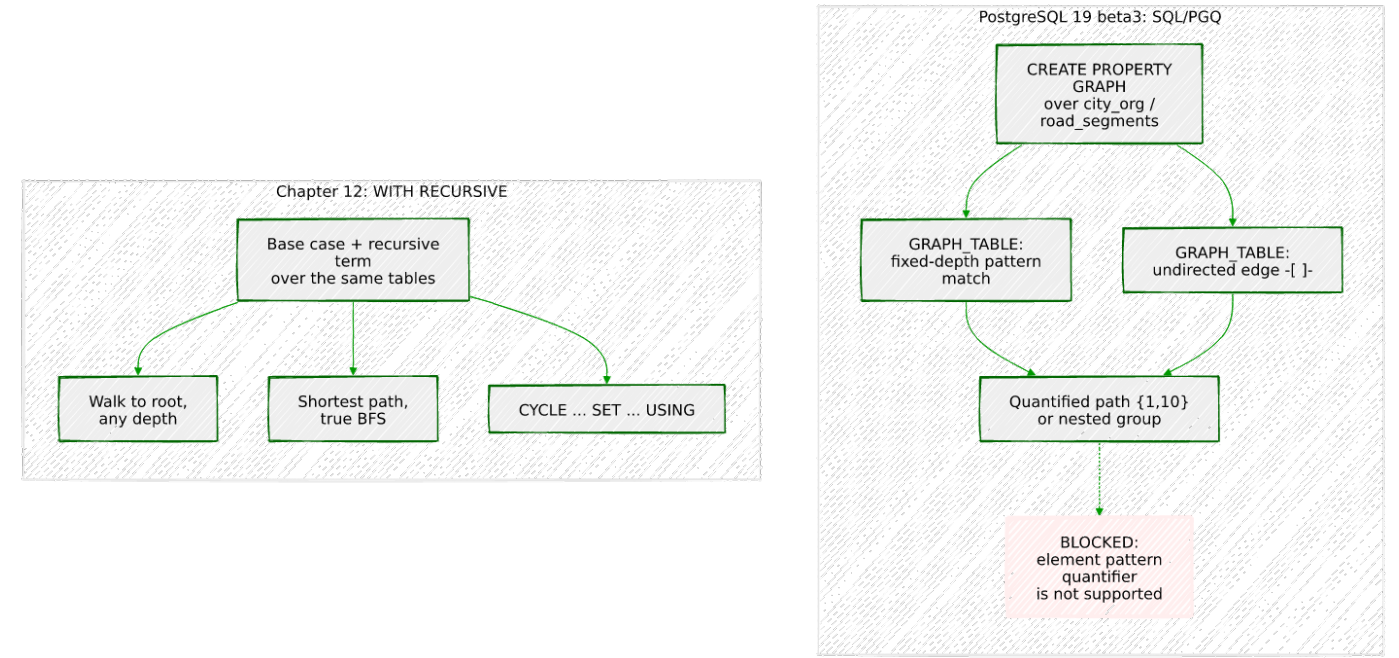
`GRAPH_TABLE` — genuinely more declarative than the equivalent self-join, and undirected (`-[ ]-`) edges are a real, clean win over hand-written `UNION`

Either — both query existing tables directly

Neither, necessarily — this is the point in the decision tree where a dedicated graph database (Neo4j and similar) starts to be worth the operational cost of running a second system, though nothing in this chapter’s small, in-memory-sized dataset actually demonstrates that threshold being crossed

The honest summary: PostgreSQL 19 ships real, working infrastructure for declaring and pattern-matching property graphs, and for the specific class of fixed-depth queries it supports, it’s a genuine readability improvement over hand-written joins. It does not yet replace recursive CTEs for anything Chapter 12 actually needed them for — true as of beta2, and, after an actual retest rather than an assumption, still true as of beta3. That could still change before general

availability — quantified path patterns are explicitly part of the SQL:2023 standard this feature implements, and “not yet supported” read from a beta’s own error message is a very different claim than “not supported,” worth re-verifying again against whatever release you’re actually running.



Summary — What You Should Now Know

Tool

CREATE PROPERTY GRAPH

GRAPH_TABLE with fixed-depth patterns

-[]- (undirected edge pattern)

Quantified path patterns ($\{m, n\}$)

Chapter 12’s WITH RECURSIVE

A second, disposable Docker cluster

What it’s actually for

Declares existing tables as a labeled graph — no new storage, a view over what you already have

A genuinely more declarative way to write a known-depth traversal or self-join

A real, clean replacement for a hand-written UNION of both directions

Standard-defined, but **not implemented as of PostgreSQL 19 beta3** — verified via two distinct real errors, unchanged from beta2 on a full retest

Still the only working tool in this release for any traversal of unknown or unbounded depth

Tool

What it's actually for

The right way to try a beta major version without touching a cluster carrying real cumulative state — and to retest cleanly against a newer beta later, since the Dockerfile has nothing pinned to un-pin

The key design insight from this chapter is less about SQL/PgQ itself than about how to evaluate a beta feature honestly: read what the standard promises, then verify against the actual release what's really there, and report the difference plainly rather than writing the chapter the outline assumed would be true. Every other chapter in this book got to lean on a stable, GA PostgreSQL; this one is a reminder that “run the real thing” sometimes means the real thing tells you “not yet.” The beta3 retest adds the other half of that discipline: “not yet” isn't a one-time verdict either — it needs re-checking against each new pre-release, and this time re-checking it cost nothing more than a container rebuild and confirmed the same answer.

Going further: Chapters 22 and 23 return to graph-shaped data from a completely different angle — RDF triples and SPARQL via `pg-ripple`, rather than SQL/PgQ's property-graph model over relational tables. It's worth holding this chapter's central finding in mind going in: a young extension or a beta feature is worth exactly what you can verify about it live, not what its README or its standard promises.

Chapter 22 — RDF Triple Stores: `pg-ripple`

Everything so far in this book has been rows, or documents shaped like rows. This chapter asks what happens when the unit of storage is a single fact — subject, predicate, object — and the query language is built around walking a graph of those facts rather than joining tables of them.

Background

RDF (the Resource Description Framework) models data as **triples**: subject predicate object, e.g. `:business_1 :locatedIn :harbour_district`. A whole database becomes one large set of these facts, queried with **SPARQL** rather than SQL. `pg-ripple` (github.com/trickle-labs/pg-ripple) brings this model into PostgreSQL as a real extension — not an ORM convention on top of ordinary tables, but genuine triple storage, a SPARQL 1.1 query engine, SHACL validation, and a Datalog-based reasoning engine, all installed with `CREATE EXTENSION`.

This is the third time this book has modeled the same kind of question — “how are things connected” — with a different tool:

- **Chapter 1 (JSONB)** models irregular *attributes* of a single document. It doesn’t model relationships between documents at all.
- **Chapter 12 (recursive CTEs) and Chapter 21 (SQL/PGQ)** model relationships as edges between *rows in tables you already have*, queried either by hand-written recursion or, as of PostgreSQL 19 beta2, fixed-depth graph pattern matching.
- **This chapter** models relationships as first-class facts, with no underlying table shape at all — a business’s category, its neighborhood, and a neighborhood’s population are all just more triples, indistinguishable in storage from the edges connecting them.

The honest throughline from Chapter 21 continues here: `pg-ripple` is real, working software, verified live against an actual build — and, same as Chapter 21, hands-on testing surfaced genuine gaps between what the README advertises and what this specific version actually does correctly. Two of this chapter’s exercises exist *because* of gaps found this way, not despite them.

The Scenario

A slice of the Portsmouth domain, recast as triples: the 48 rows of `businesses` (Chapter 1) and the 6 rows of `neighborhoods` (Chapter 2), plus a genuinely new fact with no earlier equivalent — real **neighborhood adjacency**, derived from Chapter 2’s actual polygon geometry via `ST_Touches`, not invented. This mirrors Chapter 12’s own practice of deriving graph edges from real geometry rather than making them up, and sets up a direct rerun of Chapter 21’s central finding: an “is X reachable from Y” question, asked of two different graph engines.

Like Chapter 21, this runs in its own isolated PostgreSQL container — version 18, matching what `pg-ripple`’s own README documents support for, not 19 (see the Environment Setup below for why that distinction mattered in practice).

Exercise Goals

By the end of this chapter you will be able to:

- Build a real Rust/`pgrx` PostgreSQL extension from source and know the specific version-pinning trap that broke the first attempt.
 - Export relational rows as Turtle triples and load them with `pg_ripple.load_turtle()`.
 - Write real SPARQL `SELECT` queries, including aggregation.
 - Use SPARQL property paths for unbounded-depth traversal — and know exactly why this succeeds here when the equivalent quantified path failed in Chapter 21’s PostgreSQL 19 beta2.
 - Define a SHACL shape and get a real, precise violation report — and know which part of SHACL support (scoring) works today versus which part (insert-time rejection) needs a second extension not installed here.
 - Write a custom Datalog inference rule — and know, from a reproduced, isolated test, exactly how this build’s rule engine fails to chain a two-atom rule body correctly.
-

Environment Setup — Compiling a Rust Extension

`pg-ripple` isn’t an apt package like anything earlier in this book — it’s a Rust project built against real PostgreSQL server headers via `pgrx`, PostgreSQL’s Rust extension framework. `docker/ch22/` builds it from scratch, on **PostgreSQL 18** (GA, unlike Chapter 21’s PostgreSQL 19 beta2 — `pg-ripple`’s own README documents support for 18, and there was no reason to add PostgreSQL 19’s own beta uncertainty on top of a Rust build that had plenty of its own).

Bring it up:

```
cd docker/ch22
docker compose up --build
```

Two real build failures happened getting there, both worth knowing if you build a pgrx extension yourself:

1. cargo install cargo-pgrx --version wants a full version, not a bare major.minor. `--version "0.18"` fails outright (unexpected end of input while parsing minor version number); it needs either a specific version or an explicit range qualifier.

2. cargo-pgrx's own version must exactly match the pgrx library version pinned in the extension's Cargo.toml — a loose range match isn't good enough. The natural fix for problem 1 looked like `--version "^0.18"`, which resolved to the newest 0.18.x release (0.18.1) — but pg-ripple's `Cargo.toml` pins `pgrx = 0.18.0` exactly, and `cargo-pgrx` itself refuses to proceed when its own version doesn't match, with a real, specific error naming the exact version it wants:

Error:

```
0: The installed cargo-pgrx 0.18.1 is not compatible with the dependencies in ./Cargo.toml:
    pgrx = 0.18.0, pgrx-macros = 0.18.0, pgrx-sql-entity-graph = 0.18.0, pgrx-tests = 0.18.0
    cargo-pgrx and pgrx library versions must be identical.
help: cargo install cargo-pgrx --version 0.18.0 --locked
```

The fix, exactly as the error suggests: pin the exact version, `cargo install --locked cargo-pgrx --version "0.18.0"`.

Building the cluster itself reused Chapter 21's `pg_dropcluster --stop 18 main` fix for Debian's config/data-splitting auto-created cluster, and the same `initdb --auth-local=trust` fix for the OS-user/role-name mismatch — both explained in Chapter 21's Environment Setup, not repeated here.

One more real gotcha, found only after the extension was already running: the first `CREATE EXTENSION pg_ripple` and first `sparql()` call both worked, but printed a real warning:

```
WARNING: pg_ripple: loaded without shared_preload_libraries; HTAP merge
worker, CONSTRUCT writeback, and dictionary cache are disabled. Add
pg_ripple to shared_preload_libraries in postgresql.conf.
```

Exactly the same class of gotcha Chapters 19 and 20 hit with `pg_cron/pg_stat_statements/auto_explain` — some of pg-ripple's functionality needs to be loaded at server-start time, not merely `CREATE EXTENSION`-ed into a running one. Fixed in `entrypoint.sh` by writing `shared_preload_libraries = 'pg_ripple'` into `postgresql.conf` *before* the first `pg_ctl start`, not after:


```
$ psql -h localhost -p 5434 -U chris -d portsmith22 -c "\dx pg_ripple"
pg_ripple | 0.128.0 | public | High-performance RDF triple store with SPARQL 1.1, SHACL,
Datalog, HTAP, federation, and Datalog-native PageRank
```

Exercises

Exercise 1 — Exporting Real Rows as Triples

`data/ch22_export_turtle.py` connects to the *live* PostgreSQL 16 portsmith database — not this chapter's scratch container — and writes real rows as Turtle:

```
python3 data/ch22_export_turtle.py "dbname=portsmith" data/ch22_portsmith.ttl
```

Wrote 6 neighborhoods, 10 adjacency edges, 48 businesses to `data/ch22_portsmith.ttl`

The 10 adjacency edges are genuinely derived, not invented — the exact same `ST_Touches` technique Chapter 12 used for its road-intersection graph, applied here to Chapter 2's neighborhood polygons:

```
SELECT a.name, b.name
FROM neighborhoods a
JOIN neighborhoods b ON a.id < b.id AND ST_Touches(a.geom, b.geom);
```

A sample of the output Turtle:

```
@prefix : <http://portsmith.example.org/> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .

:harbour_district a :Neighborhood ; rdfs:label "Harbour District" ; :population
4200 ; :partOf :portsmith .
:harbour_district :adjacentTo :industrial_port .
:business_1 a :Business ; rdfs:label "The Gilded
Clam" ; :locatedIn :harbour_district ; :hasCategory "restaurant" .
```

Loading it into the `pg-ripple` container — note the loader connects to the *container* (port 5434), reading the file this script just wrote:

```
with psycopg.connect("host=localhost port=5434 ... dbname=portsmith22") as conn:
    with conn.cursor() as cur:
        cur.execute("SELECT pg_ripple.load_turtle(%s, false);", (ttl,))
        print("Triples loaded:", cur.fetchone()[0])
    conn.commit()
```

Triples loaded: 261

Passing the Turtle content as a bind parameter, rather than trying to shell-escape it into a `psql -c` call, sidesteps a real amount of pain — Turtle syntax is full of the exact characters (colons, angle brackets, quotes) that are worst to quote correctly in a shell.

Exercise 2 — Real SPARQL, Including Aggregation

```
SELECT * FROM pg_ripple.sparql('
PREFIX : <http://portsmith.example.org/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?nlabel (COUNT(?b) AS ?n) WHERE {
    ?b a :Business ; :locatedIn ?nb .
    ?nb rdfs:label ?nlabel .
} GROUP BY ?nlabel ORDER BY ?nlabel
');
```

result

```
-----
{"n": 9, "nlabel": "\"Harbour District\""}
{"n": 9, "nlabel": "\"Old Town\""}
{"n": 9, "nlabel": "\"Northgate\""}
{"n": 9, "nlabel": "\"Riverside\""}
{"n": 5, "nlabel": "\"University Quarter\""}
{"n": 7, "nlabel": "\"Industrial Port\""}

```

Real, correct, and a genuine test of the query engine, not just triple storage — `GROUP BY/COUNT` over a join of three triple patterns, and the six counts sum to exactly 48, the real business count. One formatting quirk worth knowing before it surprises you: `sparql()` returns `TABLE(result jsonb)`, and string-literal bindings keep their RDF lexical quoting inside the JSON value ("`\\"Harbour District\\"",` not `"Harbour District"`) — strip the outer quote pair in application code rather than assuming a plain string.

Exercise 3 — Property Paths: The Feature Chapter 21 Didn't Have

Chapter 21 ended on a wall: PostgreSQL 19 beta2's `GRAPH_TABLE` rejected every form of variable-length path with `element pattern quantifier` is not supported. SPARQL's equivalent — the `+` (one-or-more) property path operator — is exactly the kind of unbounded traversal that broke there. Try it here, on the real adjacency data:

```
SELECT * FROM pg_ripple.sparql('
PREFIX : <http://portsmith.example.org/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT DISTINCT ?label WHERE {
    :harbour_district :adjacentTo+ ?n .
    ?n rdfs:label ?label .
}
```

```

} ORDER BY ?label
');

```

result

```

{"label": "\"Old Town\""}
{"label": "\"Northgate\""}
{"label": "\"Riverside\""}
{"label": "\"University Quarter\""}
{"label": "\"Industrial Port\""}

```

Real, unbounded, works — all five other neighborhoods, reached through however many `adjacentTo` hops it takes. This is a genuine, verified capability gap in this book’s favor for once: the exact shape of query that PostgreSQL 19 beta2 explicitly rejects, `pg-ripple`’s SPARQL engine handles correctly.

Adjacency was only stored in one direction per pair (matching `ST_Touches`’s symmetric result once, not twice), so a *directed* + path from a node that’s only ever the object of an edge would miss real neighbors — the same directionality question Chapter 21 solved with an undirected `-[ ]-` pattern. SPARQL’s answer is the inverse-path operator, `^`, combined with alternation:

```

SELECT * FROM pg_ripple.sparql('
PREFIX : <http://portsmith.example.org/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?label WHERE {
  :northgate (:adjacentTo|^:adjacentTo)+ ?n .
  ?n rdfs:label ?label .
} ORDER BY ?label
');

```

result

```

{"label": "\"Harbour District\""}
{"label": "\"Old Town\""}
{"label": "\"Northgate\""}
{"label": "\"Riverside\""}
{"label": "\"University Quarter\""}
{"label": "\"Industrial Port\""}

```

Six rows, not five — **Northgate appears in its own reachability set**. Not a bug: the directed version of this same query (`ASK { :northgate :adjacentTo+ ?n . FILTER(?n = :northgate) }`) returns `false`, but the undirected version (`ASK { :northgate (:adjacentTo|^:adjacentTo)+ :northgate }`) returns `true` — because the *undirected* adjacency graph genuinely contains a cycle: Northgate → Riverside → Harbour District → Old Town → Northgate. `+` finds it correctly. This is Chapter 12 Exercise 4’s cycle-detection lesson again, from a completely different query language: an unbounded traversal operator will walk straight into a real cycle and return the start node as its own descendant unless you explicitly guard against it (`FILTER(?n != :northgate)`, the SPARQL analog of Chapter 12’s `CYCLE ... SET ... USING`).

Exercise 4 — SHACL: Real Scoring, Gated Enforcement

A SHACL shape requiring every `:Business` to have both a category and a location:

```
@prefix sh: <http://www.w3.org/ns/shacl#> .
@prefix : <http://portsmith.example.org/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

:BusinessShape a sh:NodeShape ;
  sh:targetClass :Business ;
  sh:property [ sh:path :hasCategory ; sh:minCount 1 ; sh:datatype xsd:string ] ;
  sh:property [ sh:path :locatedIn ; sh:minCount 1 ] .
```

```
SELECT pg_ripple.load_shacl(:'shacl_text');
-- Shapes loaded: 1

SELECT pg_ripple.shacl_score('default');
-- shacl_score: 1      (fully conformant against the real, clean data)
```

Deliberately loading a malformed business — missing `:hasCategory` — and re-scoring:

```
:business_bad_1 a :Business ; rdfs:label "No Category Cafe" ; :locatedIn :harbour_district .

SELECT pg_ripple.shacl_score('default');
-- shacl_score: 0.5

SELECT pg_ripple.shacl_report_scored('default');

(http://portsmith.example.org/business_bad_1, http://portsmith.example.org/BusinessShape,
Violation, 1, "expected at least 1 value(s) for <http://portsmith.example.org/hasCategory>,
found 0")
```

Real, precise, and immediately actionable — names the exact offending entity, shape, and missing property. But this is *scoring already-stored data*, not *rejecting it at the door*. The README’s “violations caught on insert” claim is a different feature, gated behind `enable_shacl_monitors()` — which, tried here, returns a clear real answer rather than silently doing nothing:

```
WARNING: pg_trickle is not installed; SHACL violation monitors are
unavailable. Install pg_trickle and run SELECT pg_ripple.enable_shacl_monitors()
to enable.
enable_shacl_monitors
-----
f
```

A second extension, not built for this chapter. Real gap, honestly reported rather than assumed away: as installed here, `pg-ripple` validates on demand; it does not reject bad data on write.

Cleanup, using SPARQL 1.1 Update rather than reloading anything:

```

SELECT pg_ripple.sparql_update('
PREFIX : <http://portsmith.example.org/>
DELETE DATA { :business_bad_1 a :Business ; rdfs:label "No Category
Cafe" ; :locatedIn :harbour_district . }
');

```

Exercise 5 — Custom Datalog Rules: A Real, Reproduced Engine Bug

The guide’s original plan for this exercise: a transitive rule — “if X `locatedIn` Y and Y `partOf` Z, then X `partOf` Z” — inferring that every business is part of Portsmouth through its neighborhood. Getting there required first finding the real rule syntax by iterating against actual parser errors, the same technique Chapter 21 used for `CREATE PROPERTY GRAPH`:

```

SELECT pg_ripple.validate_rule('partOf(?x, ?z) :- locatedIn(?x, ?y), partOf(?y, ?z) .');
-- ERROR-shaped result: expected 3 terms in triple pattern, got 2: partOf(?x, ?z)

```

Prolog-style `predicate(args)` functors aren’t the syntax — rule heads and body atoms have to be full RDF triple patterns, subject predicate object, comma-separated in the body, one trailing period, and (found by trial) full IRIs rather than an inline `PREFIX` line:

```

SELECT pg_ripple.validate_rule(
  '?x <http://portsmith.example.org/partOf> ?z :- ?x <http://portsmith.example.org/locatedIn> ?
y , ?y <http://portsmith.example.org/partOf> ?z .'
);
-- {"valid": true, "warnings": [{"code": "UNUSED_BODY_VARIABLE", "message": "body variable ?y does
not appear in the head"}]}

```

That warning turned out to matter. Loading the rule and running it against the real 48-business dataset reported success — `pg_ripple.infer('transitive_partof')` returned 1 — but querying the actual result showed every business newly linked to its *neighborhood* by `:partOf`, not to Portsmouth. Wrong, and worth not trusting on faith — confirmed with an isolated, minimal test to rule out anything about the real dataset’s shape being the cause:

```

-- fresh, tiny dataset: :a :locatedIn :b . :b :partOf :c .
-- same rule, same predicates, isolated namespace
SELECT pg_ripple.infer('test_transitive'); -- inferred: 1
SELECT * FROM pg_ripple.sparql('PREFIX : <.../test/> SELECT ?p ?o WHERE { :a ?p ?o }');

```

```

{"o": "<.../test/b>", "p": "<.../test/partOf>"}
{"o": "<.../test/b>", "p": "<.../test/locatedIn>"}

```

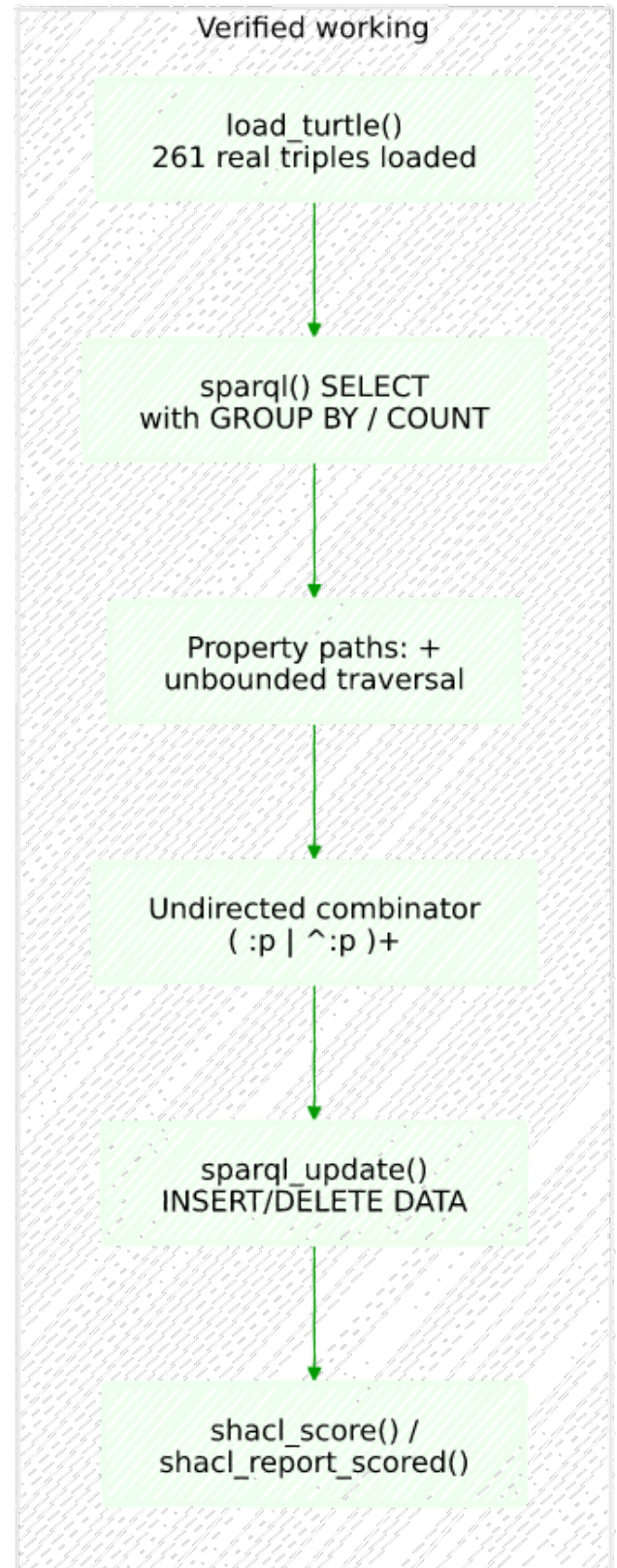
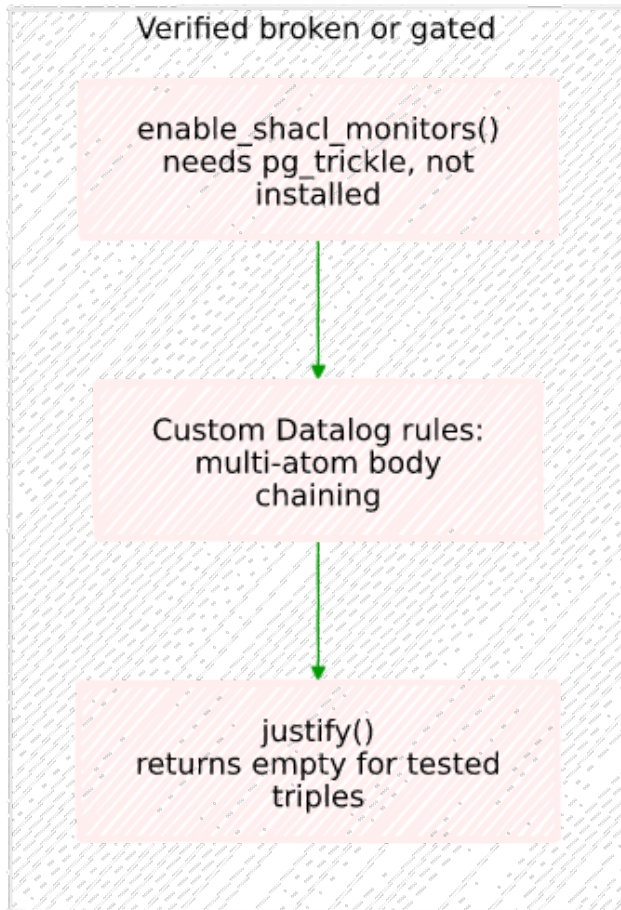
`:a :partOf :b` — not the correct `:a :partOf :c`. The rule’s second body atom (`?y partOf ?z`) isn’t actually constraining `?z`; the engine is binding the head’s `?z` to `?y` instead of chaining through to the real join result. Reordering the body atoms (`partOf` first, `locatedIn` second) produces the identical wrong answer, ruling out a simple “only evaluates the first atom” explanation — whatever the underlying cause, it’s consistent and reproducible, not a fluke of atom order.

`pg_ripple.justify()`, the function meant to return a proof tree for exactly this kind of question, returned an empty result for both the correct and the actually-inferred triple — so the “explainability” feature isn’t independently confirming or denying anything here either.

The honest conclusion, matching Chapter 21’s: multi-atom custom Datalog rule chaining does not work correctly in this build of `pg-ripple.load_rules()/validate_rule()/infer()` all function — they parse, load, and report a plausible-looking success count — but the actual inference this exercise needed is wrong, verified by checking the derived triple directly rather than trusting the success message. (The wrong triples were cleaned up with `DELETE WHERE` — which carries its own real gotcha, worth knowing before you reach for it: a `DELETE WHERE` template with *multiple* triple patterns deletes *every* pattern in the template for each matched solution, not just the one you meant. `DELETE WHERE { ?s :partOf ?o . ?s a :Business . }` correctly removed the wrong `:partOf` facts, but also silently deleted all 48 real `rdf:type :Business` declarations along with them — caught by re-checking the business count afterward, not assumed correct.)

Decision Guide: RDF/SPARQL vs. Chapter 21’s SQL/PGQ vs. Chapter 12’s Recursive CTEs

Need	Use
Unbounded-depth traversal today	<code>pg-ripple</code> ’s SPARQL property paths (<code>+</code> , <code>*</code> , undirected <code>\ ^</code>) — verified working; or Chapter 12’s recursive CTEs. Not PostgreSQL 19 beta2’s <code>GRAPH_TABLE</code> (Chapter 21) — quantified paths aren’t implemented there yet.
Data that’s naturally row-shaped, with a few graph-like relationships	Chapter 21’s <code>GRAPH_TABLE</code> over existing tables, or a plain foreign key + recursive CTE — no reason to introduce a whole second data model for this
Data that’s naturally fact-shaped — sparse, irregular attributes, relationships as first-class as the data itself	RDF/ <code>pg-ripple</code> — closer to the actual shape of the problem than forcing it into either a fixed relational schema or JSONB
Schema/data validation with a real conformance report	<code>pg-ripple</code> ’s SHACL scoring — real, working, precise. Insert-time enforcement needs <code>pg_trickle</code> , not verified here.
Rule-based inference over your own custom relationships	Not <code>pg-ripple</code> ’s custom Datalog rules, as of this version — multi-atom chaining is broken, verified by direct testing, not by assumption



Summary — What You Should Now Know

Tool

pg_ripple.load_turtle()

What it's actually for

Tool

`pg_ripple.sparql()`

SPARQL property paths (+, \|^)

`pg_ripple.shacl_score() / shacl_report_scored()`

`enable_shacl_monitors()`

`pg_ripple.load_rules() / infer()`

cargo-pgrx version pinning

What it's actually for

Bulk-loads Turtle triples — pass as a bind parameter, not shell-escaped text

Real SPARQL 1.1 SELECT, including aggregation — returns `TABLE(result jsonb)` with RDF lexical quoting preserved in string values

Working, unbounded-depth traversal — the exact capability PostgreSQL 19 beta2's `GRAPH_TABLE` doesn't have yet (Chapter 21)

Real, precise, on-demand conformance scoring
Gated behind a second extension (`pg_trickle`) not installed here — insert-time SHACL rejection isn't demonstrated working

Parse and execute without error, but **multi-atom rule body chaining produces a wrong result**, reproduced on an isolated minimal dataset independent of atom order

Must exactly match the pgrx version an extension's `Cargo.toml` pins — a semver range isn't sufficient

The key design insight from this chapter is the same discipline Chapter 21 needed, applied to a completely different piece of new software: a young extension's README describes what it's *building toward*, and the only way to know what actually works *today* is to run it and check the output, not the success message. This chapter found real wins (property paths genuinely outclass Chapter 21's current `GRAPH_TABLE`) sitting right next to real gaps (custom rule chaining, gated SHACL enforcement) — both equally worth knowing before reaching for this in anything beyond a lab exercise.

Going further: Chapter 23 builds directly on this chapter's working parts — SPARQL, property paths, and SHACL scoring — layering a real ontology (a class hierarchy, not just flat triples) on top, and pairing it with Chapter 6's `pgvector` embeddings for hybrid retrieval. It deliberately does not lean on this chapter's broken custom-rule chaining; RDFS/OWL reasoning in Chapter 23 uses `pg-ripple`'s built-in rule sets rather than hand-written Datalog, which is worth verifying independently rather than assuming it avoids the same bug.

Chapter 23 — Ontologies and Knowledge Graphs for AI Workflows

Chapter 22 gave you a way to store facts and query them. This chapter asks what it takes to make those facts machine-checkable — and why that specific property, more than raw graph storage, is what modern AI systems actually want from a “knowledge graph.”

Background

What an ontology actually is

Every earlier chapter in this book that touched structure — `CREATE TABLE`, a JSONB shape, a domain constraint, a foreign key — was really describing *storage*: where a value lives and what type it is. An **ontology** describes something narrower and more specific: the *meaning* of a domain, made explicit enough that a machine can check it and derive new facts from it. Formally, an ontology is a **shared, explicit specification of a conceptualization** — a fixed vocabulary of classes (`Restaurant`, `Neighborhood`), the relationships between them (`locatedIn`, `partOf`), and rules about how they behave (`Restaurant` is a kind of `Business`; anything `locatedIn` a `Neighborhood` is thereby `partOf` the `city`). The word comes from philosophy — the study of what exists — repurposed by computer science for a narrower question: what *categories* of thing does this system need to agree exist, and how do they relate?

This isn't a new idea invented for RDF. Library scientists have used formal classification systems (Dewey Decimal, Library of Congress Subject Headings) for over a century to make “what is this book about” checkable and consistent across a collection, not just prose in a librarian's head. Biology and medicine lean on ontologies constantly — the Gene Ontology, SNOMED CT — specifically because “is X a kind of Y” needs to mean the same thing to every system and every researcher touching the data, not just to whoever wrote the current database schema. `schema.org`, the vocabulary search engines use to understand web page markup, is an ontology in exactly this sense: a shared agreement, external to any one company's database, about what a “Recipe” or an “Event” *is*.

A useful way to place the term among near-neighbors this book has already used:

Term	What it fixes	Example from this book
Schema	Storage shape — columns, types, constraints	Chapter 1's <i>businesses</i> table, Chapter 15's domains
Taxonomy	A single hierarchy of categories, no other relationships	Chapter 12's <i>categories</i> tree, as plain parent/child rows
Knowledge graph	Facts as a graph, however structured	Chapter 21's property graph; Chapter 22's raw triples
Ontology	Formal <i>semantics</i> for a vocabulary — classes, hierarchies, and rules a machine can check and reason over	This chapter: Chapter 12's category tree, reissued as <code>rdfs:subClassOf</code> classes with checkable entailment

The distinction that matters most for this chapter: Chapter 22's triples had *predicates* (`:hasCategory "seafood"`) but no *semantics* attached to them — nothing in the store knew that "seafood" was a kind of "restaurant", or that a *Business* and its *Category* were different *kinds* of thing at all. An ontology is what turns "seafood" from an opaque string into `:Category_seafood`, a class with a real, checkable position in a hierarchy — the same category data Chapter 12 already had, given a semantics a machine can act on rather than a human reading the parent-child column pairs.

Why this matters specifically for AI workflows

Three ways this shows up in real AI system design, not just as database theory:

1. Grounding. An embedding model (Chapter 6) retrieves documents that are *semantically similar* to a query — a genuinely powerful, genuinely fuzzy notion of relevance. It has no concept of *correctness*: a document can be the closest vector match to a question and still be about the wrong neighborhood, the wrong department, or a policy that no longer applies. An ontology backs that retrieval with facts a system can actually check — not "this text sounds related," but "this specific business, in this specific neighborhood, is actually a member of the category this policy affects." Retrieval-augmented generation systems that combine both are usually called **GraphRAG** in current practice: embeddings for recall, a graph for precision.

2. Context for agents. An LLM agent given free-text "context" has to re-parse and re-infer structure from prose every time. Given a queryable ontology instead — a small, explicit model of what entities and relationships actually exist in a domain — an agent can ask a precise question ("which businesses does this ordinance affect?") and get a precise, checkable answer, rather than asking a language model to eyeball a paragraph and guess.

3. Explainability. A cosine-similarity score is a number with no narrative — it can't tell you *why* two things are considered related, only *how close*. A fact derived by rule-based reasoning over an ontology comes with a real derivation: this business is a `Restaurant` because it's a `seafood restaurant`, and `seafood rdfs:subClassOf restaurant` is an asserted fact you can point to. Chapter 22's `justify()` function gestures at exactly this — proof trees for inferred facts — even though this chapter's own testing found it doesn't reliably return one yet. The *goal* it's reaching for — inference a system can explain, not just assert — is the real reason symbolic methods keep showing up alongside purely statistical ones in current AI system design, often described as **neuro-symbolic**: neural methods for fuzzy recall, symbolic methods for checkable structure.

None of this makes vector search obsolete, and this chapter's own central exercise (below) doesn't either — it uses *both*, deliberately, because they answer different questions.

The Scenario

Two upgrades to material this book already has, both layered onto the **same** `pg-ripple` container from Chapter 22 (`docker/ch22/`, no new environment needed):

1. Chapter 12's `categories` table — 48 rows, a real 3-level tree (`All Categories` → 5 top categories → 42 specific subcategories/cuisines) — reissued as an `rdfs:subClassOf` class hierarchy, with every Chapter 1 business reclassified as an instance of its *specific* category class rather than Chapter 22's flat `:hasCategory` string.
2. Chapter 6's `pgvector` embeddings (still on the live PostgreSQL 16 `portsmith` database — nothing about Chapter 6 changes) paired with this graph for hybrid retrieval, and Chapter 5's 12 real ground-truth duplicate resident pairs (`residents.true_duplicate_of`) reused as a head-to-head entity- resolution benchmark against `pg-ripple`'s own record-linkage functions.

Same discipline as Chapters 21 and 22: every claim below was checked against a real, running instance. This chapter's single most important finding is a warning, not a feature — found exactly this way.

Exercise Goals

By the end of this chapter you will be able to:

- Design a small class hierarchy by hand and load it as `rdfs:subClassOf` facts alongside existing instance data.

- Query class hierarchies with SPARQL property paths (`rdfs:subClassOf*`) and know why this is the reliable way to ask “is X a kind of Y,” rather than PostgreSQL’s built-in RDFS rule engine.
 - Know, from direct testing, exactly what `pg_ripple.infer('rdfs')` does to existing data in this version — and why that means treating any `infer()` call as a real write against production data, never a side-effect-free query.
 - Build a real hybrid retrieval pipeline combining `pgvector` semantic search with graph-verified structured facts.
 - Evaluate a record-linkage feature against real ground truth instead of trusting a README’s description of it.
-

Exercises

Exercise 1 — Building the Ontology by Hand

Before generating anything from the database, the actual design decision worth making deliberately: which of Chapter 12’s 48 categories become *classes*, and what’s the hierarchy? A short, hand-written sample, to see the shape before scripting the full 48:

```
@prefix : <http://portsmith.example.org/> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .

:Category_all_categories a rdfs:Class ; rdfs:label "All Categories" .
:Category_restaurant a rdfs:Class ; rdfs:label "restaurant" ;
    rdfs:subClassOf :Category_all_categories .
:Category_seafood a rdfs:Class ; rdfs:label "seafood" ;
    rdfs:subClassOf :Category_restaurant .
```

The full 48-class hierarchy, generated from the real `categories` table (`data/ch23_export_ontology.py`), plus a reclassification of every business as an instance of its *specific* category rather than Chapter 22’s flat string — using `cuisine` for restaurants and `subcategory` for retail, joined carefully against the correct parent category (“pub” genuinely exists twice in Chapter 12’s data, once under `restaurant` and once under `entertainment` — the script scopes the lookup by each business’s own top-level category to resolve this correctly, not just by name):

```
python3 data/ch23_export_ontology.py "dbname=portsmith" data/ch23_ontology.ttl
```

Wrote 48 category classes, 48 business classifications (0 unmatched) to `data/ch23_ontology.ttl`

```
:business_1 a :Category_seafood .
:business_2 a :Category_pub .
:business_3 a :Category_specialty_food .
```

Loaded into the same `pg-ripple` container Chapter 22 already has running:

```
SELECT pg_ripple.load_turtle('ontology_ttl', false);
-- 191
```

Sanity check — does the class hierarchy itself resolve correctly, with no reasoning involved yet, just a direct property-path query?

```
SELECT * FROM pg_ripple.sparql('
PREFIX : <http://portsmith.example.org/>
SELECT ?super WHERE { :Category_seafood <http://www.w3.org/2000/01/rdf-schema#subClassOf>* ?
super }
');
```

```
:Category_seafood, :Category_restaurant, :Category_all_categories
```

Correct, and correctly reflexive — * (zero-or-more) includes `Category_seafood` itself as well as both real ancestors. The class hierarchy, queried directly, works exactly as an ontology should.

Exercise 2 — `infer('rdfs')`: A Real, Serious Warning

The natural next question: if `Category_seafood rdfs:subClassOf Category_restaurant` is asserted, does the RDFS-standard entailment rule — “if X is a `Category_seafood` and `Category_seafood` is a subclass of `Category_restaurant`, then X is also a `Category_restaurant`” — actually fire? PostgreSQL 19’s beta gap (Chapter 21) and the custom-rule chaining bug (Chapter 22) both taught the same lesson: test it, don’t assume it. This time, testing it mattered more than either of those cases.

`pg-ripple` ships built-in rule sets by name — discovered, as usual, from a real error message rather than documentation:

```
SELECT pg_ripple.load_rules_builtin('bogus');
-- ERROR: unknown built-in rule set 'bogus'; valid values: rdfs, owl-r1,
-- owl-e1, owl-ql, skos, skos-transitive, skosxl, dcterms,
-- dcterms-integrity, schema, schema-integrity, foaf, foaf-integrity

SELECT pg_ripple.load_rules_builtin('rdfs'); -- 13 (real RDFS entailment rules, loaded)
```

A clean, isolated test before touching the real dataset — the same discipline Chapter 22’s Datalog investigation used:

```
INSERT DATA {
  :Seafood a rdfs:Class ; rdfs:subClassOf :Restaurant .
  :Restaurant a rdfs:Class ; rdfs:subClassOf :Business .
  :thing1 a :Seafood .
}

-- BEFORE infer():
SELECT ?t WHERE { :thing1 a ?t } → :Seafood

SELECT pg_ripple.infer('rdfs'); → 8
```

```
-- AFTER infer():  
SELECT ?t WHERE { :thing1 a ?t } → rdfs:Class, :Seafood
```

Already wrong in two ways: the expected new facts (:thing1 a :Restaurant, :thing1 a :Business — the actual entailment this rule set exists to compute) never appear, and a spurious, incorrect fact does — :thing1 a rdfs:Class, conflating the *instance* with the *metaclass* of the class it belongs to.

Running the identical `infer('rdfs')` against the real 48-business ontology made the real severity clear. `pg_ripple.infer()` does not appear to be scoped to a query or a subset of data — it operates across the whole default graph — and the result was not “no new facts added,” it was **active loss of the real, correctly-asserted data:**

```
SELECT ?t WHERE { :business_1 a ?t }
```

```
:business_1, rdfs:Class
```

Every one of the 48 real `:business_N a :Category_X` classification triples was replaced by a nonsensical self-reference (`:business_1 a :business_1`) plus the same spurious `rdfs:Class` typing seen in the isolated test — reproducible, not a fluke of the larger dataset. Restoring the correct data took two steps: reloading the real `:Business` and `:Category_X` triples (`load_turtle()` is additive, so this was safe), then explicitly `DELETE DATA`-ing the exact corrupted triples (`:business_N a :business_N` and `:business_N a rdfs:Class` for all 48) — `DELETE DATA` rather than `DELETE WHERE`, deliberately, after Chapter 22’s `DELETE WHERE`-collateral-deletion lesson.

The real, load-bearing conclusion: in this version, running `pg_ripple.infer()` with a built-in rule set is not a safe, side-effect-free way to ask “what does this entail” — it’s a real write against the whole graph, and this specific rule set actively corrupted correct data rather than merely failing to add new facts. Treat it exactly like any other destructive operation: never run it directly against data you haven’t backed up, on this build.

The workaround, and the thing actually worth doing when you need “is this instance a member of that class, including subclasses”: **query the class hierarchy with a property path and join it against plain `rdf:type` yourself, entirely without calling `infer()`** — which is exactly what Exercise 1’s direct `rdfs:subClassOf*` query already proved works correctly. Chapter 3’s next exercise builds on precisely that pattern.

Exercise 3 — Hybrid Retrieval: Semantic Recall, Structural Precision

The concrete version of this chapter’s “Background” claim: combine Chapter 6’s `pgvector` semantic search (still on the original PostgreSQL 16 database) with this chapter’s ontology (on the PostgreSQL 18 / `pg-ripple` container) to answer a question neither tool fully answers alone

— “which real businesses does this policy actually affect?” `data/ch23_hybrid_retrieval.py` does both queries, joined in Python (two independent PostgreSQL instances; no cross- database SQL, deliberately the same style as Chapter 6’s own RAG scripts):

```
python3 data/ch23_hybrid_retrieval.py "food truck vendor permits near restaurants"
```

Top semantic match (0.597): 'Council Minutes – Harbour District Food Truck Permits' [City Council]

Businesses actually affected (restaurant category, Harbour District or adjacent): 16

- Anchor & Oar Tavern
- Bella Napoli
- Dragon Palace
- ... (16 total)

The semantic half found the *relevant document* — a real, clearly dominant match (0.597, well ahead of the second-best result at 0.341). The structural half — deliberately built from the two primitives Exercise 1 and Chapter 22 Exercise 3 both already verified work (`rdfs:subclassOf*` and the undirected `adjacentTo` property path), **not** `infer()` — turns “this document is relevant” into a precise, checkable, named list: every real restaurant-category business in Harbour District or an adjacent neighborhood, exactly the population a food-truck-permit policy would actually affect. Neither half replaces the other: the embedding model has no idea what “Harbour District” or “restaurant category” formally mean; the graph has no idea which document, out of thousands, is topically relevant. This is what “grounding” concretely looks like — not a metaphor, a working pipeline that composes exactly two things this book has already independently verified are correct.

Exercise 4 — Entity Resolution: A Real, Direct Comparison

Chapter 5 seeded 12 genuine duplicate resident pairs with real ground truth

(`residents.true_duplicate_of` — e.g. Eleanor Whitmore / Elenor Whitmore, Priyanka Deshmukh / Priyanka Deshmuk). `pg-ripple`’s README advertises “neuro-symbolic record linkage” and “Privacy-Preserving Record Linkage via CLK Bloom-filter encoding” — real, callable functions (`bloom_encode()`, `dice_similarity()`), worth testing against the exact same ground truth Chapter 5 already used, rather than a fresh, possibly-flattering example.

`data/ch23_entity_resolution.py` runs both approaches, head to head, on all 12 real pairs:

```
python3 data/ch23_entity_resolution.py
```

name a	name b	pg_trgm	ripple dice
Eleanor Whitmore	Elenor Whitmore	0.737	0.000
Jonathan Castellano	Jonathon Castellano	0.739	0.035
Priyanka Deshmukh	Priyanka Deshmuk	0.842	0.000
Bartholomew Okonkwo	Bartholemew Okonkwo	0.739	0.033

Marguerite Delacroix	Marguerite Delacroix	0.792	0.068
Siobhan McAllister	Siobhan MacAllister	0.773	0.034
Theodore Vance	Theodor Vance	0.812	0.034
Anastasia Volkov	Anastassia Volkov	0.842	0.035
Desmond Okafor	Desmund Okafor	0.667	0.000
Genevieve Laurent	Genevieve Lorent	0.667	0.000
Mikhail Petrenko	Mikail Petrenko	0.737	0.034
Fitzgerald Osei	Fitzgerld Osei	0.722	0.102

A complete, one-sided result: Chapter 5’s `pg_trgm` correctly scores every real duplicate pair high (0.667–0.842, comfortably above any reasonable matching threshold); `pg-ripple`’s CLK Bloom-filter `dice_similarity`, called via its documented function signature with default parameters (`bloom_encode(value, key)`, `hash_count=30`, `length=1024`), scores every single one of the same 12 real pairs near-zero (0.000–0.102) — indistinguishable from genuinely unrelated names. `dice_similarity` isn’t broken as a function — identical strings correctly score a perfect 1.0 — but the *default* parameterization is far more brittle than `pg_trgm`’s trigram approach for exactly the class of error (single-character insertions/deletions, not just substitutions) real name-typo duplicates actually have. `pg-ripple.resolve_entities()`, a higher-level orchestration function over the same primitives, ran without error against this same data but returned an empty result (`{"canonicalized": 0}`) with a best-guess options payload — its expected configuration schema wasn’t discoverable without deeper source access than this chapter’s scope, so it’s reported as untested rather than assumed working or broken.

The honest conclusion: for this specific, realistic task — fuzzy name deduplication — Chapter 5’s mature, purpose-built `pg_trgm` approach is the one that actually works, verified against real ground truth. `pg-ripple`’s privacy-preserving linkage is a genuinely different, valuable technique (Bloom-filter encoding lets you compare records *without* ever decrypting the underlying names — a real capability `pg_trgm` doesn’t have), but “genuinely different capability” and “works out of the box for this data” are two separate claims, and only checking against real ground truth tells you which one you actually have.

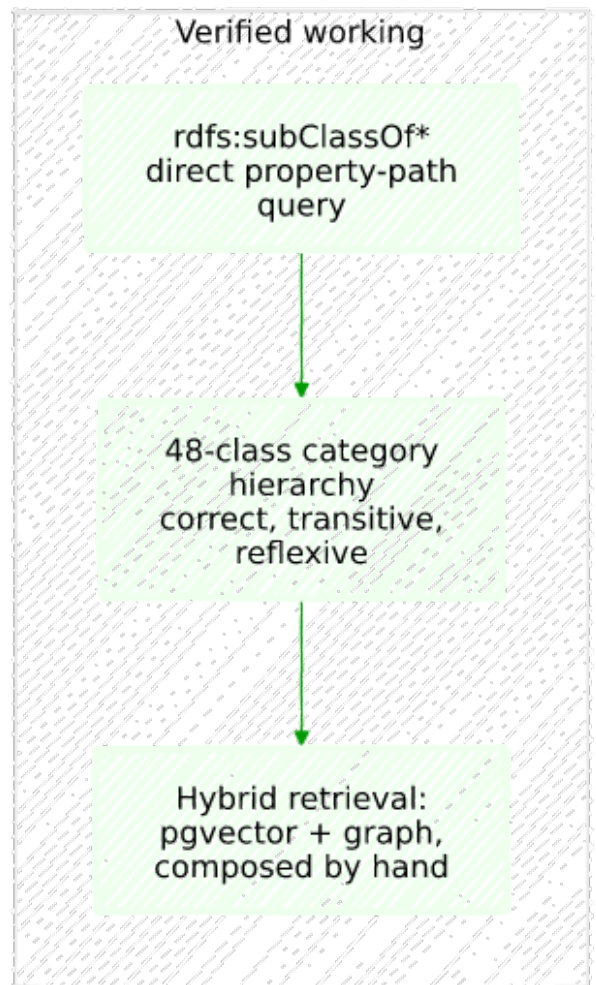
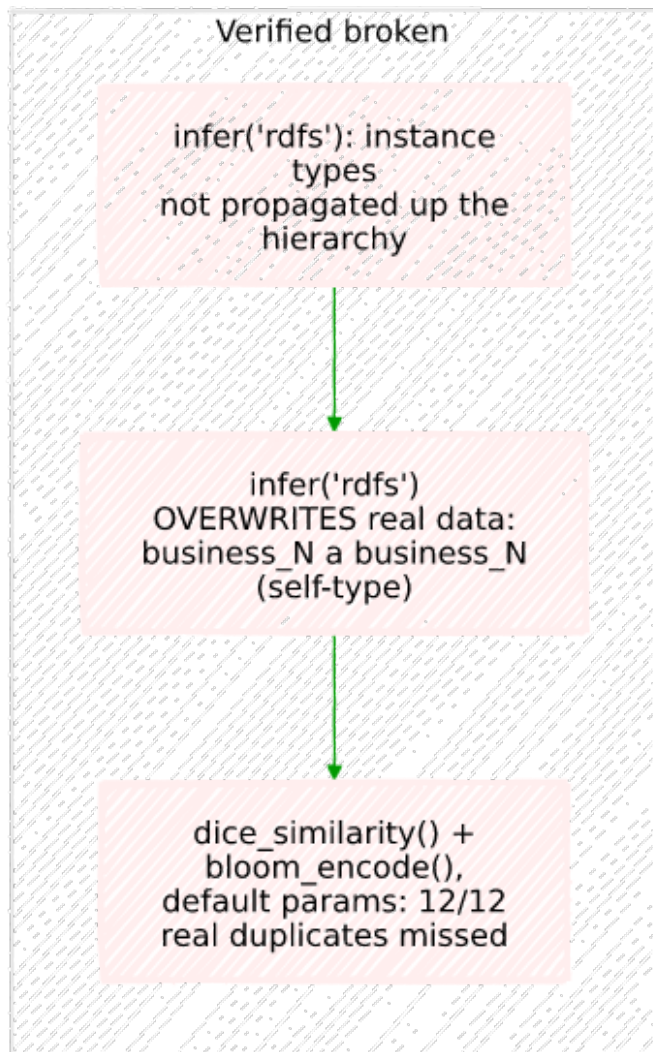
Decision Guide: Four Ways to Model “How Things Relate”

The book’s own data-modeling spectrum, in one table:

Approach	Chapter	Best for	Real limitation found in this book
JSONB	1	Irregular attributes on a single document	No relationships between documents at all
pgvector embeddings	6	Fuzzy semantic recall	

Approach	Chapter	Best for	Real limitation found in this book
Recursive CTEs	12	Unbounded-depth traversal over existing tables	No explicit facts — a similarity score, not a checkable claim Verbose for pattern-shaped (not depth-shaped) questions
SQL/PGQ property graphs	21	Fixed-depth pattern matching over existing tables	No variable-length paths yet (PostgreSQL 19 beta2)
RDF + SPARQL	22	Fact-shaped data, unbounded traversal, schema-free	Custom Datalog rule chaining is broken in this build
Ontologies (RDFS/OWL over RDF)	23	Checkable class semantics, machine-explainable structure	Built-in RDFS instance-type inference corrupts data in this build — but direct property-path queries over the same hierarchy work correctly

No single row is “the” answer — this book’s own running example needed all four models for genuinely different questions, sometimes on the very same underlying data (Chapter 1’s businesses are JSONB documents, Chapter 6’s embeddings, Chapter 21’s graph vertices, and Chapter 22’s triples, all at once, none of them wrong).



Summary — What You Should Now Know

Tool

`rdfs:subClassOf (asserted)`

`rdfs:subClassOf* / + property paths`

`pg_ripple.load_rules_builtin()`

`pg_ripple.infer('rdfs')`

What it's actually for

Real, correct, queryable class hierarchy — verified transitive and reflexive under `subClassOf*`

The **reliable** way to ask “is X a kind of Y” — direct query, no reasoning engine required

Loads real, named rule sets (`rdfs`, `owl-r1`, `skos`, ...) — loading succeeds and reports a plausible count

A real write against the whole default graph, verified to corrupt existing instance-classification data rather than just failing to

Tool	What it's actually for
	add new facts — treat as destructive, back up before running
Hybrid retrieval (embeddings + graph)	A real, working pattern: semantic search finds <i>what's relevant</i> ; a graph query, built from independently-verified primitives, turns that into a precise, checkable answer
<code>pg_ripple.dice_similarity() / bloom_encode()</code>	A real, different capability (privacy-preserving comparison) — verified, with default parameters, to miss all 12 of Chapter 5's real ground-truth duplicate pairs that <code>pg_trgm</code> catches correctly

The key design insight, closing out this book's run through PostgreSQL's extension surface: an ontology's value was never really about the storage format — RDF triples, a property graph, or plain rows all can hold "X is a kind of Y." Its value is in the *semantics* being explicit enough to check, and this chapter's own testing is the clearest demonstration of exactly why that checking matters: the class hierarchy itself, queried directly, was correct and useful throughout; it was specifically the automated reasoning layer built on top of it that silently produced wrong, damaging results until someone actually ran it against real data and looked at what came back. That is, in miniature, this entire book's argument — run the real thing, check the real output, and don't take a feature's existence as evidence of its correctness.

Going further: this closes the book's numbered chapters. Chapter 21 extended PostgreSQL's own core with a genuinely new query model still finding its footing; Chapters 22 and 23 extended it further still, into territory (RDF, SPARQL, formal ontologies) with an even younger implementation and correspondingly sharper edges. The throughline across all three, and really across this entire book: PostgreSQL's extension mechanism keeps making ambitious things installable in an afternoon — but installable was never the same claim as correct, and this book's own most useful moments came from checking the difference directly rather than assuming either one.

Chapter 24 — `pgColumnar`: Storing Data by Column Instead of by Row

Ask a heap table to add up one column, and it still has to carry every other column past you on the way to the answer. A table built column by column only ever picks up the column you asked for.

Background

Since Chapter 8, Portsmouth’s sensor network has been writing readings into `sensor_readings` — temperature, traffic counts, air quality — one row at a time, every few minutes, from well over a hundred sensors around the city. By now the table holds more than 9.6 million rows. That was always the right way to store it for *writing*: a new reading comes in, one row goes in, done.

But lately the questions being asked of that table have changed shape. Public Works doesn’t want one row anymore — they want the big picture. “How many readings do we actually have?” “What’s the average air quality reading been, city-wide?” “Show me a monthly rollup for every sensor type.” Every one of those questions only really cares about one or two of the table’s five columns, but answering them today means PostgreSQL has to read through nearly 700 megabytes of row-by-row storage — every column of every row — to get there. The queries still work. They’ve just gotten slow enough that people notice.

Here’s the underlying idea worth having in your head before anything else in this chapter:

think of `sensor_readings` as a filing cabinet of index cards, one card per reading, each card listing all five fields — sensor, type, value, time, id. Asking “what’s the average reading value?” means pulling out *every card* and reading past four fields you don’t care about just to get to the fifth. Now imagine instead you kept five separate folders — one holding nothing but every `reading_value` ever recorded, another holding nothing but every `sensor_type`, and so on. That same question now means opening exactly one folder. You never touch the other four at all. That second arrangement — organizing storage **by column instead of by row** — is what this chapter is about, and it’s a genuinely different tool from anything else in this book, not just a faster version of the same thing.

`pgColumnar` is a free, open-source extension that adds this second storage style directly inside PostgreSQL. Concretely, it adds a new **table access method** — PostgreSQL’s name for “the actual way a table’s rows are physically arranged on disk.” Every table you’ve built so far in this

book used the default access method, called `heap`, which is the filing-cabinet arrangement above: each row's values sit together, in insertion order. `pgColumnar` adds a second option, `USING pgcolumnar`, which stores the same data in the folder-by-column arrangement instead — still a real PostgreSQL table, still queried with ordinary `SELECT`, just organized differently underneath.

It also has a second, separate feature: reading and writing files in **Parquet** format, the same file format Chapter 17 exported `sensor_readings` into by hand. That looked, at first, like it might finally close a loose end Chapter 17 left open — Exercise 6 there stopped short of actually querying an exported Parquet file back *through* PostgreSQL, because the extension that would have done it was too hard to build from source. This chapter tries `pgColumnar`'s version of that instead. It's worth one section later on, with an honest result — but it isn't the main event here. The main event is the folder-by-column idea above, and whether it delivers for Portsmouth in practice. It does, clearly and measurably, and that's most of what follows.

One real setup story, worth knowing before you follow this chapter's own steps: the first attempt to install `pgColumnar` used the same `ALTER SYSTEM SET` approach Chapters 19–20 already used for `pg_cron` and `pg_stat_statements`, directly against the shared PostgreSQL 16 host every earlier chapter runs on. It went wrong in a way neither of us expected — `ALTER SYSTEM` silently wrote a corrupted value into PostgreSQL's config, and the *entire* cluster refused to start afterward, taking down every chapter's live data for about twenty minutes while it got diagnosed and fixed by hand. Nothing about `pgColumnar` itself caused that; it was a real, narrow PostgreSQL pitfall in how one specific kind of setting gets updated live. The practical upshot, and the reason this chapter runs in its own container instead: see this chapter's Environment Setup section for exactly what happened and the one-line fix that avoids it entirely.

The Scenario

Object	Lives in	Purpose
<code>sensor_readings_columnar</code>	portsmith24 (a new container)	A column-organized copy of Chapter 8/17's <code>sensor_readings</code> — same data, same 9.6M+ rows, stored the other way
<code>/tmp/sensor_readings.parquet</code>	Container filesystem	A Parquet export of that same data, produced by <code>pgColumnar</code> itself, for the “what about Chapter 17's files” section
<code>sensor_readings_pq</code>	portsmith24	A foreign table over that Parquet file — this chapter's attempt to finish

Object	Lives in	Purpose
		what Chapter 17 Exercise 6 left undone
<code>sensor_readings</code> itself never moves — it stays exactly where Chapter 8 built it, on the main PostgreSQL 16 cluster, still the table every sensor keeps writing into. Everything in this chapter is a live copy, piped across for analysis, the same “keep the operational table alone, build a separate copy for reporting” instinct Chapter 9’s materialized views already taught.		

Exercise Goals

By the end of this chapter you will be able to:

- Explain, in plain terms, the difference between row-organized and column-organized storage, and why one favors writing single records quickly while the other favors answering questions about millions of records at once.
 - Create a column-organized copy of a real table with `pgColumnar`, and measure exactly what changes — both storage size and query speed — against the same data stored the ordinary way.
 - Understand why a column-organized table isn’t automatically faster for every query, and what “sorting the data” actually buys you when it isn’t.
 - Know, from a real incident, why a new extension’s startup configuration belongs in a config file before the database ever starts, not in a live command against a running one.
 - Get an honest, tested answer to whether `pgColumnar` can also read Chapter 17’s exported Parquet files efficiently from inside PostgreSQL — and know exactly what it can and can’t do yet.
-

Environment Setup — A Fourth Container

`docker/ch24/` — the same three-file shape as Chapters 21 and 22’s containers (`Dockerfile`, `entrypoint.sh`, `docker-compose.yml`), built on PostgreSQL 18.

```
cd docker/ch24
docker compose up --build
```

Listens on host port **5435** (5432 = main cluster, 5433 = Chapter 21, 5434 = Chapters 22–23).

The real incident, in full, since it’s a useful lesson on its own. Loading a new extension into PostgreSQL requires listing it in a setting called `shared_preload_libraries` — a comma-separated

list of libraries PostgreSQL loads the moment it starts. Chapters 19–20 added to this same list for `pg_cron` and `pg_stat_statements` with `ALTER SYSTEM SET`, run against an already-running server, and it worked fine both times. Doing the same thing for `pgcolumnar` didn't:

```
shared_preload_libraries = '"pg_cron,pg_stat_statements,auto_explain,pgcolumnar"'
```

Look closely at that value: there's an extra pair of quotation marks wrapped around the *whole* list. PostgreSQL read that as one single library literally named

`pg_cron,pg_stat_statements,auto_explain,pgcolumnar` — commas and all — tried to load it, failed, and refused to start at all. The whole cluster was down until that stray pair of quotes was found and removed by hand. This is a real, narrow bug in how `ALTER SYSTEM SET` handles this particular kind of list-shaped setting — not something either of us mistyped.

The fix this chapter's container uses instead: write the setting straight into the config file *before* the database has ever started for the first time, so there's no live update to get wrong:

```
# entrypoint.sh, before the first pg_ctl start
echo "shared_preload_libraries = 'pgcolumnar'" >> "$PGDATA/postgresql.conf"

$ psql -h localhost -p 5435 -U chris -d portsmith24 -c "SELECT version();"
PostgreSQL 18.6 (Debian 18.6-1.pgdg12+2) on x86_64-pc-linux-gnu, compiled by gcc (Debian
12.2.0-14+deb12u1) 12.2.0, 64-bit

$ psql -h localhost -p 5435 -U chris -d portsmith24 -c "SELECT extname, extversion FROM
pg_extension WHERE extname='pgcolumnar';"
 extname  | extversion
-----+-----
pgcolumnar | 1.0-alpha3
```

One more small prerequisite, easy to miss: building `pgcolumnar` needs PostgreSQL's *development headers* (`postgresql-server-dev-18`), not just the plain server package — without them the build fails immediately looking for a missing file, `postgres.h`. Already handled in `docker/ch24/Dockerfile`.

Getting `sensor_readings` across. No PostGIS, no generated columns to worry about here:

```
psql portsmith -t -A -c "\copy (SELECT id, sensor_id, sensor_type, reading_value, recorded_at
FROM sensor_readings) TO STDOUT CSV" | \
psql -h localhost -p 5435 -U chris -d portsmith24 -c "\copy sensor_readings_columnar FROM
STDIN CSV"

COPY 9648001
```

A live pipe, host to container, no intermediate file — 9,648,001 rows in about 12 seconds.

Exercises

Exercise 1 — Building a Column-Organized Copy

```
CREATE TABLE sensor_readings_columnar (  
    id          bigint,  
    sensor_id   int,  
    sensor_type text,  
    reading_value double precision,  
    recorded_at timestampz  
) USING pgcolumnar;
```

That `USING pgcolumnar` is the entire difference from every `CREATE TABLE` earlier in this book — everything else about the table (its columns, its types, how you `INSERT` and `SELECT` from it) works exactly the same. No extra permission needed to create or use it, either; it's available the moment the extension exists.

Storage, measured directly:

```
SELECT pg_size_pretty(sum(pg_relation_size(inhrelid))) AS heap_table_only  
FROM pg_inherits WHERE inhparent = 'sensor_readings'::regclass;  
-- 708 MB  
  
SELECT pg_size_pretty(pg_total_relation_size('sensor_readings_columnar'));  
-- 18 MB
```

708 MB down to 18 MB — a real, measured 40x smaller, for exactly the same 9,648,001 rows. Two things make that possible: values from the same column tend to look alike (a `sensor_type` column only ever holds one of three words), which compresses well once they're grouped together — and, just as important, the folder-by-column layout means a query never has to *read* the columns it doesn't need in the first place, which is what the next exercise measures.

Exercise 2 — The Easy Win: Answering Without Reading

The simplest possible question — “how many readings do we have?” — turns out to be the clearest demonstration of what column storage buys you.

```
EXPLAIN (ANALYZE) SELECT count(*) FROM sensor_readings;           -- the original, row-organized  
table  
-- Execution Time: 562.7 ms  
  
EXPLAIN (ANALYZE) SELECT count(*) FROM sensor_readings_columnar; -- the column-organized copy  
  
Custom Scan (PgColumnarScan) (actual time=0.019..0.019 rows=1 loops=1)  
  Columnar Vectorized Aggregates: 1  
Execution Time: 0.065 ms
```

562.7 ms down to 0.065 ms — roughly 8,700 times faster. Nothing was decoded and nothing was scanned. `pgcolumnar` keeps a running row count for each stored group of data as a side note,

the way a librarian might keep a running tally of how many books are on a shelf instead of counting them fresh every time someone asks. `count(*)` with no filter is answered straight from that tally — the same reason Chapter 9’s materialized views win: the answer was already sitting there, computed once, not recomputed from scratch on every query.

Exercise 3 — A Realistic Query: Why “Faster” Isn’t Automatic

Public Works rarely wants a citywide total — they want it broken down. “What’s the average reading from our *temperature* sensors?” is a much more typical question, and this is where column storage stops being an automatic win.

```
EXPLAIN (ANALYZE, BUFFERS) SELECT count(*), avg(reading_value)
FROM sensor_readings WHERE sensor_type = 'temperature';           -- original table
-- Execution Time: 486.6 ms
```

```
EXPLAIN (ANALYZE, BUFFERS) SELECT count(*), avg(reading_value)
FROM sensor_readings_columnar WHERE sensor_type = 'temperature'; -- columnar copy
```

```
-> Custom Scan (PgColumnarScan) on sensor_readings_columnar
    Filter: (sensor_type = 'temperature'::text)
    Columnar Chunk Groups Total: 65
    Columnar Chunk Groups Removed by Filter: 0
Execution Time: 1459.5 ms
```

The columnar copy is slower here — 1459.5 ms against the original’s 486.6 ms. `pgColumnar` stores rows in large batches called **chunk groups** (think: which folder each index card’s fields ended up filed into, in the order they arrived), and it keeps a note of the lowest and highest value each chunk group holds for each column — a **zone map** — so it can sometimes skip a whole chunk group without reading it, if the value you’re filtering for can’t possibly be in that range. Here, “Columnar Chunk Groups Removed by Filter: 0” — none were skipped. Readings from every sensor type arrive continuously, all day, every day, so a chunk group built from “whatever came in this hour” ends up with temperature, traffic, *and* air-quality readings mixed together — every group’s range looks the same as every other group’s, so the zone map has nothing useful to rule out.

Exercise 4 — Fixing It, and What It Costs

`pgColumnar` can physically re-sort a table’s stored data around a chosen column, which rebuilds those chunk groups so each one holds a narrower range of values instead of an even mix:

```
SELECT pgcolumnar.vacuum_sorted('sensor_readings_columnar', 'sensor_type');
```

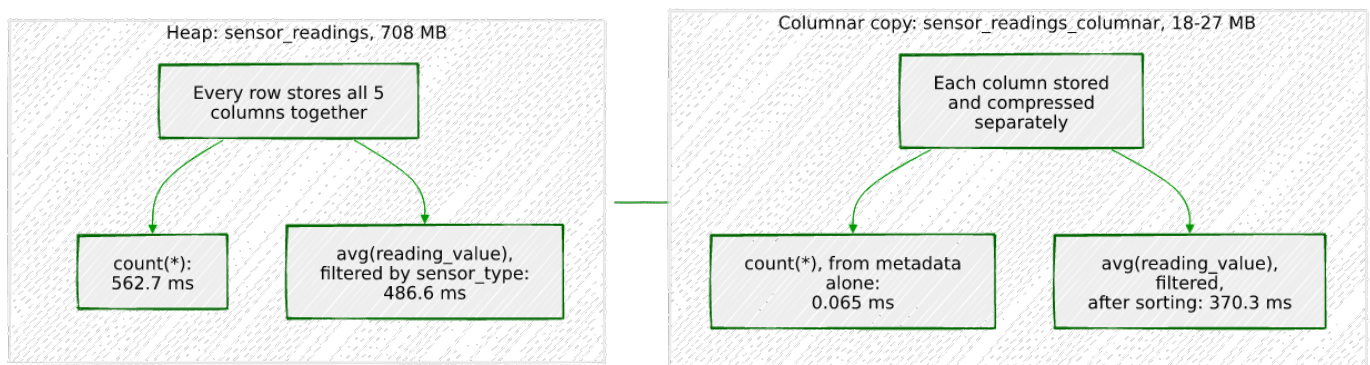
```
EXPLAIN (ANALYZE, BUFFERS) SELECT count(*), avg(reading_value)
FROM sensor_readings_columnar WHERE sensor_type = 'temperature';
```

```
-> Parallel Custom Scan (PgColumnarScan) on sensor_readings_columnar
    Columnar Chunk Groups Total: 43
    Columnar Chunk Groups Read: 11
    Columnar Chunk Groups Removed by Filter: 32
Execution Time: 370.3 ms
```

370.3 ms — now genuinely faster than the original table’s 486.6 ms, with 32 of 43 chunk groups skipped entirely. But storage moved too:

```
SELECT pg_size_pretty(pg_total_relation_size('sensor_readings_columnar'));
-- 27 MB (up from 18 MB)
```

Sorting by `sensor_type` shuffled the arrival-time order that made `recorded_at` compress so well in Exercise 1 — every column shares the same physical row order in a column-organized table, so improving one column’s ability to skip data can cost another column’s compression. **This is the same lesson Chapter 20 already taught with an ordinary index: the “obvious” fix isn’t free, and the only way to know whether it’s worth the tradeoff is to measure both sides — which is exactly what deciding whether, and how, to sort a real `pgColumnar` table for Public Works’ actual dashboard queries would require.**



Exercise 5 — What About the Parquet Files From Chapter 17?

`pgColumnar` can also export a table to Parquet, and read Parquet files back — the same file format Chapter 17 used, and worth checking honestly, since it’s the closest thing this book has found yet to finishing Chapter 17 Exercise 6’s unfinished last step.

```
SELECT pgcolumnar.export_parquet('sensor_readings_columnar', '/tmp/sensor_readings.parquet');

$ ls -la /tmp/sensor_readings.parquet
-rw----- 1 postgres postgres 405515796 sensor_readings.parquet -- 405 MB
```

That’s noticeably bigger than Chapter 17’s hand-built export of the same table (16.7 MB) — the export function here doesn’t compress the file at all, unlike the compression `pgColumnar`’s own

storage uses. Worth knowing if you’re choosing between the two: this export is fast and simple, but Chapter 17’s more deliberate, hand-tuned pyarrow script still makes the smaller file.

Reading a Parquet file back works two ways. `read_parquet()` reads the whole file every time, decoding only the columns you ask for — a real, useful savings, but it doesn’t skip rows based on a `WHERE` clause, so it isn’t a substitute for an index or a zone map. A second way, a proper foreign table (`CREATE SERVER ... FOREIGN DATA WRAPPER pgcolumnar_parquet`), is documented to skip whole chunks of the file when a filter rules them out — the exact capability that would finish Chapter 17’s story. Testing it directly, on data confirmed sorted the same way Exercise 4 sorted it, that skipping didn’t happen in the version tested here: `EXPLAIN` reported reading every chunk of the file regardless of the filter, on both PostgreSQL 16 and a fresh PostgreSQL 18 install.

The honest takeaway: `pgColumnar`’s own storage — everything Exercises 1 through 4 just measured — is the real, working half of this chapter. Its ability to efficiently query Chapter 17’s exported files is not there yet, at least not in the version tested. That’s a fine place to leave it — Chapter 17’s pattern (export to Parquet, read it with a tool built for that job, like DuckDB) is still the dependable path for that specific need.

Decision Guide: When to Reach for `pgColumnar`

Situation	What to use
A table gets written to constantly, one row at a time (sensors, permit applications, orders)	An ordinary heap table — this is what it’s built for
Big-picture questions over millions of existing rows — totals, averages, monthly rollups	A <code>pgColumnar</code> copy: real, measured wins here (40x smaller, up to ~8,700x faster on the right query)
You know which column you’ll usually filter by	Sort the columnar copy on that column (<code>vacuum_sorted</code>) — but expect a real compression tradeoff, and measure before assuming it’s worth it
Reading a Parquet file exported elsewhere, efficiently, from inside PostgreSQL	Not yet dependable here — use Chapter 17’s export-and-read-with-DuckDB pattern instead

Summary — What You Should Now Know

Concept	What it does
Row storage vs. column storage	

Concept	What it does
	Row storage (the default) keeps a record's fields together — good for writing one row at a time. Column storage keeps each field together across every record — good for questions that only touch a few columns out of many
<code>CREATE TABLE ... USING pgcolumnar</code>	Adds a column-organized copy of a table, right inside PostgreSQL — no separate warehouse system
Real, measured wins	708 MB → 18 MB (40x smaller); <code>count(*)</code> 562.7 ms → 0.065 ms (~8,700x)
Chunk groups and zone maps	Data is stored in batches, each with a note of its min/max values per column — lets a query skip a whole batch it can't match, but only if the filtered column is actually grouped that way
<code>vacuum_sorted</code>	Physically re-sorts a table around one column, so its zone maps become useful — at a real cost to how well <i>other</i> columns compress
<code>shared_preload_libraries</code> before first start	The safe way to load a new extension — a live <code>ALTER SYSTEM SET</code> on this particular kind of setting can silently corrupt the whole list
Reading Chapter 17's Parquet files back	Not yet a strength of this extension — its own native storage is the part worth using today

The key design insight from this chapter is the one Public Works actually needed answered: keep writing sensor readings the ordinary way, one row at a time, and keep a second, column-organized copy around specifically for the big questions. Neither storage style is better in general — a heap table is still the right choice for `sensor_readings` itself, exactly as Chapter 8 built it, and a `pgColumnar` copy is the right choice for the reporting queries layered on top of it. That's the same instinct behind Chapter 9's materialized views, applied one level deeper: sometimes the fastest way to answer a different *kind* of question isn't a smarter query — it's storing the same data a second way.

Going further: the Parquet-reading gap noted in Exercise 5 is worth revisiting later, since `pgColumnar` is young, actively developed software — what didn't work during this chapter's testing may well work in a newer release. If you pick this chapter back up, it's worth checking whether a newer `pgColumnar` version closes that gap before assuming Chapter 17's DuckDB-based pattern is still the only option.

Appendix A — Environment Setup

This book does not run on one PostgreSQL installation. It runs on **four**, and that split is itself a real finding worth understanding before you set anything up, not an accident of how the book was written: Chapters 1–20 share one long-lived cluster; Chapters 21–24 each needed a separate, disposable one, for reasons specific to what each chapter was testing.

Environment	Chapters	Version	Where
Main cluster	1–20	PostgreSQL 16	Installed directly on the host (apt)
SQL/PGQ container	21	PostgreSQL 19 beta2	docker/ch21/
pg-ripple container	22–23	PostgreSQL 18	docker/ch22/
pgColumnar container	24	PostgreSQL 18	docker/ch24/

Why four, not one

The main cluster accumulates real, cumulative state across twenty chapters — roles, grants, two databases, rows mutated by earlier exercises that later chapters depend on. Chapter 21 needed PostgreSQL 19, which was still in beta at the time of writing; Chapters 22–23 needed a Rust extension (`pg-ripple`) built from source against PostgreSQL 18. Running either of those against the main cluster would have meant either upgrading a host carrying real state onto beta software, or bolting a from-source Rust build onto an installation everything else depends on staying stable. Both risks were judged not worth it for two chapters’ worth of exercises — isolate instead, and throw the container away if something goes wrong. This turned out to matter in practice: Chapter 21’s PostgreSQL 19 needed rebuilding from scratch after a packaging misconfiguration, and Chapter 22’s container was restarted mid-chapter to fix a `shared_preload_libraries` setting. Neither touched the main cluster at all.

Chapter 24 is the sharpest version of this lesson, because it isn’t hypothetical: `pgColumnar` was tried against the main cluster *first* — reasonable, since it needs nothing beta or built from source — and a genuine `ALTER SYSTEM SET` bug (a `shared_preload_libraries` update silently mis-quoted into one broken value) took the entire cluster down for real, mid-book, before the fourth container replaced that attempt. Isolation here isn’t precautionary; it’s a direct response to an outage that already happened once on this exact extension.

The main cluster (Chapters 1–20)

Installed directly, not in a container — every extension in the table below is a plain `apt install` against a single PostgreSQL 16 server.

```
sudo apt install -y postgresql-16 postgresql-client-16
```

Two databases exist on it by the end of Chapter 20:

- `portsmith` — the primary database essentially every chapter uses.
- `portsmith_legacy` — introduced in Chapter 17 as a genuinely separate `postgres_fdw` target, reused in Chapter 18 as the logical-replication subscriber.

The full extension list, required `postgresql.conf` settings, and the role/grant history this cluster accumulated chapter by chapter are in **Appendix C**, not repeated here — that appendix is the canonical reference for “what needs to be installed,” this one is about the overall shape of the setup.

The PostgreSQL 19 beta2 container (Chapter 21)

`docker/ch21/` — `Dockerfile`, `entrypoint.sh`, `docker-compose.yml`. Bring it up:

```
cd docker/ch21
docker compose up --build
```

Listens on host port **5433** (5432 is the main cluster). Two real build problems, both explained in full in Chapter 21’s own Environment Setup section, worth knowing before you rebuild this yourself:

1. PGDG publishes pre-release major versions as an additional `main <version>` **component** inside the normal `-pgdg` suite, not a separate suite — and that component alone isn’t sufficient; the plain `main` component is also needed, for `postgresql-common/libpq5`.
2. Debian’s `postgresql-19` package auto-creates a cluster on install via `pg_createcluster`, splitting `config (/etc/postgresql/)` from `data (/var/lib/postgresql/)` — incompatible with this image’s hand-rolled `initdb`, fixed with `pg_dropcluster --stop 19 main` before `initdb` runs.

The PostgreSQL 18 / `pg-ripple` container (Chapters 22–23)

`docker/ch22/` — same three-file shape as Chapter 21’s container, built on PostgreSQL 18 (GA, not 19 — `pg-ripple`’s own documentation targets 18, and there was no reason to stack PostgreSQL 19’s own beta uncertainty on top of an already-real-risk Rust build).

```
cd docker/ch22
docker compose up --build
```

Listens on host port **5434**. `pg-ripple` is compiled from source via Rust/`pgrx`, not installed as a package — the real, non-obvious trap here: `cargo-pgrx`'s own version must **exactly** match the `pgrx` library version an extension's `Cargo.toml` pins (0.18.0 for `pg-ripple`), not just satisfy a semver range. `cargo install cargo-pgrx --version "^0.18"` resolves to whatever the newest 0.18.x happens to be, and `cargo-pgrx` refuses to build against a mismatched version with a clear, specific error — the fix is pinning the exact version:

```
RUN cargo install --locked cargo-pgrx --version "0.18.0"
```

Chapter 22's environment setup covers the rest, including the `shared_preload_libraries = 'pg_ripple'` setting this container needs from its very first start to get `pg-ripple`'s background workers running (the same class of gotcha Chapters 19–20's `pg_cron`/`pg_stat_statements` needed on the main cluster).

The second PostgreSQL 18 container (Chapter 24)

`docker/ch24/` — same three-file shape again, a second, separate PostgreSQL 18 container rather than reusing Chapter 22–23's (kept apart so a `pgColumnar` crash or a bad `docker compose down -v` can't take `pg-ripple`'s state with it, and vice versa).

```
cd docker/ch24
docker compose up --build
```

Listens on host port **5435**. No beta packaging dance and no Rust build here — `pgColumnar` is a plain C extension built via PGXS against `postgresql-server-dev-18` — but two things still matter, both folded into this image directly: the dev headers are easy to forget (a bare `postgresql-18` package doesn't include them), and `shared_preload_libraries` is written into `postgresql.conf` *before* the first `pg_ctl` start, never as a live `ALTER SYSTEM SET` — see this appendix's own "Why four, not one" section for exactly what that setting broke when it was tried the other way.

Connecting to all four

```
psql portsmith
psql -h localhost -p 5433 -U chris -d portsmith19
psql -h localhost -p 5434 -U chris -d portsmith22
psql -h localhost -p 5435 -U chris -d portsmith24
```

```
# main cluster, PG16
# Chapter 21, PG19 beta2
# Chapters 22-23, PG18
# Chapter 24, PG18
```

All container passwords are set directly in their respective `docker-compose.yml` files (`ch21-scratch`, `ch22-scratch`, `ch24-scratch`) — these are throwaway scratch instances, not meant to hold anything you'd mind losing to a `docker compose down -v`.

A note for anyone continuing this book

If you're picking Chapters 22–23's container back up, one thing is worth knowing before you run anything against it: Chapter 23 found that `pg_ripple.infer()`, run with a built-in rule set, does not behave as a safe read-only reasoning query — it was verified, twice, to overwrite real classification data with incorrect facts. Don't run it against data in that container you haven't already exported, and see Chapter 23 Exercise 2 for the full, reproduced finding before relying on it for anything.

Chapter 24's container is worth a quick recheck if you pick it back up. Its central exercises (the native columnar storage — real, measured compression and speed wins) don't depend on anything likely to change. Its shorter closing section, on reading Chapter 17's exported Parquet files back through `pgColumnar` itself, tested against a specific, young alpha release — worth confirming against whatever version you're actually running before assuming that part's conclusion still holds, the same “verify again” habit Chapter 21's own findings needed for its beta.

Appendix B — Synthetic Data Generation Scripts

Every chapter in this book follows the same rule: real, synthetic-but- realistic data first, exercises against it second — nothing in the exercises is invented on the spot or asserted without a live query behind it. This appendix indexes every script that builds that data, in `data/`, run against `dbname=portsmith` unless noted otherwise.

Seed scripts (build a chapter’s tables from scratch)

Script	Chapter	Builds
<code>ch01_seed.py</code>	1 — JSONB	<code>businesses</code> , with a heterogeneous details JSONB column varying by category
<code>ch02_seed.py</code>	2 — PostGIS	<code>neighborhoods</code> , <code>city_infrastructure</code> , <code>parks</code> , and <code>business_point_geometry</code>
<code>ch03_seed.py</code>	3 — Job Queues	<code>jobs</code> , synthetic permit-application work items
<code>ch04_seed.py</code>	4 — Full-Text Search	<code>city_documents</code> (council minutes, zoning ordinances, public notices)
<code>ch05_seed.py</code>	5 — Fuzzy Matching	<code>residents</code> (with 12 real seeded typo/ duplicate pairs — see below) and <code>business_names</code>
<code>ch06_seed.py</code>	6 — pgvector	<code>city_photos</code> and synthetic embedding scaffolding
<code>ch07_seed.py</code>	7 — IP/Network	<code>network_events</code> , <code>blocklists</code>
<code>ch08_seed.py</code>	8 — Partitioning & BRIN	<code>sensor_readings</code> , partitioned IoT data, ~9.6M rows
<code>ch11_seed.py</code>	11 — Window Functions	<code>business_revenue</code> (48 businesses × 4 quarters)
<code>ch12_seed.py</code>	12 — Recursive CTEs	<code>city_org</code> (invented org chart), <code>intersections/road_segments</code> (derived from Chapter 2’s real geometry via

Script	Chapter	Builds
		ST_Intersects), categories (derived from Chapter 1's real category values)

Chapters 9, 10, 13–20 deliberately seed nothing new — each reuses tables earlier chapters already built, per the book's running principle of building on real prior state rather than starting fresh every chapter.

Data-processing and demo scripts (not seeding — used within exercises)

Script	Chapter	Purpose
ch03_worker.py	3	Simulates a concurrent job-queue worker claiming rows with <code>FOR UPDATE SKIP LOCKED</code>
ch03_reclaim.py	3	Reclaims stalled jobs past a timeout — later ported to SQL as Chapter 19's <code>sweep_stalled_jobs()</code>
ch06_embed_documents.py	6	Computes real sentence-transformers embeddings for <code>city_documents</code>
ch06_semantic_search.py	6	Semantic and hybrid (semantic + keyword) search over embedded documents
ch06_rag_ingest.py / ch06_rag_chat.py	6 (bonus)	A small local RAG pipeline — chunk/embed/retrieve, then generate via Ollama
ch13_listen.py	13	A psycopg LISTEN client used across the LISTEN/NOTIFY exercises
ch14_leader_election.py	14	Multiprocessing leader-election race over <code>pg_try_advisory_lock</code>
ch17_export_to_parquet.py	17	Exports <code>sensor_readings</code> to Parquet in MinIO, partitioned by month
ch17_query_parquet.py	17	

Script	Chapter	Purpose
		Verifies the Parquet export independently via DuckDB, no PostgreSQL involved
ch17_census.csv	17	Real CSV used for the <code>file_fdw</code> exercise
ch18_replication_stream.py	18	Consumes the logical replication stream directly via <code>psycopg</code> 's low-level <code>pgconn</code> API
ch22_export_turtle.py	22	Exports businesses/neighborhoods (plus real <code>ST_Touches</code> -derived adjacency) as Turtle triples
ch23_export_ontology.py	23	Exports Chapter 12's categories tree as an <code>rdfs:subClassOf</code> class hierarchy
ch23_hybrid_retrieval.py	23	Combines Chapter 6's <code>pgvector</code> search with the Chapter 22/23 graph — two databases, joined in Python
ch23_entity_resolution.py	23	Head-to-head: <code>pg_trgm</code> VS. <code>pg-ripple</code> 's <code>dice_similarity()</code> against real ground-truth duplicates

Worth knowing before you run any of these

- **Run seed scripts in chapter order.** Several later chapters' scripts assume earlier chapters' tables already exist (Chapter 12's road graph reads Chapter 2's real geometry live; Chapter 22's export reads Chapter 1/2's live, possibly-mutated data).
- **The live database reflects cumulative mutation, not just seed output.** Chapter 1 Exercise 5 bumps a rating via `jsonb_set`; Chapter 15 converts `jobs.status` from TEXT to an enum in place. Querying the seed script alone won't show you the database's actual current state — query the database.
- **Always set `SET timezone = 'UTC'` ;** before any verification query touching a timestamp column. This environment's default session timezone is not UTC, and several chapters (8, 9, 11) bucket rows by day on `TIMESTAMPTZ` columns — a non-UTC session silently groups a different set of rows per bucket.

- **Chapter 22/23 scripts connect to two different PostgreSQL instances** (the main cluster and the Chapter 22 container) — check the DSN each script uses before assuming “the database” means the same thing every time in this book.

Appendix C — Extension Installation Reference

Every `CREATE EXTENSION` this book actually ran, in one place. Package names follow the PGDG `postgresql-<version>-<name>` convention; adjust the version number for whichever cluster you’re installing into (see Appendix A for which chapters use which of the three).

Main cluster (PostgreSQL 16, Chapters 1–20)

Chapter	Extension	apt package	CREATE EXTENSION	Notes
2	PostGIS	postgresql-16-postgis-3	postgis	Needs superuser
5	pg_trgm	bundled (core/contrib)	pg_trgm	No separate package
6	pgvector	postgresql-16-pgvector	vector	Needs superuser
7	ip4r	postgresql-16-ip4r	ip4r	Needs superuser
17	postgres_fdw	bundled (core/contrib)	postgres_fdw	Needs superuser; <code>USAGE</code> grant needed for non-roles
17	file_fdw	bundled (core/contrib)	file_fdw	Needs superuser; reading needs <code>pg_read_seq_membership</code> too
17 (Ex6, sketch only)	parquet_s3_fdw	build from source	parquet_s3_fdw	Built against Apache Arrow C++; never actually in the book — see Chapter 17’s own honest account
19	pg_cron	postgresql-16-cron	pg_cron	Needs <code>shared_preload_libraries</code> ; schema created in <code>postgres</code>
19/20	pg_stat_statements	bundled (core/contrib)	pg_stat_statements	Needs <code>shared_preload_libraries</code> ; <code>pg_stat_statements_reset()</code> revoked from public default
19/20	auto_explain	bundled (core/contrib)	(none — loaded via config, not <code>CREATE EXTENSION</code>)	<code>shared_preload_libraries</code> only; GUCs are superuser-only (<code>PGC_SUSET</code>)

Required postgresql.conf settings, all needing a full restart, not just a reload — set once, in Chapter 19, and left in place for the rest of the book:

```
shared_preload_libraries = 'pg_cron,pg_stat_statements,auto_explain'
wal_level = logical
cron.database_name = 'portsmith'
```

PostgreSQL 19 beta2 container (Chapter 21)

No extensions — CREATE PROPERTY GRAPH/GRAPH_TABLE (SQL/PGQ) are core PostgreSQL 19 features, not an installable extension. The only package needed beyond the server itself is PostGIS, and even that turned out unnecessary — Chapter 21 sidesteps it by exporting intersections.geom as plain lon/lat columns rather than pulling PostGIS into a still-beta major version.

PostgreSQL 18 / pg-ripple container (Chapters 22–23)

Extension	Install	CREATE EXTENSION	Notes
pg_ripple	build from source (Rust + pgrx 0.18.0	pg_ripple	See Appendix A for the exact version-pinning trap; needs
	exactly, against PostgreSQL 18 server headers)		shared_preload_libraries = 'pg_ripple' from first start for its background merge worker

A note on GRANTS, not just CREATE EXTENSION

Installing an extension is frequently the *easy* privilege gate in this book, not the only one. Real walls hit and documented, chapter by chapter, on the main cluster:

- CREATE EXTENSION itself always needs superuser.
- postgres_fdw/file_fdw need GRANT USAGE ON FOREIGN DATA WRAPPER to a non-superuser role separately from installing the extension.
- file_fdw additionally needs pg_read_server_files role membership — and even then, PostgreSQL reads server-side as the postgres OS user, so file permissions on disk matter independently of the SQL grant (Chapter 17’s sharpest gotcha: a file under a 700 home directory was unreadable regardless of the file’s own mode).
- pg_cron’s schema (cron) and pg_stat_statements’ reset/reload functions are all revoked from PUBLIC by default — GRANT EXECUTE/GRANT USAGE explicitly, per database (both are per-database extensions, easy to grant into the wrong one by accident).

- `pg_cron` jobs run as the job-owning role, authenticated over a **fresh libpq connection** opened by the `postgres` OS user — a real, non-obvious consequence: that OS user needs its own `~postgres/.pgpass` entry for the job-owning role's password, or every scheduled job fails with `connection failed` and no corresponding entry in the server log.
- Chapter 24's `pgColumnar` needs its own `USAGE` grants beyond installing the extension, twice: `GRANT USAGE ON SCHEMA pgcolumnar + GRANT EXECUTE ON ALL FUNCTIONS IN SCHEMA pgcolumnar` before a non-superuser role can call any `pgcolumnar.*` maintenance function (creating and querying a table `USING pgcolumnar` needs neither), and a second, separate `GRANT USAGE ON FOREIGN DATA WRAPPER pgcolumnar_parquet` before that role can `CREATE SERVER` against it — the same shape of gate Chapter 17 already documented for `postgres_fdw/file_fdw`.

None of this is unique to this book's exact setup — it's the general shape of PostgreSQL's privilege model, and the book's own experience is that each of these gates was found by testing, one at a time, not by reading the whole list in advance. Expect the same if you're installing on a fresh cluster of your own.

Appendix D — Index Decision Guide

A decision tree for which index type to reach for, built from the real index choices this book actually made — and, in one important case (Chapter 20), a real index choice that turned out to be wrong until measured.

The decision tree

1. Is the column being matched with plain equality or a range (`=`, `<`, `BETWEEN`, `ORDER BY`)? → B-tree — PostgreSQL’s default, and correct by default for most columns. Every primary key and foreign key in this book uses one without a second thought. The one real trap: an **implicit type cast** silently defeats it. Chapter 20 Exercise 3 found `businesses.id = 5` (integer column, integer literal) using `Index Cond` correctly, while `businesses.id = 5::numeric` used `Filter` instead — a full scan comparing every row, invisible unless you actually read the plan.

2. Is the column a `JSONB` document, and are you querying with containment (`@>`) or existence (`?`)? → GIN, directly on the `jsonb` column. Chapter 1’s `businesses.details` uses this for exactly this reason — Chapter 1 Exercise 3 confirms via `EXPLAIN ANALYZE` that the GIN index is actually used, not just present. `jsonb_path_ops` is worth a second GIN index alongside the default operator class if your queries are containment-only — Chapter 1 builds both, on the same column, and compares.

3. Is the column full-text search (`tsvector`) or a trigram-matched string (`pg_trgm`)? → GIN in both cases, for the same underlying reason: both are matching against a large, variable-length set of tokens (lexemes for text search, trigrams for fuzzy matching) per row, which is what GIN is actually built for. Chapter 4’s `city_documents.search_vector` and Chapter 5’s trigram indexes on `residents/business_names` both use GIN. `pg_trgm` also supports a **GiST** trigram index (`gist_trgm_ops`) — Chapter 5 builds one specifically to compare: GiST trigram indexes are typically smaller and faster to build, GIN indexes are typically faster to query, and the right choice depends on your write-vs-read ratio more than any fixed rule.

4. Is the column geometry, a network range (`ip4r`), or otherwise a “does this overlap/contain that” question rather than plain equality? → GiST. Chapter 2’s `neighborhoods.geom`/
`businesses.geom` and Chapter 7’s `blocklists` CIDR ranges both use GiST for the same underlying reason: both are answering containment/overlap questions over 2D or range-shaped data, which B-tree’s linear ordering can’t represent and GIN’s token-set model doesn’t fit either.

5. Is the table huge, append-mostly, and naturally correlated with insertion order (a timestamp, a sequential ID)? → BRIN, and *only* if that correlation is real. Chapter 8's `sensor_readings` (9.6M rows, inserted roughly in `recorded_at` order) is the textbook case — a BRIN index stores block-range summaries, not per-row entries, so it's tiny compared to a B-tree on the same column, at the cost of only being useful when physical row order genuinely tracks the indexed value. Combined with partitioning (also Chapter 8), most queries never need the index at all — partition pruning already eliminates the irrelevant months before any index gets consulted.

6. Is the column a vector embedding, queried by approximate nearest neighbor? → HNSW or IVFFlat, both `pgvector`-specific, both genuinely different trade-offs rather than one being strictly better: - **IVFFlat**: faster to build, needs `lists/probes` tuning, recall degrades further from exact as the dataset grows unless retuned. - **HNSW**: slower to build, no equivalent tuning parameter needed, generally better recall at query time.

Chapter 6 builds both against the same data specifically to compare build time and recall directly rather than taking either trade-off on faith.

7. Does the query only ever touch a small, predictable subset of rows — most rows never relevant to any query that matters? → A partial index, on top of whichever type above fits the column. Chapter 3's `idx_jobs_claim_order` only indexes queued jobs — a completed job is never going to be claimed again, so indexing it is pure waste. This isn't a separate index *type*, it's a `WHERE` clause on any of the types above, and it's worth considering by default for any status-flag-shaped column, not just this one.

8. Do queries always filter on a column combination together, not independently? → A compound index, ordered with the most selective/most-commonly- filtered column first (or the one narrowing the range in a partitioned table, so the query planner can combine it with partition pruning). Chapter 20's most important, counter-intuitive real finding belongs here: a naive single-column index on `sensor_readings.sensor_id` (a low-cardinality, evenly-scattered column, roughly 1-in-120 rows matching) made a real query **slower** — 338ms → 377ms — than the original parallel sequential scan, because a Bitmap Heap Scan at that scatter still has to visit nearly every heap page, while giving up the seq scan's free parallelism entirely. The actual fix needed both a realistic, time-bounded query (letting Chapter 8's partition pruning narrow to one month first) *and* a compound index, (`sensor_id`, `recorded_at`), matching that query's actual shape — a real, measured ~4.5× win over pruning alone, ~43× over the original query. **Row count and column cardinality alone don't tell you whether an index will help — the physical scatter of matching rows across pages does, and the only way to know is EXPLAIN (ANALYZE, BUFFERS) against the real query shape, not an assumption about "add an index" being automatically correct.**

Quick-reference table

Index type	Best for	Chapter(s)	Real caveat found in this book
B-tree	Equality, range, ORDER BY	1, 3, 15, 20	Implicit casts silently defeat it — check Index Cond VS. Filter Two operator classes worth having side by side (default + jsonb_path_ops)
GIN (jsonb)	JSONB containment/existence	1	Pair with a <i>generated</i> tsvector column, not a trigger, where possible (Chapter 16)
GIN (tsvector)	Full-text search	4, 16	Faster queries, larger index/slower writes than the GiST alternative
GIN (pg_trgm)	Fuzzy/substring matching	5	Smaller/faster to build; slower queries than GIN
GiST (pg_trgm)	Fuzzy/substring matching	5	The only correct choice for st_* operators — B-tree/GIN don't apply
GiST (geometry)	Spatial containment/overlap	2	Same shape of problem as spatial GiST, different domain
GiST (ip4r)	CIDR/range containment	7	Only useful if physical order genuinely correlates — verify, don't assume
BRIN	Huge, insertion-order-correlated columns	8	Slower to build than IVFFlat
HNSW	Vector ANN, best recall	6	Needs lists/probes tuning; recall degrades more as data grows
IVFFlat	Vector ANN, faster build	6	

Index type	Best for	Chapter(s)	Real caveat found in this book
Partial (any type)	Narrow, predictable subset of rows	3, 18	Free win wherever a status flag makes most rows permanently irrelevant
Compound (any type)	Multi-column filter shapes	20	Match the index to the <i>actual query shape</i> , not just “the columns in the WHERE clause” — verified via EXPLAIN, not assumed

Appendix E — Further Reading

Official documentation and project sources for every chapter — not blog posts or opinion, deliberately. Each chapter’s own exercises are the place for “how it actually behaves”; these are the primary sources for the parts a single lab exercise can’t cover.

Chapter	Topic	Source
1	JSONB	postgresql.org/docs/current/datatype-json.html
2	PostGIS	postgis.net/documentation
3	FOR UPDATE/SKIP LOCKED	postgresql.org/docs/current/sql-select.html — locking clause section
4	Full-text search	postgresql.org/docs/current/textsearch.html
5	pg_trgm	postgresql.org/docs/current/pgtrgm.html
6	pgvector	github.com/pgvector/pgvector
6 (bonus)	Sentence embeddings / Ollama	sbert.net , ollama.com
7	ip4r	github.com/RhodiumToad/ip4r
8	Partitioning	postgresql.org/docs/current/ddl-partitioning.html
8	BRIN indexes	postgresql.org/docs/current/brin-intro.html
9	Materialized views	postgresql.org/docs/current/rules-materializedviews.html
10	PostgREST	postgrest.org
11	Window functions	postgresql.org/docs/current/tutorial-window.html
12	Recursive CTEs	postgresql.org/docs/current/queries-with.html
13	LISTEN/NOTIFY	postgresql.org/docs/current/sql-notify.html
14	Advisory locks	

Chapter	Topic	Source
		postgresql.org/docs/current/explicit-locking.html — advisory locks section
15	Types, domains, enums	postgresql.org/docs/current/sql-createtype.html , sql-createdomain.html
16	Generated columns	postgresql.org/docs/current/ddl-generated-columns.html
17	postgres_fdw / file_fdw	postgresql.org/docs/current/postgres-fdw.html , file_fdw.html
17	DuckDB (independent Parquet verification)	duckdb.org
17	MinIO	min.io
18	Logical replication	postgresql.org/docs/current/logical-replication.html
19	pg_cron	github.com/citusdata/pg_cron
20	pg_stat_statements	postgresql.org/docs/current/pgstatstatements.html
20	auto_explain	postgresql.org/docs/current/auto-explain.html
21	CREATE PROPERTY GRAPH / SQL/PGQ	postgresql.org/docs/current/sql-create-property-graph.html — check the version-specific docs, not just “current,” since this feature is new as of PostgreSQL 19
22	pg-ripple	github.com/trickle-labs/pg-ripple
22	SPARQL 1.1	w3.org/TR/sparql11-query
22	Turtle (RDF syntax)	w3.org/TR/turtle
22	SHACL	w3.org/TR/shacl
23	RDF Schema (RDFS)	w3.org/TR/rdf-schema
23	OWL 2 Web Ontology Language	w3.org/TR/owl2-overview
24	pgColumnar	

Chapter	Topic	Source
		github.com/commandprompt/pgcolumnar
24	Apache Parquet format	parquet.apache.org/docs

A note on reading these versus running the exercises

This book's own recurring finding, most sharply in Chapters 21–23: official documentation and a specification describe what a feature is *meant* to do, not necessarily what a specific version actually does today. PostgreSQL 19's own SQL/PGQ documentation describes quantified path patterns as part of the feature; PostgreSQL 19 beta2 itself rejects them. `pg-ripple`'s README describes working RDFS reasoning and CLK Bloom-filter record linkage; this book's own testing found the former corrupts data and the latter misses real duplicates at default settings. None of that is a reason to skip the documentation — it's the reason every exercise in this book is written to run something real and check the actual output against it, rather than trust either source alone.

Appendix F — Syntax Quick Reference

One page per chapter. Meant to be flipped to, not read start to end — the syntax this book actually used, nothing more. Where this book found something doesn't work as documented, it's flagged [△ verified broken/unsupported](#) right here, not just buried in that chapter's prose — don't copy those forms expecting them to work.

Chapter 1 — JSONB

```
details -> 'key'           -- get JSON value (as jsonb)
details ->> 'key'          -- get JSON value (as text)
details #>> '{a,b}'        -- get nested value by path (as text)
details @> '{"k":"v"}'     -- containment
details ? 'key'           -- key exists
jsonb_set(details, '{k}', '"v"')
jsonb_insert(details, '{arr,0}', '"v"')
jsonb_array_elements(details -> 'tags')
jsonb_path_query(details, '$.hours[*] ? (@.day == "sun"')
CREATE INDEX ... USING GIN (details);
CREATE INDEX ... USING GIN (details jsonb_path_ops);
```

Chapter 2 — PostGIS

```
ST_GeomFromText('POINT(-1.1 50.8)', 4326)
ST_DWithin(a.geom, b.geom, 500)           -- meters, if geography-cast
ST_Within(point_geom, polygon_geom)
ST_Contains(polygon_geom, point_geom)
ST_Area(geom::geography)
ST_Distance(a.geom, b.geom)
CREATE INDEX ... USING GIST (geom);
```

Chapter 3 — Job Queues

```
SELECT * FROM jobs
WHERE status = 'queued'
ORDER BY priority DESC, created_at
FOR UPDATE SKIP LOCKED
LIMIT 1;
```

Chapter 4 — Full-Text Search

```
to_tsvector('english', body)
to_tsquery('english', 'budget & housing')
```

```
plainto_tsquery('english', 'budget housing')
ts_rank(search_vector, query)
ts_rank_cd(search_vector, query)
ts_headline('english', body, query)
CREATE TEXT SEARCH CONFIGURATION portsmith (COPY = english);
CREATE INDEX ... USING GIN (search_vector);
```

Chapter 5 — Fuzzy Matching

```
CREATE EXTENSION pg_trgm;
similarity('a', 'b')
word_similarity('needle', 'haystack containing needle')
'query' % column -- similarity above pg_trgm.similarity_threshold
CREATE INDEX ... USING GIN (name gin_trgm_ops);
CREATE INDEX ... USING GIST (name gist_trgm_ops);
```

Chapter 6 — pgvector

```
CREATE EXTENSION vector;
embedding vector(384)
embedding <-> query_vec -- L2 distance
embedding <#> query_vec -- negative inner product
embedding <=> query_vec -- cosine distance
CREATE INDEX ... USING ivfflat (embedding vector_cosine_ops) WITH (lists = 100);
CREATE INDEX ... USING hnsw (embedding vector_cosine_ops);
```

Chapter 7 — IP/Network (ip4r)

```
CREATE EXTENSION ip4r;
ip '10.0.0.5' <=> cidr '10.0.0.0/24' -- contained by
cidr >> ip -- contains
network(addr), masklen(addr)
CREATE INDEX ... USING GIST (cidr_range);
```

Chapter 8 — Partitioning & BRIN

```
CREATE TABLE sensor_readings (...) PARTITION BY RANGE (recorded_at);
CREATE TABLE sensor_readings_2024_01 PARTITION OF sensor_readings
FOR VALUES FROM ('2024-01-01') TO ('2024-02-01');
ALTER TABLE sensor_readings DETACH PARTITION sensor_readings_2024_01;
CREATE INDEX ... USING BRIN (recorded_at);
```

Chapter 9 — Materialized Views

```
CREATE MATERIALIZED VIEW mv_sensor_daily AS SELECT ...;
REFRESH MATERIALIZED VIEW mv_sensor_daily;
REFRESH MATERIALIZED VIEW CONCURRENTLY mv_sensor_daily; -- needs a unique index first
```


Chapter 10 — PostgREST

```
CREATE ROLE web_anon NOLOGIN;
CREATE ROLE authenticator NOINHERIT LOGIN PASSWORD '...';
GRANT web_anon TO authenticator;
GRANT SELECT ON api.businesses TO web_anon;
CREATE POLICY resident_self_only ON residents FOR SELECT
    USING (id = current_setting('request.jwt.claims', true)::json ->> 'resident_id');
```

postgrest postgrest.conf

Chapter 11 — Window Functions

```
SELECT sensor_id, recorded_at,
    AVG(value) OVER (PARTITION BY sensor_id ORDER BY recorded_at
        ROWS BETWEEN 4 PRECEDING AND CURRENT ROW) AS rolling_avg,
    RANK() OVER (PARTITION BY sensor_id ORDER BY value DESC) AS rnk
FROM sensor_readings;
```

Chapter 12 — Recursive CTEs

```
WITH RECURSIVE reports AS (
    SELECT id, manager_id, 0 AS depth FROM city_org WHERE id = 1
    UNION ALL
    SELECT c.id, c.manager_id, r.depth + 1
    FROM city_org c JOIN reports r ON c.manager_id = r.id
)
SELECT * FROM reports;

WITH RECURSIVE walk AS (
    ...
    CYCLE id SET is_cycle USING path
)
SELECT * FROM walk;
```

Chapter 13 — LISTEN/NOTIFY

```
LISTEN permit_updates;
NOTIFY permit_updates, '{"job_id": 42}';
pg_notify('permit_updates', payload);
UNLISTEN permit_updates;
```

Chapter 14 — Advisory Locks

```
pg_try_advisory_lock(hashtext('demolition_permit'))
pg_advisory_lock(key)           -- session-level, must be explicitly unlocked
pg_advisory_unlock(key)
```

```
pg_advisory_xact_lock(key)      -- transaction-level, auto-released on commit/rollback
SELECT * FROM pg_locks WHERE locktype = 'advisory';
```

Chapter 15 — Custom Types, Domains, Enums

```
CREATE TYPE job_status AS ENUM
('queued', 'on_hold', 'in_progress', 'completed', 'failed', 'cancelled');
ALTER TYPE job_status ADD VALUE 'on_hold' AFTER 'queued';
CREATE DOMAIN positive_integer AS INTEGER CHECK (VALUE > 0);
CREATE TYPE contact_info AS (phone TEXT, postcode uk_postcode);
```

Chapter 16 — Generated Columns

```
ALTER TABLE sensor_readings
  ADD COLUMN reading_date DATE
  GENERATED ALWAYS AS ((recorded_at AT TIME ZONE 'UTC')::date) STORED;
-- bare ::date on a timestamptz fails: "generation expression is not immutable"
```

Chapter 17 — Foreign Data Wrappers

```
CREATE EXTENSION postgres_fdw;
CREATE SERVER legacy_srv FOREIGN DATA WRAPPER postgres_fdw
  OPTIONS (host 'localhost', dbname 'portsmith_legacy');
CREATE USER MAPPING FOR chris SERVER legacy_srv
  OPTIONS (user 'chris', password 'fdw-demo-password');
IMPORT FOREIGN SCHEMA public FROM SERVER legacy_srv INTO public;

CREATE EXTENSION file_fdw;
CREATE FOREIGN TABLE census_raw (...) SERVER file_srv
  OPTIONS (filename '/tmp/census.csv', format 'csv', header 'true');
```

Chapter 18 — Logical Replication

```
ALTER SYSTEM SET wal_level = 'logical'; -- own -c call, needs restart
CREATE PUBLICATION portsmith_pub FOR TABLE businesses (id, name, ...) WHERE (active = true);
SELECT pg_create_logical_replication_slot('portsmith_slot', 'pgoutput');
CREATE SUBSCRIPTION portsmith_sub CONNECTION '...' PUBLICATION portsmith_pub
  WITH (create_slot = false, slot_name = 'portsmith_slot');
ALTER TABLE businesses REPLICA IDENTITY USING INDEX idx_businesses_replident;
```

Chapter 19 — pg_cron

```
CREATE EXTENSION pg_cron;
SELECT cron.schedule('refresh-mv-sensor-daily', '0 * * * *', $$CALL
refresh_and_log('mv_sensor_daily')$$);
SELECT cron.schedule_in_database('legacy-analyze', '0 4 * * *', 'ANALYZE businesses_archive',
```

```
'portsmith_legacy');
SELECT * FROM cron.job_run_details ORDER BY start_time DESC NULLS LAST;
```

Chapter 20 — pg_stat_statements / EXPLAIN

```
EXPLAIN (ANALYZE, BUFFERS) SELECT ...;
SELECT query, calls, total_exec_time, mean_exec_time
FROM pg_stat_statements ORDER BY total_exec_time DESC LIMIT 10;
SELECT pg_stat_statements_reset();
ALTER SYSTEM SET auto_explain.log_min_duration = 200;
SELECT pg_reload_conf();
```

Chapter 21 — SQL/PGQ (PostgreSQL 19 beta2)

```
CREATE PROPERTY GRAPH city_org_graph
  VERTEX TABLES ( city_org KEY (id) LABEL employee PROPERTIES ALL COLUMNS )
  EDGE TABLES (
    city_org AS reports_to KEY (id)
      SOURCE KEY (id) REFERENCES city_org (id)
      DESTINATION KEY (manager_id) REFERENCES city_org (id)
      LABEL reports_to NO PROPERTIES
  );

SELECT * FROM GRAPH_TABLE (city_org_graph
  MATCH (a IS employee) -[IS reports_to]-> (b IS employee)
  COLUMNS (a.name, b.name)
);
```

⚠ verified broken/unsupported (beta2):

```
-- quantified path patterns -- both forms rejected:
MATCH (a) (-[IS reports_to]->(b)){1,10} (root)      -- "unsupported element pattern kind"
MATCH (a) -[IS reports_to]->{1,10} (root)           -- "element pattern quantifier is not supported"
```

No working substitute exists in this release — use Chapter 12's WITH RECURSIVE for anything of unbounded/unknown depth.

Chapter 22 — RDF / pg-ripple

```
SELECT pg_ripple.load_turtle(turtle_text, false);
SELECT * FROM pg_ripple.sparql('SELECT ?s ?o WHERE { ?s :locatedIn ?o }');
SELECT pg_ripple.sparql_update('INSERT DATA { ... }');
SELECT pg_ripple.load_shacl(shacl_text);
SELECT pg_ripple.shacl_score('default');
SELECT pg_ripple.shacl_report_scored('default');
```

```
-- property paths (these work correctly):
:a :adjacentTo+ ?n          -- one-or-more, directed
:a (:adjacentTo|^:adjacentTo)+ ?n  -- one-or-more, either direction
```

⚠ **verified broken:** custom Datalog rules (`pg_ripple.load_rules()/infer()`) — multi-atom rule bodies don't chain correctly; a rule head's second variable binds to the wrong value regardless of body atom order. Reproduced on an isolated 3-triple example, not just the chapter's larger dataset.

⚠ **gotcha:** `DELETE WHERE { pat1 . pat2 }` deletes *every* pattern per matched solution, not just the one you meant — use `DELETE DATA` with exact triples for anything you can't afford to lose collaterally.

Chapter 23 — Ontologies

```
-- correct, reliable way to query a class hierarchy (no infer() needed):
SELECT * FROM pg_ripple.sparql(
  'SELECT ?super WHERE { :Category-seafood <...#subClassOf>* ?super }'
);
SELECT pg_ripple.load_rules_builtin('rdfs');  -- valid names discovered via a bogus-name error:
                                              -- rdfs, owl-rl, owl-el, owl-ql, skos, skos-
transitive,
                                              -- skosxl, dcterms, dcterms-integrity, schema,
                                              -- schema-integrity, foaf, foaf-integrity
```

⚠ **verified broken, and dangerous:** `pg_ripple.infer('rdfs')` does not correctly propagate `rdfs:type` up a class hierarchy, and was verified — twice, isolated test and full dataset — to **overwrite real classification triples** with an incorrect self-type (`:business_N a :business_N`) and a spurious `a rdfs:Class`. Treat any `infer()` call with a built-in rule set as a real write against the whole default graph, not a safe query. Back up before running it.

Chapter 24 — pgColumnar

```
CREATE TABLE t (...) USING pgcolumnar;
-- columnar storage, real measured wins
SELECT pgcolumnar.vacuum_sorted('t', 'col' [, 'col2', ...]);  -- physically sort, improves
chunk-group skip
SELECT * FROM pgcolumnar.sort_status('t');                    -- verify a table is actually
sorted
SELECT pgcolumnar.export_parquet('t', '/path/file.parquet');  -- no compression option, see
note below
SELECT * FROM pgcolumnar.read_parquet('/path/file.parquet')
  AS t(col1 type1, col2 type2, ...);                            -- decodes requested columns
only, see note below
```

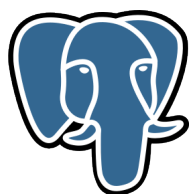
Note: the storage engine above (`USING pgcolumnar` and `vacuum_sorted`) is this extension's strength — real, measured compression and speed wins in Chapter 24. Its Parquet-reading side is weaker

as tested: `export_parquet()` takes no compression argument (larger files than a deliberately compressed export), `read_parquet()` only saves decoding unwanted columns, not reading unwanted rows, and a `pgcolumnar_parquet` foreign table didn't skip file chunks based on a `WHERE` filter in the version tested here. Worth rechecking against a newer release before leaning on that part.

Appendix G — Acknowledgments

Twenty-four chapters of a fictional city rest on a very real stack of software this book didn't write a line of. PostgreSQL itself, every extension bolted onto it, the languages and libraries that generated Portsmouth's synthetic data, and the tools that turned two dozen Markdown files into the HTML, PDF, and EPUB you're reading — none of it was built for this book, and all of it made this book possible. This page is a plain, direct thank-you to the people and projects behind that stack.

PostgreSQL



Every chapter in this book is, at bottom, a chapter about **PostgreSQL** — free, open-source, and developed by a global group of volunteers and companies coordinated through the PostgreSQL Global Development Group. Thirty years of that work is the reason a single database can hold JSON documents, geospatial polygons, vector embeddings, RDF triples, and a columnar table side by side, and still feel like one coherent system. [postgresql.org](https://www.postgresql.org)

Extensions and Companion Databases

Nearly every chapter after the first turned on one extension or another. In the order this book met them:

- **PostGIS**  — geospatial types and operators (Chapter 2), maintained by the PostGIS Project Steering Committee. postgis.net
- **pg_trgm** — trigram fuzzy matching (Chapter 5), bundled with PostgreSQL core.
- **pgvector** — vector similarity search (Chapter 6), created by Andrew Kane and maintained with the broader pgvector community. github.com/pgvector/pgvector
- **ip4r** — IPv4/IPv6 range and CIDR types (Chapter 7), maintained by Andrew Gierth (RhodiumToad). github.com/RhodiumToad/ip4r
- **postgres_fdw** and **file_fdw** — foreign data wrappers (Chapter 17), bundled with PostgreSQL core.

- **parquet_s3_fdw** — the Parquet-over-S3 foreign data wrapper Chapter 17 discussed honestly rather than pretending to have finished building.
- **pg_cron** — in-database job scheduling (Chapter 19), originally created at Citus Data, now maintained under `citusdata/pg_cron`. github.com/citusdata/pg_cron
- **pg_stat_statements** and **auto_explain** — query statistics and automatic plan logging (Chapter 20), bundled with PostgreSQL core.
- **pg-ripple** — RDF triples, SPARQL, and SHACL inside PostgreSQL (Chapters 22–23), from Trickle Labs. github.com/trickle-labs/pg-ripple
- **pgColumnar** — columnar table storage (Chapter 24), from CommandPrompt, Inc. github.com/commandprompt/pgcolumnar. Particular thanks to **OffgridwithJD**, one of the project’s developers, for real, direct help while this book was working through Chapter 24.

Chapter 21’s `CREATE PROPERTY GRAPH` and `GRAPH_TABLE` needed no extension at all — thanks there go to the PostgreSQL contributors who landed SQL/PGQ in core.

Tools Built Around PostgreSQL

- **PostgREST** — turns a PostgreSQL schema into a REST API (Chapter 10), created by Joe Nelson and maintained by the PostgREST community. postgrest.org
 - **DuckDB**  **DuckDB** — the independent, in-process analytical engine used to verify Chapter 17’s Parquet export actually worked. duckdb.org
 - **MinIO** **MINIO** — the self-hosted, S3-compatible object storage Chapter 17’s Parquet files lived in. min.io
-

Python and Its Libraries



Every synthetic data generator, migration script, and RAG demo in this book is **Python** — created by Guido van Rossum and maintained by the Python Software Foundation. python.org

The libraries doing the real work underneath those scripts: **psycopg** (the PostgreSQL driver every script in this book connects through), **pyarrow** and **boto3** (Chapter 17’s Parquet export and S3 upload), **pgvector**’s Python client and **sentence-transformers** (Chapter 6), **duckdb**’s Python bindings (Chapter 17’s independent verification), **PyJWT** (Chapter 10’s token minting),

NumPy, **Pillow**, and **requests**. Thank you to every maintainer of every one of them — this book leaned on all of it without a second thought, which is exactly what good infrastructure earns.

Ollama, used in Chapter 6’s bonus local RAG section to run an open-weight model entirely on-device, deserves its own mention — ollama.com.

Building This Book Itself

None of the above gets read without a second stack, the one that turns Markdown source into the book in your hands:

- **Markdown**, the plain-text format every chapter is written in, designed by John Gruber with Aaron Swartz.
 - **Pandoc**, the “universal document converter” that turns this book’s Markdown into HTML, PDF, and EPUB in one pass — created by John MacFarlane. pandoc.org
 - **Mermaid.js**, which renders every diagram in this book — flowcharts, sequence diagrams, the graph contrasts in Chapters 17 through 24 — from plain text descriptions, created by Knut Sveidqvist and maintained by the Mermaid team. Reached in this book’s build through **mermaidx**, a CLI wrapper around it. mermaid.js.org
 -  **GNU Make** — the build tool coordinating every step above, part of the GNU Project. gnu.org/software/make
 - **WeasyPrint**, which lays out this book’s PDF from styled HTML. weasyprint.org
 - **Inkscape**, converting hand-made diagrams to the PNGs the PDF build needs.
 -  **Debian** and **Docker**  **docker** — the operating system and containerization underneath every isolated environment in Chapters 21, 22, and 24, letting three different PostgreSQL majors coexist on one machine without touching each other. debian.org, docker.com
-

Specifications and Open Standards



Apache Arrow and **Apache Parquet**, the in-memory and on-disk columnar formats behind Chapters 17 and 24, both projects of the Apache Software Foundation. arrow.apache.org, parquet.apache.org

Chapters 22 and 23 rest on a stack of World Wide Web Consortium standards — **SPARQL**, **Turtle**, **SHACL**, **RDF Schema**, and **OWL 2** — the product of years of working-group effort to make data meaning machine-checkable, long before “AI-ready data” was a phrase anyone used.

AI Assistance

ANTHROPIC

This book was written with **Claude Code**, from **Anthropic**, as a genuine collaborator throughout — drafting exercises, standing up and tearing down disposable PostgreSQL environments, running the real queries whose output fills these pages, and, more than once, finding and diagnosing the book’s own mistakes (a broken build script, a mis-set config value that took down the working cluster mid-chapter) before they became someone else’s problem. anthropic.com

If you maintain something on this page and think it deserves more than a line here, that’s fair — open an issue or a pull request. This list exists because none of Portsmouth’s twenty-four chapters would have been possible without every project named on it. I am deeply grateful for the years of dedicated effort into all these foundational tools.